TS1000
TS2000
TS2100
Robot Controller

INSTRUCTION MANUAL

TS1000/TS2000 ROBOT CONTROLLER

OPERATOR'S MANUAL

Notice

- Make sure that this instruction manual is delivered to the final user of Toshiba Machine's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

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TOSHIBA MACHINE CO., LTD.

TOKYO, JAPAN

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Preface

Structure of TS series robot controller instruction manuals:
These instruction manuals were published in parts according to the application and purpose, and the name and outline of each manual are as follows:

[Safety Manual]

This manual contains the important information to use the robot safety and correctly. Be sure to read through and understand this manual before operating the robot. Also, strictly observe the descriptions made there.

[Startup Manual]

This manual describes the procedures covering from power ON of the robot after installation to robot operation by manual guide. When you use the TS series robot controller for the first time, be sure to read through this manual.

[Operator's Manual]

This manual deals with the TS series robot controller operating procedures. Read through this manual before operating the robot, and refer to it as necessary.

[Robot Language Manual]

This manual refers to the robot language called "SCOL". When you have to create a program based on this language, read through the manual.

[Interface Manual]

This manual describes the external signals for the robot. Concerning the interface conditions between the robot and peripheral equipment, specifications, timing, etc., refer to the manual when necessary.

[Installation & Transport Manual]

This manual describes the transport, unpacking and installation of the robot and controller. Be sure to read through this manual before unpacking the shipment containing the robot.

[Maintenance Manual]

This manual deals with the daily and regular inspections to be made on the robot and controller. Read through this manual to use the robot safely over long years to come.

[Communication Manual]

This manual describes the serial communication between the robot controller and other equipment. Refer to this manual when connecting the robot controller with a host computer, optical sensor, etc., via a serial cable.

[User Parameter Manual]

This manual describes the setting procedures of the robot controller. Refer to this manual when you perform the setting of communication, I/O and motion conditions.

This manual describes how to use various functions of the robot system to manually guide the robot through the teach pendant, edit programs, compile data and perform automatic operation.

This manual is intended for the robot operator. Before reading this manual, he or she should first read the Safety Manual and Startup Manual.

Cautions on Safety

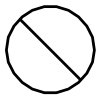
This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2).

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.
*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. Details of the actual caution are indicated with pictures or words in or near the symbol.

[Operation]

⚠ DANGER	
 Prohibited	• During operation, NEVER enter the dangerous area of the robot. Otherwise, you will be injured seriously. • DO NOT leave in the working range any machinery or materials which will hinder the operation. If the equipment went wrong, a person nearby will be injured or involved in an accident. • Anyone other than the operator MUST NOT approach the equipment. Should he negligently touch the dangerous part of the equipment, he will get injured or involved in a serious accident. • NEVER perform an inappropriate operation which is not described in the instruction manual. Otherwise, the equipment will start by mistake, resulting in personal injury or serious accident.
 Danger	• If you feel even a little that you are exposed to danger or that the equipment works abnormally, press the EMERGENCY stop pushbutton switch to stop the equipment. If the equipment is used as it is, you will be injured or involved in a serious accident. When this happens, ask our after-sale service agent for repair. • During operation, be sure to close the equipment cover. Should the cover be opened during operation, you will be struck by an electric shock or get injured. • Only a well-trained and qualified person is allowed to perform the operation. Should the equipment be operated improperly, it will start by mistake, causing a personal injury or serious accident. • If the equipment has malfunctioned, turn the power off, identify and remove the cause of the abnormality, maintain the peripheral equipment and completely restore the malfunctioned equipment. Then start the equipment at a slow speed. If the equipment starts, leaving the abnormality, you will be involved in a serious accident.

CAUTION	
 Prohibited	• DO NOT change the data of the system structure file. Otherwise, the robot will move abnormally, resulting in damage or an accident.
 Mandatory	• In principle, teaching operation should be performed outside the dangerous area of the robot. If it should be performed inevitably within the dangerous area, strictly observe the following matters. (1) The teaching operation should always be performed by two (2) persons. One person performs the job and the other person watches outside the dangerous area. Also, both persons should try to prevent mis-operation with each other. (2) The operator should do the job in an attitude ready to press the EMERGENCY stop pushbutton switch at any time. Also, he should perform the job at a position from which he can evacuate immediately at the time of an emergency after confirming the robot working range and shields nearby. (3) The supervisor should keep watch on the job at a position where he can see the entire robot system and operate the EMERGENCY stop pushbutton switch at the time of an emergency. Also, he should keep anyone from entering the dangerous area. Unless the operator or other person follows the instructions of the supervisor, an accident will be caused. • If an abnormality has generated or the POWER LED lamp on the control panel remains off after the main power switch of the equipment was turned on, turn off the main power immediately and confirm the wiring. Otherwise, you will be struck by an electric shock or a fire will break out. • Unless the robot operates toward a designated direction at manual guide, turn off the servo power. Otherwise, the robot will be damaged or you will be involved in an accident. When this happens, call us at the after-sale service agent. • Pushbutton operations of the control panel and teach pendant should be confirmed visually. Otherwise, you will be involved in an accident due to mis-operation. • After the power is turned on, be sure to reset a program to start an automatic operation. If the program is executed continuously, the robot will interfere with the peripheral equipment, resulting in damages or accidents.

CAUTION	
 Mandatory	• Before operating the equipment, perform the following inspection. (1) Make sure that visual appearance of the robot, controller, peripheral equipment and cables is in the good condition. (2) Make sure that no obstacle stands in or near the working range of the robot and peripheral equipment. (3) Make sure that the emergency stop and other safety devices operate properly. (4) Make sure that no abnormal noise or vibration is involved in the robot operation. If the above prior inspection is skipped, the equipment will be damaged or you will be involved in an accident.
 Caution	• The speed of test operations is initially set at 25 % of the maximum robot speed. • The speed of automatic operation is initially set at 100 % of the maximum robot speed.

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Section 1 Before Operating Your Robot

1.1 Overview of Operation Functions

Function	Descriptions	Control panel	Teach pendant	External control signal	HOST commu-nication
Main power ON/OFF	Used to turn on and off the main power.	❍			
Servo power ON/OFF	Used to turn on and off the servo driver main circuit power.	❍	Δ (ON only)	❍	❍
Master mode selection	Used to select either the external automatic mode, internal automatic mode or teaching mode.	❍			
Start and stop	Used to start and stop automatic operation.	❍	❍	❍	❍
Cycle stop	Used to stop automatic operation. (When 1-cycle operation has finished)	❍	❍	❍	
Feed hold	Used to temporarily stop automatic operation.		❍		❍
Emergency stop	Used to immediately stop automatic operation.	❍	❍	❍	❍
Teaching operation	Used to manually guide the robot by jogging or inching.		❍		❍
Auxiliary signal operation	Used to operate the pneumatic drive shaft for grip, etc.)		❍		Δ
Program editing	Used to create and edit programs.		❍		❍
Data editing	Used to create and edit positional data, coordinate data and load data.		❍		❍
File operation	Used to manage files by copying, renaming, deleting, etc.		❍		❍
Reset	Used to reset output signals and execution environment.		❍	❍	❍
Date setting	Used to set the date used by the system.		❍		
Time setting	Used to set the time used by the system.		❍		
Error display	Used to display errors which are yet to be reset.	❍	❍		❍
Error reset	Used to reset errors that have been corrected.	❍	❍	❍	
Error history display	Used to display a historical list of errors which have been corrected.		❍		❍

Function	Descriptions	Control panel	Teach pendant	External control signal	HOST commu-nication
Work coordinate system selection	Used to select the work coordinate system.		❍		
Tool coordinate system selection	Used to select the tool coordinate system.		❍		
Present position monitor	Used to display the present position of the robot in each coordinate system.		❍		❍
External I/O signal monitor	Used to display the status of digital I/O signals.		❍		❍
Working time monitor	Used to display the power ON time and run hours for each operation.		❍		❍
File selection	Used to select a file to be executed.		❍	❍	❍
Move	Used to move the robot to a point which has been taught.		❍		❍

❍: Operation is possible.

∆: Partial operation is possible.

1.2 Mode Structure

Master mode	External automatic mode (EXT.HOST)	Used to perform automatic programmed operation, using the host computer.
	External automatic mode (EX. SIGNAL)	Used to perform automatic programmed operation by means of external control signals.
	Internal automatic mode (INTERNAL)	Used to perform automatic programmed operation through the control panel or teach pendant.
	Teaching mode (TEACHING)	Used to perform such operations as teaching, guiding, file management and program test run, using the teach pendant.
	The master mode should be selected by means of the MODE select key equipped on the control panel.	
Utility		Used to perform such operations as coordinate selection, position monitor and signal monitor.
Error display		Used to display and reset an error through the teach pendant, irrespective of the master mode currently selected.

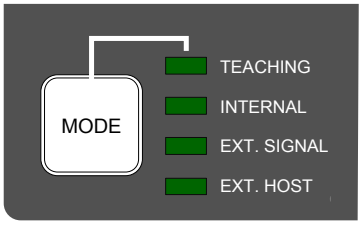
The emergency stop and servo OFF operations are possible, irrespective of the mode currently selected.

1.2.1 External Automatic Mode

External Automatic Mode (EXT)	External control signal mode (SIGNAL)	Used to perform automatic programmed operation by means of external control signals. For details, see the Interface Manual.
	Host mode (HOST)	Used to perform automatic programmed operation, file uploading and downloading. All operations are carried out by the host computer through serial communication channel(s). For details, see the Communication Manual.

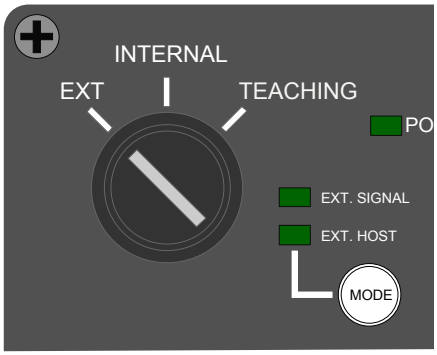
The external automatic mode comes in the two (2) sub modes as shown above. Mode selection is performed by means of the MODE select key on the control panel.

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Select "EXT. SIGNAL" or "EXT. HOST" by means of the MODE select key.

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After selecting "EXT", select "EXT. SIGNAL" or "EXT. HOST", using the MODE key.

File selection, run mode selection and speed override setting in the internal automatic operation mode or test operation mode are also valid in the external automatic mode.

1.2.2 Internal Automatic Mode

Internal Automatic Mode (INTERNAL)	File selection (SEL)	Used to select a file to be executed.
	Run mode selection (MODE)	Used to select a run mode.
	Speed override (OVRD)	Used to set the speed override.
	Start	Used to start program execution from the control panel.
	Stop (STOP, BREAK)	Used to stop or interrupt program execution.
	Direct run (DO)	Used to directly execute the SCOL command.
	RESET (RESET)	Used to reset the program execution step, variables and digital output signals.

1.2.3 Teaching Mode (Including Test Operation)

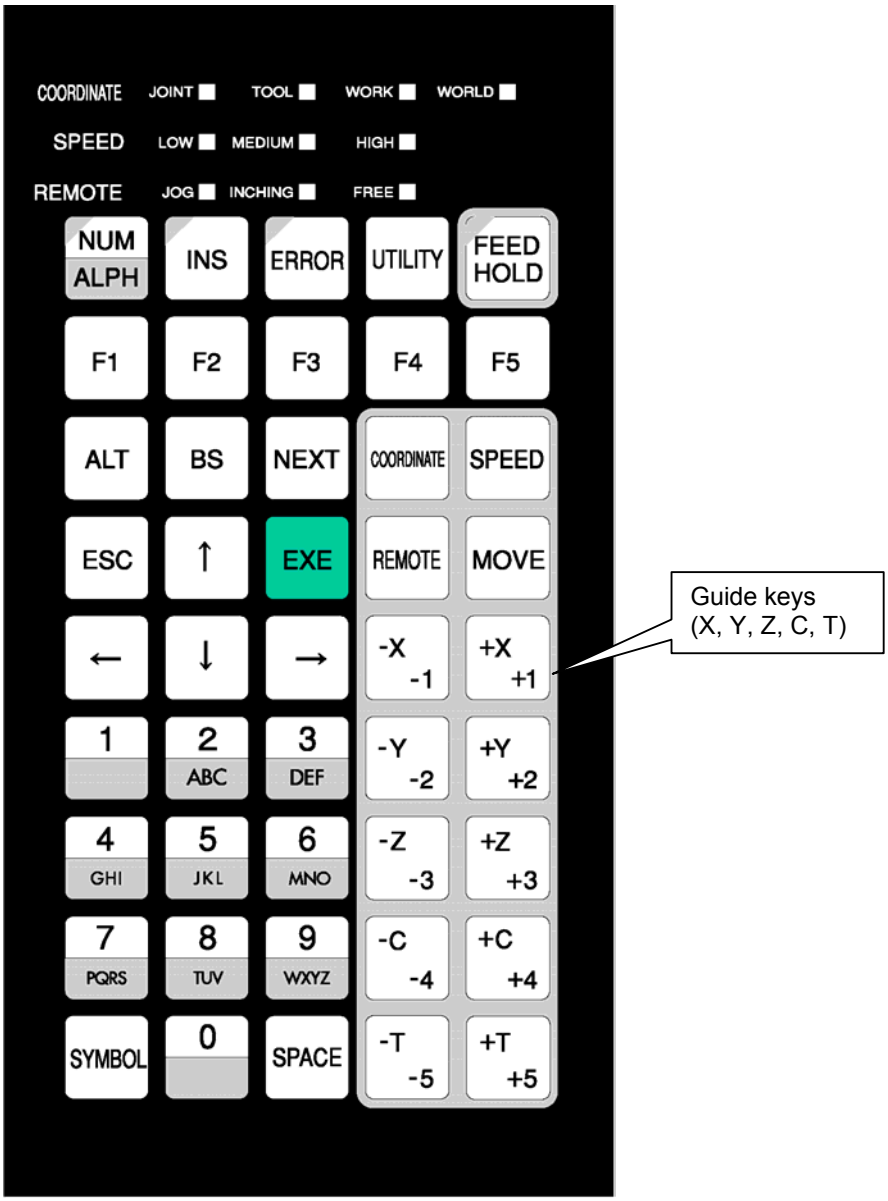
Teaching Mode (TEACHING)	Program edit (EDIT)	Used to edit programs.
	Data edit (DEDIT)	Used to edit various data (position, coordinate and load).
	File selection (SEL)	Used to select a file to be executed.
	Start (RUN)	Used to execute programmed test operation.
	File operation (FILE)	Used to display a file directory, copy, delete and rename files.
	Run mode selection (MODE)	Used to select a run mode.
	Speed override (OVRD)	Used to set the speed override.
	Direct run (DO)	Used to directly execute the SCOL command.
	RESET (RESET)	Used to reset the program execution step, variables and digital output signals.
	Guide	Used to select the guide mode, guide coordinate and guide speed, then move the end effector by means of the guide keys. This mode is ineffective during test operation.

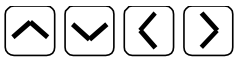
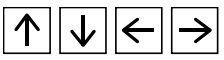
1.2.4 Edit

Edit	Program editor	Program editing. Used to edit programs and global data by character input, command menu selection and SCOL menu selection.
	Data editor	Position data editing. Used to edit position data by teaching and numerical value input. The work coordinate system is selected. Coordinate data editing. Used to edit coordinate data by teaching and numerical value input. The work or tool coordinate system is selected. Load data editing. Used to edit load data by numerical value input.

1.3 General Operation

The key seat array of the teach pendant is as shown below.



Cursor keys  are expressed by  in this manual.

1.3.1 Entry of Alphanumeric Characters and Symbols

The alphabet input mode and numeric input mode can be changed over alternately by pressing the [NUM / ALPH] key. While the numeric input mode is selected, the key lamp is illuminated.

(1) Numeric key input
Select the numeric input (NUM) mode and press a desired key.

Example: When entering number "9":

Key operation	Display
[9 / WXYZ]	9

(2) Alphabet and symbol key input

(2)–1 Alphabet key input
Select the alphabet input (ALPH) mode. Alphabets are classified into eight (8) groups. Press an appropriate key until a desired alphabet is displayed. After the alphabet is displayed, press the EXE key, right cursor key or any other group key, and it is established.
Even if the ALPH mode is selected, pressing the key once calls a number and you can input the number thus called.

Example: When entering alphabet "B":

Key operation	Display
[2 / ABC]	B

Press the key three (3) times.
When alphabet "B" is displayed,
press the EXE key.

(2)–2 Symbol (special character) key input
Every time the SYMBOL key is pressed, a special character is displayed. Though input is possible in both the NUM and ALPH modes, codes that can be input are grouped according to each mode.

(2)–3 Keys relating to unsettled characters

[ESC]	Used to cancel an unsettled character or characters.
[BS]	Used to cancel an unsettled character or characters.
[EXE]	Used to determine an unsettled character or characters.
[↑]	Used to select a choice before current character string.
[↓]	Used to select a choice after current character string.
[←]	Used to cancel an unsettled character or characters.
[→]	Used to determine an unsettled character or characters.

1.3.2 Combined Key Operation

Next page	Press the [ALT] key and [↓] key.
Previous page	Press the [ALT] key and [↑] key.
Delete (DEL)	Press the [ALT] key and [BS] key.

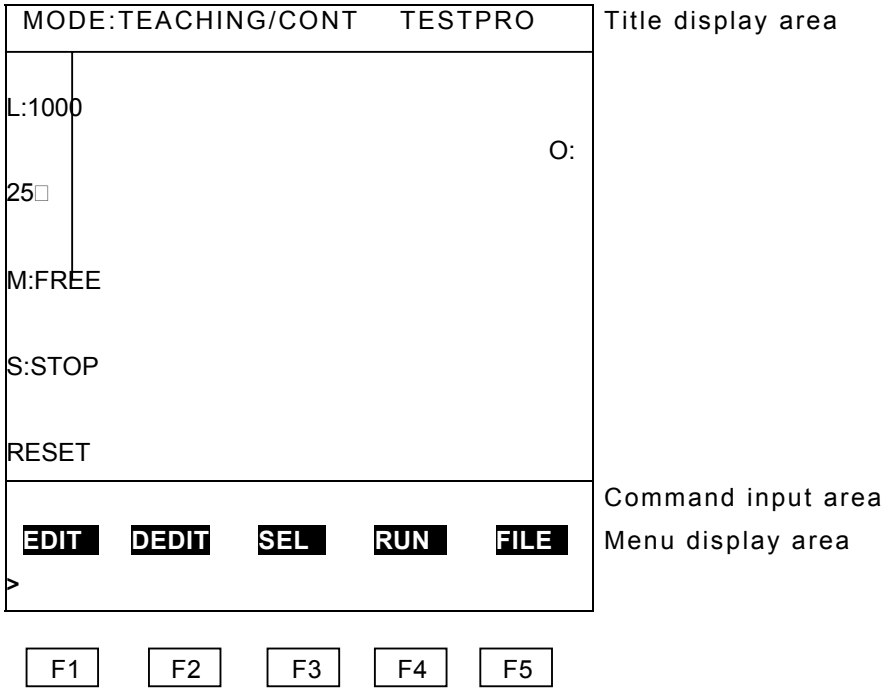
[Supplementary descriptions]

1. When the cursor is not located on main display though it has a plural number of pages, page changeover is basically performed by pressing the [ALT] key and [↑] key or [ALT] key and [↓] key. Page changeover is also possible by pressing the single [↑] key or [↓] key. (This is mainly related to the utility monitor display, alarm display and variable display.)

1.3.3 Use of Function Keys

The teach pendant is provided with five (5) function keys (F1 ~ F5) to facilitate command input.
The functions to which F1 to F5 correspond will change with the mode selected.
The command menu corresponding to each function key is displayed at the bottom of the display. Select a desired command by means of the function key.
When six (6) or more commands are available in the menu, mark ">" appears on the right end of the menu line. Press the NEXT key then, and the remaining command menu is displayed.

Example 1: When selecting the "FILE" command:



Press the F5 key, and the FILE display is called with new menu displayed. Select a desired function key consecutively.

[Supplementary descriptions]

1. When the function key is pressed in turn, the hierarchy deepens with relevant new menu displayed. To return to the original hierarchical menu, press the ESC key.

2. In principle, the NEXT key is used to select the MEXT command menu. Unless marks ">" is available on the display, however, it can serve as the NEXT PAGE key.

1.3.4 Repeating an Entry

The repeat function allows entry of repeated operation by pressing the same key continuously.
The following keys are available for the repeat operation.

- Cursor keys (The keys can be moved rapidly by the repeat function.)
- Guide keys (A selected axis moves continuously as long as this key is kept pressed.)
- MOVE key (A selected axis moves continuously as long as this key is kept pressed.)

1.3.5 Correcting Incorrect Entries

Use the BS key to erase any characters which were incorrectly entered.

Example:

(Correct)	OVERRIDE : 80	(Incorrect)	OVERRIDE : 65__
	Press the BS key.		↓
			OVERRIDE : 6__
	Press the BS key.		↓
			OVERRIDE : __

Keep pressing the BS key until the incorrect characters are erased. Then try again.

Note: In the editor operation, press the BS key while pressing the ALT key, and a character at the cursor position is erased. It functions the same way as the DEL key.

! CAUTION

The INS, ↑, ↓, ←, → keys cannot be used at the time of command input.

1.3.6 Cancelling an Entry

If you wish to cancel the command you are entering, press the ESC key.
This cancels the command you have entered and allows you to input the next command on the following line.
Should an error message or other such message appear during command input, that command line is cancelled.

1.3.7 Use of Wild Cards

You can use wild cards (*, ?) when specifying files with the directory (DIR) or delete (DEL) command.
The wild cards are useful to simultaneously specify multiple files. Using the wild cards correctly can save time and increase efficiency when specifying files.

(1) Wild card "*"
The asterisk (*) wild card can take the place of a file name or extension. For instance, designation of "A*.*" refers to all files having a name starting with A and having any type of extension (or no extension at all). All of the following files would apply:

ADATA.BAT
ABC.PAR
AFILE

(2) Wild card "?"
The question mark (?) wild card can take the place of any single character in name or extension. For instance, designation of "A??.*" refers to all files that have a name starting with A and are three (3) characters long. A space or null character is also regarded as one (1) character. All of the following files would apply:

ABC.PAR
ACD.DAT
AAA

To use the wild cards effectively, it is recommended to predetermine a rule for assigning a file name and file extension.

1.3.8 Format for Operating Instructions

The following format is used in this manual to explain how to use commands.

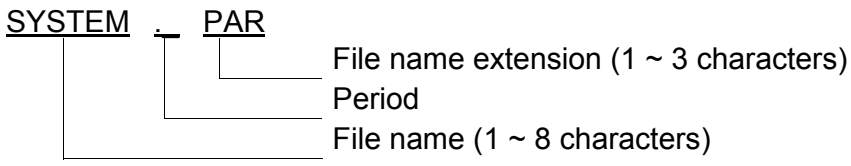
Title [Command name]

(1)	Function	Explains the function and application.
(2)	Operating procedures	Explains the procedures and system response for various operations.
(3)	Caution	Explains the cautions and restrictions.
(4)	Related functions	Explains the related functions, if any.

1.4 Assigning Names

1.4.1 Assigning a File Name

Every file registered in the RAM drive has its own name. This is called the file name. Generally, the file name has the following format.



The file name consists of a basic file name (SYSTEM), a file name extension (PAR) and a period (.) discriminating them.

The following characters can be used for the file name and its extension.

Alphabetic characters : A ~ Z
Numerals : 0 ~ 9

The file name extension is used to identify the nature (type) of the file. By using different extensions for the same file name, you can express different types of files.

The following extensions are used for this controller.

File name extension	Descriptions
SYS	Signifies a system file.
BAT	Signifies a batch file.
PAR	Signifies a parameter file.
LIB	Signifies a library file.
LOG	Signifies a maintenance file.

The file name extension can be omitted. When this happens, the period is unnecessary.
When specifying a file having a file name extension, you cannot omit that extension. If you do, the controller interprets it as a completely different file.

1.4.2 Assigning a Point Name

When entering positional data for a point, you have to assign a name to that point. This name is used later in the program when teaching the robot where to go.

Generally, the point name has the following format.

PNT100
Point name (1 ~ 10 characters)

The following characters can be used for the point name. The first letter of a point name must be an alphabetical character.

Alphabetic characters	:	A ~ Z
Numerals	:	0 ~ 9

Caution: You cannot use an identifier (reserved word) used for the SCOL language to name a point. For details, see Appendix B of the Robot Language Manual.

The point name entered should consist of up ten (10) characters. If it exceeds this limit, the entry is invalid with an error saying "Illegal name" displayed.

If a character other than the alphabetical characters has been entered for the first character of the point name, the entry is invalid with an error saying "Illegal name" displayed.

1.4.3 Assigning a Name to Coordinate Data

When entering coordinate data for a the work coordinate system, tool coordinate system or base coordinate system, you have to assign a coordinate name to that data. A coordinate of each coordinate system is specified by the coordinate name.

Generally, the coordinate name has the following format.

TABLE1
Coordinate name (1 ~ 10 characters)

The following characters can be used for the coordinate name. The first letter of a coordinate name must be an alphabetical character.

Alphabetic characters	:	A ~ Z
Numerals	:	0 ~ 9

Caution: You cannot use an identifier (reserved word) used for the SCOL language to name a point. For details, see Appendix B of the Robot Language Manual.

The coordinate name entered should consist of up ten (10) characters. If it exceeds this limit, the entry is invalid with an error saying "Illegal name" displayed.

If a character other than the alphabetical characters has been entered for the first character of the coordinate name, the entry is invalid with an error saying "Illegal name" displayed.

1.4.4 Assigning a Name to Load Data

When entering load data, you have to assign a load name to that data. A load in the program is specified by the load name.

Generally, the load name has the following format.

SAMPLE1
Load name (1 ~ 10 characters)

The following characters can be used for the coordinate name. The first letter of a load name must be an alphabetical character.

Alphabetic characters	:	A ~ Z
Numerals	:	0 ~ 9

Caution: You cannot use an identifier (reserved word) used for the SCOL language to name a point. For details, see Appendix B of the Robot Language Manual.

The load name entered should consist of up ten (10) characters. If it exceeds this limit, the entry is invalid with an error saying "Illegal name" displayed.

If a character other than the alphabetical characters has been entered for the first character of the load name, the entry is invalid with an error saying "Illegal name" displayed.

1.5 File

To move the robot, robot language programs, position data and parameters should be registered beforehand in the controller RAM drive (memory). All of these programs and data are handled on the basis of a file.

1.5.1 Kind of File

The following files are used in the robot system.

(1) Machine parameter file (MACHINE. PAR)
This file contains parameters used by the maker, such as parameters for robot option setting and axis origin offset.
The user MUST NOT change the values of these parameters.

(2) System parameter file (SYSTEM. PAR)
This file includes parameters characteristic of the robot, such as parameters for the number of axes, servo data, inching amount, speed limit for test operation.
The user MUST NOT change the values of these parameters.

(3) User parameter file (USER. PAR)
This file includes parameters relating to limit on the working envelope, setting of communication mode, etc., which can be changed by the user.

(4) Library file (SCOL. LIB)
Some instruction words used for the robot language are contained in this file. The instruction words in the library file are written by the robot language. They include the instruction words for operating the robot hand (OPEN, CLOSE, OPENI, CLOSEI), and these instruction words can be used only when the library file is present. For further information, see the Robot Language Manual.

(5) Palletize library file (PALLET. LIB)
This library file is required when using palletize commands (INITPLT, MOVEPLT) in the program created by the robot language. For further information, see the Robot Language Manual.

(6) Automatic start file (AUTOSTR. BAT)
A kind of batch file, it is automatically executed when the controller power is turned on.

(7) Program file
A file stating the robot motions. It saves position data and programs created by the robot language. Each file corresponds to one job.

(8) External select file (EXTRNSEL. SYS; for TS2000 only)
This file is used when you select a program you wish to execute by means of an input signal to the controller. It contains the program names which can be selected by means of the input signal.
In the TS1000 system, as the program names are specified in the user parameter file, this file is unnecessary.

The following files are mandatory in the RAM drive.

MACHINE. PAR
SYSTEM. PAR
USER. PAR

The following files can be omitted unless necessary.

AUTOSTR. BAT
SCOL. LIB
PALLET. LIB
EXTRNSEL. SYS (for TS2000 only)

1.5.2 Batch File

The controller is provided with a function executing a series of commands in the pack, which is called the "batch processing." To execute the batch processing, commands should be written in the file in the order of execution. This file is generally referred to as the "batch file." The batch file extension is ".BAT".

A file executed automatically just after the controller power is turned on is called the "automatic start file." The name of this file is "AUTOSTR. BAT."
The automatic start file can be used for file selection in the external automatic mode. For the file editing operation, see Para. 10.2.

Example: When executing files to be executed just after the power ON in the external automatic mode:
File names : AUTOSTR. BAT
SELECT PRO1

Save the above automatic start files in the controller RAM drive beforehand. When the controller has started up after its power was turned on, the file PRO1 is selected as the file for execution.

The commands which are effective in the automatic start file are SELECT, OVRD, DATE and TIME.

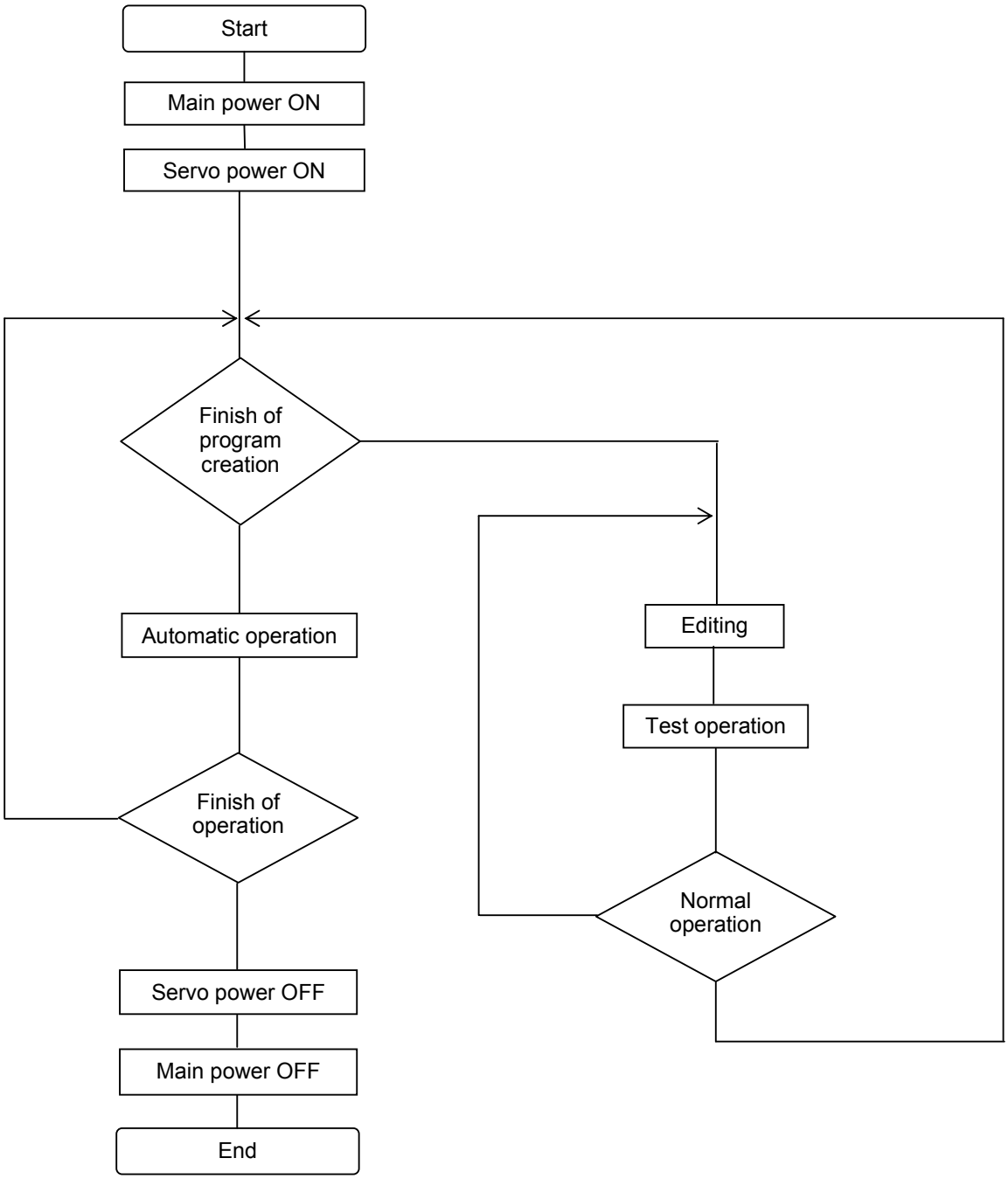
1.5.3 Backup

Though programs using the robot language, position data, changed user parameters, etc., are saved in the RAM drive, it is recommended to back them up to prepare for the worst. For this purpose, connect the serial port host and personal computer and use our personal computer support software. This is also applicable when loading the programs and data in the RAM drive.
For the personal computer support software, see the separate manual.

Section 2 Basic Operation

2.1 Basic Operating Procedures

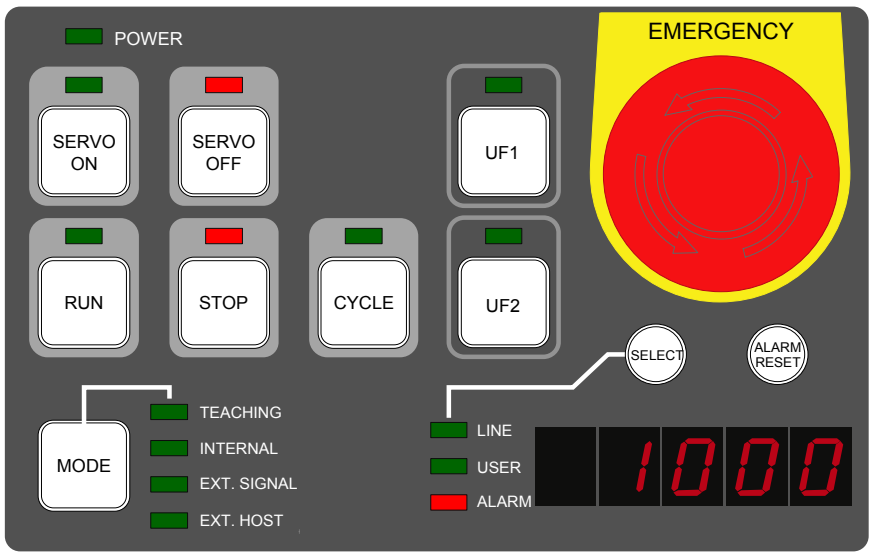
The following flow chart shows the basic operating procedures for automatic operation. It covers all operations from turning on the main power to turning off the main power.



2.2 Control Panel

The control panel is shown below.

TS1000



(1) SERVO ON, SERVO OFF : Used to turn on and off the servo power.

(2) RUN : Used to start the internal automatic mode operation.

(3) STOP : Used to stop the operation after the active block has been executed.

(4) CYCLE : Used to select the cycle operation mode during the internal automatic mode.

(5) MODE : Used to select the operation master mode.

(6) SELECT : Every time this switch is pressed, content shown on the display changes.

LINE: No. of program execution steps.

USER: A value set by PLCDATAW of the program.

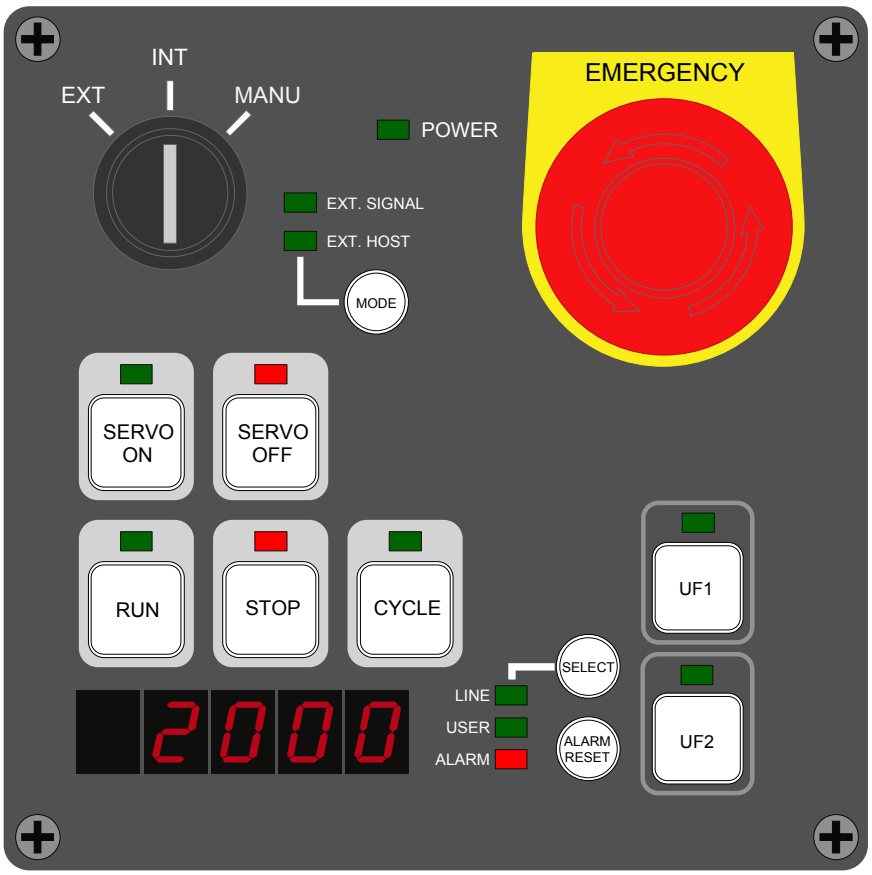
ALARM: Alarm number currently generated in the controller.
If two (2) or more alarms have occurred, such alarm numbers are displayed at intervals of two (2) seconds.

(7) UF1, UF2 : User function switches.
This switch input can be referred to by DIN and the lamp output by DOUT.

(8) ALARM RESET : Used to reset all alarms.

(9) EMERGENCY : EMERGENCY stop pushbutton switch.

TS2000



(1) Key switch : Used to select the operation master mode.

(2) MODE : Used to change over the EXT. SIGNAL and HOST modes while the EXT master mode is selected.

(3) SERVO ON, SERVO OFF : Used to turn on and off the servo power.

(4) RUN : Used to start the internal automatic mode operation.

(5) STOP : Used to stop the operation after the active block has been executed.

(6) CYCLE : Used to select the cycle operation mode during the internal automatic mode.

(7) SELECT : Every time this switch is pressed, content shown on the display changes.

LINE: No. of program execution steps.

USER: A value set by PLCDATAW of the program.

ALARM: Alarm number currently generated in the controller. If two (2) or more alarms have occurred, such alarm numbers are displayed at intervals of two (2) seconds.

(8) UF1, UF2 : User function switches. This switch input can be referred to by DIN and the lamp output by DOUT.

(9) ALARM RESET : Used to reset all alarms.

(10) EMERGENCY : EMERGENCY stop pushbutton switch.

2.3 Turning ON the Main Power

CAUTION

If an abnormality has generated or the POWER LED lamp on the control panel remains off after the main power switch of the controller was turned on, turn off the main power immediately and confirm the wiring. Otherwise, you will be struck by an electric shock or a fire will break out.

When the power is turned on, the controller enters a program reset status.

(1) Function

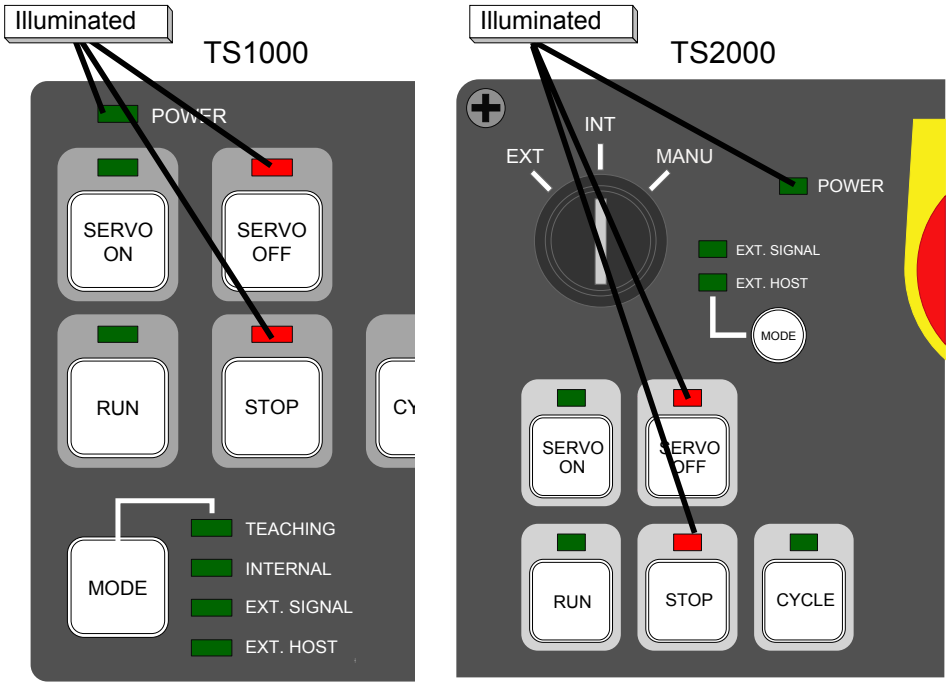
The controller is started up by the system programs saved in the flash memory.

(2) Procedures

(a) Step 1: Turn on the main power.

Set ON the main power switch equipped on the controller, and the POWER LED and SERVO OFF LED are illuminated.

The controller starts initializing the system. The STOP LED on the control panel is illuminated and the system is ready for an operation with the initial display of a selected mode called on the teach pendant.



(3) Caution

The following reserved files are required in the RAM drive.

MACHINE. PAR	SCOL. LIB *
SYSTEM. PAR	AUTOSTR. BAT *
USER. PAR	PALLET. LIB *
	EXTRNSEL. SYS (for TS2000 only)

The asterisk (*)-marked files are not absolutely necessary to run the system. However, an error occurs when you try to execute a function dependent on a missing file.

2.4 Turning ON the Servo Power

CAUTION

- If the robot moves abnormally at servo power ON, immediately press the EMERGENCY stop pushbutton switch to turn the servo power off.
- If you enters the dangerous area of the robot, turn off the servo power beforehand.

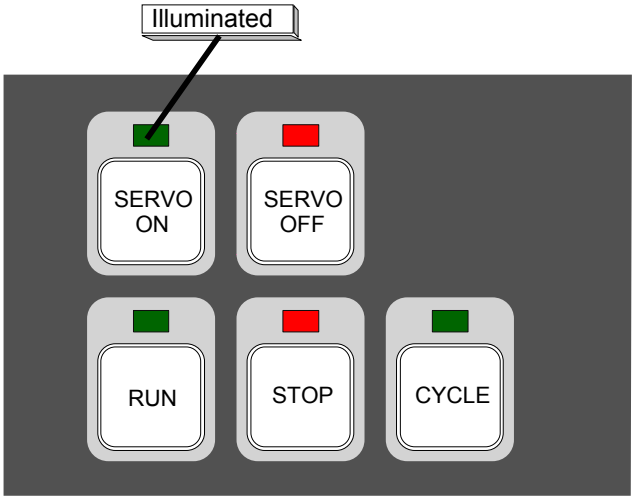
(1) Function

Turns on the main circuit power of the servo driver. The servo-controlled axes are locked and any brake is released.

(2) Procedures

(a) Step 1: Turn on the servo power.

Press the SERVO ON switch equipped on the control panel or teach pendant. When the main circuit power of the servo driver is turned on, the SERVO ON LED equipped on the control panel is illuminated with the SERVO OFF LED turned off. The SERVO ON LED on the teach pendant is also illuminated.



(3) Caution

While the EMERGENCY stop switch is pressed, the servo power will not turn on. Be sure to cancel an emergency stop before turning on the servo power. The servo power can be turned on through the teach pendant only when the teaching mode is selected.

To turn on the servo power by means of an external control signal, the teach pendant connector (or dummy connector) should be connected to the TP connector of the control panel . Also, the contacts for the emergency stop and servo OFF external signals should be open. For more information, see the Interface Manual.

2.5 Turning OFF the Servo Power

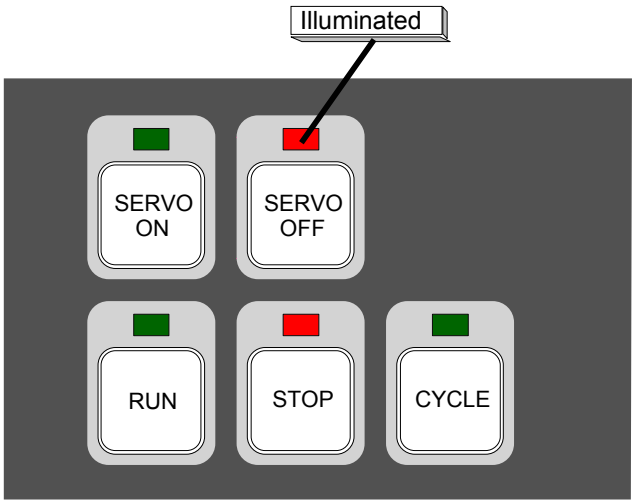
(1) Function

Turns off the main circuit power of the servo driver. The servo-controlled axes are set free and any brake is activated.

(2) Procedures

(a) Step 1: Turn off the servo power.

When the main circuit power of the servo driver has turned off, the SERVO OFF LED equipped on the control panel is illuminated with the SERVO ON LED turned off. The SERVO ON LED on the teach pendant is also illuminated.



(3) Caution

DO NOT turn off the servo power while the robot is operating. When you have to stop the robot urgently, press the EMERGENCY stop switch instead.

Before you have to enter the robot working envelope, or when no operator is to be present at other than automatic operation, turn off the servo power for safety reasons.

For the servo power OFF by external control signal, see the Interface Manual.

2.6 Turning OFF the Main Power

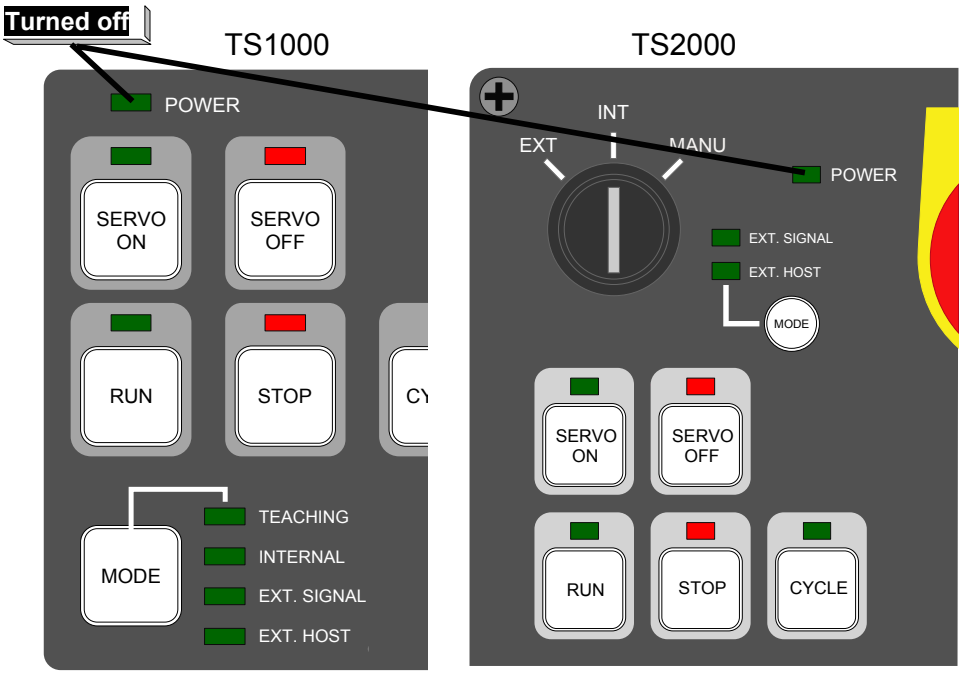
(1) Function

Turns off the main power and stops controller processing.

(2) Procedures

(a) Step 1: Turn off the main power.

Set OFF the main power switch equipped on the controller, and the POWER LED on the control panel goes out.



(3) Caution

DO NOT turn off the main power while the servo power is on or during file operation or any other processing. Before turning off the main power, make sure that the servo power is off and that no active processing is present.

Section 3 Teaching Operation

3.1 Selecting the Teaching Mode

(1) Function

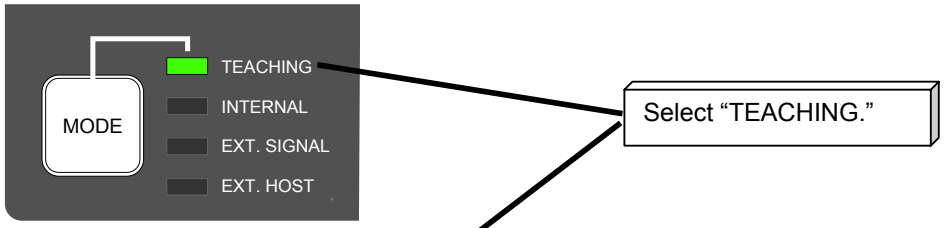
Changes over to the teaching mode in which the robot is manually guided, points are taught, files are operated and the system is controlled. All of these operations are performed through the teach pendant.

(2) Procedures

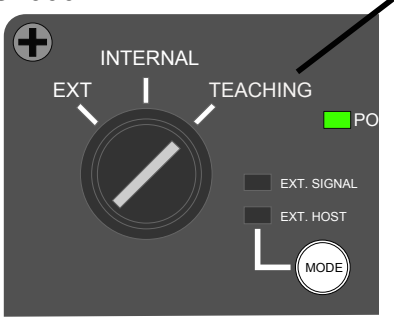
(a) Step 1: Selection of teaching mode.

Change over the master mode select [MODE] switch on the control panel to select "TEACHING."

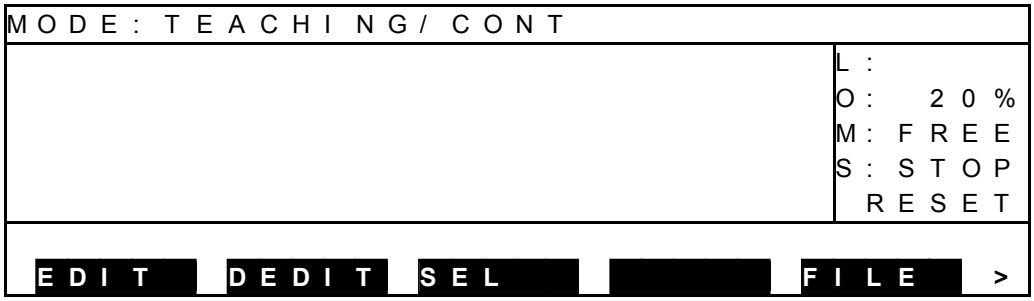
TS1000



TS2000



When the teaching mode is selected, the following display appears.



- The function key menu is displayed on the bottom of the display.
  The menu shown in the teaching mode is as follows:

EDIT	Program editing
DEDIT	Data editing
SEL	Selection of program to be executed
RUN	Test operation start
FILE	File mode
MODE	Run mode selection
OVRD	Override setting
RESET	Reset
DO	Direct execution
WATCH	Variable monitor

3.2 Guidance Coordinates [COORDINATE]

(1) Function

Selects a coordinate system in which the robot is guided, from among the joint, tool, work and world coordinate systems.

(2) Procedures

(a) Step 1: Selection of tool or work coordinate system.

Select the work or tool coordinate system used for [TRANS] of the utility function. For details, see Section 12.

When using the joint or world coordinate system this operation is unnecessary.

(b) Step 2: Coordinate selection.

Each time the COORDINATE key on the teach pendant is pressed, JOINT, TOOL, WORK and WORLD is selected in turn with appropriate LED illuminated. The initial setting is JOINT.

(3) Cautions

For each coordinate system of the robot, see the Introductory Manual.

Unless a tool, base or work coordinate has been taught to the robot, the robot interprets in the following manner.

- The tool coordinate system coincides with the surface of the hand mounting flange.
- The work and world coordinate systems coincide with the base coordinate system.

  When using the tool or work coordinate system, make sure that you have selected the right one.

3.3 Guide Mode [REMOTE]

The guide mode comes in three (3) types; JOG, INCHING and FREE.

In the JOG mode, the robot moves in a specified direction at a specified speed in a selected coordinate system as long as the corresponding guide key is kept pressed. This mode is useful for rough positioning of the robot to get it near the target position.

In the INCHING mode, the robot moves by a specified travel distance in a specified direction in a selected coordinate system each time the corresponding guide key is pressed once. This mode is useful for fine positioning of the robot from near the target position to the target position.

In the FREE mode, you can cancel the servo lock of a desired axis to move it by hand. This mode is useful for fine positioning of the robot near a workpiece, etc.

(1) Function
Selects the manual guide mode.

(2) Procedures

(a) Step 1: Teaching mode selection.
Press the MODE switch on the control panel to select "TEACHING."

(b) Step 2: Guide mode selection.
Each time the REMOTE key on the teach pendant is pressed, JOG, INCHING and FREE is selected in turn with appropriate LED illuminated. The initial setting is JOG.

(3) Caution
When another mode is selected from the FREE mode, servo lock is effected on all axes.

3.4 Guide Speed [SPEED]

(1) Function

Selects the speed or travel distance for manual guide.
When the JOG mode is selected, speed selection becomes effective. When the INCHING mode is selected, travel distance selection prevails.
Select the speed or travel distance according to the distance to the target position.

(2) Procedures

(a) Step 1: Guide speed selection.
Each time the SPEED key on the teach pendant is pressed, LOW, MEDIUM and HIGH is selected in turn with appropriate LED illuminated. The initial setting is LOW.

3.5 Guidance

! CAUTION

- When guiding the robot manually, DO NOT enter the dangerous area of the robot.
- If the robot moves abnormally, immediately effect an emergency stop.

(1) Function

While the JOG or INCHING mode is selected, the robot can be moved in a desired direction by pressing appropriate guide key.

(2) Procedures

(a) Step 1: Teaching mode selection.

Press the MODE switch on the control panel to select "TEACHING."

(b) Setting of guidance conditions.

Set the guidance conditions (guide coordinates, guide mode and guide speed) as necessary.

The guide coordinates, guide mode and guide speed can be selected in any order. Select them before pressing the guide key.

(c) Step 3: Guidance

While pressing by one (1) step the enable switch provided on the right side of the teach pendant, press the guide key for a desired axis. Then the robot moves as directed in the specified coordinate system.

(3) Caution

When the FREE mode is selected, the guide keys will not specify the direction in which the robot is to move. Rather, they serve to cancel each axis servo lock.

The guide keys are operative only when pressed while the enable switch on the teach pendant is kept pressed by one (1) step. This is to prevent accidents.

3.6 Servo Free

(1) Function

In the FREE mode, it is possible to cancel a servo lock of any axis to make it servo-free. The axis thus set servo-free can be moved by hand. This is convenient when moving the robot near a workpiece, etc.

(2) Procedures

(a) Step 1: Teaching mode selection.
Press the MODE switch on the control panel to select "TEACHING."

(b) Step 2: Guide mode selection.
Press the REMOTE key on the control panel to select "FREE". Make sure that the FREE LED is illuminated.

(c) Step 3: Servo lock cancel.
While pressing by one (1) step the enable switch equipped on the right side of the teach pendant, press the minus (–) key for an axis whose servo locked condition is to be cancelled.

(d) Step 4: Guidance.
The axis whose servo locked condition was cancelled enters a servo-free status. Now the robot can be guided by hand.

(e) Step 5: Servo lock.
While pressing by one (1) step the enable switch equipped on the right side of the teach pendant, press the plus (+) key for the axis whose servo locked condition is cancelled. Then the servo-free axis enters a servo-locked status.

(3) Caution

When you select the RUN mode, DO mode, internal automatic mode or external automatic mode while the servo lock is cancelled, servo lock is effected on all axes.

The guide keys are operative only when pressed while the enable switch on the teach pendant is kept pressed by one (1) step.

If you move the robot by hand in the servo free status, other joint axes will move also due to axis interference.

You cannot cancel the servo free status during feed hold.

You can cancel the servo free status during error history display.

Section 4 Auxiliary Signals

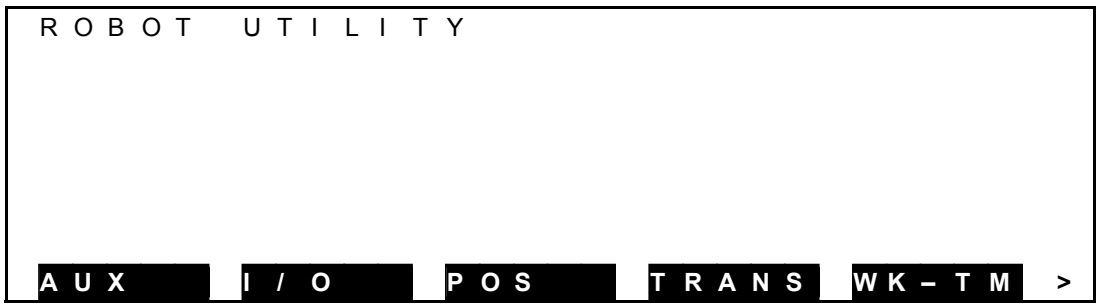
(1) Function

Operates auxiliary signals used to control the hand, air-controlled axes, etc.

(2) Procedures

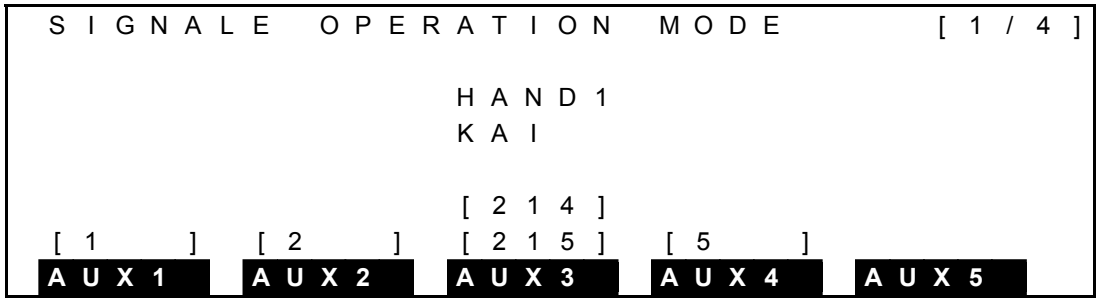
(a) Step 1: Utility mode selection.

Press the UTILITY key on the teach pendant, and the utility display as shown below appears.



(b) Step 2: Auxiliary signal operation mode selection.

Press [AUX] from the utility menu, and the auxiliary signal operation display as shown below is called.

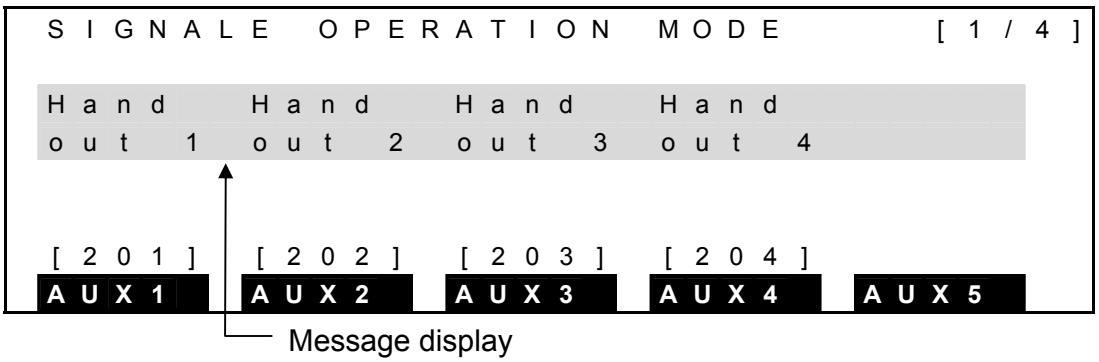


The numbers on the display correspond to the output signal ports. Auxiliary signals having two (2) output signal ports are set for the double solenoid, and those having one (1) output signal port are intended for the single solenoid. Those having no output signal ports are not intended for anything.

Ports inversely displayed are ON and ports not inversely displayed are OFF.

(c) Step 3: Function key ON/OFF.
Each time a function key corresponding to a port is pressed, the solenoid status changes from ON to OFF or vice versa. For the double solenoid, the status of the two (2) ports is transposed.

(3) Message display
Messages for respective signals can be displayed on the 2nd to the 3rd lines of the display. They should be defined beforehand in the user parameter file in the controller RAM drive.



Example)
Contents of "USER. PAR" file

```
•
= ❑0❑❑201❑❑"Hand❑out❑1"
•  (Messages for 19 lines)
•
```

Note: ❑ : Space
(Insert a space to a blank line.)

(4) Caution
The output signal port and solenoid type corresponding to each function key should be defined in the user parameter file in advance.
When the ESC key is pressed, the auxiliary signal operation mode terminates and the system returns to the previous display.
During emergency stop, auxiliary signals can be turned on and off.
The initial state at power ON, including signals which are set for the double solenoid, is OFF.

Section 5 Program Editing

5.1 Starting Up the Program Editor [EDIT]

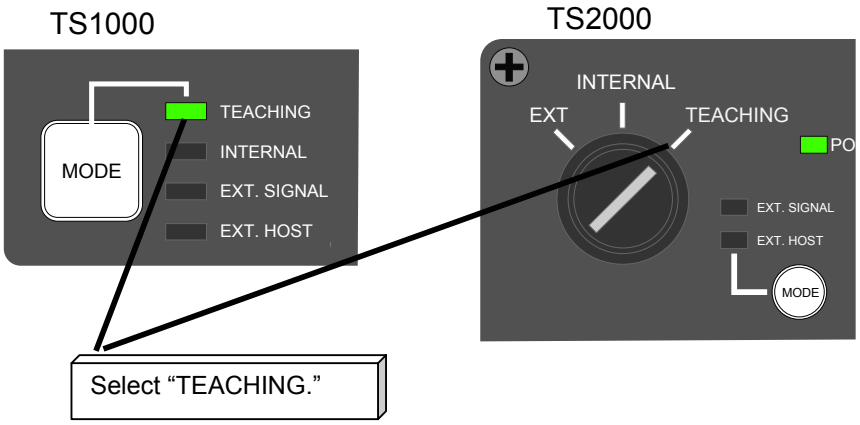
(1) Function

Starts up the program editor to edit a program.

(2) Procedures

(a) Step 1: Mode selection.

Change over the MODE switch equipped on the control panel to select "TEACHING."



(b) Step 2: Menu selection.

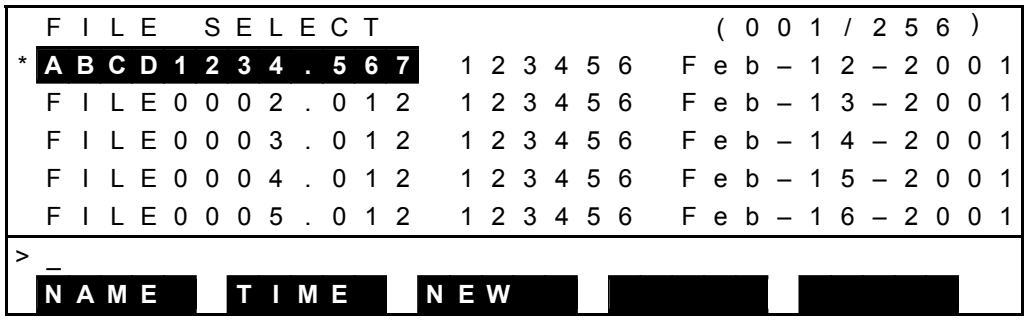
Keep pressing the NEXT key until the EDIT menu appears on the display. Then press the function key corresponding to [EDIT].

```
MODE : TEACHING/CONT      12345678.123
                                L:9999
                                O:100%
                                M:FREE
                                S:STOP
                                  RESET
 EDIT  DEDIT  SEL  RUN  FILE  >
```

(c) Step 3: File selection.

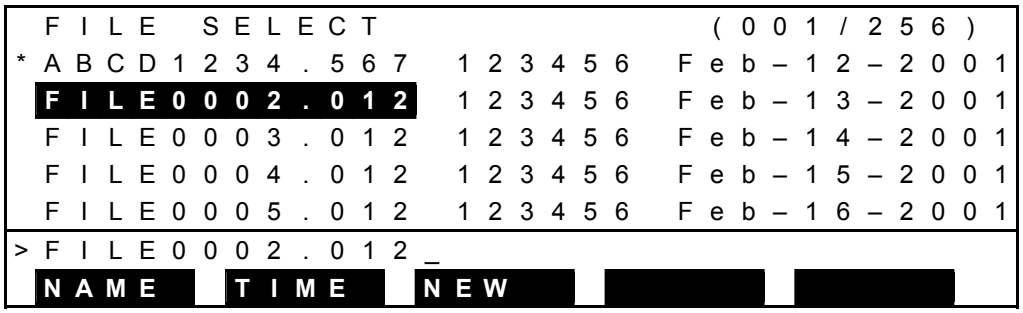
Select the file name by means of the up and down cursor keys.

To create a new file, press [NEW].

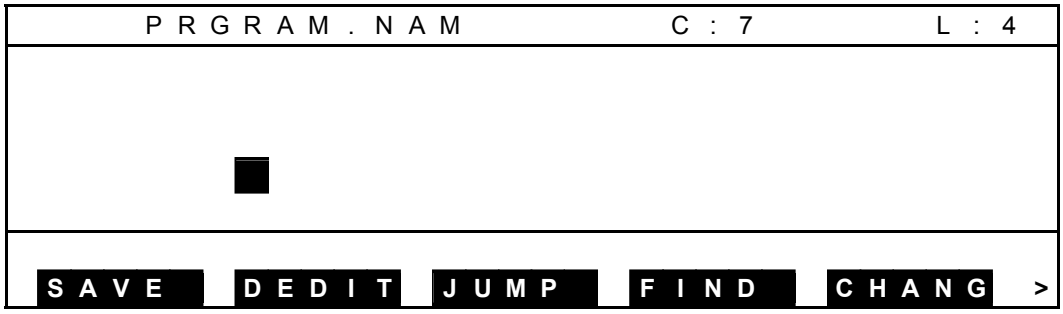


Unless the F3 [NEW] key was pressed:

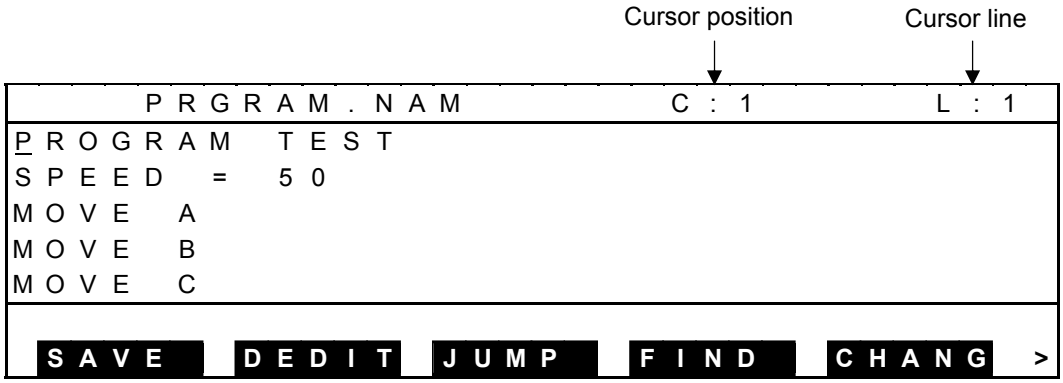
Press the EXE key, and the selected file name is displayed on the command line. It is also possible to rename the file. For this purpose, return the cursor by means of the BS key and enter a desired file name.



(d) Step 4: Check of file name to be edited.
Check for the name of a file to be edited to make sure that you are dealing with the right file. If OK, press the EXE key. Unless the specified file is present in the RAM drive, you can create a new file.



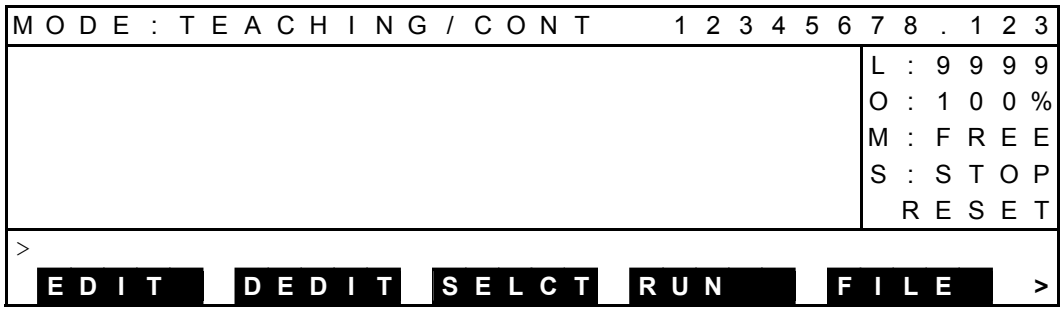
If the specified file exists in the RAM drive, the character input mode is selected with the contents of the specified file shown on the editing display. Now you can edit the file.



If the memory size of the file you specified is especially large, it may take about ten (10) seconds until "Opening [file name]" is displayed and you can start editing. Wait for a while.

If the file name you specified is illegal, the system returns to the initial display.

If there is not enough available memory space in the RAM drive and you cannot create a new file, or if the number of registered files is 256 and you cannot create a new file, the system returns to the initial display as well. Delete an unnecessary file or files, then execute [EDIT] again.



(3) Caution

A program is edited in the RAM drive. Unless a specified program exists in the RAM drive, the system interprets that you are going to create a new file and creates a new file of that name. If you wish to edit an existing file, you have to first load it into the RAM drive.

If you omit designation of a file name, the file selected as the one for execution is taken as the file to be edited.

Even if the MODE key switch is changed from "TEACHING" during program editing, the editing mode will not terminate.

5.2 Character Input

5.2.1 Character Input Mode Selection

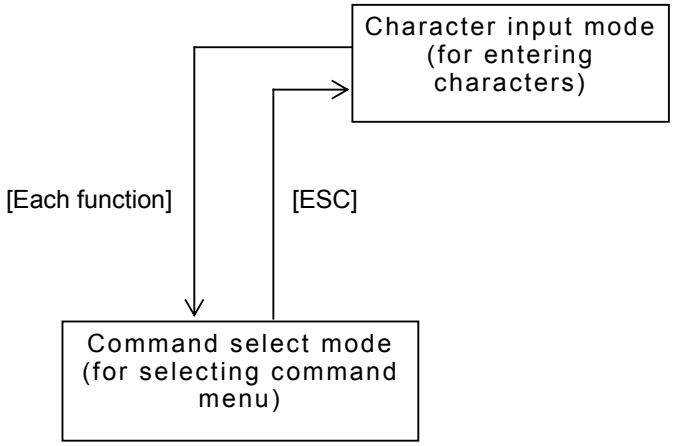
(1) Function

This mode allows characters to be entered at the cursor position on the editing display through the keyboard. Edit commands can be input directly.

(2) Procedures

(a) Step 1: Mode selection.

If the command select mode is selected, press the ESC, and system returns to the character input mode. Use this operation to return to the character input mode without menu selection.



(3) Caution

When the program editor has started up, the character input mode become effective. After the command menu has been selected, the system automatically returns to the character input mode.

In the character input mode, the cursor can be moved up and down, right and left on the editing display, using the cursor keys.

5.2.2 Insert Mode

(1) Function

Used to select the insert mode for character input. A character entered through the keyboard is inserted to the cursor position.

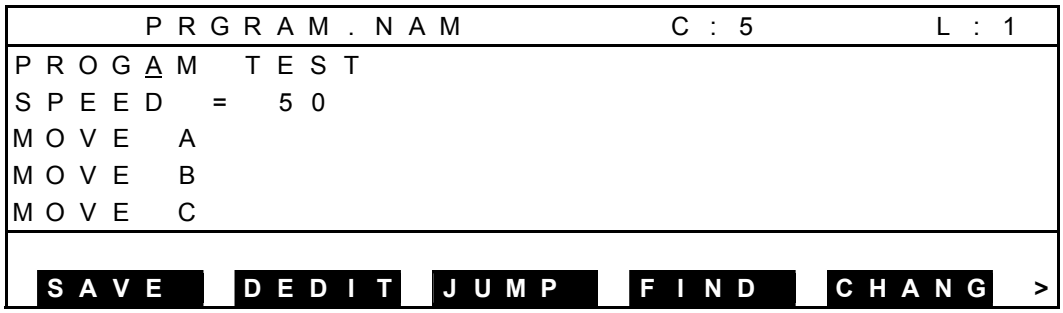
(2) Procedures

(a) Step 1: INS key operation.

Each time the INS key is pressed, the input mode or replace mode is selected alternately. When the insert mode is already selected, this operation is unnecessary.

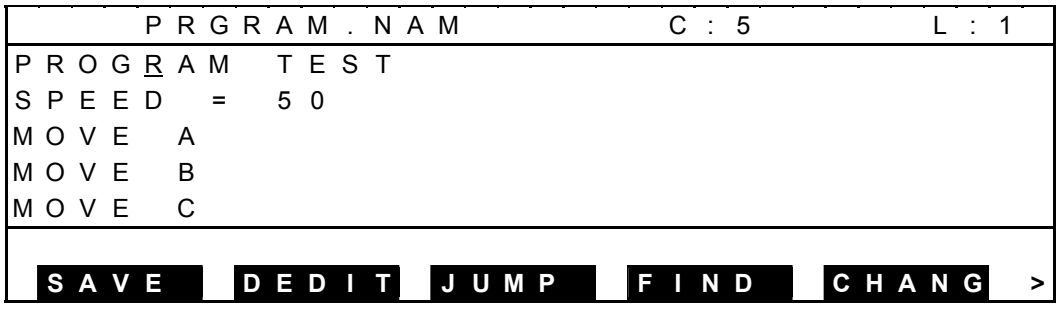
(b) Step 2: Designation of insert position.

Move the cursor to a position where you wish to insert characters, using the cursor keys (↑, ↓, ←, →).



(c) Step 3: Insertion.

Press a key for a desired character, and the character thus input is inserted at the cursor position.



(3) Related functions

- Line insertion
  While the insert mode is selected, move the cursor to the top of a line where a character or characters are to be added, using the cursor keys (↑, ↓, ←, →).

```
        PRGRAM.NAM          C:5          L:1
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  B
MOVE  D

 SAVE  DEDIT  JUMP  FIND  CHANG  >
```

Enter a line to be added through the keyboard, and it is inserted just before the line where the cursor was located.

```
        PRGRAM.NAM          C:7          L:4
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  BMOVE  C
MOVE  D

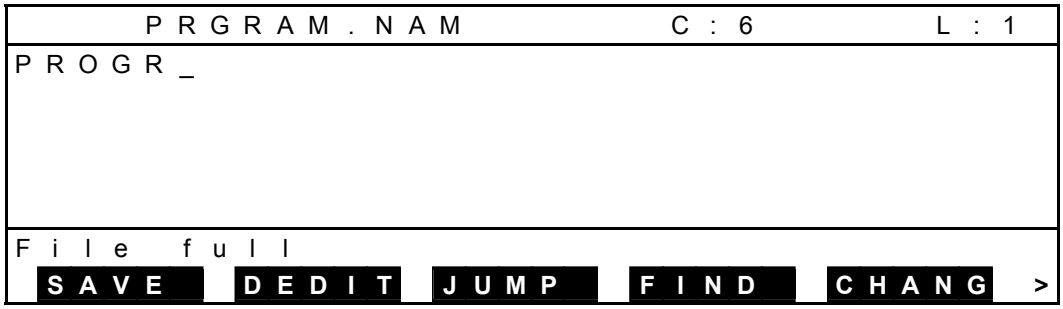
 SAVE  DEDIT  JUMP  FIND  CHANG
```

Press the EXE key, and the line where the cursor was located is shifted to the next line.

```
        PRGRAM.NAM          C:1          L:5
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  B
MOVE  C

 SAVE  DEDIT  JUMP  FIND  CHANG
```

When the available memory space becomes small during editing operation, a warning saying "A few memory remains" is displayed. When this happens, quit the program editing mode, delete any unnecessary file or files, then resume the program editing operation.
If character input continues with the warning of "A few memory remains" displayed, a warning of "File full" is displayed and the subsequent character input is invalid. If you try to quit the program editing, a warning saying "Delete some data until rem >=0" is displayed, and you cannot finish the program editing. Delete program contents, then terminate the editing.



(4) Caution
When the program editor has started up, the insert mode takes effect. The insert mode is discriminated from the replace mode by the shape of the cursor.

Insert mode : _
Replace mode : ■

When the EXE key is pressed while the cursor is located at the top of a line in the insert mode, a blank line is inserted at the cursor position.

5.2.3 Replace Mode

(1) Function

Used to select the replace mode for character input. A character entered through the keyboard is replaced with the one at the cursor position.

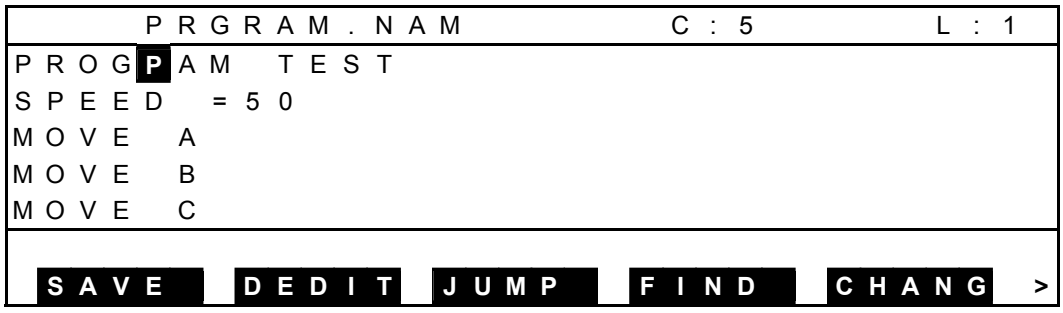
(2) Procedures

(a) Step 1: INS key operation.

Each time the INS key is pressed, the input mode or replace mode is selected alternately. When the replace mode is already selected, this operation is unnecessary.

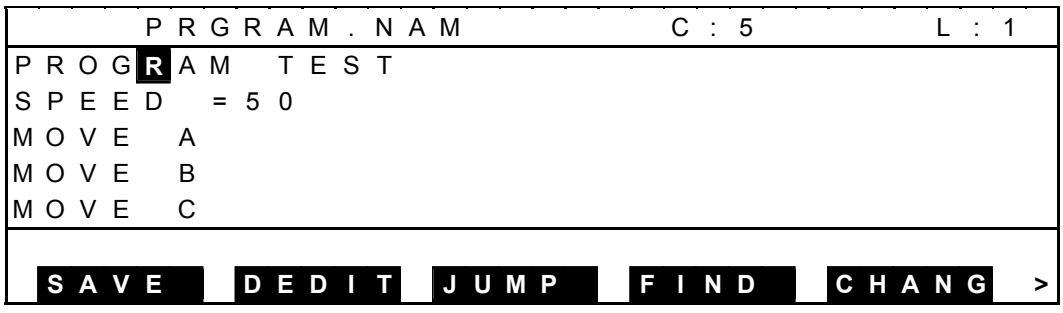
(b) Step 2: Designation of replaced character.

Move the cursor to a character to be replaced, using the cursor keys (↑, ↓, ←, →).



(c) Step 3: Replacement.

Press a key for a desired replacing character, and the character thus input is replaced with the one at the cursor position.



(3) Caution

When the program editor has started up, the insert mode takes effect. The insert mode is discriminated from the replace mode by the shape of the cursor.

Insert mode : _

Replace mode : ■

5.2.4 Input Correction

(1) Function

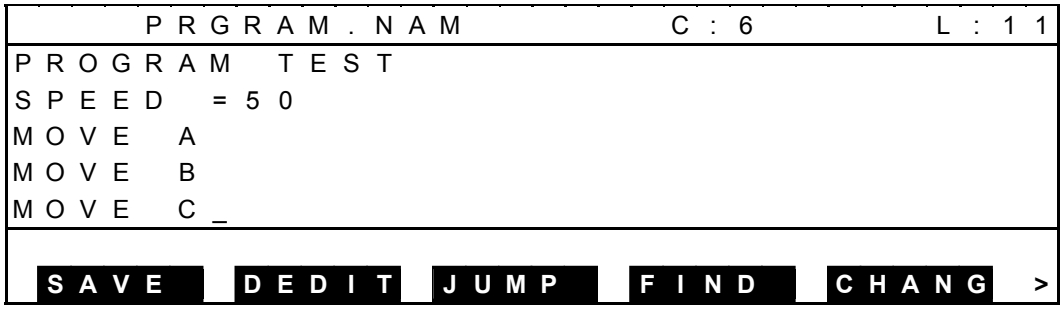
While the character input mode is selected, one (1) character just before the cursor position can be deleted on the editing display.

This function is useful for deleting an error just after key entry.

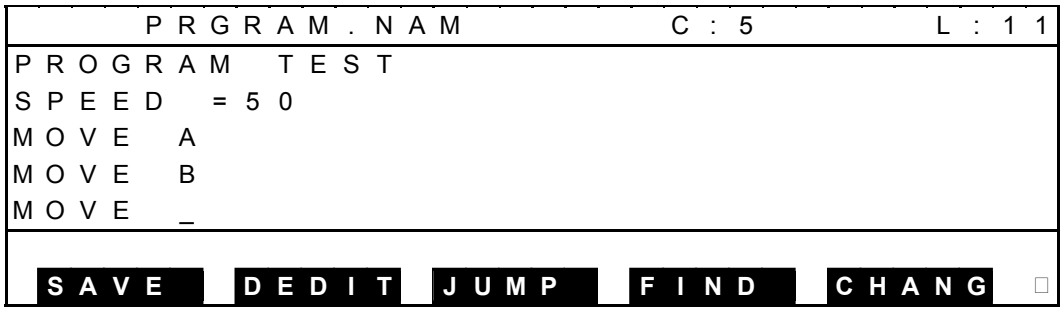
(2) Procedures

(a) Step 1: Input correction.

Press the BS key, and one (1) character just before the cursor position is deleted with successive characters on the same line shifted left in turn.



↓ BS



(3) Caution: None

5.2.5 Character Deletion

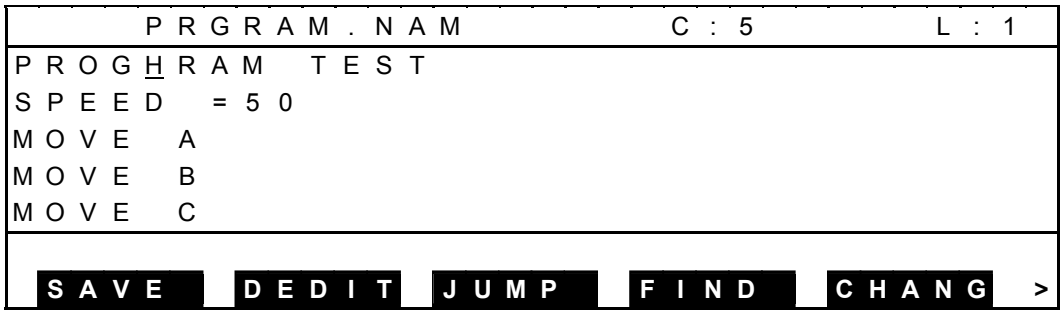
(1) Function

Deletes one (1) character at the cursor position.

(2) Procedures

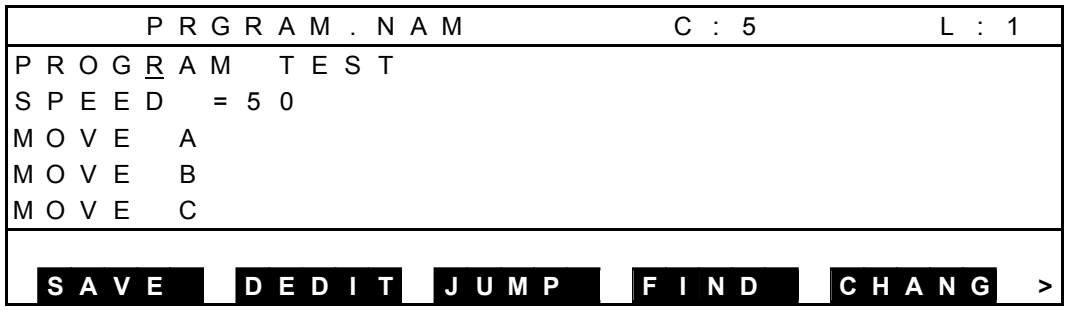
(a) Step 1: Designation of deleted character.

Move the cursor to a character to be deleted.



(b) Step 2: Deletion.

Press the ALT and BS keys, and the character at the cursor position is deleted with successive characters on the same line shifted left in turn.



(3) Caution

When the character deletion is executed at the end of a line, a line feed is deleted with the next line succeeding the current line.

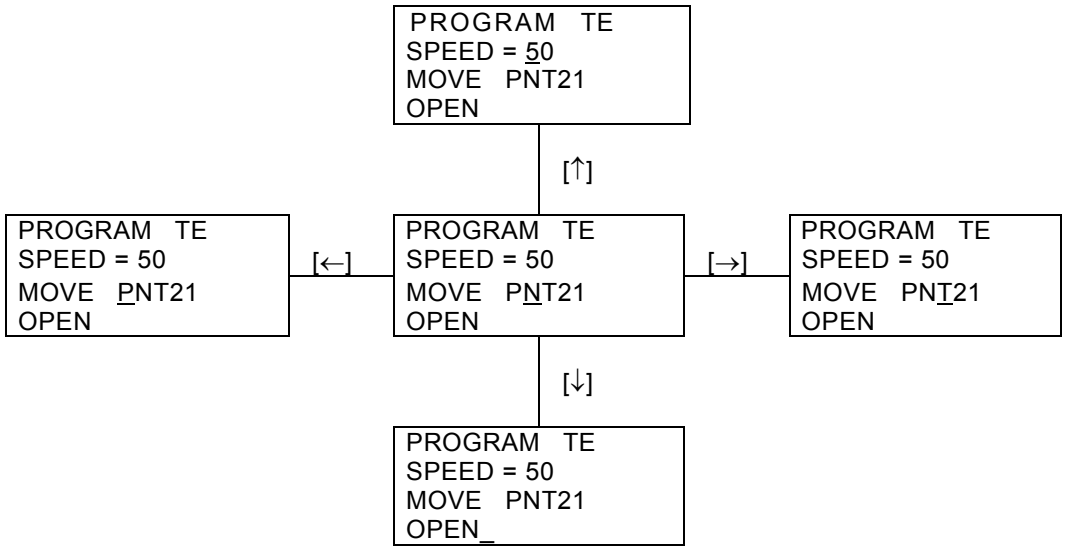
5.2.6 Moving the Cursor [↑] [↓] [←] [→]

(1) Function

Moves the cursor by one (1) character in the arrow-marked direction.

(2) Procedures

(a) Step 1: Cursor movement



(3) When the cursor is moved from the top, bottom, right or left end of the display, it scrolls the screen in an appropriate direction. However, the cursor cannot be moved past the top line, bottom line, top of a line or end of a line.
If the cursor is located after the line feed position, it is moved the line feed position.

5.3 Edit Command Input

Edit commands can be input, using function keys.

5.3.1 List of Edit Commands

Function	Command	Descriptions
1st page		
F1: SAVE	Save	Quits the editor (saves program).
F2: DEDIT	Dedit	Edits data.
F3: JUMP	Jump	Moves the cursor to specified line.
F4: FIND	Find	Searches a character string.
F5: CHANG	Change	Replaces a character string.
2nd page		
F1: NOSAV	Nosave	Aborts and quits the editor.
F2: COPY	Copy	Saves a block in buffer.
F3: CUT	Cut	Cuts a block (saves in buffer).
F4: PASTE	Paste	Pastes the buffer contents.

5.3.2 Command Menu Selection

(1) Function

Selects and inputs an edit command by menu selection.

(2) Procedures

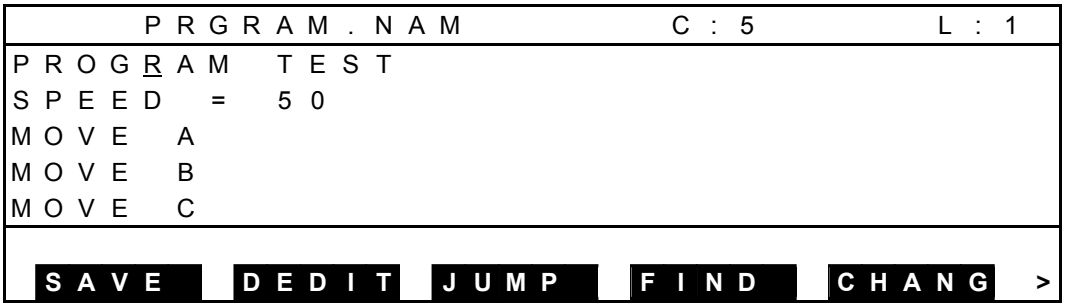
(a) Step 1: Designation of command execution position.

Move the cursor to a position on the editing display where you wish to execute an edit command, using the cursor keys (↑, ↓, ←, →).

(b) Step 2: Command selection.

Press a desired function key to select and execute a command.

When the NEXY key is pressed, other commands not currently displayed are called.



(3) Caution

For some commands, you have to specify a position where you wish to execute a command, using the cursor on the editing display, before menu selection. For details, see Para. 5.4. To return to the character input mode by cancelling the menu selection, press the ESC key.

5.3.3 Input Correction [BS]

(1) Function

At execution of an edit command, one (1) character just before the cursor position can be deleted.

(2) Procedures

(a) Step 1: Deletion of input data.

(In the example below, the jump command is being input.)

Press the BS key repeatedly, and one (1) character just before the cursor position is deleted in turn.

```
        PRGRAM.NAM              C:5          L:1
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  B
MOVE  C
  Line:  10_
 SAVE  DEDIT  JUMP  FIND  CHANG  >
```

↓ BS

```
        PRGRAM.NAM              C:5          L:1
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  B
MOVE  C
  Line:  1
 SAVE  DEDIT  JUMP  FIND  CHANG  >
```

Input a legal value through the keyboard.

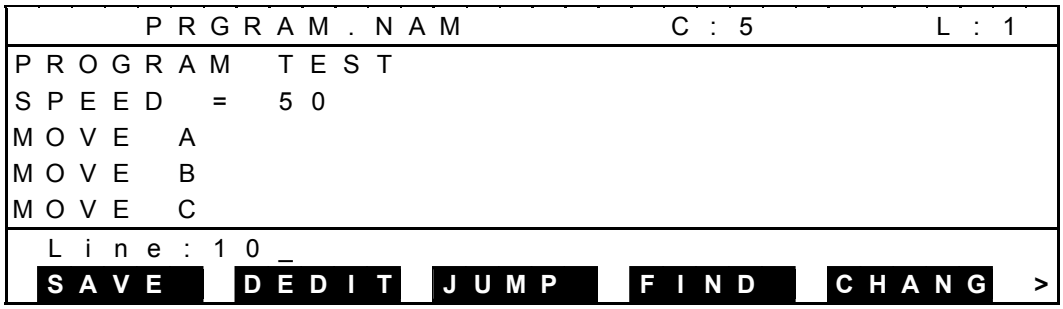
(3) Caution

The cursor keys (←, →) are inoperative.

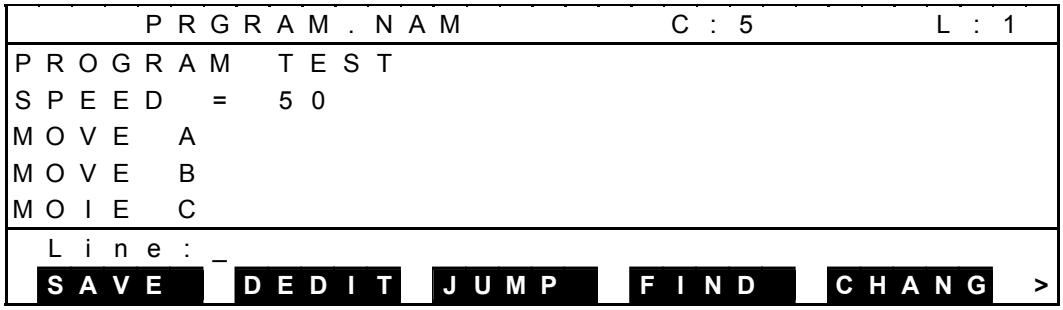
5.3.4 Cancel [ESC]

(1) Function
Clears the command line and cancels the execution of an edit command.

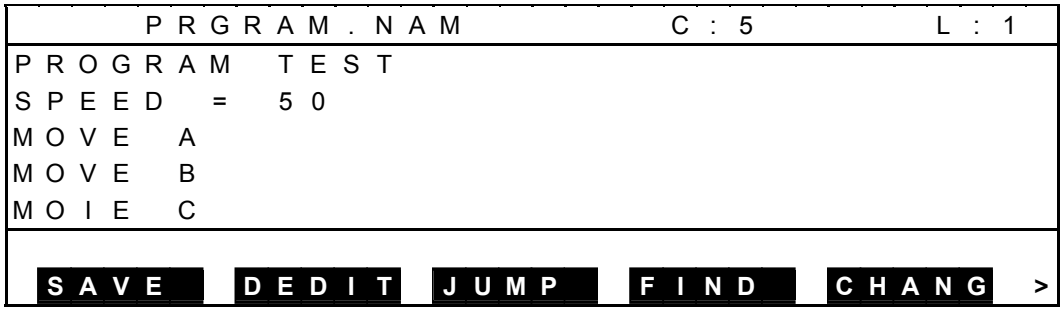
(2) Procedures
During execution of a command, press the ESC key before pressing the EXE key.



↓ ESC



↓ ESC



5.4 Descriptions on Edit Commands

5.4.1 Data Editing [DEDIT]

(1) Function
Edits data.

(2) Procedures

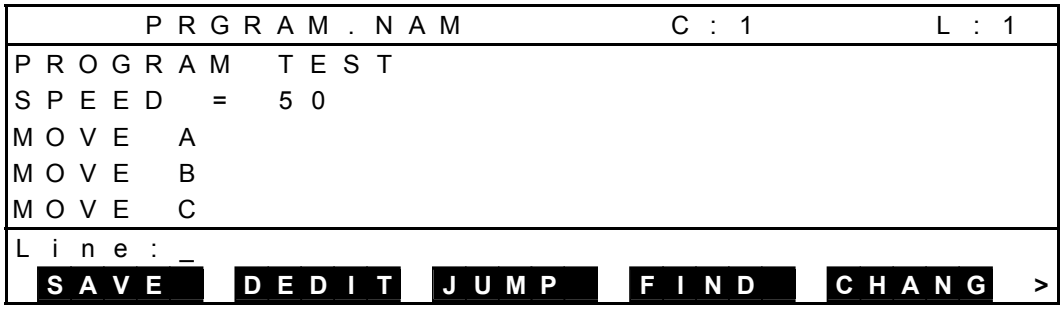
(a) Step 1: Startup of data editor.
Press [DEDIT] to start up the data editor. For details, see Section 6.

5.4.2 Jump to Specified Line [JUMP]

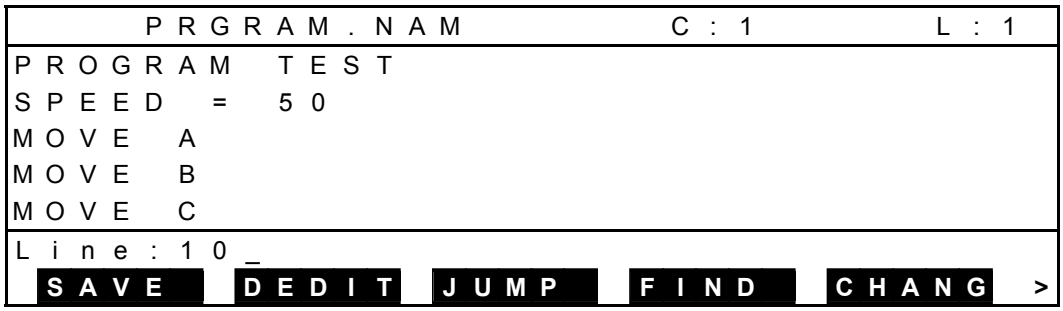
(1) Function
Calls the editing display including a specified line with the cursor moved to the top of the specified line.

(2) Procedures

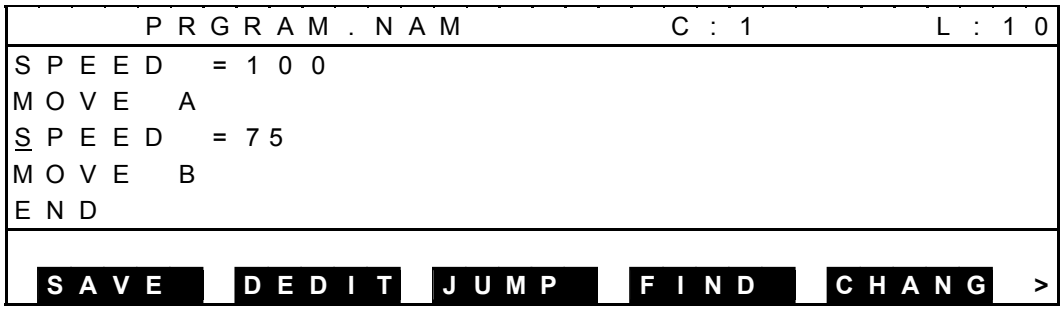
(a) Step 1: Execution of jump command.
Press [JUMP] to execute the jump command.



(b) Step 2: Input of specified line.
Make sure that "Line:" is displayed on the last line of the display, then enter a line number where you wish to jump.



(c) Step 3: Input of EXE key.
Press the EXE key, and a screen including the specified line is displayed with the cursor moved to the head of the specified line.



(3) Caution
If you have specified a line number greater than the number of the last line, the cursor moves to the head of the last line.

(4) Reference
If you specify the "+" or "–" sign, the cursor jumps to the relative position from the current line.

5.4.3 Deleting Block [CUT]

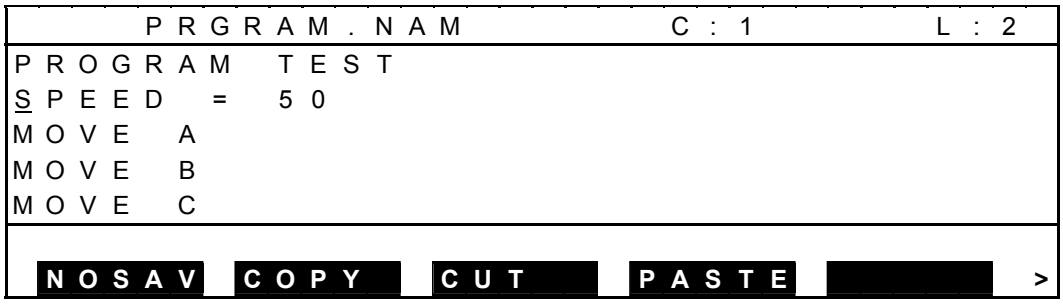
(1) Function

Deletes specified all lines (or blocks) and shifts up the subsequent lines in turn.

(2) Procedures

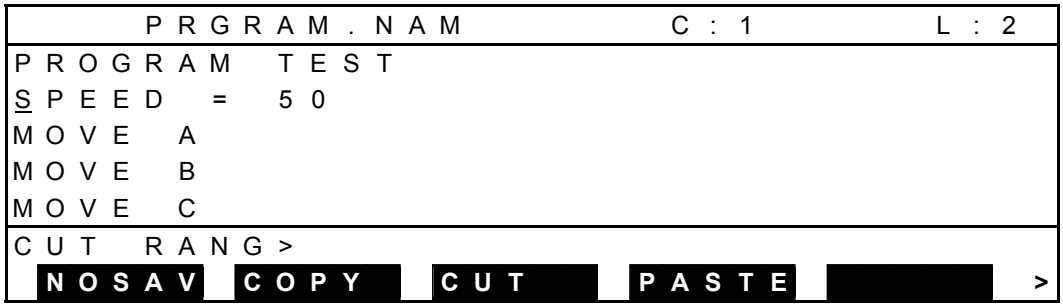
(a) Step 1: Designating the start block of a range to be deleted.

Move the cursor to the first character of a desired start block.



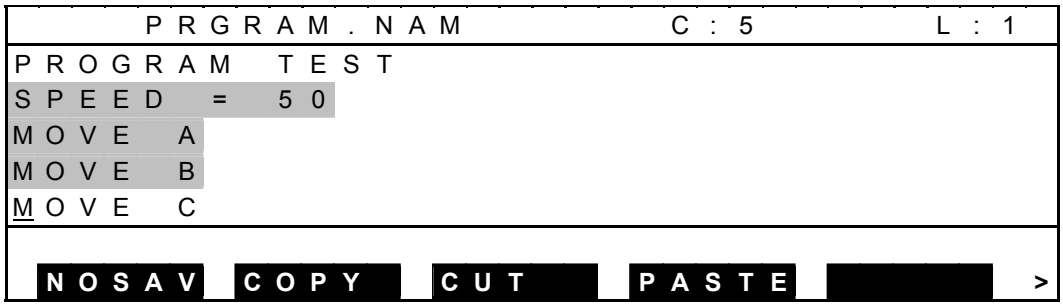
(b) Step 2: Execution of cut command.

Press [CUT] to execute the cut command.



(c) Step 3: Designation of end position.

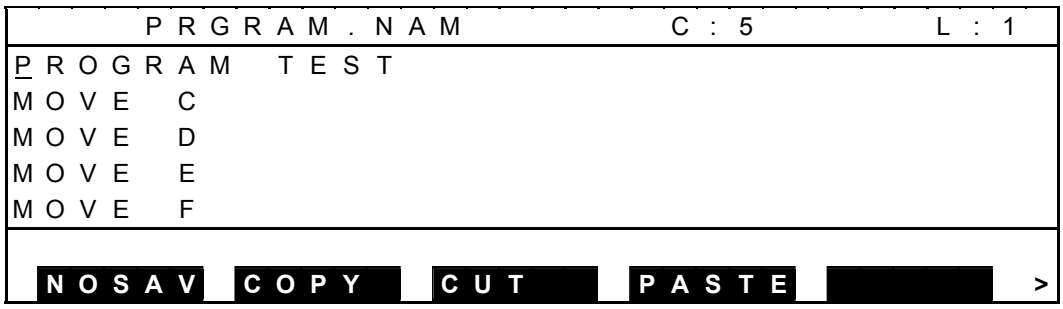
Move the cursor so that the final character of the specified range is indicated by reversed image. The specified range of blocks is inversely displayed and you can delete the blocks under this condition.



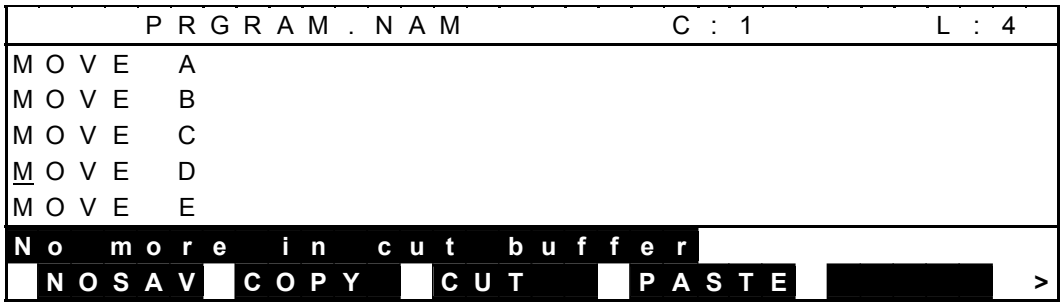
(d) Step 4: Deletion of specified range.

Press [CUT], and the contents of the specified range of blocks are saved in the buffer with the specified range of blocks deleted.

The contents thus saved can be pasted at any desired position by pressing [PASTE].



If the memory becomes full during setting, a warning saying "No more space in cut buffer" is displayed.



(3) Caution

The deleted blocks are saved in the internal buffer.

5.4.4 Copying [COPY]

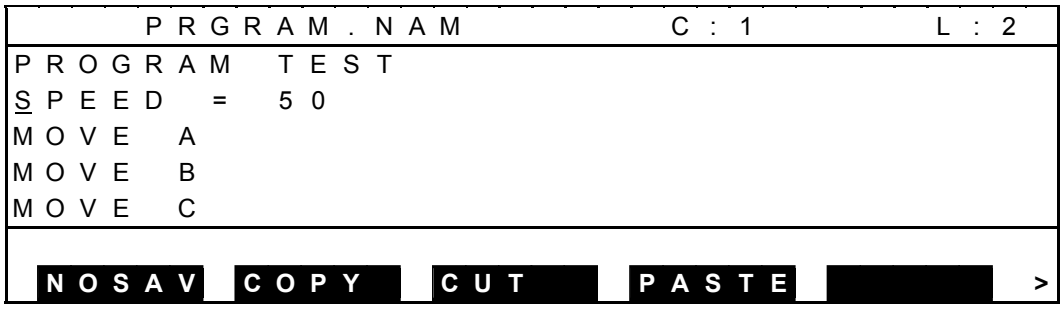
(1) Function

Saves a specified range of blocks in the buffer.

(2) Procedures

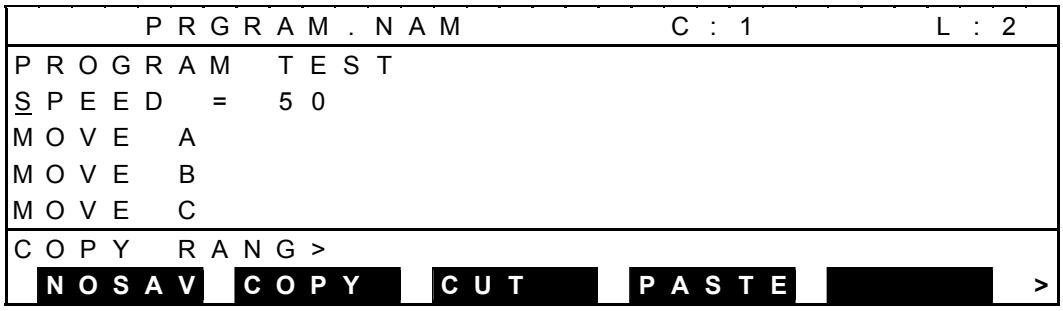
(a) Step 1: Designating the start block of a range to be deleted.

Move the cursor to the first character of a desired start block.



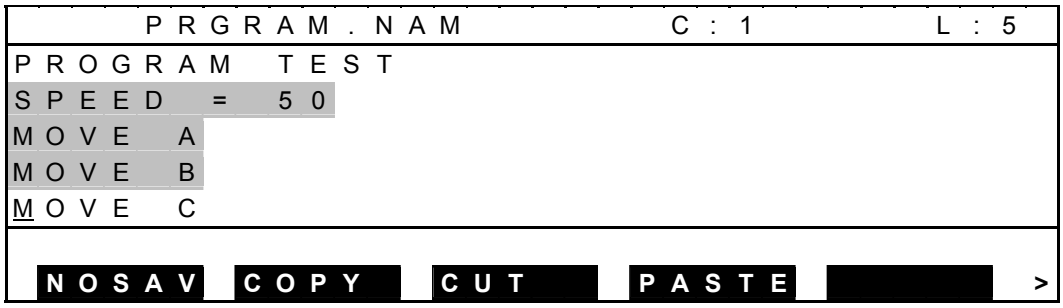
(b) Step 2: Execution of copy command.

Press [COPY to execute the copy command.

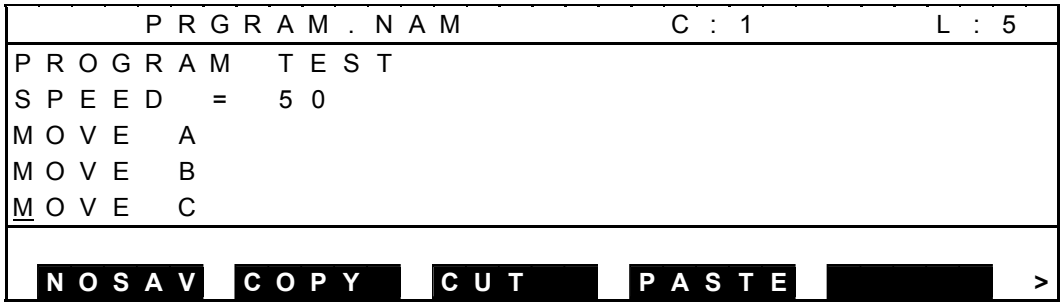


(c) Step 3: Designation of end position.

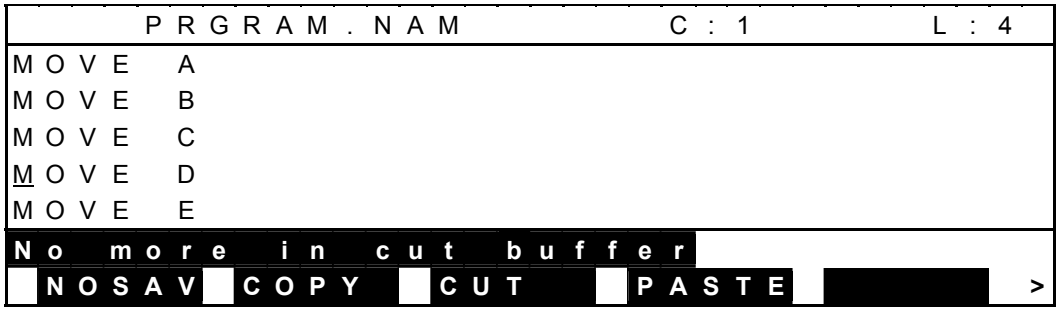
Move the cursor so that the final character of the specified range is indicated by reversed image. The specified range of blocks is inversely displayed and you can copy the blocks under this condition.



(d) Step 4: Copy of specified range.
Press [COPY], and the contents of the specified range of blocks are saved in the buffer. The contents thus saved can be pasted at any desired position by pressing [PASTE].



If the memory becomes full during copying, a warning saying "No more space in cut buffer" is displayed.



(3) Caution
If the memory becomes full during copying, the copying is not possible.

5.4.5 Pasting [PASTE]

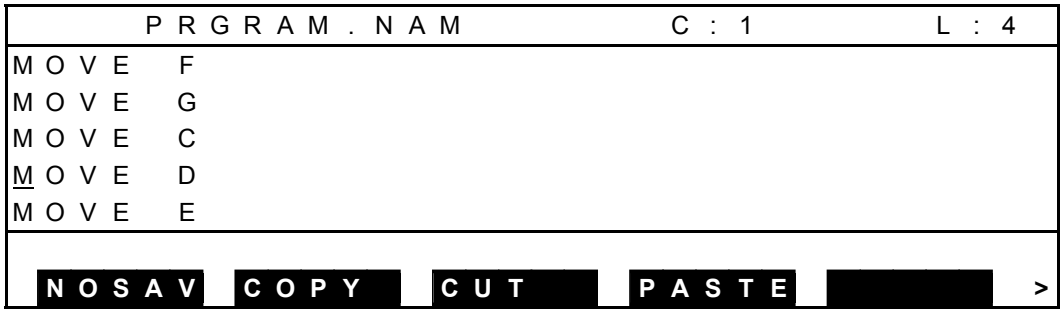
(1) Function

Pastes the contents saved by the copy or cut command.

(2) Procedures

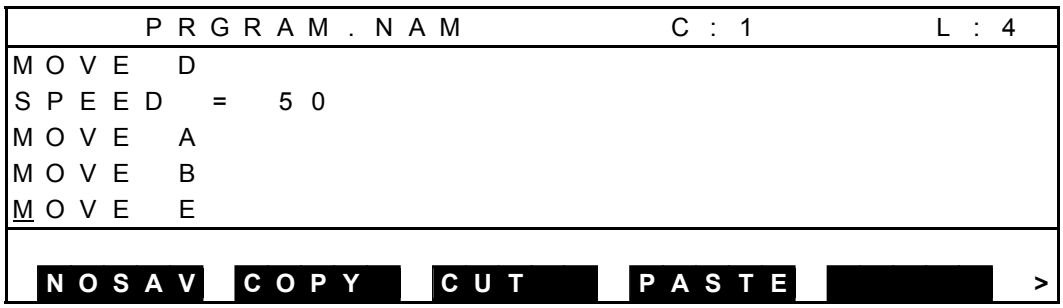
(a) Step 1: Designation of pasted position.

Move the cursor to just before the position where you wish to insert a block or blocks.

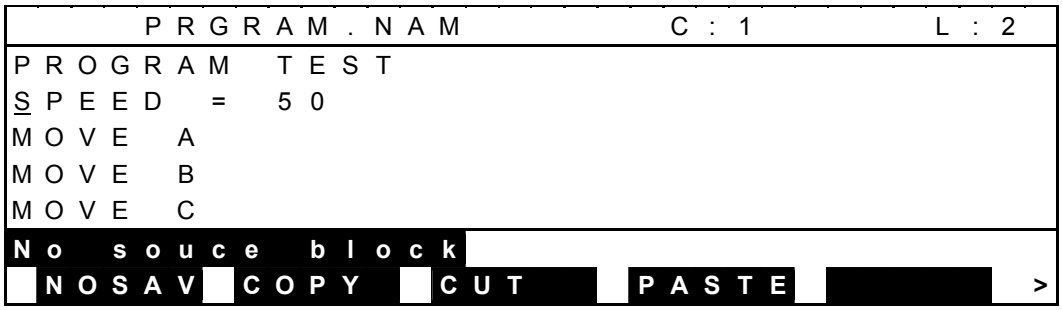


(b) Step 2: Execution of paste command.

Press [PASTE] to execute the paste command. After pasting, the cursor moves after the finally pasted block.



Unless a block or blocks are specified, copying is invalid. When this happens, a message saying "No source block" is displayed to terminate copying.



(3) Caution

Unless a block or blocks are specified in the internal buffer, pasting is invalid.

5.4.6 Searching Character String [FIND]

> **CAUTION**
>
> Two-byte characters contained in a program cannot be searched.

(1) Function

Searches a specified character string, starting from the position next to the cursor position, displays the editing screen including the initially searched character string and moves the cursor to the top of the character string.

(2) Procedures

(a) Step 1: Execution of find command.

Press [FIND] to execute the find command.

```
        PRGRAM.NAM              C:5          L:1
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  B
MOVE  C
Pattern:_
  SAVE   DEDIT   JUMP   FIND   CHANG   >
```

(b) Step 2: Input of searched character string.

Make sure that "Pattern:" is displayed on the last line of the display, then enter a character string to be searched.

```
        PRGRAM.NAM              C:5          L:1
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  B
MOVE  C
Pattern:MOVES_
  SAVE   DEDIT   JUMP   FIND   CHANG   >
```

(c) Step 3: Input of EXE key.
Press the EXE key, and the search starts. The cursor moves to the top of the character string initially found.

```
        PRGRAM.NAM          C:1          L:8
MOVE  A
SPEED  =100
MOVES  C
SPEED  =75
MOVE  D

 SAVE  DEDIT  JUMP  FIND  CHANG  >
```

Unless the specified character string exists, a message of "Not found" is displayed and the character string search terminates.

```
        PRGRAM.NAM          C:5          L:1
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  B
MOVE  C

 Not found.
 SAVE  DEDIT  JUMP  FIND  CHANG  >
```

(3) Caution
Unless the specified character string has been found until the final line, the search continues from the top line to the cursor position.

5.4.7 Replacing Character String [CHANG]

> **! CAUTION**
>
> Two-byte characters contained in a program cannot be searched.

(1) Function

Searches a specified character string, starting from the cursor position and replaces it with another character string.

(2) Procedures

(a) Step 1: Execution of change command.

Press [CHANG] to execute the change command.

```
        PRGRAM.NAM              C:5          L:1
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  B
MOVE  C
Pattern:_
 SAVE  DEDIT  JUMP  FIND  CHANG  >
```

(b) Step 2: Input of searched character string.

Make sure that "Pattern:" is displayed on the last line of the display, enter a character string to be searched and press the EXE key.

```
        PRGRAM.NAM              C:5          L:1
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  B
MOVE  C
Pattern:MOVE_
 SAVE  DEDIT  JUMP  FIND  CHANG  >
```

(c) Step 3: Input of replacing character string.
Make sure that "Replacement:" is displayed on the last line of the display, enter a replacing character string and press the EXE key.

```
        PRGRAM.NAM          C:5          L:1
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  B
MOVE  C
Replacement:MOVES_
 SAVE  DEDIT  JUMP  FIND  CHANG  >
```

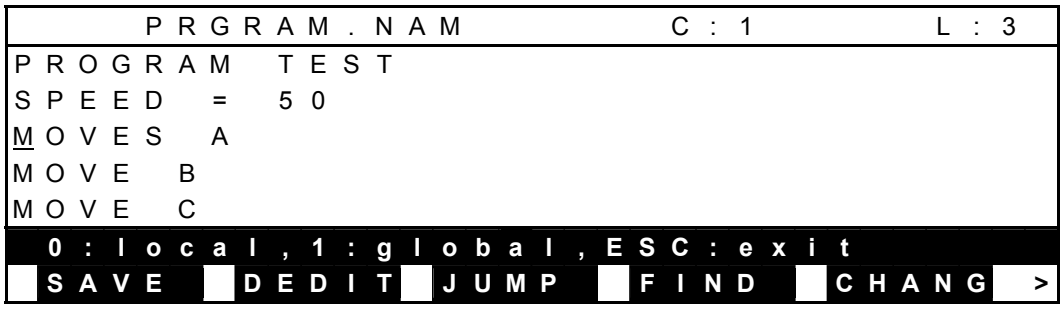
Unless the specified character string exists, a message of "Not found" is displayed and the character string change is cancelled.

```
        PRGRAM.NAM          C:5          L:1
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  B
MOVE  C
 Not found.
 SAVE  DEDIT  JUMP  FIND  CHANG  >
```

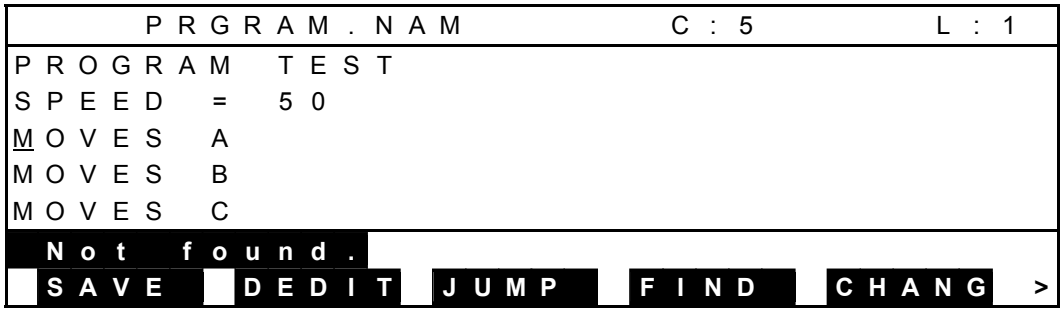
(d) Step 4: Confirmation of replacement.
The cursor moves to the top of the character string initially found with the following message displayed.

```
        PRGRAM.NAM          C:1          L:3
PROGRAM  TEST
SPEED  =  50
MOVE  A
MOVE  B
MOVE  C
 0:local,1:global,ESC:exit
 SAVE  DEDIT  JUMP  FIND  CHANG  >
```

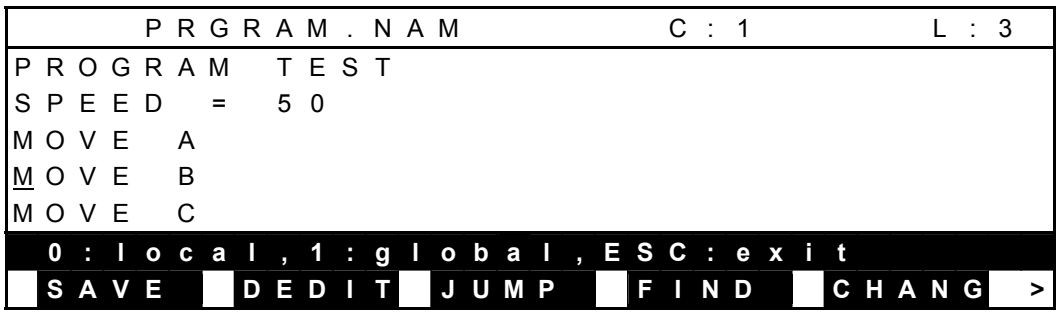
When the [0] key is pressed, the character string at the cursor position is replaced with another character string. The cursor moves to the top of the character string found next with the confirmation message displayed again.



When the [1] key is pressed, all character strings identical with the specified character string are replaced with the new character string. If the specified character string does not exist any more, a message of "Not found" is displayed to cancel the character string replacement.



Press the ESC key to quit the replacement.
If any other key is pressed, the system skips the character string at the cursor position and moves the cursor to the top of the character string found next with the confirmation message displayed again.



(3) Caution

Unless the specified character string has been found until the final line, the search continues from the top line to the cursor position.

If the memory becomes full at character string replacement, the character string replacement is invalid. When this happens, a message of "File full" is displayed to cancel the character string replacement.

5.4.8 Quit [SAVE]

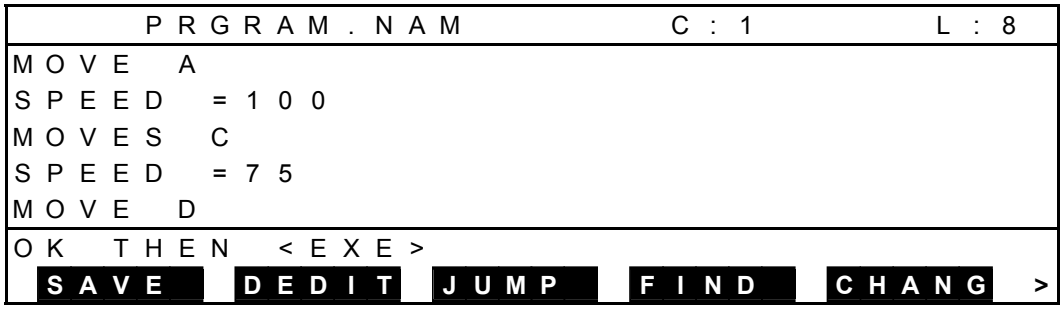
(1) Function

Saves a program and exits the editor.

(2) Procedures

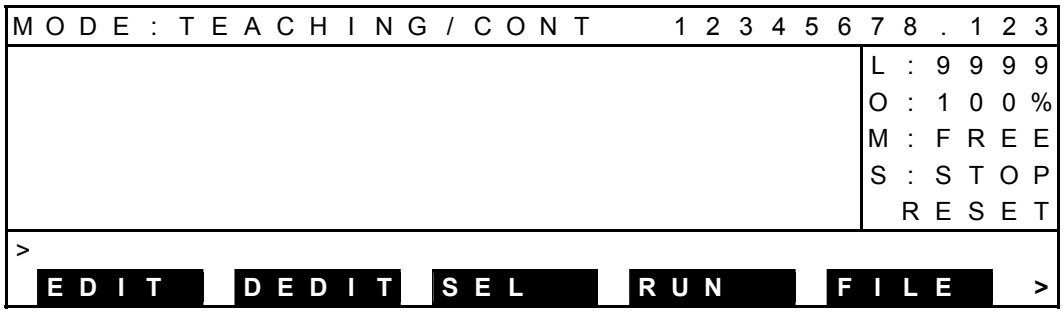
(a) Step 1: Execution of save command.

Press [SAVE] to execute the save command. The confirmation message is displayed.



(b) Step 2:

Press the EXE key, and the program is saved and the system returns to the teaching mode display.



To cancel the save function, press the ESC key, and you can continue editing.

5.4.9 Aborting and Quitting Edit [NOSAV]

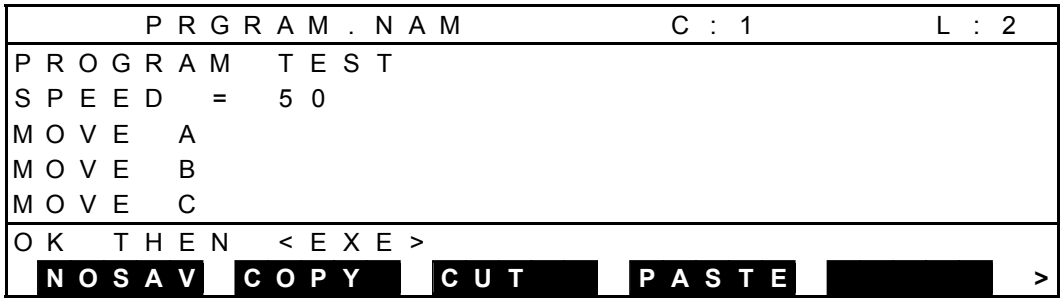
(1) Function

Aborts and quit the editor.

(2) Procedures

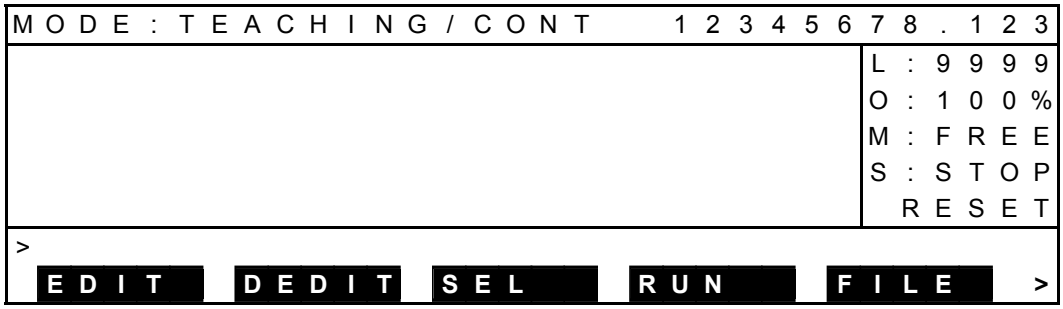
(a) Step 1: Execution of "no save" command.

Press [NOSAV] to execute the "no save" command. The confirmation message is displayed.



(b) Step 2:

Press the EXE key, and editing of the program is aborted and the system returns to the teaching mode display.



To cancel the "no save" function, press the ESC key, and you can continue editing.

(3) Caution

The contents edited are cancelled to keep the original program.

If a data format error occurs at the start of editing, a backup file is not created.

Even if the "no save" command is executed, the original file cannot be maintained.

Section 6 Data Editing

The data editing function allows input and editing of position data, coordinate data and load data. Each data is saved in a file together with a program, and can be edited on the file basis.

6.1 Starting Up the Data Editor

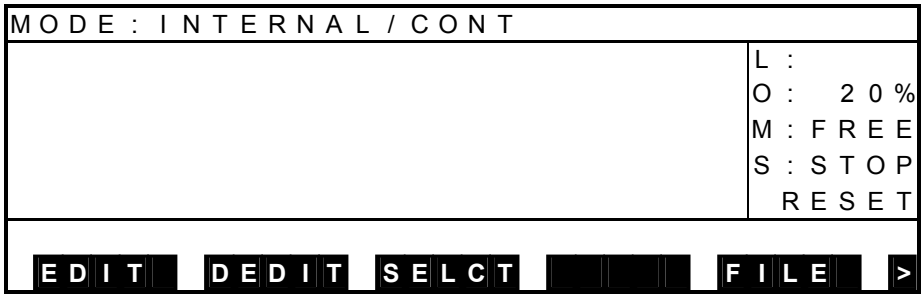
The data editor can be started up in one of the following two (2) manners.

1. [DEDIT] of basic menu
2. [DEDIT] of program editing menu

For the startup procedures from the program editing menu, see Section 5. This section describes the startup procedures from the basic menu.

(1) Procedures

1. Call the basic display as shown below.

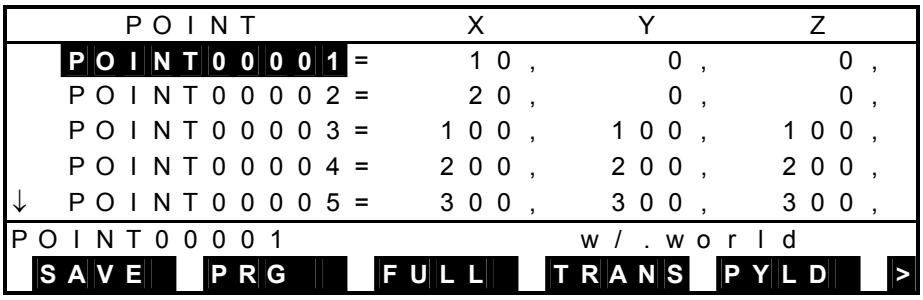


2. Press [DEDIT], and the following file select display appears.

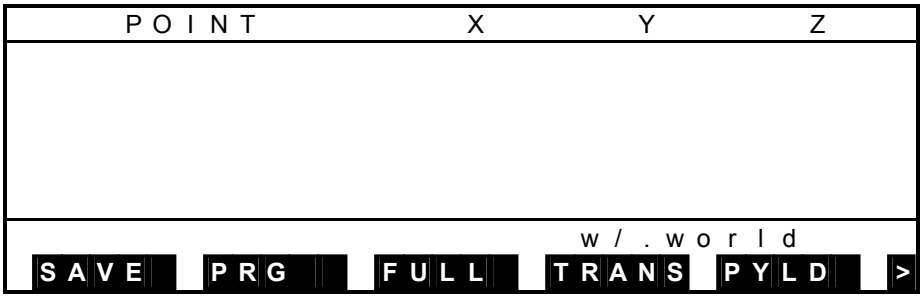


For details of the file select display, see Section 5.

3. Select or input a file you are going to edit, and press the EXE key. The data editor starts up. When the data editor has started up initially, the following world coordinate position data directory is displayed.



For the new file, no data is available, and only the title is displayed as shown below.



- New data can be created, using the INS key. For details, see Para. 6.2.3.
- When the data editor has been started up by [DEDIT] of the basic menu, the [PRG] menu as shown above is not displayed. It is displayed only when the data editor has been started up by [DEDIT] of the program editing menu. (See Caution below.)

(2) Caution

Though some menu keys are not displayed unless they work, all menu keys are displayed for convenience of explanations.

6.2 Position Data Directory Display Mode

The position data directory display mode displays one (1) position data of each coordinate system on the same line.
Data displayed immediately after the startup are those in the world coordinate system. In this mode, it is possible to edit the position data.
An example of display is given below.

```
      POINT              X          Y          Z
↑   POINT00003 =     100 ,      100 ,      100 ,
    POINT00004 =     200 ,      200 ,      200 ,
    POINT00005 =     300 ,      300 ,      300 ,
    POINT00006 =     400 ,      400 ,      400 ,
↓   POINT00007 =     500 ,      500 ,      500 ,
POINT00005                          w/ .world
  SAVE   PRG    FULL   TRANS  PYLD    >
```

Each time the NEXT key is pressed, the menu is changed over as follows:

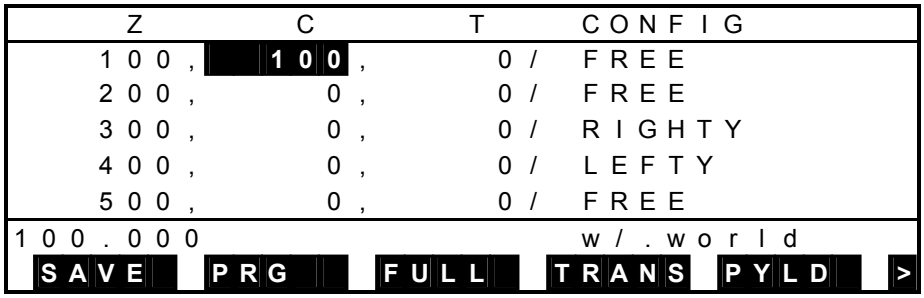
```
< NOSAV  FIND   TEACH  M-TC   CUT     >

< PASTE                                 
```

- An inversely displayed section in the data is called the "main cursor" which indicates that this data is selected for editing.
- Symbol "↑" shown in the 1st column signifies that there is position data before the top data.
- Symbol "↓" shown in the 1st column signifies that there is position data after the final data.
- C and T data, and configuration data are displayed when you move the main cursor to the right.
- Data following "w/" on the right side of the command line identifies the coordinate data to which the selected position data belongs.
- Though the name of array position data, etc., can consist of up to twenty (20) characters, only ten (10) characters from the top are displayed in this mode. For the data indicated by the main cursor, however, all data are displayed on the command line.

- For the X, Y, Z, C and T data, a value of four (4) digits and three (3) digits above and below the decimal point, respectively, can be input. In this mode, however, only a value of four (4) digits above the decimal point is displayed. For the data indicted by the main cursor, a full value is displayed, including decimal fractions.

When you have moved the main cursor to the right and the cursor is located at the C data, the display is as shown below.



Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[←]	Used to move the main cursor to the left. If the main cursor is located under the name column, it will not move any further. When the cursor has moved from the Z data column to Y-data column while the C data and after were displayed, the screen scrolls right.
[→]	Used to move the main cursor to the right If the main cursor is located under the "CONFIG" column, it will not move any further. When the cursor has moved from the Z data column to C-data column, the screen scrolls left.
[↑]	Used to move up the main cursor. If the main cursor is located on the top line of screen, the screen scrolls down. If the main cursor is on the top data, it will not move any further.
[↓]	Used to move down the main cursor. If the main cursor is located on the last line of screen, the screen scrolls up. If the main cursor is on the final data, it will not move any further.
[ALT]+[←]	Used to display a position data directory in a coordinate system defined before.
[ALT]+[→]	Used to display a position data directory in a coordinate system defined after.
[ALT]+[↑]	Used to move up the main cursor on the screen basis.
[ALT]+[↓]	Used to move down the main cursor on the screen basis.
[INS]	Used to create a new position data just after the main cursor line.
[ALT]+[BS]	Used to delete a position data indicated by the main cursor.
[EXE]	Used to edit a data indicated by the main cursor.
[NEXT]	Used to select the next menu.
[ALT]+[NEXT]	Used to select the previous menu.

Operations of respective soft keys are shown below.

Key	Descriptions
[SAVE]	Used to save a data and quit the data editor. (When the data editor was started up from the program editor, the program editor terminates also.)
[PRG]	Used to select the program editor display. If the data editor was started up from the basic menu, however, the display will not change over.
[FULL]	Used to select the detailed position data display. For details, see Para. 6.3.
[TRANS]	Used to select the coordinate data editor display. For details, see Para. 6.4.
[PYLD]	Used to select the load data editor display. For details, see Para. 6.6.
[NOSAV]	Used to quit the editor without saving data. All changed data are aborted. (When the data editor was started up from the program editor, the program editor terminates also.)
[FIND]	Used to search a position name. For details, see Para. 6.2.4.
[TEACH]	Used to register a current position of the robot in a position data indicated by the main cursor. For details, see Para. 6.2.5.
[M–TO]	Used to move the robot to a position indicated by the main cursor. For details, see Para. 6.2.6.
[CUT]	Used to select the range select mode. For details, see Para. 6.2.7.
[PASTE]	Used to paste the contents saved in the cut buffer (i.e., data deleted by the previous cut menu) after a position data indicated by the main cursor to clear the cut buffer. For details, see Para. 6.2.8.

For details of each soft key, see the descriptions given later.

6.2.1 Editing of Position Data Name

(1) Function

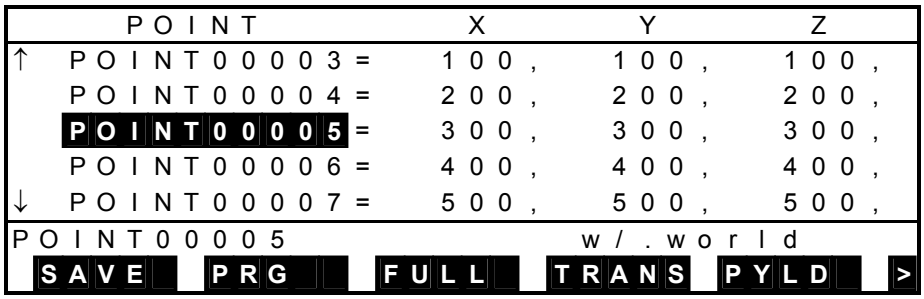
Edits a position data name.

(2) Procedures

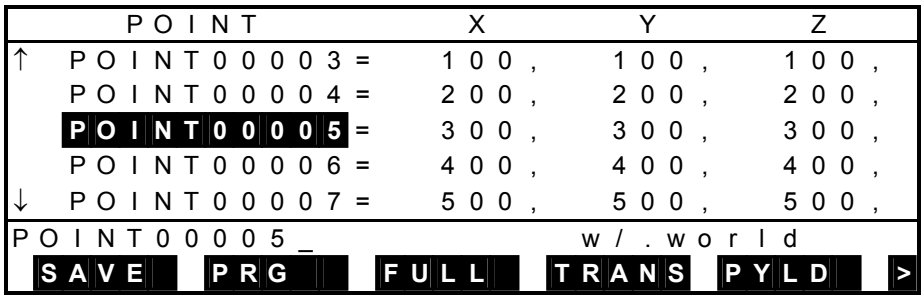
When you press the EXE key while the main cursor is located at the position name, you can edit the position name.

In the example below, assume that you are going to change the name of "POINT00005" to "POINT55555."

1. Move the main cursor to "POINT00005" by means of the cursor keys (↑, ↓, →, ←), and the following display appears.

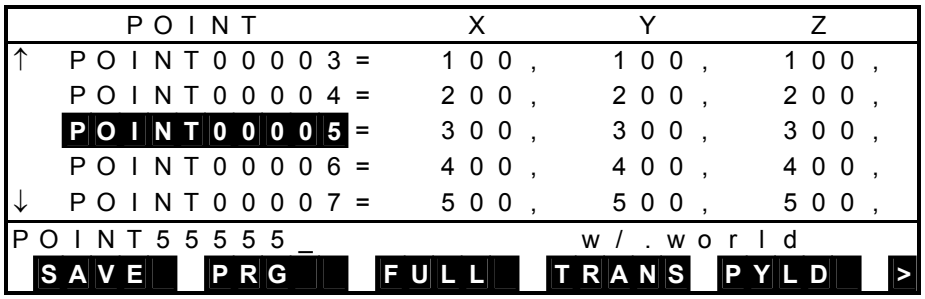


2. Press the EXE key, and the following display is called. (The cursor is displayed on the command line.)



- Current data is displayed as the initial value on the command line.
- The cursor is displayed just after the data indicated as the initial value, and you can edit the data now.
- When the cursor is shown by mark "■", the replace mode takes effect. When the cursor is shown by mark "_", the insert mode is effective.

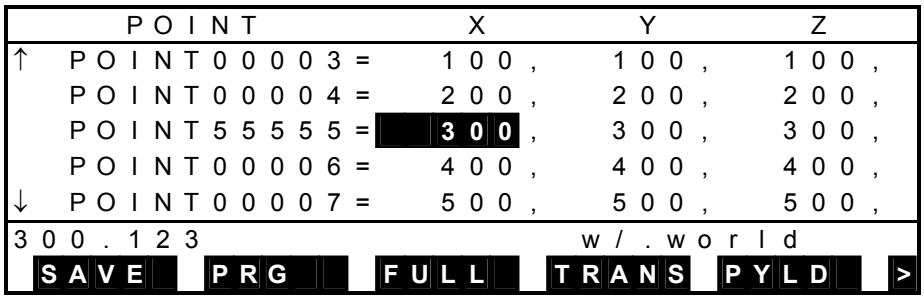
3. Enter a desired name to the command line, using alphanumeric keys, →, ← cursor keys. Then the following display appears.



Keys operative in the command line input mode and their operations are tabled below.

Key	Descriptions
←	Used to move left the cursor on the command line.
→	Used to move right the cursor on the command line.
Numeral Alphabets	Used to input characters to the command line.
EXE	Used to change current position name to an entered one.
INS	Used to select the insert or replace mode for the command line.
BS	Used to delete one (1) character just before the cursor position on the command line.
ALT + BS	Used to delete one (1) character at the cursor position on the command line.
ESC	This function works in a different manner according to the presence or absence of input character(s). Presence of input character(s): All input character(s) on the command line are cleared. Absence of input character(s): The system returns to the position data directory display.

4. Press the EXE key, and the position name is replaced by the newly specified name. The following display appears.



- The main cursor moves to the X data area.
- If the input name is illegal, a message saying "Illegal name" is displayed on the command line and no data is changed. Press any key, and the system returns to the name editing mode.
- If the specified name already exists, a message of "Duplicate name" is displayed on the command line and no data is changed. Press any key, and the system returns to the name editing mode.

6.2.2 Editing of Position Element

(1) Function

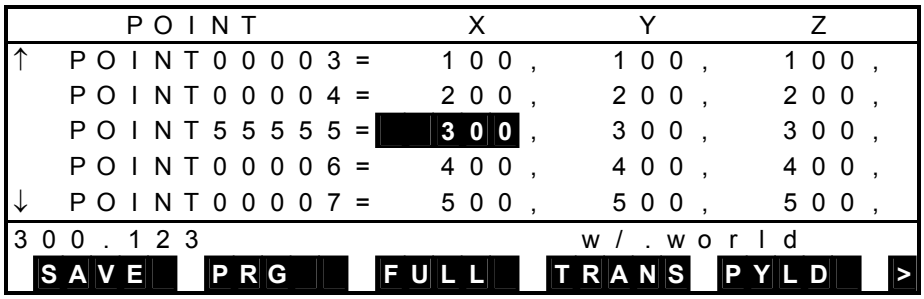
Used to edit a position element.

(2) Procedures

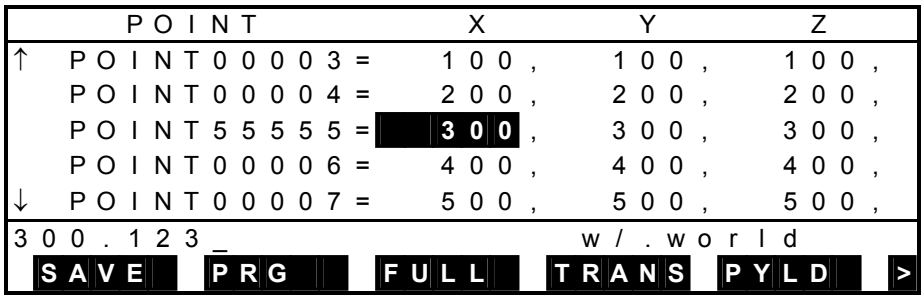
When you press the EXE key while the main cursor is located at each position element, you can edit that data.

In the example below, assume that you are going to change the X data of "POINT00005" from "300.123" to "333."

1. Move the main cursor to the data you wish to edit by means of the cursor keys (↑, ↓, →, ←), and the following display appears.

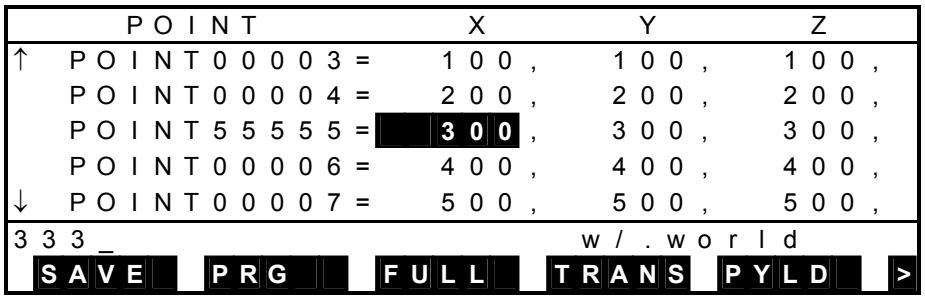


2. Press the EXE key, and the following display is called. (The cursor is displayed on the command line.)



- Current data is displayed as the initial value on the command line.
- The cursor is displayed just after the data indicated as the initial value, and you can edit the data now.
- When the cursor is shown by mark "■", the replace mode takes effect. When the cursor is shown by mark "_", the insert mode is effective.

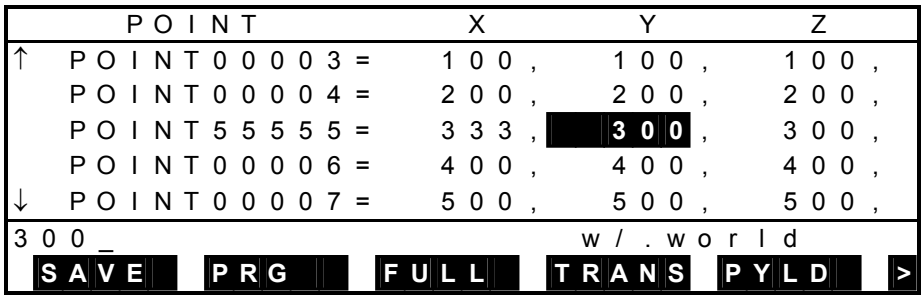
3. Enter a desired value to the command line, using alphanumeric keys, →, ← cursor keys. Then the following display appears.



Keys operative in the command line input mode and their operations are tabled below.

Key	Descriptions
[←]	Used to move left the cursor on the command line.
[→]	Used to move right the cursor on the command line.
[Numeral] [.] [+] [–]	Used to input characters to the command line.
[EXE]	Used to change current position element to an entered one. The system returns to the position directory display mode with the main cursor moved to the next data column.
[INS]	Used to select the insert or replace mode for the command line.
[BS]	Used to delete one (1) character just before the cursor position on the command line.
[ALT]+[BS]	Used to delete one (1) character at the cursor position on the command line.
[ESC]	This function works in a different manner according to the presence or absence of input character(s). Presence of input character(s): All input character(s) on the command line are cleared. Absence of input character(s): The system returns to the position data directory display.

4. Press the EXE key, and the position element is replaced by the newly specified data. The system returns to the position data directory display mode with the main cursor moved to the next data column. The following display appears.



- The main cursor moves to the next data column. When you have edited the CONFIG data, however, the cursor will not move.
- If the input data is illegal, a message saying "Illegal data" is displayed on the command line and no data is changed. Press any key, and the system returns to the element editing mode.

(3) Caution

When editing the CONFIG data, all you have to do is enter only one (1) top character. The relationship between input character and value to be set is as follows:

Input character	Value to be set
F or 0	FREE
R or 1	RIGHTY
L or 2	LEFTY

6.2.3 Addition of New Position Data [INS]

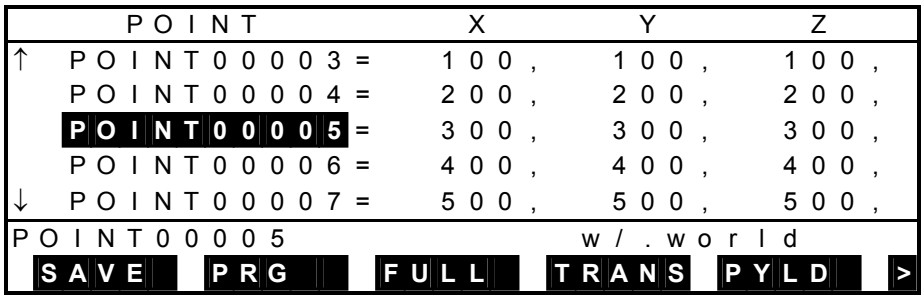
(1) Function

Adds a new position data.

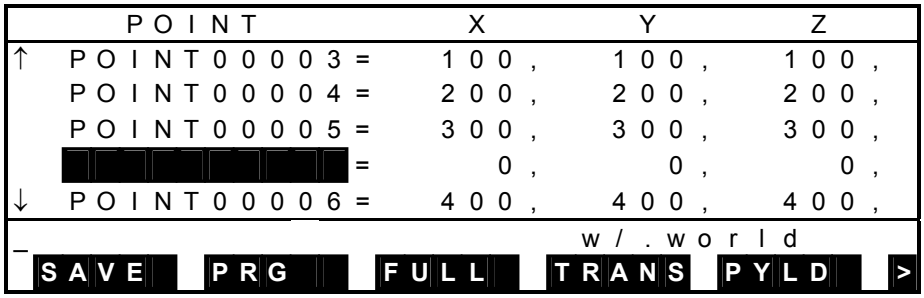
(2) Procedures

Assume that you are going to add a new position data ("POINT00011") just after "POINT00005."

1. Move the main cursor to "POINT00005", and the following display is called.



2. Press the INS key, and the display changes as shown below with the name input mode selected. (The cursor is displayed on the command line.)



3. Enter a desired name ("POINT00011" in this example) and press the EXE key, as already described in Para. 6.2.1. The display changes as follows:

```
      POINT          X          Y          Z
↑  POINT00003=    100,       100,       100,
   POINT00004=    200,       200,       200,
   POINT00005=    300,       300,       300,
   POINT00011=      0,         0,         0,
↓  POINT00006=    400,       400,       400,
0                                  w/.world
 SAVE  PRG  FULL  TRANS  PYLD  >
```

(3) Caution

- For the newly added position data, zero (0) is set for all elements.
- When you have pressed the ESC key without specifying the name, addition of the new data is cancelled and the system returns.

6.2.4 Deletion of Position Data ([ALT] + [BS])

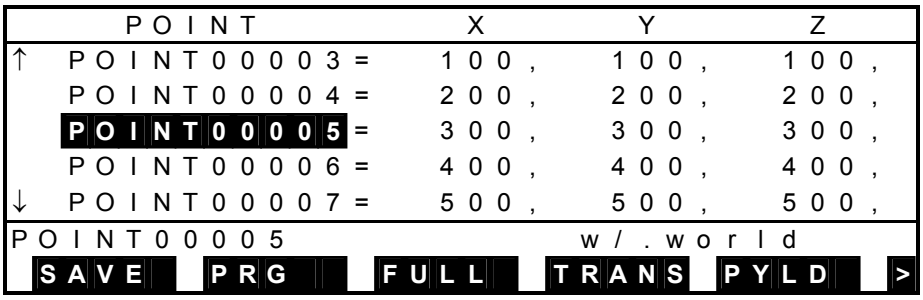
(1) Function

Deletes a position data.

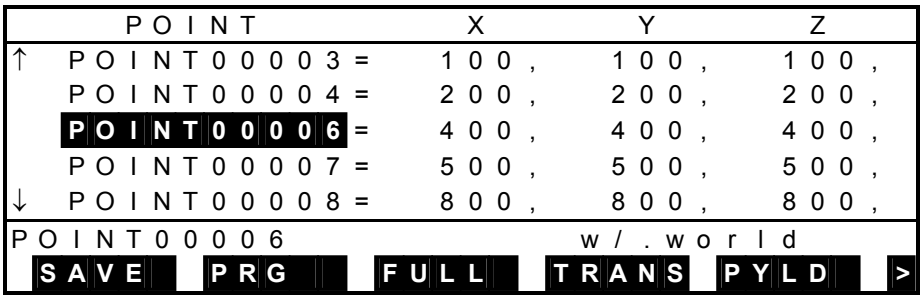
(2) Procedures

Assume that you are going to delete the data of "POINT00005".

1. Move the main cursor to "POINT00005", and the following display is called.



2. Press the ALT and BS keys at the same time, and the following display appears with the specified data deleted.



(3) Caution

Different from the [CUT] function, you cannot undo a data deleted by this operation.

6.2.5 Quit of Data Editing [SAVE]

(1) Function

Saves an edited data in a file and quits editing data.

(2) Procedures

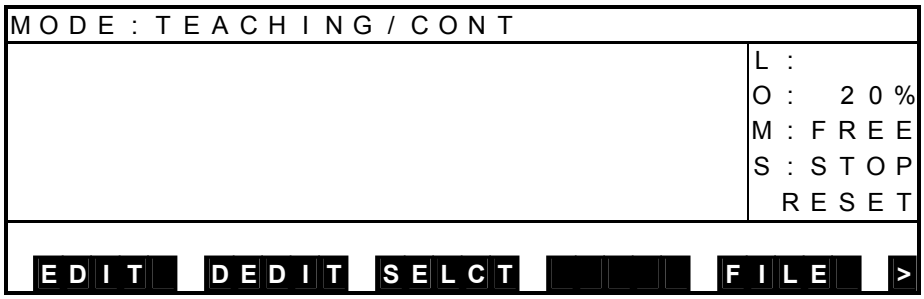
1. Select the data display mode as shown below (i.e., the cursor is not present on the command line).

```
      POINT              X          Y          Z
↑   POINT00003 =     100 ,      100 ,      100 ,
    POINT00004 =     200 ,      200 ,      200 ,
    POINT00005 =     300 ,      300 ,      300 ,
    POINT00006 =     400 ,      400 ,      400 ,
↓   POINT00007 =     500 ,      500 ,      500 ,
POINT00005                          w/.world
 SAVE  PRG  FULL  TRANS  PYLD  >
```

2. Press [SAVE], and the system asks with a confirmation message if you wish to save the edited data.

```
      POINT              X          Y          Z
↑   POINT00003 =     100 ,      100 ,      100 ,
    POINT00004 =     200 ,      200 ,      200 ,
    POINT00005 =     300 ,      300 ,      300 ,
    POINT00006 =     400 ,      400 ,      400 ,
↓   POINT00007 =     500 ,      500 ,      500 ,
OK  THEN <EXE>                      w/.world
 SAVE  PRG  FULL  TRANS  PYLD  >
```

3. Press the EXE key, and the system returns to the basic display as shown below.



To cancel this function, press the ESC key. You can continue editing then.

(3) Caution

- When the data editor was started up from the program editor, the latter editor (program editor) also terminates after saving the program in the file.
- In the data change mode (i.e., state where the cursor is on the command line), the keys may be inoperative even if pressed. When this happens, select the data display mode first, then press the keys.

6.2.6 Abortion and Quit of Data Editing [NOSAV]

(1) Function

Aborts an edited data without saving it in a file and quits editing data.

(2) Procedures

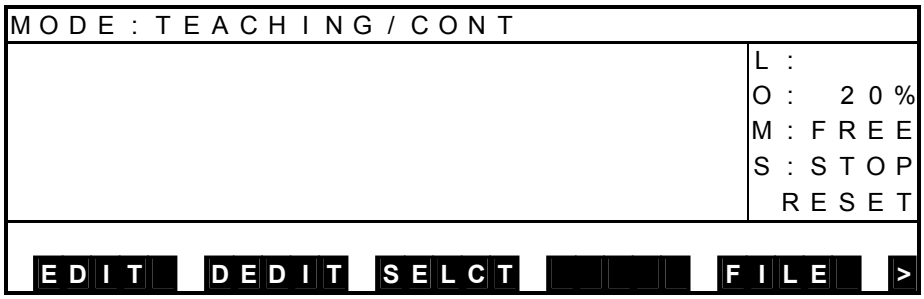
1. Select the data display mode as shown below (i.e., the cursor is not present on the command line), and press the NEXT key repeatedly until [NOSAV] is displayed in the soft key menu.

```
      POINT          X        Y        Z
↑  POINT00003 =   100 ,    100 ,    100 ,
   POINT00004 =   200 ,    200 ,    200 ,
   POINT00005 =   300 ,    300 ,    300 ,
   POINT00006 =   400 ,    400 ,    400 ,
↓  POINT00007 =   500 ,    500 ,    500 ,
POINT00005                  w/.world
<NOSAV FIND  TEACH M-TO  CUT   >
```

2. Press [NOSAV], and the system asks with a confirmation message if you wish to abort the edited data.

```
      POINT          X        Y        Z
↑  POINT00003 =   100 ,    100 ,    100 ,
   POINT00004 =   200 ,    200 ,    200 ,
   POINT00005 =   300 ,    300 ,    300 ,
   POINT00006 =   400 ,    400 ,    400 ,
↓  POINT00007 =   500 ,    500 ,    500 ,
OK THEN <EXE>               w/.world
<NOSAV FIND  TEACH M-TO  CUT   >
```

3. Press the EXE key, and the system returns to the basic display as shown below.



To cancel this function, press the ESC key. You can continue editing then.

(3) Caution

- When the data editor was started up from the program editor, the latter editor (program editor) also terminates without saving the program in the file.
- In the data change mode (i.e., state where the cursor is on the command line), the keys may be inoperative even if pressed. When this happens, select the data display mode first, then press the keys.

6.2.7 Return to Program Editor [PRG]

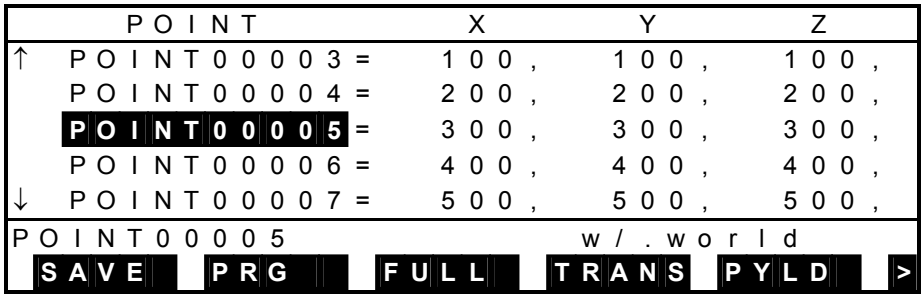
(1) Function

When the data editor was started from the program editor, the system returns to the program editing mode. At this point of time, edited data is not saved in the file. (The edited data remains.)

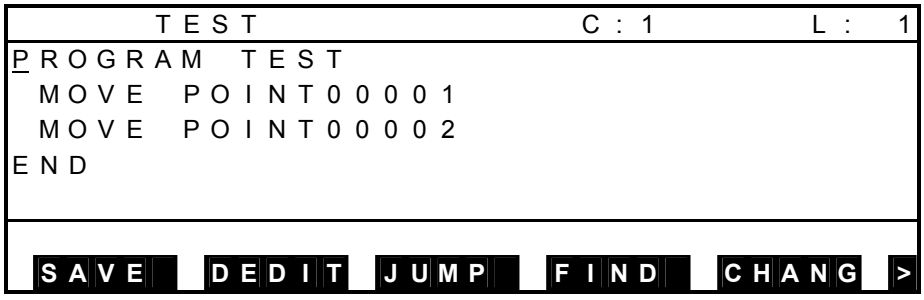
If the data editor was started from the basic menu, this function is inoperative.

(2) Procedures

1. Select the data display mode as shown below (i.e., the cursor is not present on the command line).



2. Press [PRG], and the system returns to the following program editing display.



(3) Caution

- When the data editor was started up from the basic menu, this function becomes invalid.
- In the data change mode (i.e., state where the cursor is on the command line), the keys may be inoperative even if pressed. When this happens, select the data display mode first, then press the keys.

6.2.8 Call of Detailed Position Data Display [FULL]

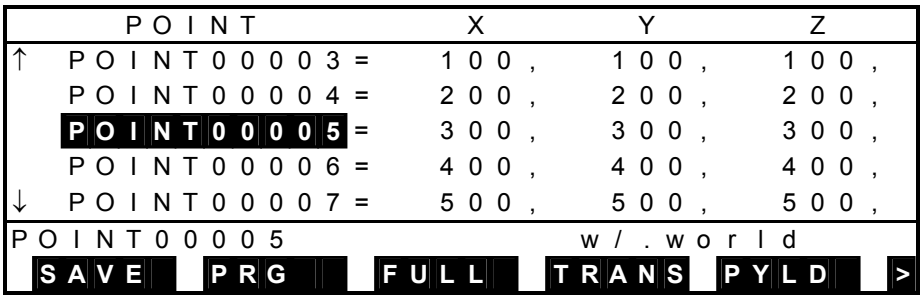
(1) Function

Displays the detailed position data display.

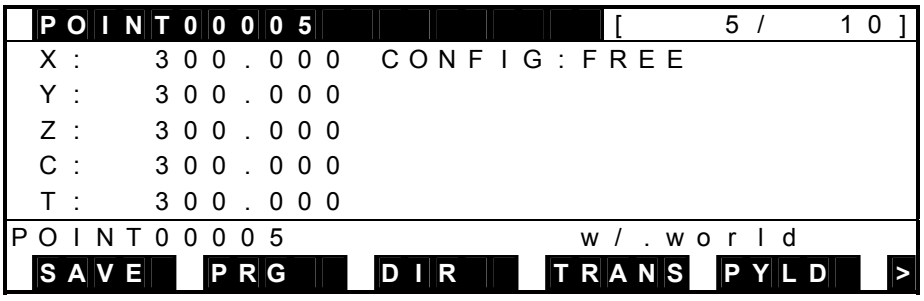
(2) Procedures

Assume that you are going to display the data of "POINT00005" in detail.

1. Move the main cursor to "POINT00005", and the following display appears.



2. Press [FULL] and the following detailed display is called.



(3) Caution

Even if the main cursor is located under the column of a data element you wish to display in detail, the detailed display appears as well.

For details of operations to be done on the detailed display, see Para. 6.3.

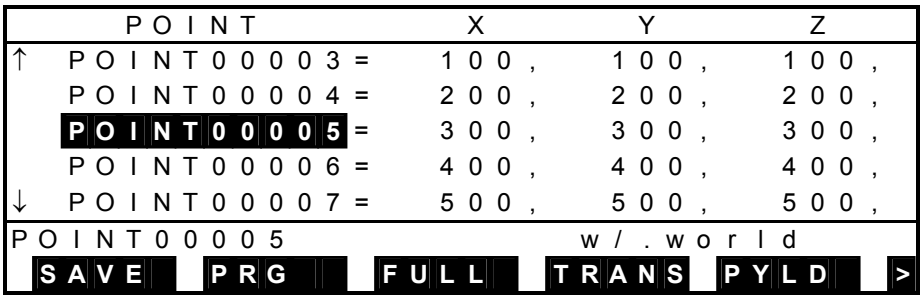
6.2.9 Call of Coordinate Data Directory Display [TRANS]

(1) Function

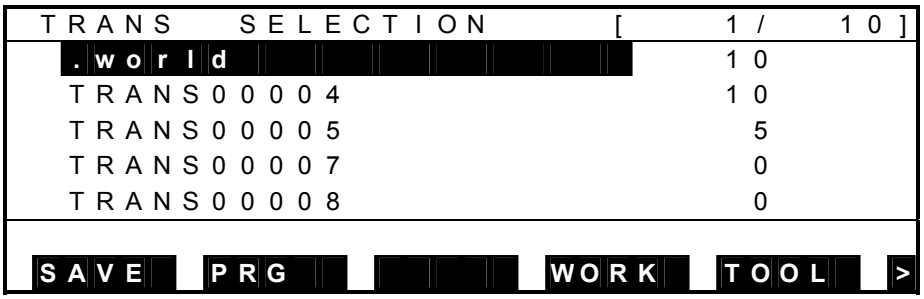
Calls the coordinate data directory display.

(2) Procedures

1. Select the data display mode as shown below (i.e., the cursor is not present on the command line).



2. Press [TRANS], and the following coordinate data directory display appears.



(3) Caution

On the coordinate data directory display, the main cursor indicates a coordinate system to which the position data belongs.

For details of operations to be done on the coordinate data display, see Para. 6.4.

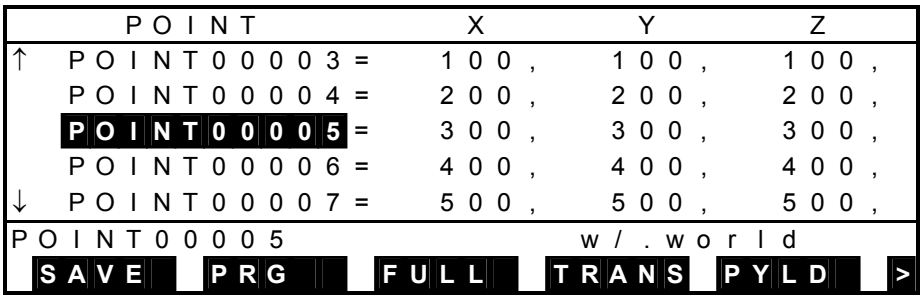
6.2.10 Call of Load Data Directory Display [PYLD]

(1) Function

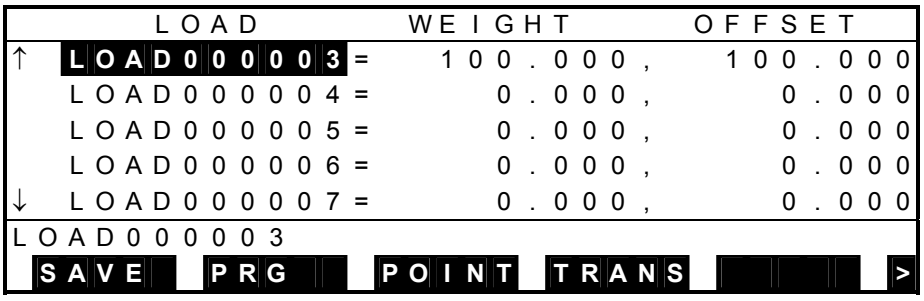
Calls the load data directory display.

(2) Procedures

1. Select the data display mode as shown below (i.e., the cursor is not present on the command line).



2. Press [PYLD], and the following load data directory display appears.



(3) Caution

On the load data directory display, the main cursor indicates the position of the data displayed before.

For details of operations to be done on the load data display, see Para. 6.6.

6.2.11 Search of Position Data [FIND]

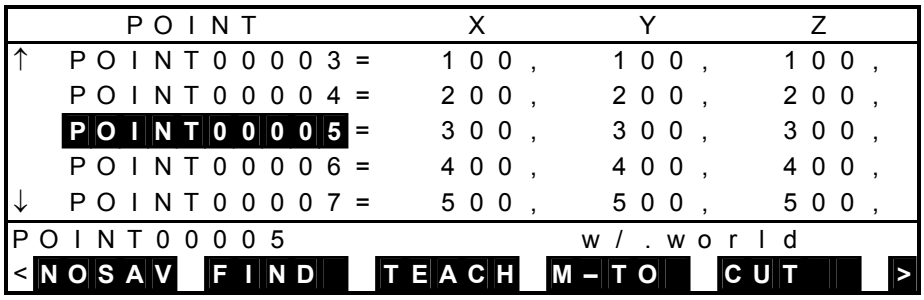
(1) Function

Searches the data bearing the specified position name.

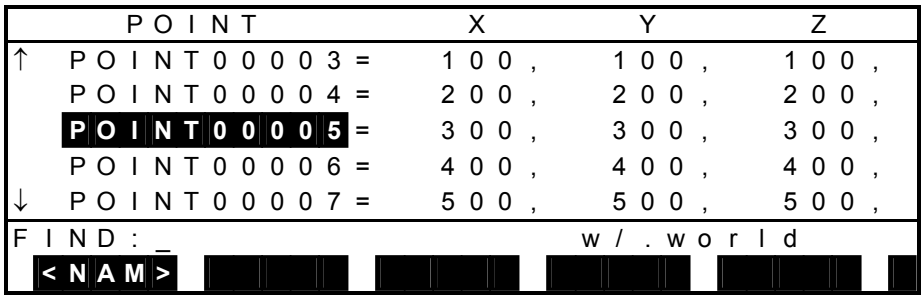
(2) Procedures

Assume that you are going to search the position data named "POINT00005."

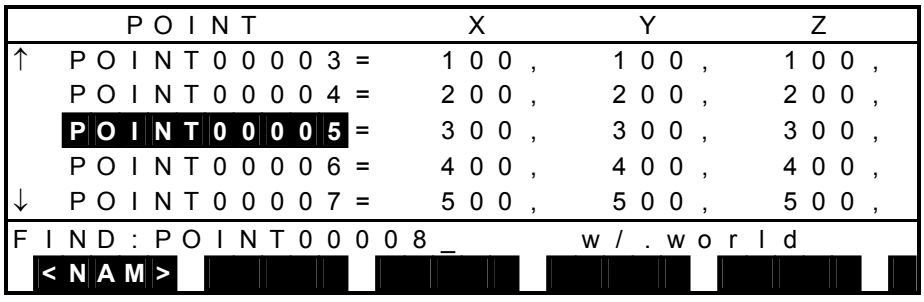
1. Press the NEXT key repeatedly until [FIND] is displayed in the soft key menu.



2. Press [FIND], and the following display appears.



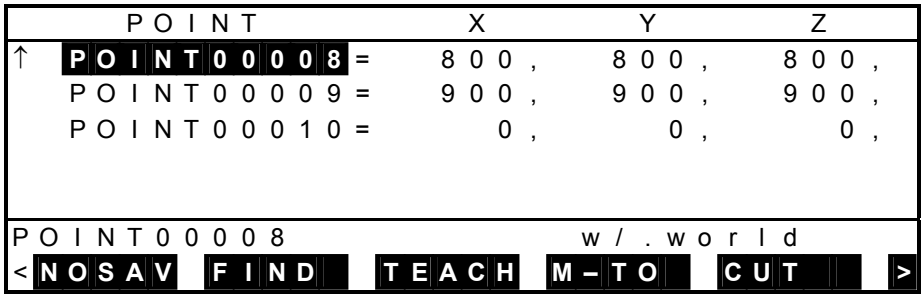
3. Enter "POINT00008", and the display is changed over as shown below.



Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[←]	Used to move left the cursor on the command line.
[→]	Used to move right the cursor on the command line.
[Numeral] [Alphabets]	Used to input characters to the command line.
[EXE]	Used to search the specified position name and return the system to the position data directory display. Unless the specified data is present, a message saying "Not found" is displayed on the command line.
[INS]	Used to select the insert or replace mode for the command line.
[BS]	Used to delete one (1) character just before the cursor position on the command line.
[ALT]+[BS]	Used to delete one (1) character at the cursor position on the command line.
[ESC]	This function works in a different manner according to the presence or absence of input character(s). Presence of input character(s): All input character(s) on the command line are cleared. Absence of input character(s): The system returns to the position data directory display.

3. Press the EXE key, and search of the specified position name starts. If found, the position data directory is displayed with the specified data at the head of the screen and the main cursor moved there. The display is as shown below.



(3) Caution

- Unless the specified name exists, the system returns to the previous directory display with a message of "Not found" shown on the command line.
- As the search is performed throughout the file, the search start position is unrelated to the search operation.

6.2.12 Teach of Position Data [TEACH]

(1) Function

Guides the robot tool tip to the teach point, then registers the current position of the robot in the position data indicated by the main cursor.

(2) Procedures

Assume that you are going to teach and register the current position of the robot in the position data of "POINT00005."

1. Guide the robot tool tip to the teach point.

! CAUTION

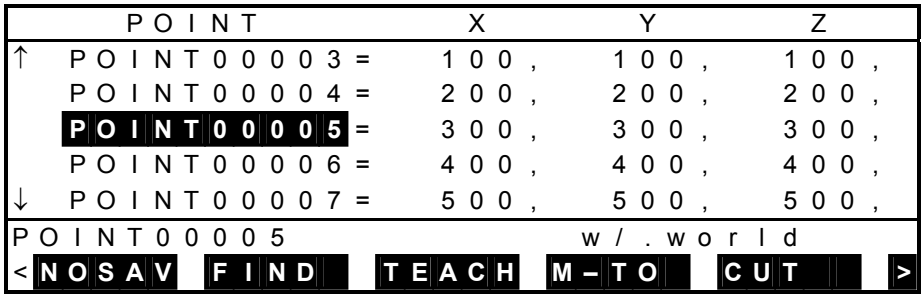
In principle, teaching operation should be performed outside the dangerous area of the robot. If it should be performed inevitably within the dangerous area, strictly observe the following matters.

(1) Teaching operation should always be performed by two (2) persons. One person performs the job and the other person watches outside the dangerous area. Also, both persons should try to prevent mis-operation with each other.

(2) The operator should do the job in an attitude ready to press the EMERGENCY stop pushbutton switch at any time. Also, he should perform the job at a position from which he can evacuate immediately at the time of emergency after confirming the robot working range and shields nearby.

(3) The supervisor should keep watch on the job at a position where he can see the entire robot system and operate the EMERGENCY stop pushbutton switch at the time of an emergency. Also, he should keep anyone from entering the dangerous area.

2. Move the main cursor to "POINT00005", then press the NEXT key repeatedly until [TEACH] is displayed in the soft key menu.



3. Press [TEACH]. When the current position data of the robot is "X100.0, Y0, Z0, C0, T0, CONFIG FREE", the display is as shown below.

```
      POINT              X          Y          Z
↑  POINT00003 =      100 ,      100 ,      100 ,
   POINT00004 =      200 ,      200 ,      200 ,
   POINT00005 =      100 ,        0 ,        0 ,
   POINT00006 =      400 ,      400 ,      400 ,
↓  POINT00007 =      500 ,      500 ,      500 ,
TEACH:PRESS <EXE>      w/.world
```

To input a position data, press the EXE key. (To stop, press the ESC key.) When the robot current position is as shown below, the following display appears.

X	100.0
Y	0
Z	0
C	0
T	0
CONFIG	FREE

```
      POINT              X          Y          Z
↑  POINT00003 =      100 ,      100 ,      100 ,
   POINT00004 =      200 ,      200 ,      200 ,
   POINT00005 =      100 ,        0 ,        0 ,
   POINT00006 =      400 ,      400 ,      400 ,
↓  POINT00007 =      500 ,      500 ,      500 ,
POINT00005             w/.world
<NOSAV FIND TEACH M-TO CUT >
```

(3) Caution

- The main cursor will not move even after finish of teaching.
- Unless actually used tool coincides with tool coordinate data selected, we cannot guarantee the tool tip position at program execution.

6.2.13 Move to Teach Point [M–TO]

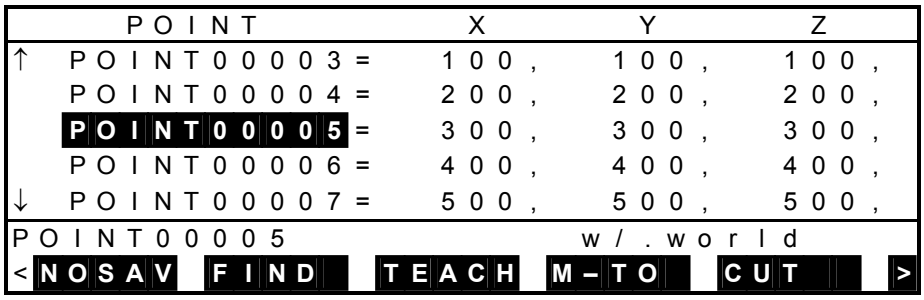
(1) Function

Moves the robot to a position of the position data indicated by the main cursor by synchronous operation.

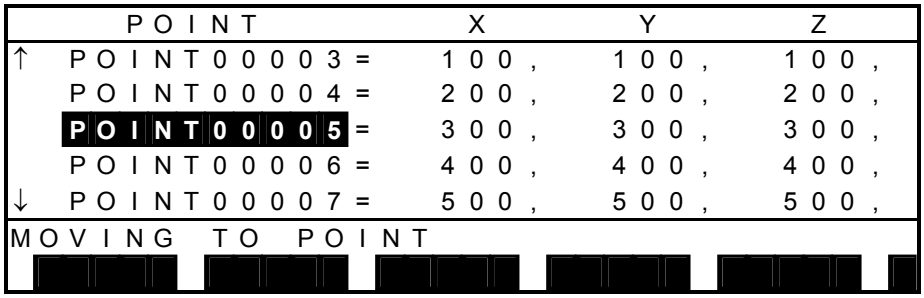
(2) Procedures

Assume that you are going to move the robot to the position of "POINT00005."

1. Move the main cursor to "POINT00005" and press the NEXT key repeatedly until [M–TO] is displayed in the soft key menu.



2. Press [M–TO], and the following display appears.



3. Press the MOVE key while pressing the enable switch, and the robot moves to the teach point in synchronism. When the robot has reached the teach point, the system returns to the previous display.

(3) Caution

- This function is useful when you set the next teach point based on the current teach point. If you have released the MOVE key or enable switch while the robot was moving, or if you have pressed the FEED HOLD key, the robot will decelerate and stop.
- If the feed hold is cancelled while you were pressing the MOVE key in the feed hold mode, the robot will resume an operation.

6.2.14 Cut of Position Data [CUT]

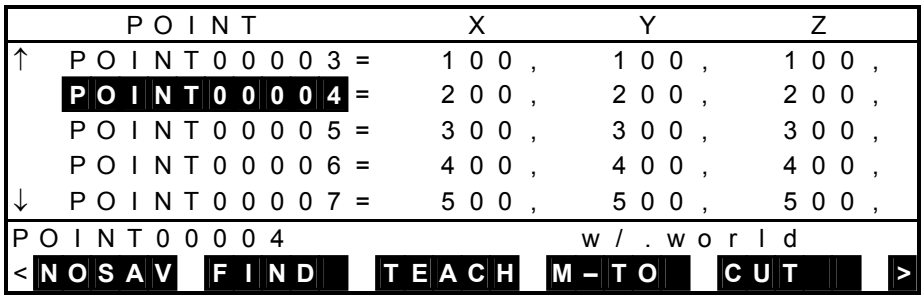
(1) Function

Cuts multiple position data simultaneously.

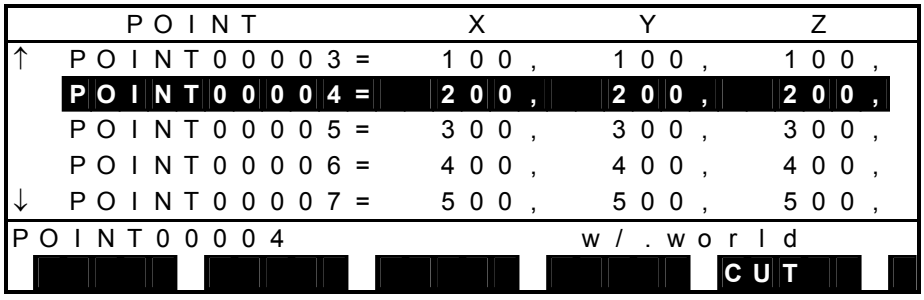
(2) Procedures

Assume that you are going to cut the position data of "POINT00004" ~ "POINT00005".

1. Move the main cursor to "POINT00004" and press the NEXT key repeatedly until [CUT] is displayed in the soft key menu.



2. Press [CUT], and the entire data at the main cursor position is displayed by reversed image to indicate that it is selected. The display is as shown below.



- The entire data selected is displayed inversely.

Keys operative in this mode and their operations are tabled below.

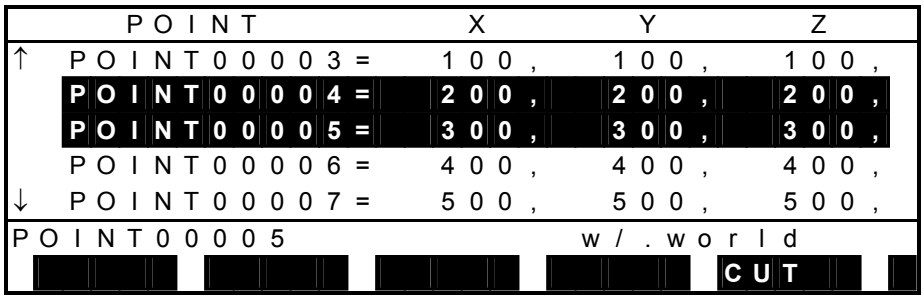
Key	Descriptions
[↑]	Used to widen the selection range upward.
[↓]	Used to widen the selection range downward.
[ALT]+[↑]	Used to widen the selection range upward on the screen basis.
[ALT]+[↓]	Used to widen the selection range downward on the screen basis.
[ESC]	Cancels the range selection and returns the system to the position data directory display mode.

Operation of the soft key is shown below.

Key	Descriptions
[CUT]	Used to save the data in the selection range in the cut buffer and delete that data.

3. Move the main cursor by means of the [↓] key to change the selection range.

   The following display appears then.



4. Press [CUT] to cut the selected data at once. The display changes over as shown below.

	POINT	X	Y	Z
↑	POINT00003 =	100,	100,	100,
	POINT00006 =	400,	400,	400,
	POINT00007 =	500,	500,	500,
	POINT00008 =	0,	0,	0,
↓	POINT00009 =	0,	0,	0,

POINT00006 w/.world

< NOSAV FIND TEACH M-TO CUT >

(3) Caution

When this cut function is used, cut data can be saved in the cut buffer and pasted at another position by means of the paste function stated in the next paragraph.

6.2.15 Paste of Position Data [PASTE]

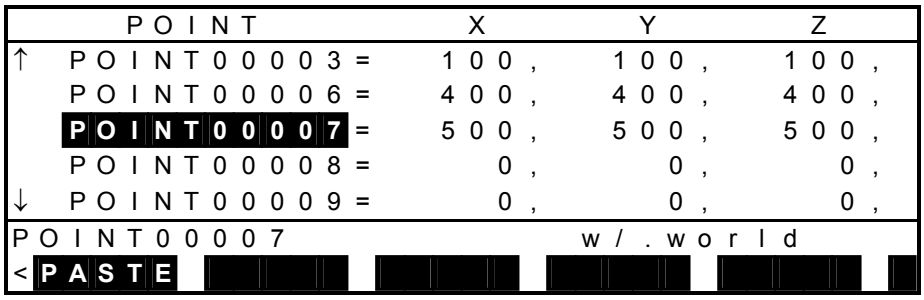
(1) Function

Pastes multiple position data cut by the cut function at another position.

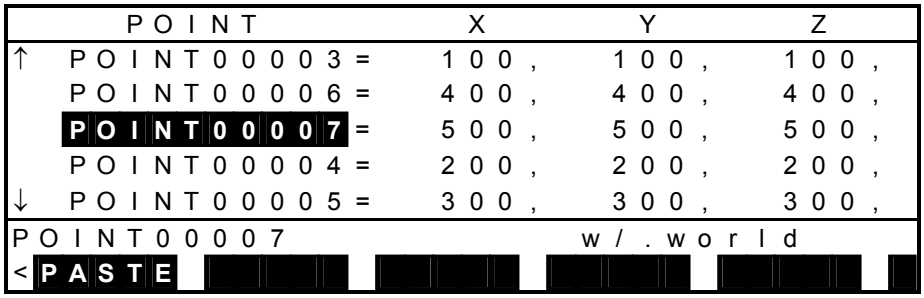
(2) Procedures

Assume that you are going to paste the cut data (POINT00004 ~ POINT00005) after "POINT00007."

1. Move the main cursor to "POINT00007" and press the NEXT key repeatedly until [PASTE] is displayed in the soft key menu.



2. Press [PASTE], and the cut data is pasted. The display is as shown below.

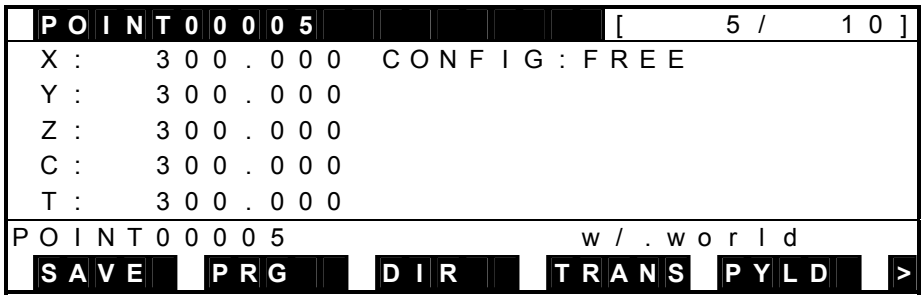


(3) Caution

Cut data can be pasted only at a position after the main cursor.

6.3 Detailed Position Data Display Mode

The detailed position data directory display mode displays one (1) position data on the full screen. In this mode, it is possible to edit the position data.
An example of display is given below.



Each time the NEXT key is pressed, the menu is changed over as follows:

- An inversely displayed section in the data is called the "main cursor" which indicates that this data is selected for editing.
- For the data of [nnnn/mmmm] displayed on the 1st line, "nnnn" signifies the number of order where it is defined in a selected coordinate system. "mmmm" represents the total number of position data in a selected coordinate system.
- Data following "w/" on the right side of the command line identifies the coordinate data to which the selected position data belongs.

Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[↑]	Used to move the main cursor to each data in the order of [Name ← X ← Y ← Z ← C ← T ← CONFIG ← Name].
[↓]	Used to move the main cursor to each data in the order of [Name → X → Y → Z → C → T → CONFIG → Name].
[ALT]+[←]	Used to display detailed position data in a coordinate system defined before.
[ALT]+[→]	Used to display detailed position data in a coordinate system defined after.
[ALT]+[↑]	Used to display the defined detailed position data before current position data.
[ALT]+[↓]	Used to display the defined detailed position data after current position data.
[INS]	Used to create a new position data just after a current position data.
[ALT]+[BS]	Used to delete a current position data and display in detail the position data defined next.
[EXE]	Used to edit a data indicated by the main cursor.
[NEXT]	Used to select the next menu.
[ALT]+[NEXT]	Used to select the previous menu.

Operations of respective soft keys are shown below.

Key	Descriptions
[SAVE]	Used to save a data and quit the data editor.
[PRG]	Used to select the program editor display. If the data editor was started up from the basic menu, however, the display will not change over.
[DIR]	Used to select the position data directory display. For details, see Para. 6.2.
[TRANS]	Used to select the coordinate data editor display. For details, see Para. 6.4.
[PLOAD]	Used to select the load data editor display. For details, see Para. 6.6.
[NOSAV]	Used to quit the editor without saving data. All changed data are aborted.
[FIND]	Used to select the position name search mode.
[TEACH]	Used to register a current position of the robot in a position data indicated by the main cursor.
[M–TO]	Used to move the robot to a position indicated by the main cursor.

Each menu of [SAVE], [NOSAV], [PRG], [TRANS], [PLOAD], [FIND], [TEACH] and [M–TO] is the same as in the position data directory display mode and is not described in this paragraph.

6.3.1 Editing of Detailed Position Data Name

(1) Function

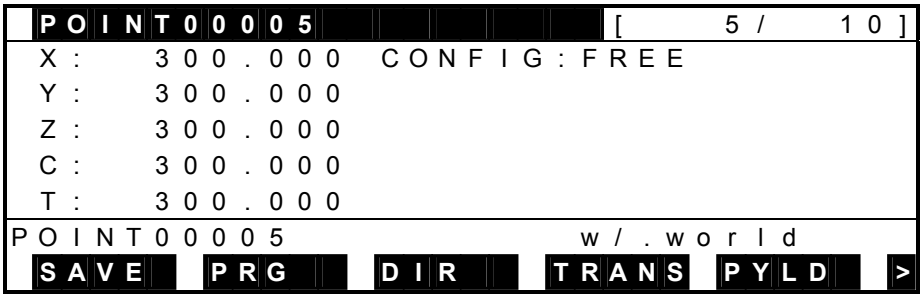
Edits a position data name.

(2) Procedures

When you press the EXE key while the main cursor is located at the position name, you can edit the position name.

In the example below, assume that you are going to change the name of "POINT00005" to "POINT55555."

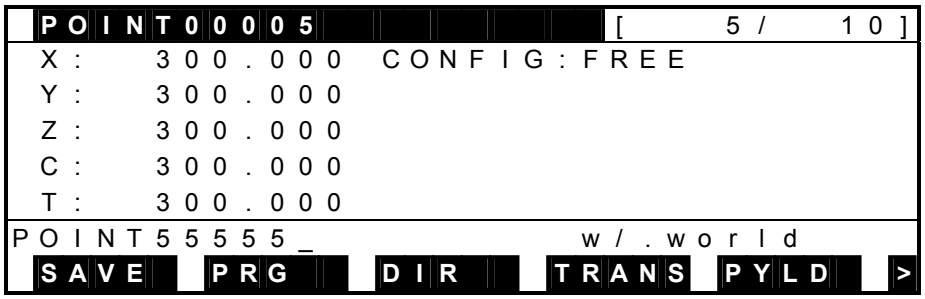
1. Move the main cursor to "POINT00005" by means of the cursor keys (↑, ↓, →, ←), and the following display appears.



2. Press the EXE key, and the following display is called. (The cursor is displayed on the command line.)

- Current data is displayed as the initial value on the command line.
- The cursor is displayed just after the data indicated as the initial value, and you can edit the data now.
- When the cursor is shown by mark "■", the replace mode takes effect. When the cursor is shown by mark "_", the insert mode is effective.

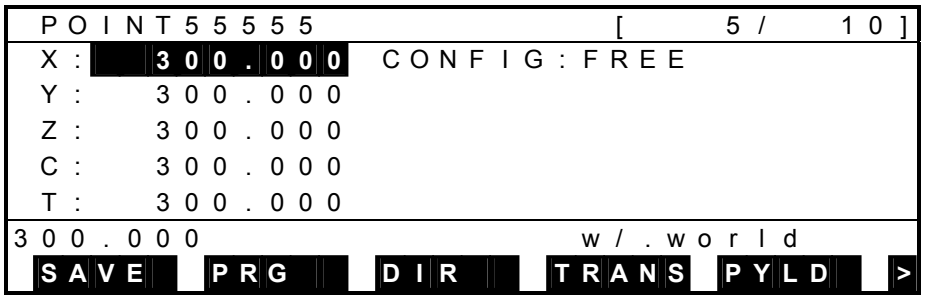
3. Enter a desired name to the command line, using alphanumeric keys, →, ← cursor keys. Then the following display appears.



Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[←]	Used to move left the cursor on the command line.
[→]	Used to move right the cursor on the command line.
[Numeral] [Alphabets]	Used to input characters to the command line.
[EXE]	Used to change current position name to an entered one. The system returns to the detailed position data display mode with the cursor moved to the X-data area.
[INS]	Used to select the insert or replace mode for the command line.
[BS]	Used to delete one (1) character just before the cursor position on the command line.
[ALT]+[BS]	Used to delete one (1) character at the cursor position on the command line.
[ESC]	This function works in a different manner according to the presence or absence of input character(s). Presence of input character(s): All input character(s) on the command line are cleared. Absence of input character(s): The system returns to the detailed position data directory display.

4. Press the EXE key, and the position name is replaced by the newly specified name. The following display appears.



- The main cursor moves to the X data area.
- If the input name is illegal, a message saying "Illegal name" is displayed on the command line and no data is changed.
- If the specified name already exists, a message of "Duplicate name" is displayed on the command line and no data is changed.

6.3.2 Editing of Detailed Position Element

(1) Function

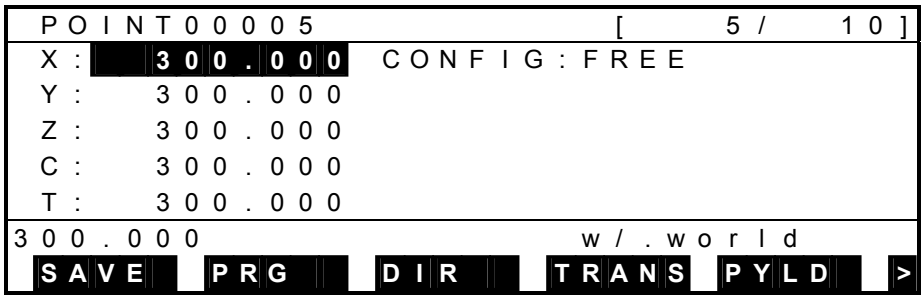
Used to edit a position element.

(2) Procedures

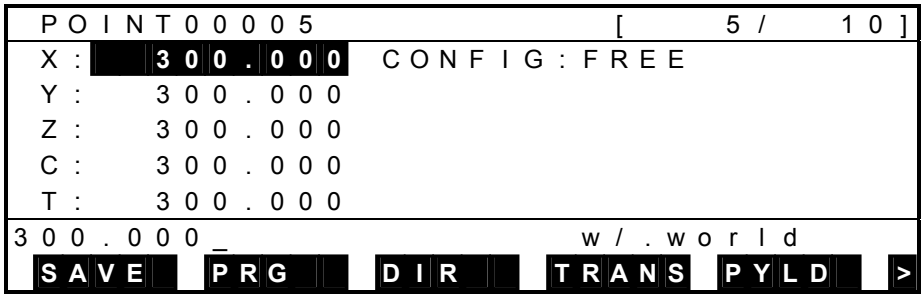
When you press the EXE key while the main cursor is located at each position element, you can edit that data.

In the example below, assume that you are going to change the X data from "300.000" to "100".

1. Move the main cursor to the data you wish to edit by means of the cursor keys (↑, ↓), and the following display appears.

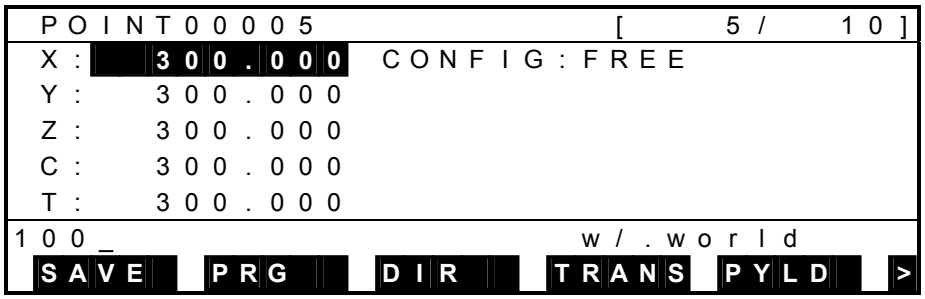


2. Press the EXE key, and the following display is called. (The cursor is displayed on the command line.)



- Current data is displayed as the initial value on the command line.
- The cursor is displayed just after the data indicated as the initial value, and you can edit the data now.
- When the cursor is shown by mark "■", the replace mode takes effect. When the cursor is shown by mark "_", the insert mode is effective.

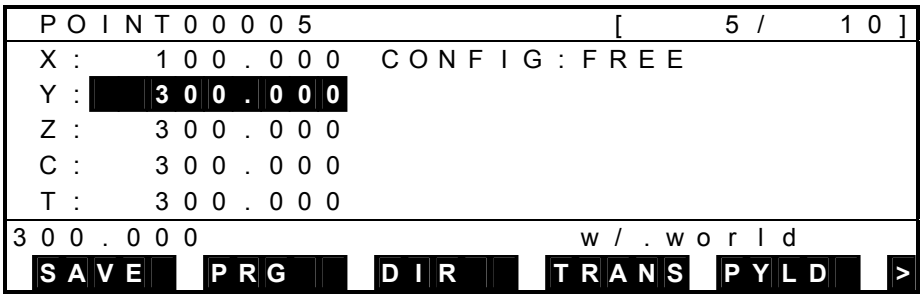
3. Enter a desired value to the command line, using alphanumeric keys, →, ← cursor keys. Then the following display appears.



Keys operative in this mode and their operations are tabled below.

Key	Descriptions
←	Used to move left the cursor on the command line.
→	Used to move right the cursor on the command line.
Numeral . + –	Used to input characters to the command line.
EXE	Used to change current position element to an entered one. The system returns to the detailed position directory display mode with the main cursor moved to the next data area.
INS	Used to select the insert or replace mode for the command line.
BS	Used to delete one (1) character just before the cursor position on the command line.
ALT + BS	Used to delete one (1) character at the cursor position on the command line.
ESC	This function works in a different manner according to the presence or absence of input character(s). Presence of input character(s): All input character(s) on the command line are cleared. Absence of input character(s): The system returns to the detailed position data directory display.

4. Press the EXE key, and the position element is replaced by the newly specified data. The system returns to the detailed position data directory display mode with the main cursor moved to the next data area. The following display appears.



(3) Caution

When editing the CONFIG data, all you have to do is enter only one (1) top character. The relationship between input character and value to be set is as follows:

Input character	Value to be set
F or 0	FREE
R or 1	RIGHTY
L or 2	LEFTY

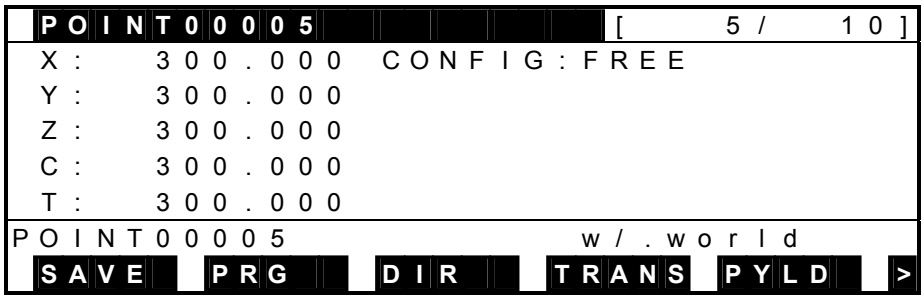
6.3.3 Call of Position Data Directory Display [DIR]

(1) Function

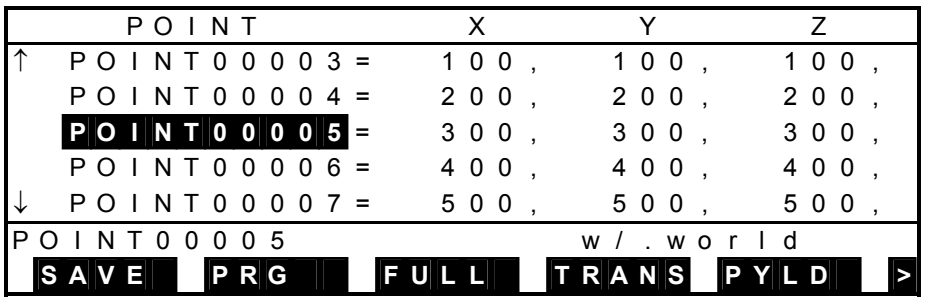
Displays the position data directory display.

(2) Procedures

1. Select the data display mode as shown below (i.e., the cursor is not present on the command line).



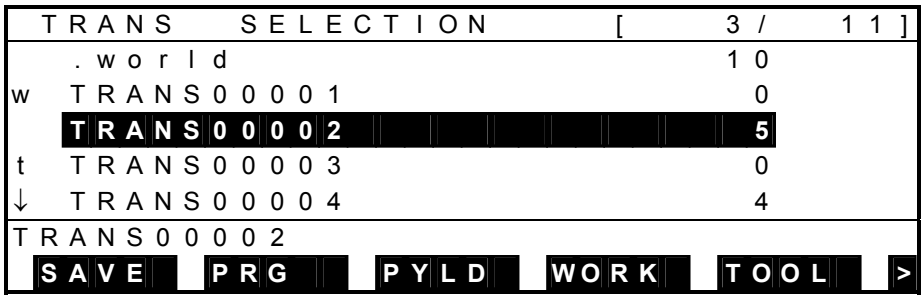
2. Press [DIR] and the following directory display is called.



For details of operations to be done on the detailed display, see Para. 6.2.

6.4 Coordinate Data Directory Display Mode

The coordinate data directory display mode displays one (1) coordinate on one (1) line at a time. In this mode, it is possible to edit the coordinate data.
An example of display is given below.



Each time the NEXT key is pressed, the menu is changed over as follows:



- An inversely displayed section in the data is called the "main cursor" which indicates that this data is selected for editing.
- Symbol "↑" shown in the 1st column signifies that there is coordinate data before the top data.
- Symbol "↓" shown in the 1st column signifies that there is coordinate data after the final data.
- Alphabet "t" displayed in the 1st column represents that the coordinate data on the same line is based on the tool coordinate system.
- Alphabet "w" displayed in the 1st column represents that the coordinate data on the same line is based on the work coordinate system.
- On the right side of the coordinate name, the number of position data belonging to that coordinate is displayed.
- Regarding the data of [nnnn/mmmm] on the 1st line, "nnnn" identifies the selected coordinate index, and "mmmm" the number of coordinate data.

Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[↑]	Used to move up the main cursor. If the main cursor is located on the top line of screen, the screen scrolls down.
[↓]	Used to move down the main cursor. If the main cursor is located on the last line of screen, the screen scrolls up.
[ALT]+[↑]	Used to move up the main cursor on the screen basis.
[ALT]+[↓]	Used to move down the main cursor on the screen basis.
[INS]	Used to create a new coordinate data.
[ALT]+[BS]	Used to delete a coordinate data indicated by the main cursor. All position data contained this coordinate are also deleted.
[EXE]	The system proceeds to the detailed coordinate display where you can edit a coordinate data indicated by the main cursor.
[NEXT]	Used to select the next menu.
[ALT]+[NEXT]	Used to select the previous menu.

Operations of respective soft keys are shown below.

Key	Descriptions
[SAVE]	Used to save a data and quit the data editor.
[PRG]	Used to select the program editor display. If the data editor was started up from the basic menu, however, the display will not change over.
[PYLD]	Used to select the load data editor display.
[WORK]	Used to select the position data directory display where you can edit a position data belonging to a coordinate system indicated by the main cursor.
[TOOL]	Used to specify the tool coordinate system for a coordinate system indicated by the main cursor, and cancel the designation.
[NOSAV]	Used to quit the editor without saving data. All changed data are aborted.
[FIND]	Used to search a coordinate name.
[CUT]	Used to select the range select mode.
[PASTE]	Used to paste the contents saved in the cut buffer (i.e., data deleted by the previous cut menu) after a position data indicated by the main cursor to clear the cut buffer.

Each menu of [SAVE], [NOSAV] and [PRG] is the same as in the position data directory display mode and is not described in this paragraph.

6.4.1 Addition of New Coordinate Data [INS]

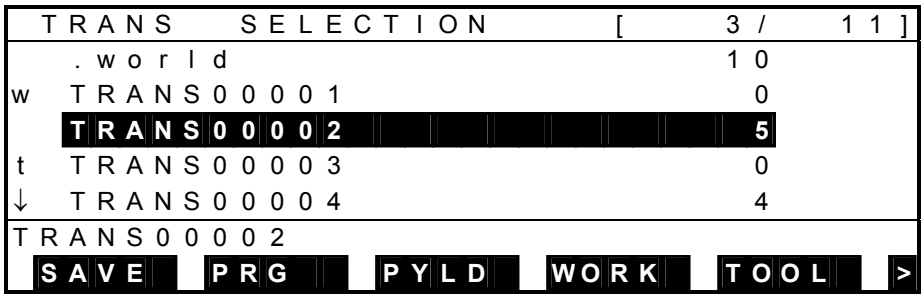
(1) Function

Adds a new coordinate data.

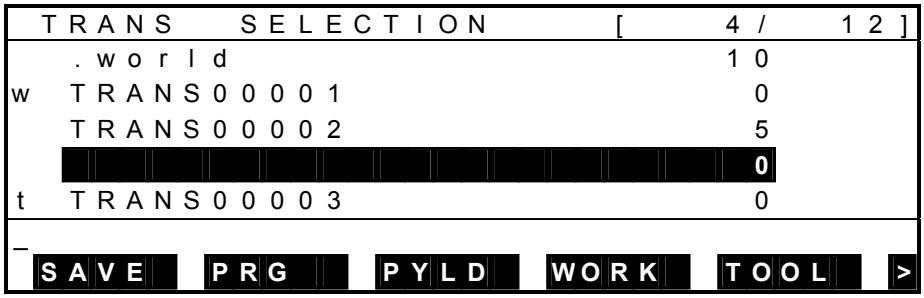
(2) Procedures

Assume that you are going to add a new coordinate data ("TRANS00020") just after "TRANS00002."

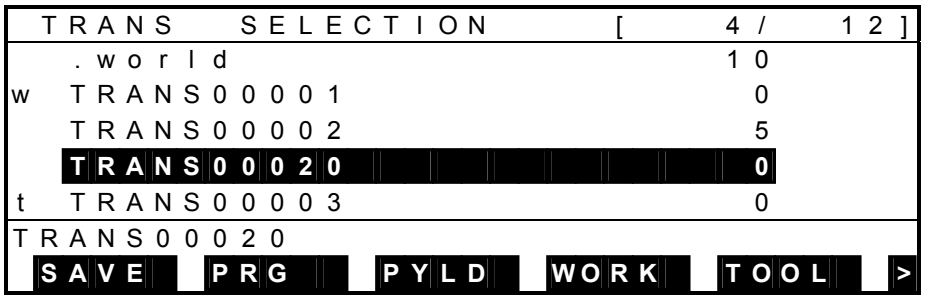
1. Move the main cursor to "TRANS00002", and the following display is called.



2. Press the INS key, and the display changes as shown below with the name input mode selected. (The cursor is displayed on the command line.)



3. Enter a desired name ("TRANS00020" in this example) and press the EXE key. The display changes as follows:



(3) Caution

- For the newly added coordinate data, zero (0) is set for all elements.
- A data is added to just below the main cursor.
- When you have pressed the ESC key without specifying the name, addition of the new data is cancelled and the system returns.

6.4.2 Deletion of Coordinate Data ([ALT] + [BS])

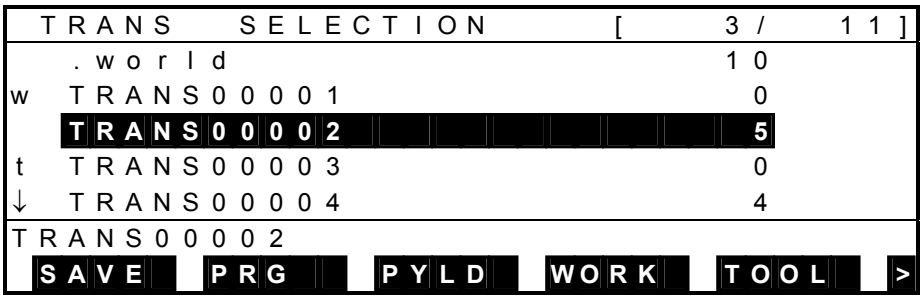
(1) Function

Deletes a coordinate data.

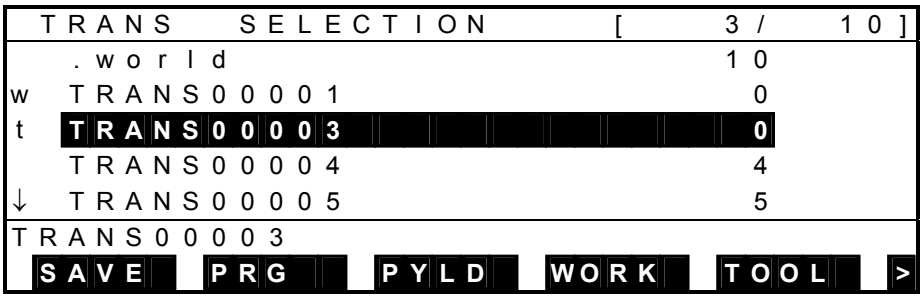
(2) Procedures

Assume that you are going to delete the data of "TRANS00002".

1. Move the main cursor to "TRANS00002", and the following display is called.



2. Press the ALT and BS keys at the same time, and the following display appears with the specified data deleted.



(3) Caution

- When a coordinate data is deleted, all position data belong to that coordinate system are also deleted.
- Different from the [CUT] function, you cannot undo a data deleted by this operation.
- A world coordinate cannot be deleted.
- A coordinate selected as the tool or work coordinate cannot be deleted.

6.4.3 Call of Detailed Coordinate Data Display

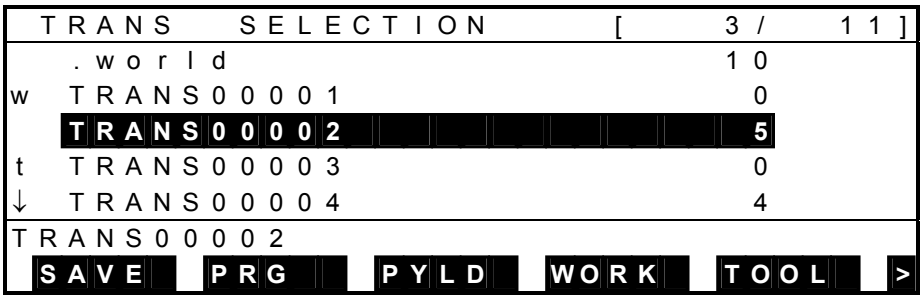
(1) Function

Displays the detailed coordinate data display.

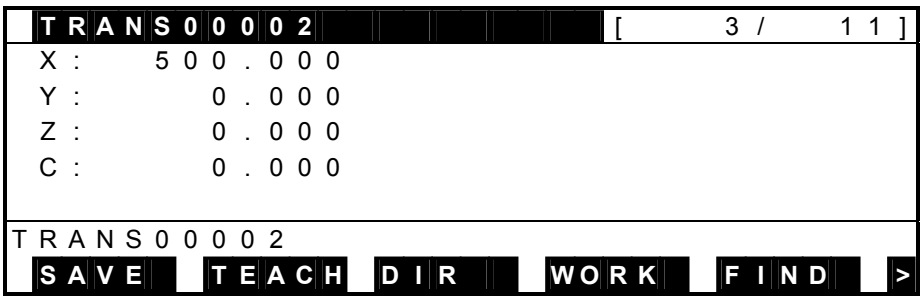
(2) Procedures

Assume that you are going to display the data of "TRANS00002" in detail.

1. Move the main cursor to "TRANS00002", and the following display appears.



2. Press [EXE] and the following detailed display is called.



(3) Caution

- A world coordinate cannot be edited.
- A coordinate selected as the tool coordinate or work coordinate cannot be edited.

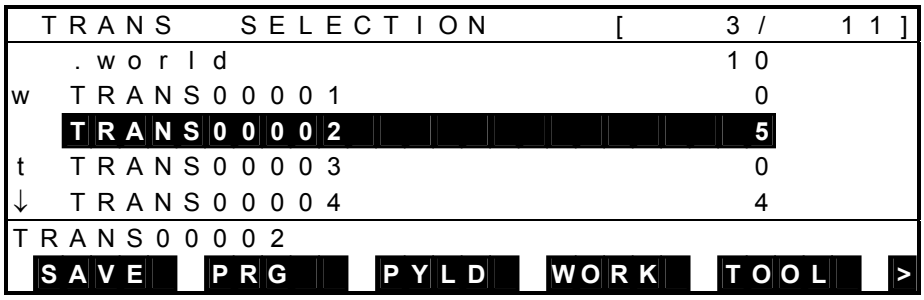
For details of operations to be done on the detailed display, see Para. 6.5.

6.4.4 Call of Load Data Directory Display [PYLD]

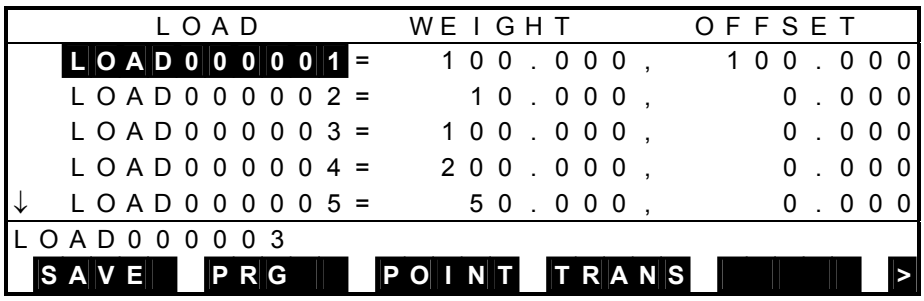
(1) Function
Calls the load data directory display.

(2) Procedures

1. Select the data display mode as shown below (i.e., the cursor is not present on the command line).



2. Press [PYLD], and the following load data directory display appears.



(3) Caution
On the load data directory display, the main cursor indicates the position of the data displayed before. At initial display, the data are displayed from the beginning.

For details of operations to be done on the load data display, see Para. 6.6.

6.4.5 Setting of Work Coordinate System (Call of Position Data Directory Display) [WORK]

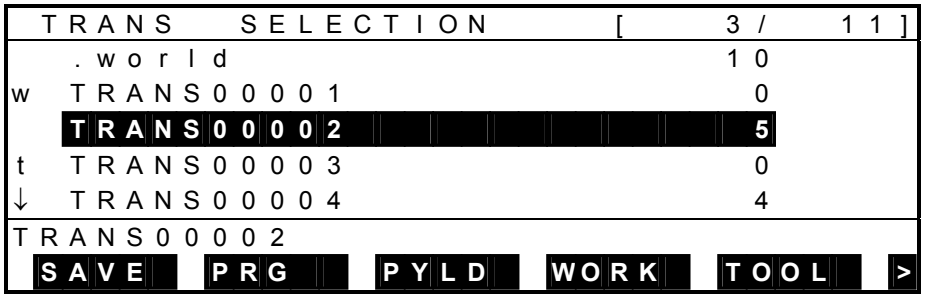
(1) Function

Sets a specified coordinate in the work coordinate system and calls the directory display of a position data belonging to that coordinate system.

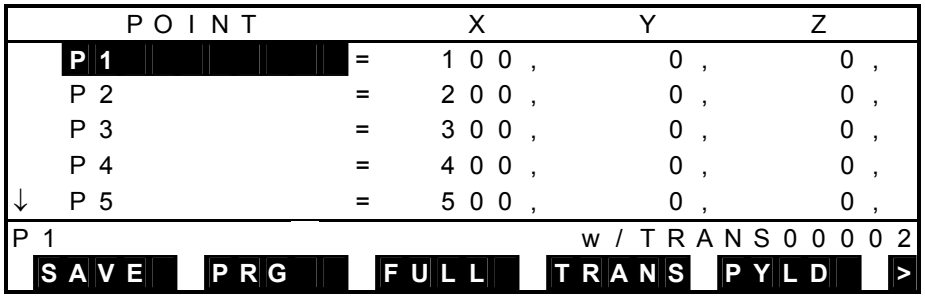
(2) Procedures

Assume that you are going to set "TRANS00002" in the work coordinate system and call the relevant position data directory display.

1. Move the main cursor to "TRANS00002", and the following display appears.



2. Press [WORK], and the data of "TRANS00002" is set in the work coordinate system with the position data directory display belonging to the "TRANS00002" coordinate system called, as shown below.



(3) Caution

- If this operation is done for the world coordinate system, setting of the work coordinate system is cancelled.
- A coordinate set as the work coordinate is identified by "w" in the 1st column on the coordinate data directory display.

6.4.6 Setting and Cancel of Tool Coordinate System [TOOL]

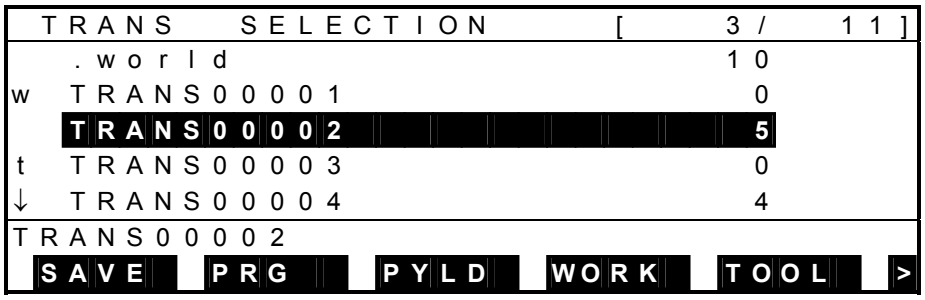
(1) Function

Sets a specified coordinate in the tool coordinate system. If this operation is done for a coordinate already set as the tool coordinate, setting of the tool coordinate system is cancelled.

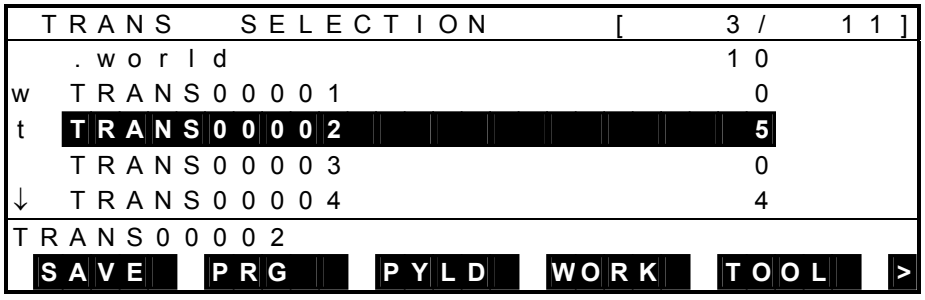
(2) Procedures

Assume that you are going to set "TRANS00002" in the tool coordinate system.

1. Move the main cursor to "TRANS00002", and the following display appears.



2. Press [TOOL], and the data of "TRANS00002" is set in the tool coordinate system and the following display appears.



(3) Caution

- If another coordinate was already set as the tool coordinate, it is cancelled by this function.
- A coordinate set as the tool coordinate is identified by "t" in the 1st column of the display.

6.4.7 Search of Coordinate Data [FIND]

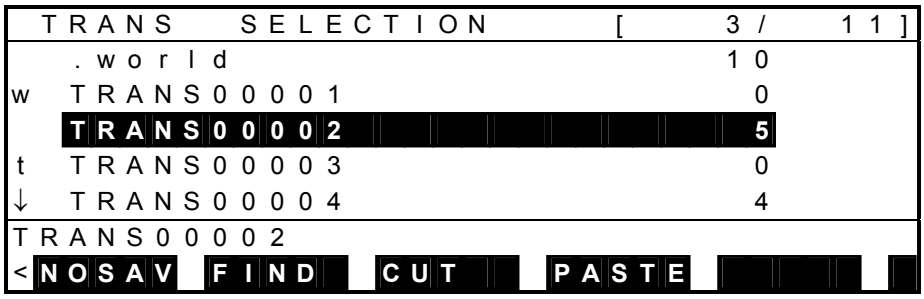
(1) Function

Searches the data bearing the specified coordinate name.

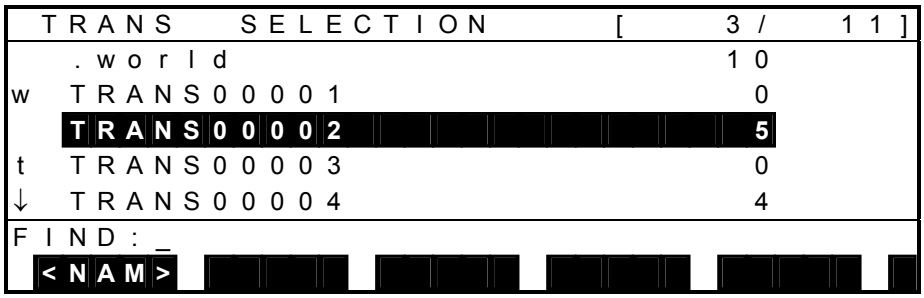
(2) Procedures

Assume that you are going to search the coordinate data named "TRANS00008."

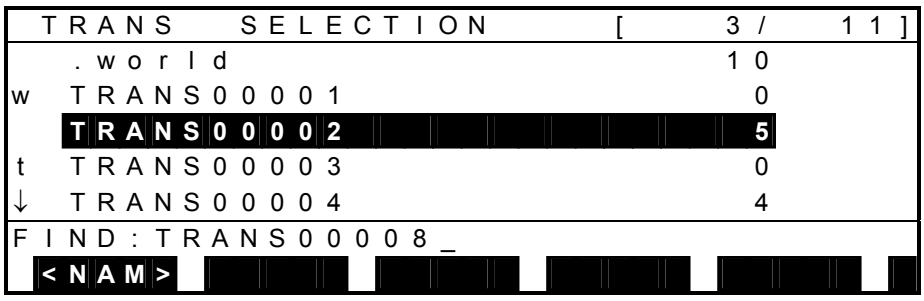
1. Press the NEXT key repeatedly until [FIND] is displayed in the soft key menu.



2. Press [FIND], and the following display appears.



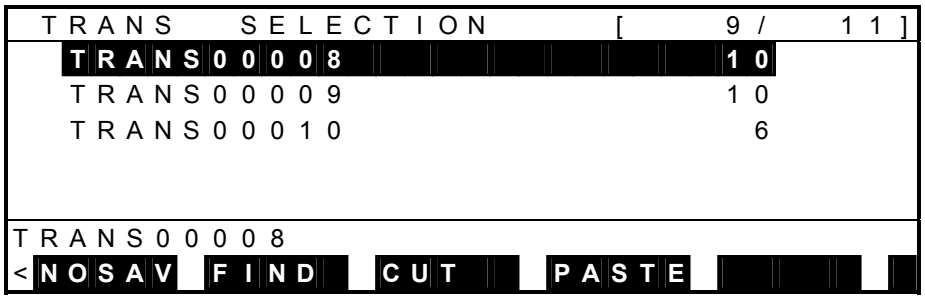
3. Enter "TRANS 00008", and the display is changed over as shown below.



Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[←]	Used to move left the cursor on the command line.
[→]	Used to move right the cursor on the command line.
[Numeral] [Alphabets]	Used to input characters to the command line.
[EXE]	Used to search the specified coordinate name and return the system to the coordinate data directory display. Unless the specified data is present, a message saying "Not found" is displayed on the command line.
[INS]	Used to select the insert or replace mode for the command line.
[BS]	Used to delete one (1) character just before the cursor position on the command line.
[ALT]+[BS]	Used to delete one (1) character at the cursor position on the command line.
[ESC]	This function works in a different manner according to the presence or absence of input character(s). Presence of input character(s): All input character(s) on the command line are cleared. Absence of input character(s): The system returns to the coordinate data directory display.

4. Press the EXE key, and search of the specified coordinate name starts. If found, the coordinate data directory is displayed with the specified data at the head of the screen and the main cursor moved there. The display is as shown below.



(3) Caution

- Unless the specified name exists, the system returns to the previous directory display with a message of "Not found" shown on the command line.
- As the search is performed throughout the file, the search start position is unrelated to the search operation.

6.4.8 Cut of Coordinate Data [CUT]

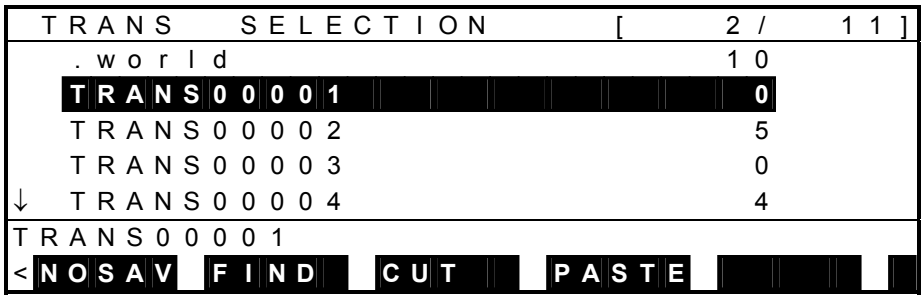
(1) Function

Cuts multiple coordinate data simultaneously.

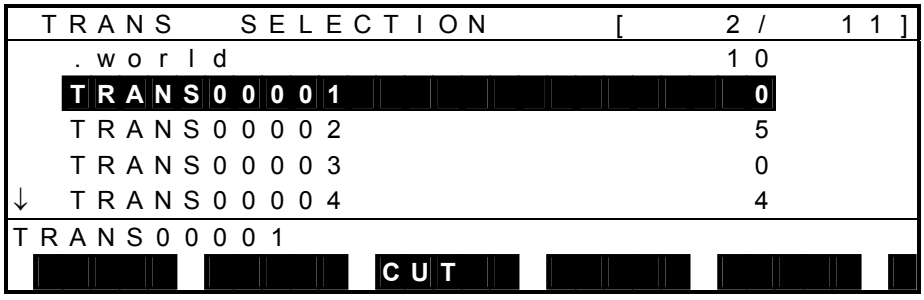
(2) Procedures

Assume that you are going to cut the coordinate data of "TRANS00001" ~ "TRANS00003".

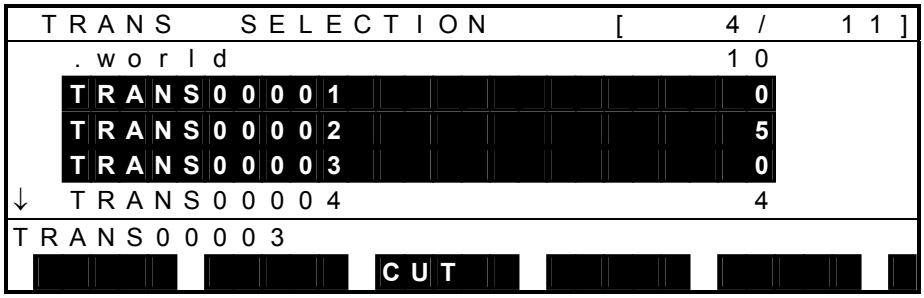
1. Move the main cursor to "TRANS00001" and press the NEXT key repeatedly until [CUT] is displayed in the soft key menu.



2. Press [CUT], and the entire data at the main cursor position is displayed by reversed image to indicate that it is selected. The display is as shown below.



3. Move the main cursor by means of the ↓ key to change the selection range. The following display appears then.



- The entire data selected is displayed inversely.
- Either world coordinate, tool coordinate or work coordinate is not included in the selection range.
- The main cursor indicates the data next to the selection range.

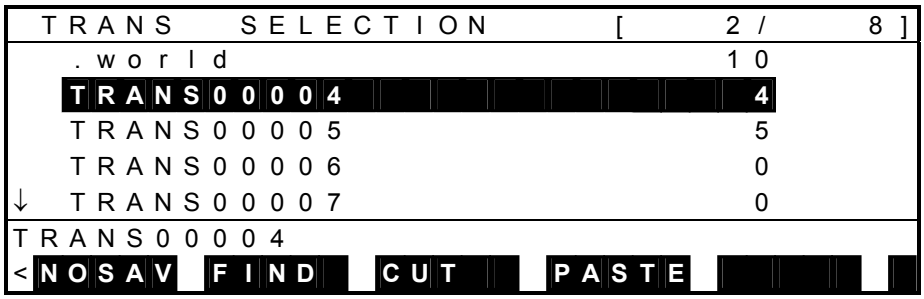
Keys operative in this mode and their operations are tabled below.

Key	Descriptions
↑	Used to widen the selection range upward.
↓	Used to widen the selection range downward.
ALT + ↑	Used to widen the selection range upward on the screen basis.
ALT + ↓	Used to widen the selection range downward on the screen basis.
ESC	Cancels the range selection and returns the system to the coordinate data directory display mode.

Operation of the soft key is shown below.

Key	Descriptions
[CUT]	Used to save the data in the selection range in the cut buffer and delete that data.

4. Press [CUT] to cut the selected data at once. The display changes over as shown below.



(3) Caution

- All position data belonging to relevant coordinate system are also deleted.
- A world coordinate cannot be selected or deleted.
- A data set in the tool coordinate system cannot be selected or deleted.
- A data set in the work coordinate system cannot be selected or deleted.
- When this cut function is used, cut data can be saved in the cut buffer and pasted at another position by means of the paste function stated in the next paragraph.

6.4.9 Paste of Coordinate Data [PASTE]

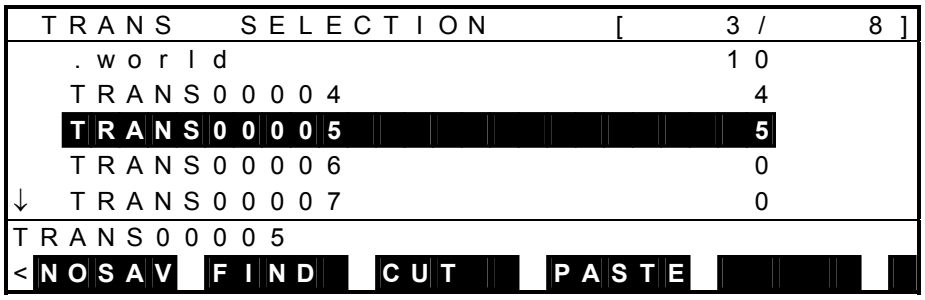
(1) Function

Pastes multiple coordinate data cut by the cut function at another position.

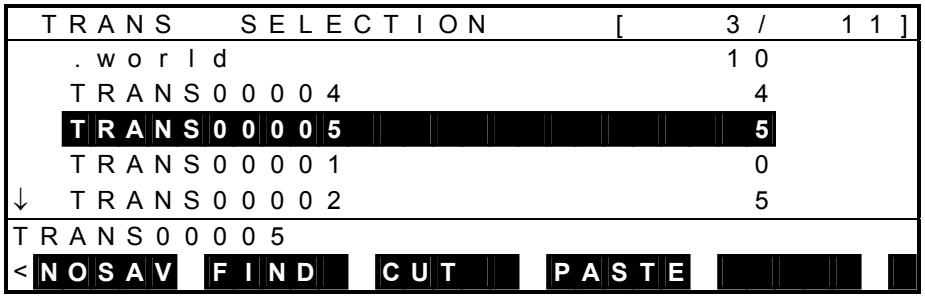
(2) Procedures

Assume that you are going to paste the cut data (TRANS00001 ~ TRANS00003) after "TRANS00005."

1. Move the main cursor to "TRANS00005" and press the NEXT key repeatedly until [PASTE] is displayed in the soft key menu.



2. Press [PASTE], and the cut data is pasted. The display is as shown below.

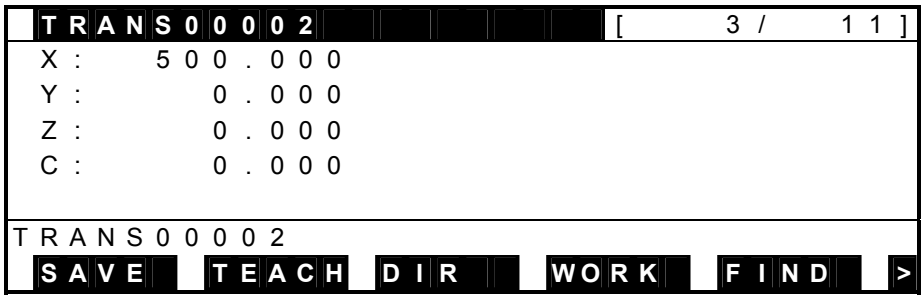


(3) Caution

- Cut data can be pasted only at a position after the main cursor.
- Position data belonging to each coordinate data is also pasted.

6.5 Detailed Coordinate Data Display Mode

The detailed coordinate data display mode displays one (1) coordinate data on the full screen. In this mode, it is possible to edit the coordinate data.
An example of display is given below.



Each time the NEXT key is pressed, the menu is changed over as follows:



- An inversely displayed section in the data is called the "main cursor" which indicates that this data is selected for editing.
- For the data of [nnnn/mmmm] displayed on the 1st line, "nnnn" signifies the number of positions contained in the selected coordinate system.
  "mmmm" represents the total number of position data in all coordinate systems.

Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[↑]	Used to move the main cursor to each data in the order of [Name ← X ← Y ← Z ← C ← Name].
[↓]	Used to move the main cursor to each data in the order of [Name → X → Y → Z → C → Name].
[ALT]+[↑]	Used to display the defined detailed coordinate data before current coordinate data.
[ALT]+[↓]	Used to display the defined detailed coordinate data after current coordinate data.
[INS]	Used to create a new coordinate data.
[ALT]+[BS]	Used to delete a current coordinate data and display in detail the coordinate data defined next.
[EXE]	Used to edit a data indicated by the main cursor.
[NEXT]	Used to select the next menu.
[ALT]+[NEXT]	Used to select the previous menu.

Operations of respective soft keys are shown below.

Key	Descriptions
[SAVE]	Used to save a data and quit the data editor.
[TEACH]	Used to register a current position of the robot in a coordinate data indicated by the main cursor.
[DIR]	Used to select the coordinate data directory display.
[WORK]	Used to select the position data directory display where you can edit a position data belonging to a coordinate system indicated by the main cursor.
[FIND]	Used to select the coordinate name search mode.
[NOSAV]	Used to quit the editor without saving data. All changed data are aborted.

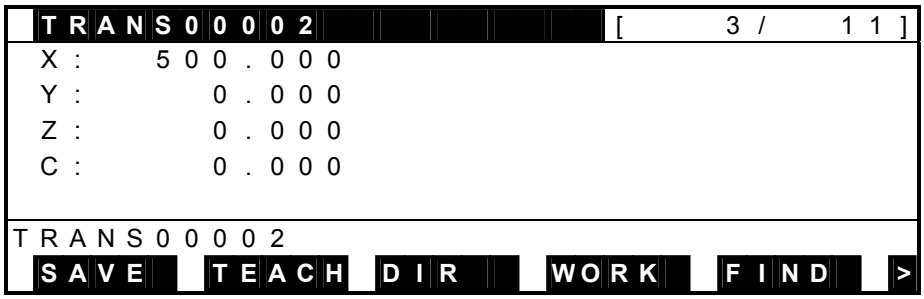
Each menu of [SAVE] and [NOSAV] is the same as in the position data directory display mode and is not described in this paragraph.

6.5.1 Editing of Coordinate Data Name

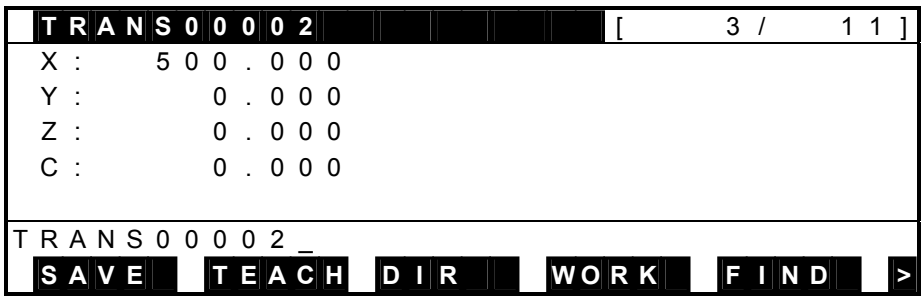
(1) Function
Edits a coordinate data name.

(2) Procedures
When you press the EXE key while the main cursor is located at the coordinate name, you can edit the coordinate name.
In the example below, assume that you are going to change the name of "TRANS00002" to "TRANS00022."

1. Move the main cursor to "TRANS00002" by means of the cursor keys (↑, ↓, →, ←), and the following display appears.

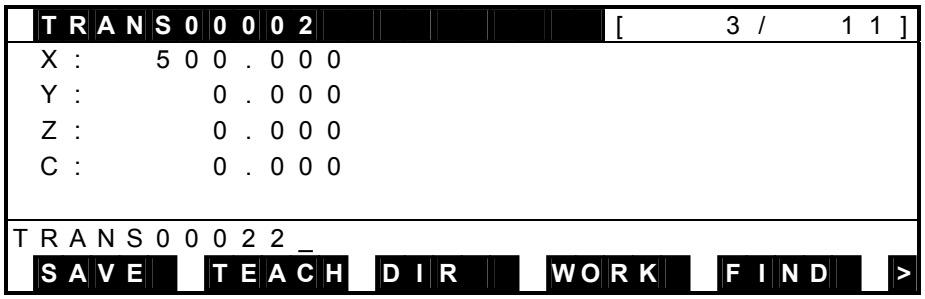


2. Press the EXE key, and the following display is called. (The cursor is displayed on the command line.)



- Current data is displayed as the initial value on the command line.
- The cursor is displayed just after the data indicated as the initial value, and you can edit the data now.
- When the cursor is shown by mark "■", the replace mode takes effect. When the cursor is shown by mark "_", the insert mode is effective.

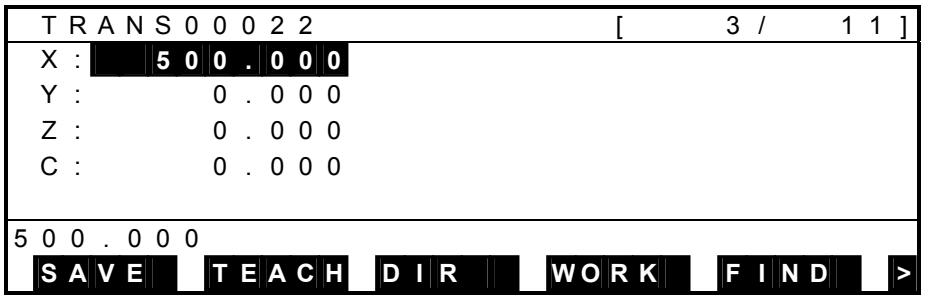
3. Enter a desired name to the command line, using alphanumeric keys, →, ← cursor keys. Then the following display appears.



Keys operative in this mode and their operations are tabled below.

Key	Descriptions
←	Used to move left the cursor on the command line.
→	Used to move right the cursor on the command line.
Numeral Alphabets	Used to input characters to the command line.
EXE	Used to change current coordinate name to an entered one. The system returns to the detailed coordinate data display mode with the cursor moved to the X-data area.
INS	Used to select the insert or replace mode for the command line.
BS	Used to delete one (1) character just before the cursor position on the command line.
ALT + BS	Used to delete one (1) character at the cursor position on the command line.
ESC	This function works in a different manner according to the presence or absence of input character(s). Presence of input character(s): All input character(s) on the command line are cleared. Absence of input character(s): The system returns to the detailed coordinate data directory display.

4. Press the EXE key, and the coordinate name is replaced by the newly specified name. The following display appears.



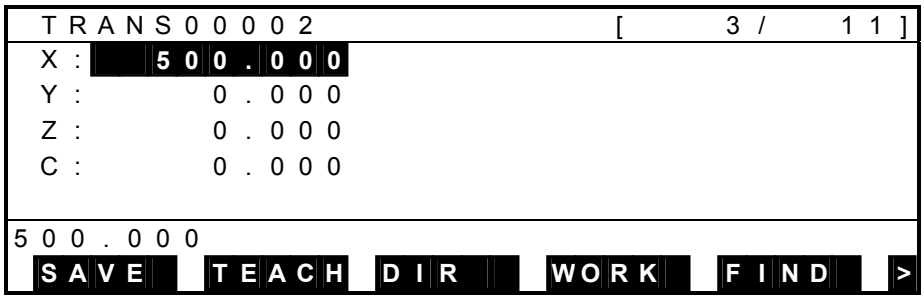
- The main cursor moves to the X data area.
- If the input name is illegal, a message saying "Illegal name" is displayed on the command line and no data is changed.
- If the specified name already exists, a message of "Duplicate name" is displayed on the command line and no data is changed.

6.5.2 Editing of Coordinate Element

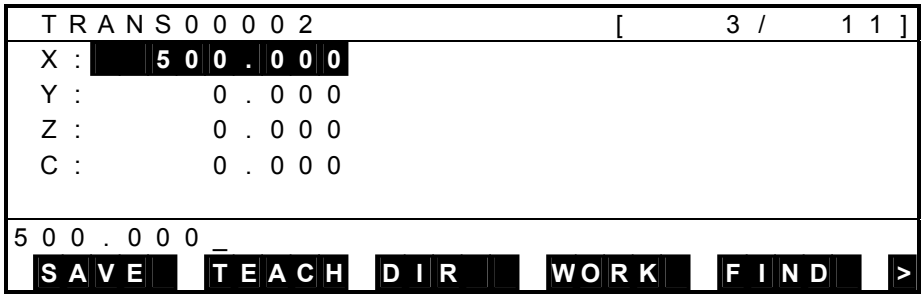
(1) Function
Used to edit a coordinate element.

(2) Procedures
When you press the EXE key while the main cursor is located at each coordinate element, you can edit that data.
In the example below, assume that you are going to change the X data from "500.000" to "100".

1. Move the main cursor to the data you wish to edit by means of the cursor keys (↑, ↓), and the following display appears.

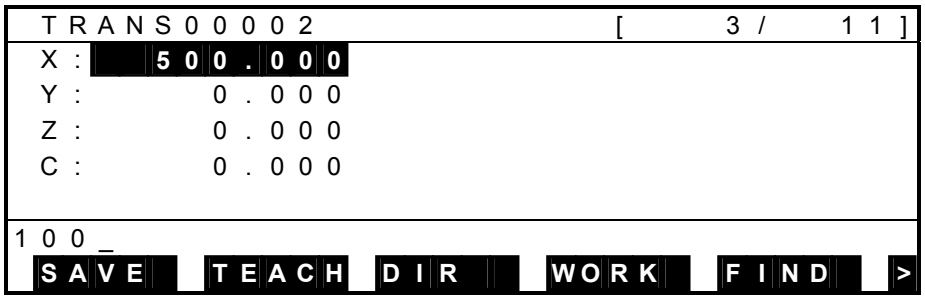


2. Press the EXE key, and the following display is called. (The cursor is displayed on the command line.)



- Current data is displayed as the initial value on the command line.
- The cursor is displayed just after the data indicated as the initial value, and you can edit the data now.
- When the cursor is shown by mark "■", the replace mode takes effect. When the cursor is shown by mark "_", the insert mode is effective.

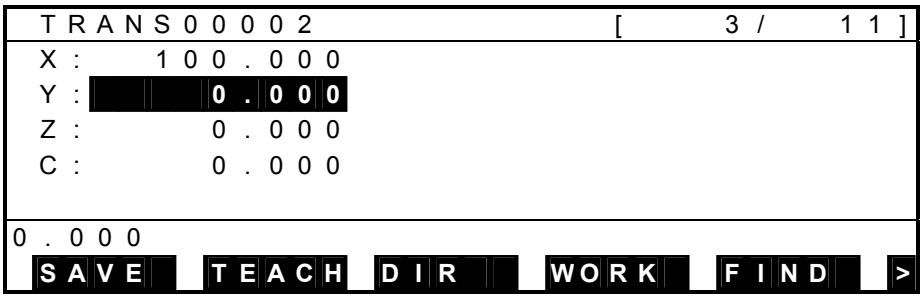
3. Enter a desired value to the command line, using alphanumeric keys, →, ← cursor keys. Then the following display appears.



Keys operative in this mode and their operations are tabled below.

Key	Descriptions
←	Used to move left the cursor on the command line.
→	Used to move right the cursor on the command line.
Numeral . + –	Used to input characters to the command line.
EXE	Used to change current coordinate element to an entered one. The system returns to the display mode with the main cursor moved to the next data area.
INS	Used to select the insert or replace mode for the command line.
BS	Used to delete one (1) character just before the cursor position on the command line.
ALT + BS	Used to delete one (1) character at the cursor position on the command line.
ESC	This function works in a different manner according to the presence or absence of input character(s). Presence of input character(s): All input character(s) on the command line are cleared. Absence of input character(s): The system returns to the detailed coordinate data directory display.

4. Press the EXE key, and the coordinate element is replaced by the newly specified data. The system returns to the detailed coordinate data directory display mode with the main cursor moved to the next data area. The display is as shown below.



6.5.3 Teach of Coordinate Data [TEACH]

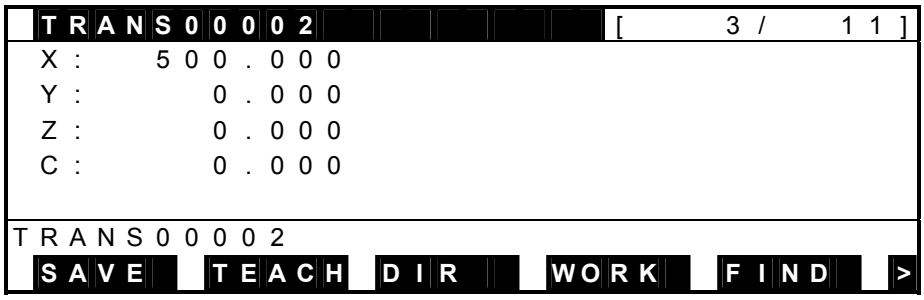
(1) Function

Guides the robot tool tip to the teach point, then registers the current position coordinate data of the robot in the coordinate data indicated by the main cursor.

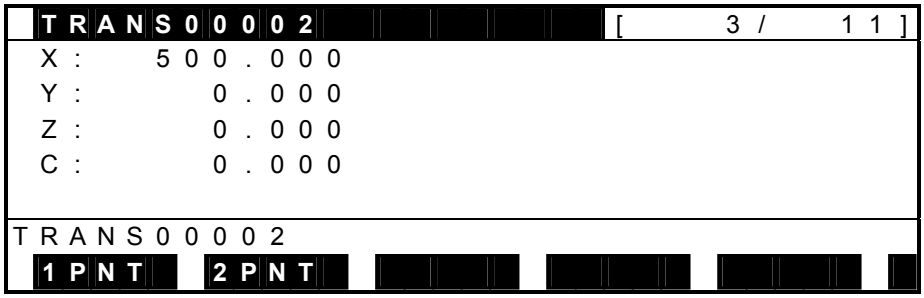
(2) Procedures

Assume that you are going to teach and register the current position coordinate data of the robot in the coordinate data of "TRANS00002."

1. Call the detailed coordinate data display of "TRANS00002."



2. Press [TEACH], and the display as shown below appears.



Teaching of coordinate data comes in the following two (2) types.

[1] 1-point teaching mode in which the origin and direction are taught at the same time.

[2] 2-point teaching mode in which the origin and direction are taught separately in two (2) steps.

Select a desired teaching mode now.

The key operative in this mode and its operation are tabled below.

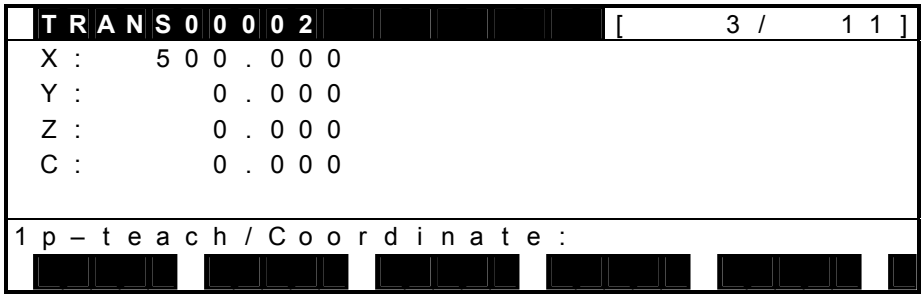
Key	Descriptions
ESC	Used to cancel the detailed coordinate teaching mode and return the system to the detailed display.

Operations of respective soft keys are shown below.

Key	Descriptions
[1PNT]	Used to select the 1-point teaching mode.
[2PNT]	Used to select the 2-point teaching mode.

[A] 1-point teaching mode

3. Press [1PNT], and the following display appears.



4. Guide the robot tool tip to the teach point and set the direction of the tool coordinate to the direction of the coordinate you wish to teach.

   Keys operative in this mode and their operations are tabled below.

Key	Descriptions
EXE	Used to register the current position of the robot as the coordinate data.
ESC	Used to cancel the 1-point teaching mode and return the system to the teaching mode select display.

! CAUTION

In principle, teaching operation should be performed outside the dangerous area of the robot. If it should be performed inevitably within the dangerous area, strictly observe the following matters.

(1) Teaching operation should always be performed by two (2) persons. One person performs the job and the other person watches outside the dangerous area. Also, both persons should try to prevent mis-operation with each other.

(2) The operator should do the job in an attitude ready to press the EMERGENCY stop pushbutton switch at any time. Also, he should perform the job at a position from which he can evacuate immediately at the time of emergency after confirming the robot working range and shields nearby.

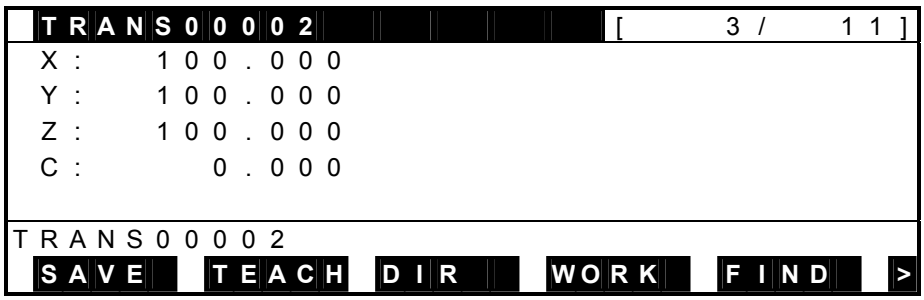
(3) The supervisor should keep watch on the job at a position where he can see the entire robot system and operate the EMERGENCY stop pushbutton switch at the time of an emergency. Also, he should keep anyone from entering the dangerous area.

5. Press the EXE key, and the current position of the robot is registered as the coordinate data.

   For instance, when the current position of the robot is,

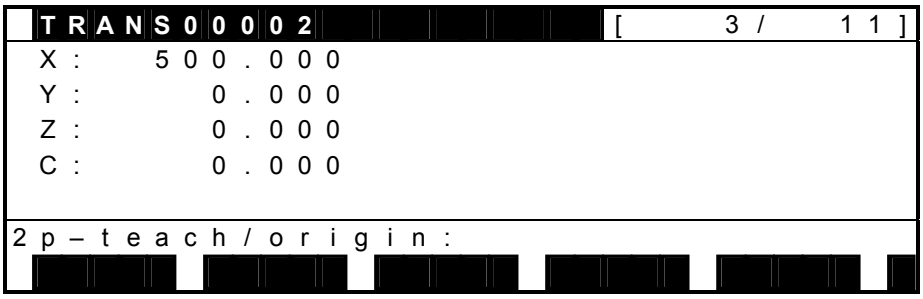
   X 100.0
   Y 100.0
   Z 100.0
   C 0

   the display is as shown below.



[B] 2-point teaching mode

3. Press [2PNT], and the following display appears, where you can teach the origin.



4. Guide the robot tool tip to the origin of the coordinate system you wish to teach.

   Keys operative in this mode and their operations are tabled below.

Key	Descriptions
EXE	Used to register the current position of the robot as the origin of a new coordinate system.
↓	Used to select the current coordinate data for the origin.
ESC	Used to cancel the 2-point teaching mode and return the system to the teaching mode select display.

! CAUTION

In principle, teaching operation should be performed outside the dangerous area of the robot. If it should be performed inevitably within the dangerous area, strictly observe the following matters.

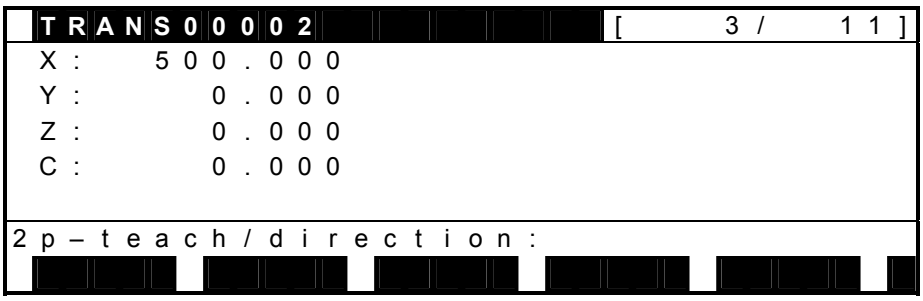
(1) Teaching operation should always be performed by two (2) persons. One person performs the job and the other person watches outside the dangerous area. Also, both persons should try to prevent mis-operation with each other.

(2) The operator should do the job in an attitude ready to press the EMERGENCY stop pushbutton switch at any time. Also, he should perform the job at a position from which he can evacuate immediately at the time of emergency after confirming the robot working range and shields nearby.

(3) The supervisor should keep watch on the job at a position where he can see the entire robot system and operate the EMERGENCY stop pushbutton switch at the time of an emergency. Also, he should keep anyone from entering the dangerous area.

5. Press the EXE key or [↓] key, and the following display appears.

   Assume that you are going to teach the X-axis plus (+) direction on this display.

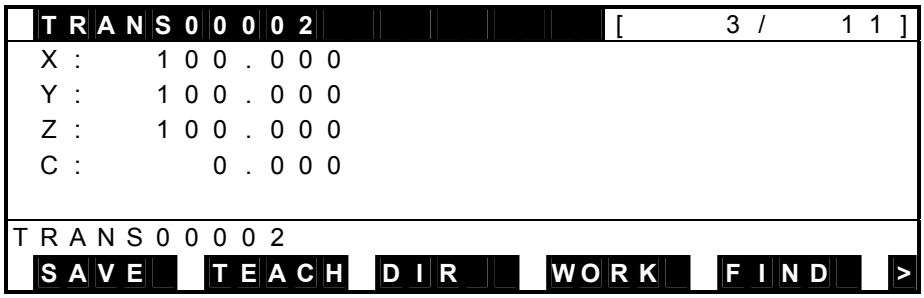


6. Guide the robot tool tip so that it can be located in the X-axis plus (+) direction of the coordinate system you wish to teach.

Keys operative in this mode and their operations are tabled below.

Key	Descriptions
EXE	Used to register a coordinate indicating the X-axis plus (+) direction, which is located on a straight line connecting the origin of a new coordinate system as defined by origin setting with a current position of the robot.
ESC	Used to cancel the 2-point teaching mode and return the system to the teaching mode select display.

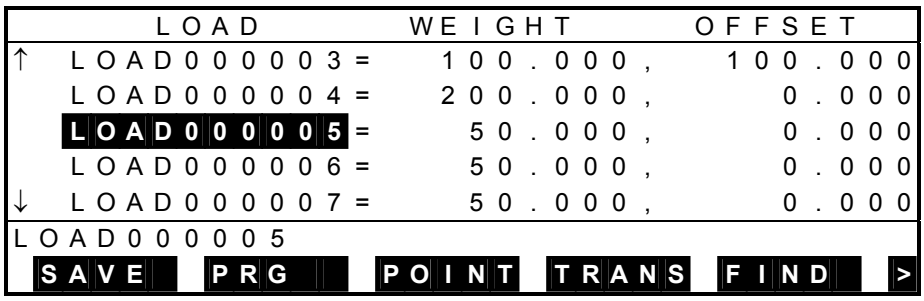
7. Press the EXE key, and the coordinate indicating the X-axis plus (+) direction, which is located on the straight line connecting the origin of the new coordinate system as defined by origin setting with the current position of the robot, is registered.
   For instance, when the current coordinates of the robot are,
   X 100.0
   Y 100.0
   Z 100.0
   C 0
   the display is as shown below.



6.6 Load Data Display Mode

The load data display mode displays the mass and gravity center offset of a load used to control the gain and acceleration. In this mode, it is possible to edit the load data.

An example of display is given below.



Each time the NEXT key is pressed, the menu is changed over as follows:

- An inversely displayed section in the data is called the "main cursor" which indicates that this data is selected for editing.
- Symbol "↑" shown in the 1st column signifies that there is load data before the top data.
- Symbol "↓" shown in the 1st column signifies that there is load data after the final data.
- "WEIGHT" represents the mass of a load (in kg).
- "OFFSET" represents the horizontal length (in mm) between the gravity center of a load and the tool shaft center.
- The load data should include the tool own load (i.e., tare weight).

Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[←]	Used to move left the main cursor.
[→]	Used to move right the main cursor.
[↑]	Used to move up the main cursor.
[↓]	Used to move down the main cursor.
[ALT]+[↑]	Used to move up the main cursor on the screen basis.
[ALT]+[↓]	Used to move down the main cursor on the screen basis.
[INS]	Used to create a new load data just after the main cursor line.
[ALT]+[BS]	Used to delete a load data indicated by the main cursor.
[EXE]	Used to edit a load data indicated by the main cursor.
[NEXT]	Used to select the next menu.
[ALT]+[NEXT]	Used to select the previous menu.

Operations of respective soft keys are shown below.

Key	Descriptions
[SAVE]	Used to save a data and quit the data editor. (When the data editor was started up from the program editor, the program editor terminates also.)
[PRG]	Used to select the program editor display. If the data editor was started up from the basic menu, however, the display will not change over.
[TRANS]	Used to select the coordinate data display.
[POINT]	Used to select the position data display.
[FIND]	Used to search a load name.
[NOSAV]	Used to quit the editor without saving data. All changed data are aborted. (When the data editor was started up from the program editor, the program editor terminates also.)
[CUT]	Used to select the range select mode.
[PASTE]	Used to paste the contents saved in the cut buffer (i.e., data deleted by the previous cut menu) after a position data indicated by the main cursor to clear the cut buffer.

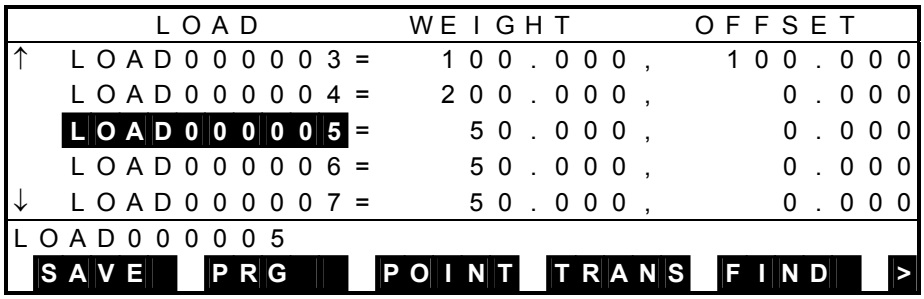
Each menu of [SAVE], [PRG], [FIND], [NOSAV], [CUT] and [PASTE] is the same as in the position data directory display mode and is not described in this paragraph.

6.6.1 Editing of Load Data Name

(1) Function
Edits a load data name.

(2) Procedures
When you press the EXE key while the main cursor is located at the load name, you can edit the load name.
In the example below, assume that you are going to change the name of "LOAD000005" to "LOAD000055."

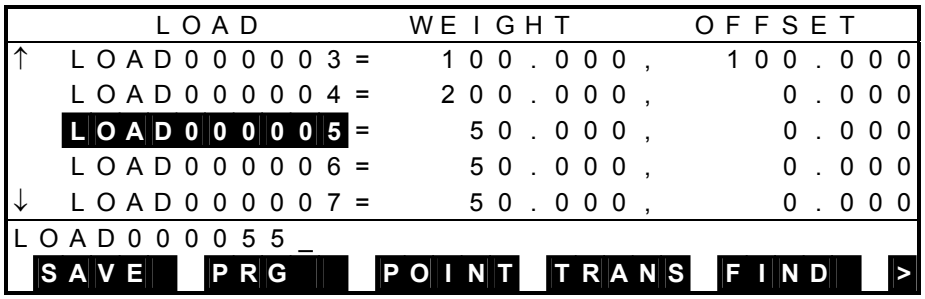
1. Move the main cursor to "LOAD000005" by means of the cursor keys (↑, ↓, →, ←), and the following display appears.



2. Press the EXE key, and the following display is called. (The cursor is displayed on the command line.)

- Current load name is displayed as the initial data on the command line.
- The cursor is displayed just after the name indicated as the initial data, and you can edit the data now.
- When the cursor is shown by mark "■", the replace mode takes effect. When the cursor is shown by mark "_", the insert mode is effective.

3. Enter a desired name to the command line, using alphanumeric keys, →, ← cursor keys. Then the following display appears.



Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[←]	Used to move left the cursor on the command line.
[→]	Used to move right the cursor on the command line.
[Numeral] [Alphabets]	Used to input characters to the command line.
[EXE]	Used to change current load name to an entered one. The system returns to the load data display mode with the cursor moved to "WEIGHT".
[INS]	Used to select the insert or replace mode for the command line.
[BS]	Used to delete one (1) character just before the cursor position on the command line.
[ALT]+[BS]	Used to delete one (1) character at the cursor position on the command line.
[ESC]	This function works in a different manner according to the presence or absence of input character(s). Presence of input character(s): All input character(s) on the command line are cleared. Absence of input character(s): The system returns to the load data display.

4. Press the EXE key, and the load name is replaced by the newly specified name. The following display appears.

```
        LOAD          WEIGHT          OFFSET
↑   LOAD0000003 =    100.000 ,     100.000
    LOAD0000004 =    200.000 ,       0.000
    LOAD0000055 =     50.000 ,       0.000
    LOAD0000006 =     50.000 ,       0.000
↓   LOAD0000007 =     50.000 ,       0.000
50.000
 SAVE  PRG  POINT TRANS FIND  >
```

- The main cursor moves to the "WEIGHT" data area.
- If the input name is illegal, a message saying "Illegal name" is displayed on the command line and no data is changed.
- If the specified name already exists, a message of "Duplicate name" is displayed on the command line and no data is changed.

6.6.2 Editing of Load Element

(1) Function
Used to edit a load element.

(2) Procedures
When you press the EXE key while the main cursor is located at each load element, you can edit that data.
In the example below, assume that you are going to change the weight data of "LOAD000005" from "50.000" to "100".

1. Move the main cursor to the data you wish to edit by means of the cursor keys (↑, ↓), and the following display appears.

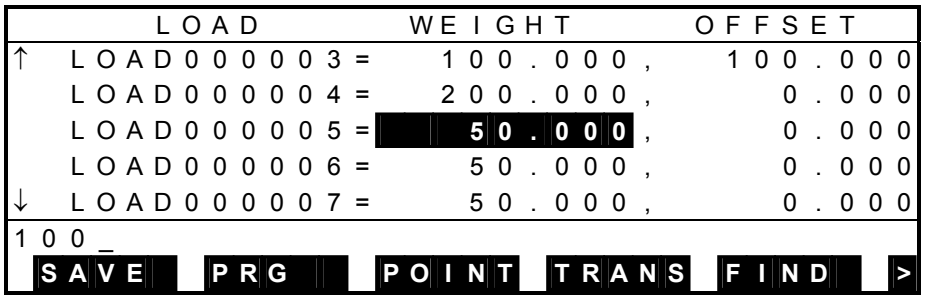
```
        LOAD          WEIGHT        OFFSET
↑   LOAD000003 =     100.000,     100.000
    LOAD000004 =     200.000,       0.000
    LOAD000005 =      50.000,       0.000
    LOAD000006 =      50.000,       0.000
↓   LOAD000007 =      50.000,       0.000
50.000
 SAVE  PRG  POINT TRANS FIND  >
```

2. Press the EXE key, and the following display is called. (The cursor is displayed on the command line.)

```
        LOAD          WEIGHT        OFFSET
↑   LOAD000003 =     100.000,     100.000
    LOAD000004 =     200.000,       0.000
    LOAD000005 =      50.000,       0.000
    LOAD000006 =      50.000,       0.000
↓   LOAD000007 =      50.000,       0.000
50.000_
 SAVE  PRG  POINT TRANS FIND  >
```

- Current data is displayed as the initial value on the command line.
- The cursor is displayed just after the data indicated as the initial value, and you can edit the data now.
- When the cursor is shown by mark "■", the replace mode takes effect. When the cursor is shown by mark "_", the insert mode is effective.

3. Enter a desired value to the command line, using alphanumeric keys, →, ← cursor keys. Then the following display appears.



Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[←]	Used to move left the cursor on the command line.
[→]	Used to move right the cursor on the command line.
[Numeral] [.] [+] [–]	Used to input characters to the command line.
[EXE]	Used to change current load element to an entered one. The system returns to the display mode with the main cursor moved to the next data area.
[INS]	Used to select the insert or replace mode for the command line.
[BS]	Used to delete one (1) character just before the cursor position on the command line.
[ALT]+[BS]	Used to delete one (1) character at the cursor position on the command line.
[ESC]	This function works in a different manner according to the presence or absence of input character(s). Presence of input character(s): All input character(s) on the command line are cleared. Absence of input character(s): The system returns to the load data display.

4. Press the EXE key, and the load element is replaced by the newly specified data. The system returns to the load data display mode with the main cursor moved to the next data area. The display is as shown below.

	LOAD	WEIGHT	OFFSET
↑	LOAD0000003=	100.000,	100.000
	LOAD0000004=	200.000,	0.000
	LOAD0000005=	100.000,	0.000
	LOAD0000006=	50.000,	0.000
↓	LOAD0000007=	50.000,	0.000

0.000

SAVE PRG POINT TRANS FIND >

6.6.3 Addition of New Load Data [INS]

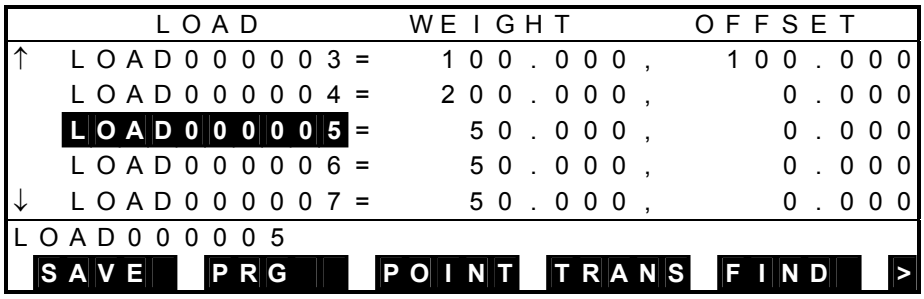
(1) Function

Adds a new load data.

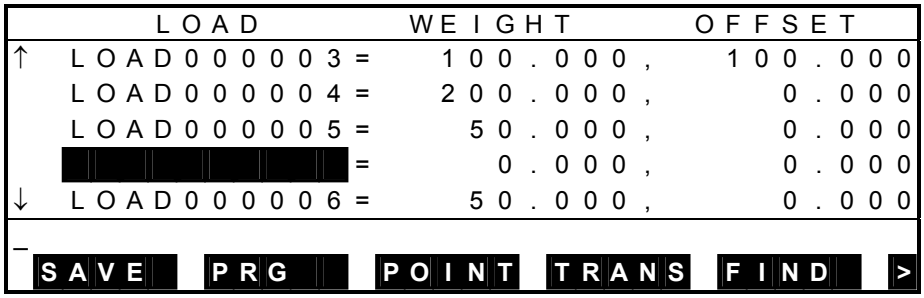
(2) Procedures

Assume that you are going to add a new load data ("LOAD000010") just after "LOAD000005."

1. Move the main cursor to "LOAD000005", and the following display is called.



2. Press the INS key, and the display changes as shown below with the name input mode selected.



3. Enter a desired name ("LOAD000010" in this example) and press the EXE key, as shown in Para. 6.4.1. The display changes as follows:

```
          LOAD          WEIGHT          OFFSET
↑  LOAD000003 =       100.000,       100.000
   LOAD000004 =       200.000,         0.000
   LOAD000005 =        50.000,         0.000
   LOAD000010 =         0.000,         0.000
↓  LOAD000006 =        50.000,         0.000
0.000
 SAVE  PRG  POINT TRANS FIND  >
```

(3) Caution

- For the newly added load data, zero (0) is set for all elements.
- When you have pressed the ESC key without specifying the name, addition of the new data is cancelled and the system returns.

6.6.4 Deletion of Load Data ([ALT] + [BS])

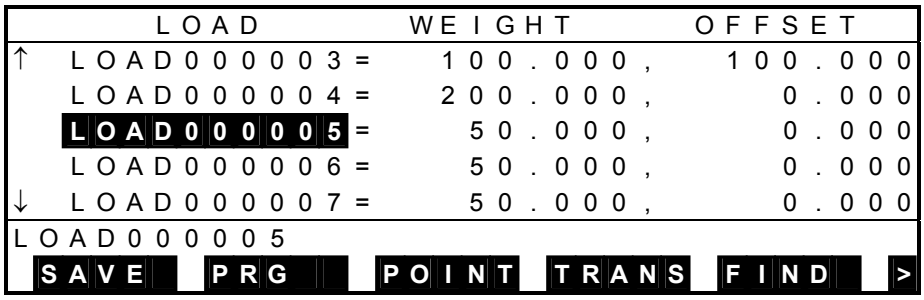
(1) Function

Deletes a load data.

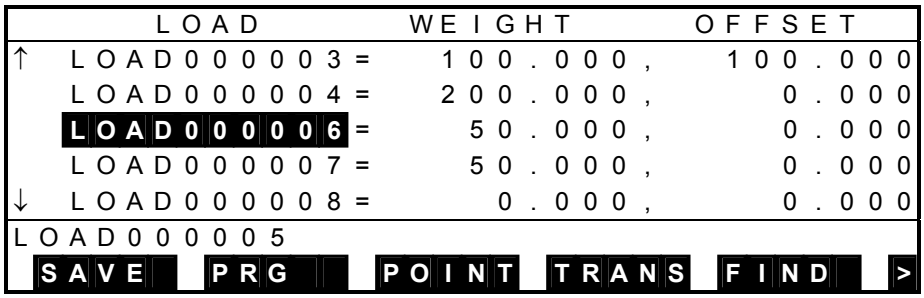
(2) Procedures

Assume that you are going to delete the data of "LOAD000005".

1. Move the main cursor to "LOAD000005", and the following display is called.



2. Press the ALT and BS keys at the same time, and the following display appears with the specified data deleted.



(3) Caution

- You cannot undo a data deleted by this operation.

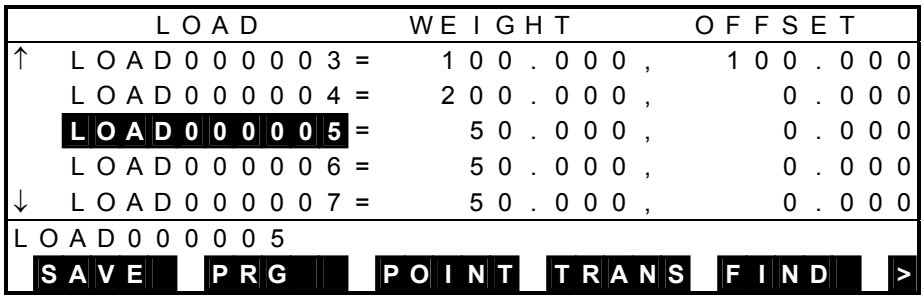
6.6.5 Call of Position Data Directory Display [POINT]

(1) Function

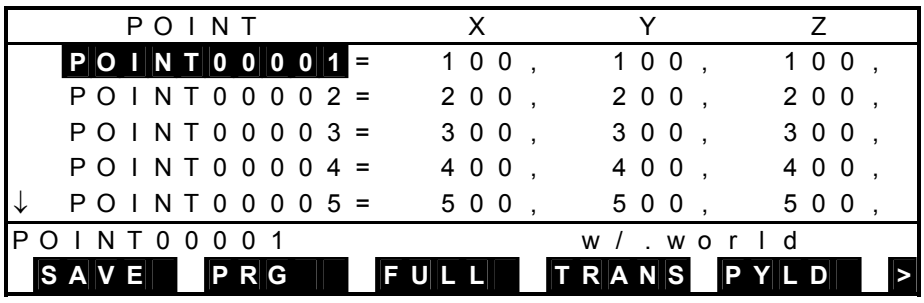
Displays the position data directory display.

(2) Procedures

1. Select the data display mode as shown below (i.e., the cursor is not present on the command line).



2. Press [POINT] and the following position data directory display is called.



(3) Caution

- On the position data directory display, the main cursor indicates the position of the data displayed before.

For details of operations to be done on the position data directory display, see Para. 6.2.

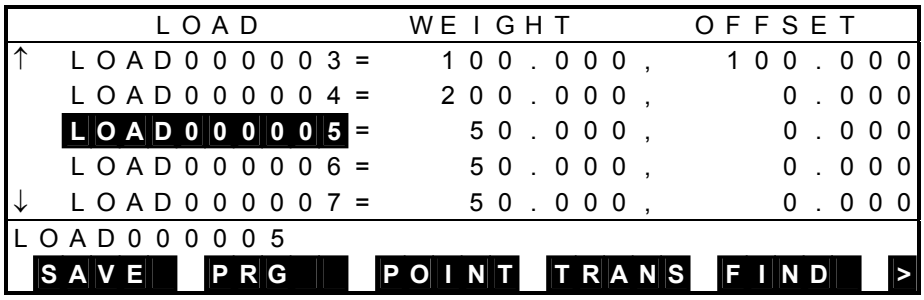
6.6.6 Call of Coordinate Data Directory Display [TRANS]

(1) Function

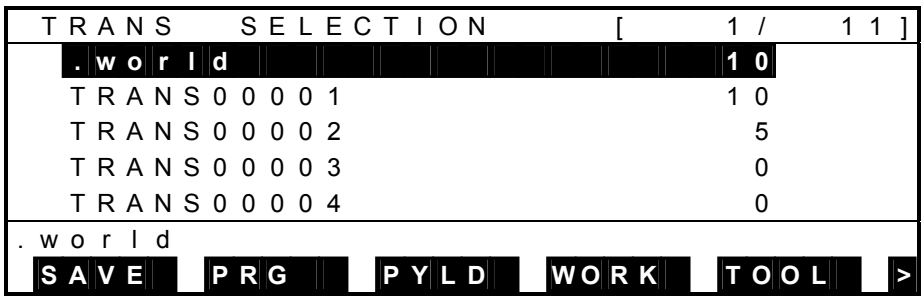
Calls the coordinate data directory display.

(2) Procedures

1. Select the data display mode as shown below (i.e., the cursor is not present on the command line).



2. Press [TRANS], and the following coordinate data directory display appears.



(3) Caution

- On the coordinate data directory display, the main cursor indicates the position of the data displayed before.

For details of operations to be done on the coordinate data directory display, see Para. 6.4.

Section 7 Test Operation

DANGER

NEVER enter the dangerous area of the robot during automatic operation. If you still have to enter the dangerous area, be sure to terminate the automatic operation and turn the servo power off beforehand.

CAUTION

1. Make sure that no obstacle stands in the robot working range. If any obstacle exists, perform automatic operation at a slow speed and make sure that the robot will not collide with the obstacle at all.
2. If the robot motion becomes abnormal during program execution, press the EMERGENCY stop pushbutton switch immediately to stop the robot. NEVER restart the automatic operation until the cause of the abnormality is identified.
3. The speed of automatic operation is initially set at 100 % of the maximum robot speed.
4. When performing automatic operation after power ON, reset the program for the automatic operation in advance. If the program is executed continuously, the robot will collide with the peripheral equipment, resulting in damage of the equipment and accident.

7.1 Selecting Test Operation Mode

(1) Function

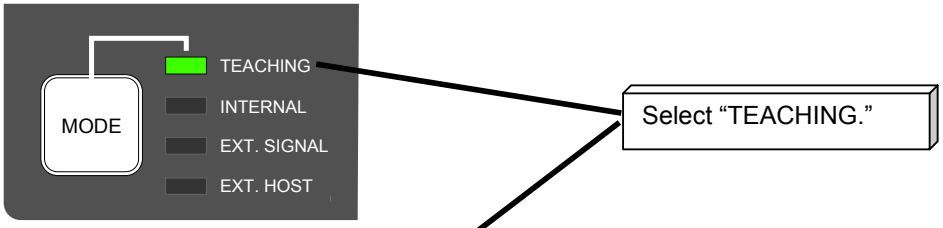
Selects the test operation mode where you can execute a program partially or wholly to verify the program.

(2) Procedures

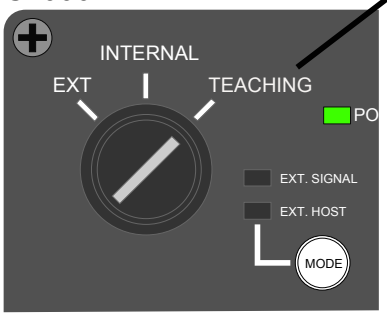
(a) Step 1: Selection of teaching mode.

Change over the MODE switch equipped on the control panel to "TEACHING."

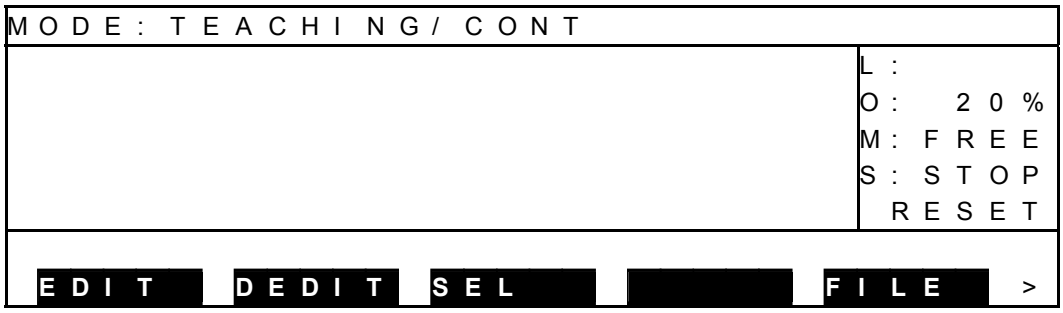
TS1000



TS2000



(b) Step 2: Selection of test operation mode.
When the teaching mode is selected, the test operation display of the teaching mode is called, as shown below.



Descriptions on display

[1] MODE: Indicates the currently selected master mode and run mode.
- CONT : Continuous operation
- CYCLE : Cycle operation
- SEG : Segment operation
- STEP : Step operation

[2] Right side on the 1st line: Identifies active file name.

[3] L: No. of execution steps

[4] O: Speed override

[5] M: Machine status
- FREE : Machine lock OFF status

[6] S: Execution status
- RUN : A program is currently executed.
- STOP RESET : A program is stopped in the initialize mode. This is the same as the program reset.
- STOP RETRY : A program is stopped in the restart mode. It is restarted from the interrupted motion.
- STOP CONT : A program is stopped in the continue mode. It is restarted from the current program step.

[7] Bottom line: Function key menu display field

EDIT	:	Program editing
DEDIT	:	Data editing
SEL	:	Selection of program to be executed
RUN	:	Test operation start
FILE	:	File mode
MODE	:	Run mode selection
OVRD	:	Override setting
RESET	:	Reset
DO	:	Direct execution
WATCH	:	Variable monitor
STOP	:	Stop
BREAK	:	Interruption

7.2 Selecting File (or Program) [SEL]

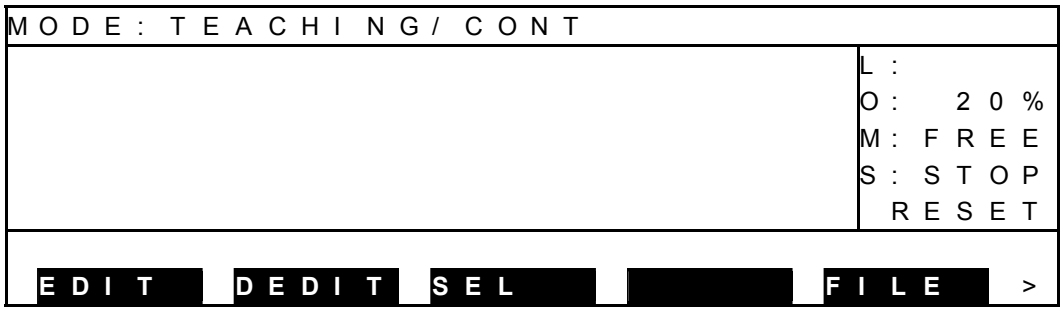
(1) Function

Selects a file (or program) to be executed.

(2) Procedures

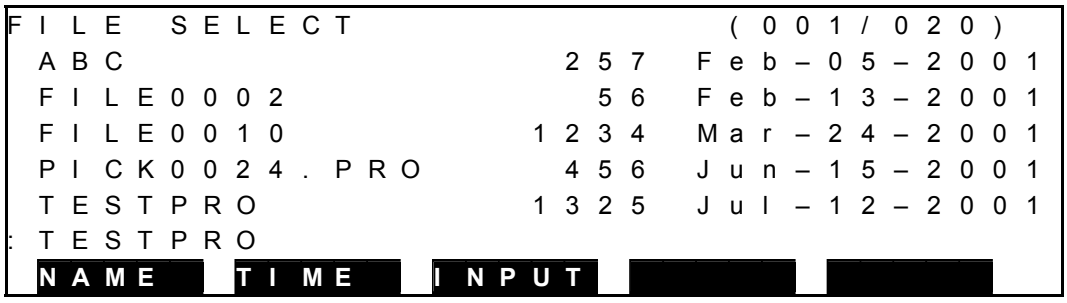
(a) Step 1: Command selection.

Press the function key corresponding to the [SEL] menu. Unless the SEL menu is available, press the NEXT key repeatedly until [SEL] is displayed in the soft key menu.



(b) Step 2: Input of file name.

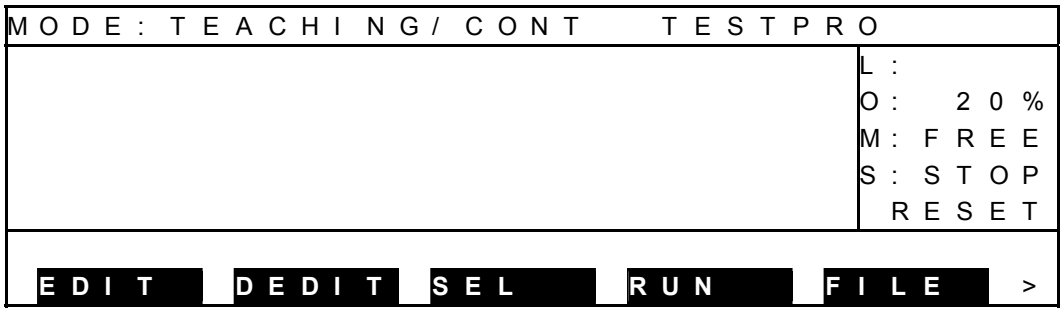
When the file directory is displayed, inversely display and select a desired file name, using the up and down cursor keys, then press the EXE key.
NAME and TIME in the menu are used to sort the file directory by name and time, respectively. To directly enter the file name through the keyboard, press the INPUT function key and input the file name you wish to execute. If the file name contains an extension, be sure to input the extension also.



Note: If mark * precedes a file name and this file name is inversely displayed on the file select screen, it is the execution file already selected.

(c) Step 3: Input of EXE key.

Make sure that your entry is correct, and press the EXE key. Then the specified file name is displayed in the execution file display field on the 1st line at the same time that [RUN] is displayed in the soft key menu to indicate that you can start a test operation.



(3) Caution

Once the execution file has been selected, it remains effective until another file is selected. If a compile error has occurred, however, selection of the execution file is cancelled.

When another file has been selected, the execution environment of automatic operation is reset. To reset the execution environment without changing the execution file, press the NEXT key followed by [RESET] and [PROG].

7.3 Selecting Run Mode [MODE]

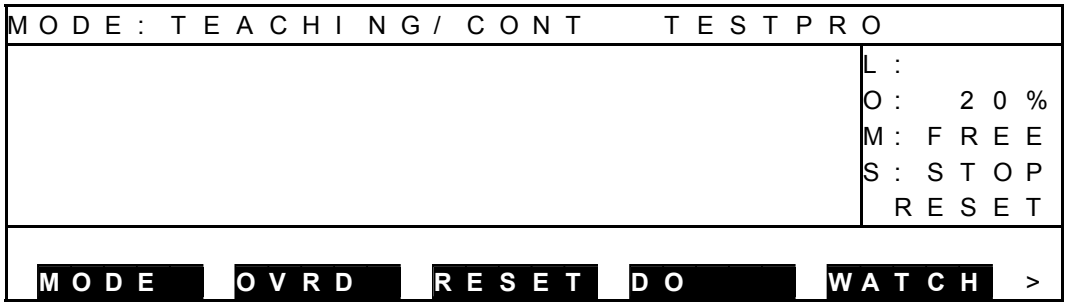
(1) Function

Changes over the run mode of automatic operation.

(2) Procedures

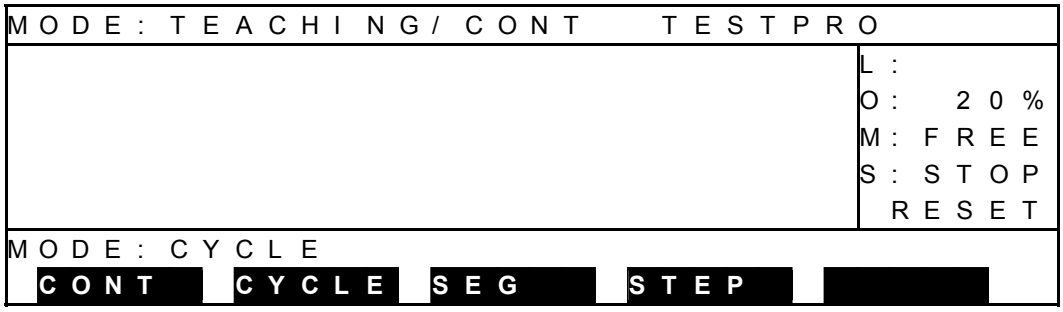
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [MODE] is displayed in the soft key menu. Press the corresponding function key then.



(b) Step 2: Mode selection.

Select a desired run mode from the menu.

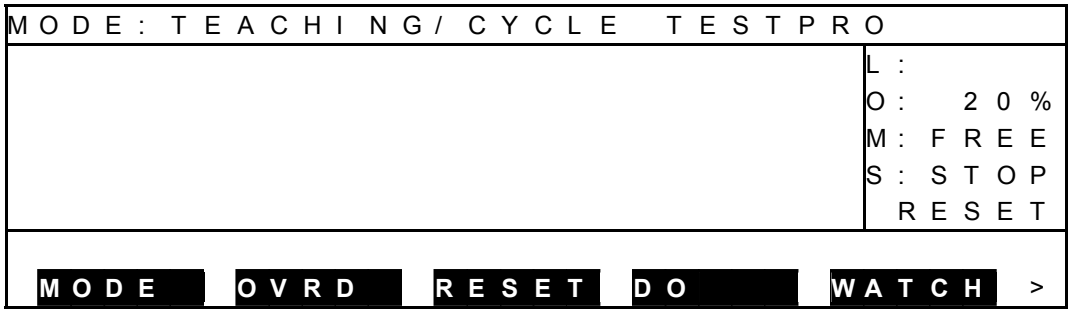


CONT : Repeats program execution continuously.
CYCLE : Stops program execution at the end of the program.
SEG : Stops program execution just before the next motion command.
STEP : Executes a program each time one program step has been executed.

(c) Step 3: Input of EXE key.

Make sure that your entry is correct, and press the EXE key. If you have entered a wrong data, press the ESC key to cancel the entry and select again. Then the specified run mode is shown on the display.

When the CYCLE mode is selected, the CYCLE switch LED on the control panel is illuminated also.



(3) Caution

The run mode can be selected even while the program is being executed. The execution mode selected is also operative in both the internal automatic mode and external automatic mode. The selected run mode remains valid until another run mode is selected.

If the stop command is executed in the program, program execution stops, irrespective of the run mode currently selected.

In the teaching mode, the CYCLE switch provided on the control panel is inoperative.

7.4 Speed Override [OVRD]

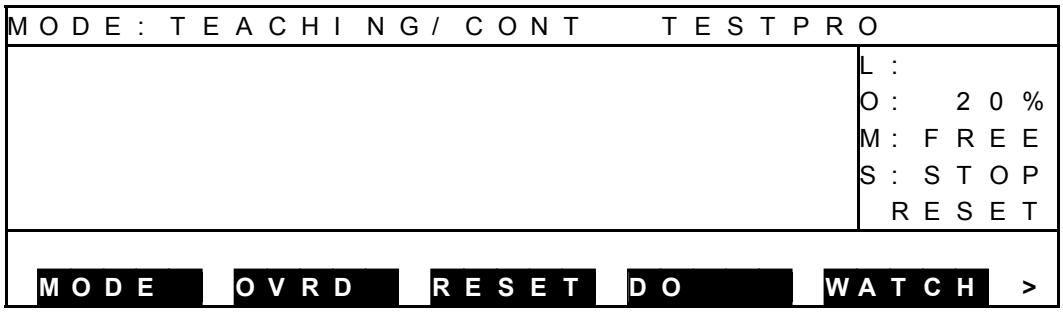
(1) Function

Sets the speed override for automatic operation. This function is useful to effect an additional override, taking the programmed speed as 100 %.

(2) Procedures

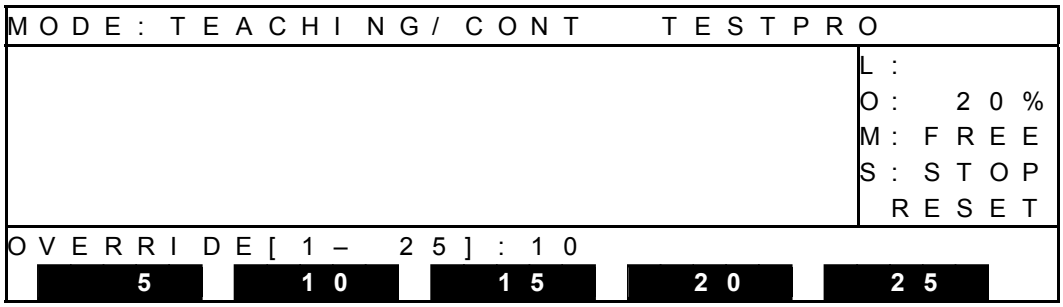
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [OVRD] is displayed in the soft key menu. Press the corresponding function key then.



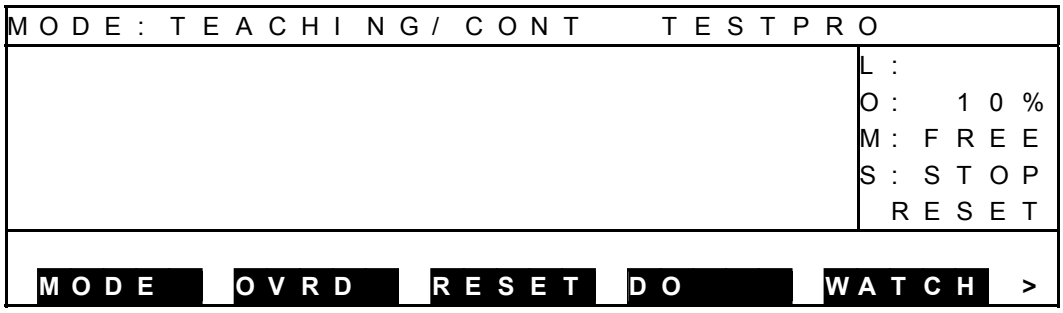
(b) Step 2: Input of speed override.

Set an override of 1 ~ 100 on the speed by means of appropriate function key.



Note: The maximum override for the teaching mode can be specified by the user parameter. In the teaching mode, a set value MUST NOT exceed this maximum value.

(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key. Then the specified override value is shown under "O" on the display.



(3) Caution
The specified speed override is also operative in both the internal automatic mode and external automatic mode. The selected override remains valid until another override is specified.

(4) Related matter
The speed can be limited by the external signal ("low speed command"). When this happens, the motion speed can be specified by the user parameter. If the override speed set exceeds this low speed, it is clamped there. The external signal (low speed command) also takes effect in both the internal automatic mode and external automatic mode.

7.5 Executing Program Step [STEP]

(1) Function

Executes a test operation in the STEP mode in which a program is stopped each time a step has been executed.

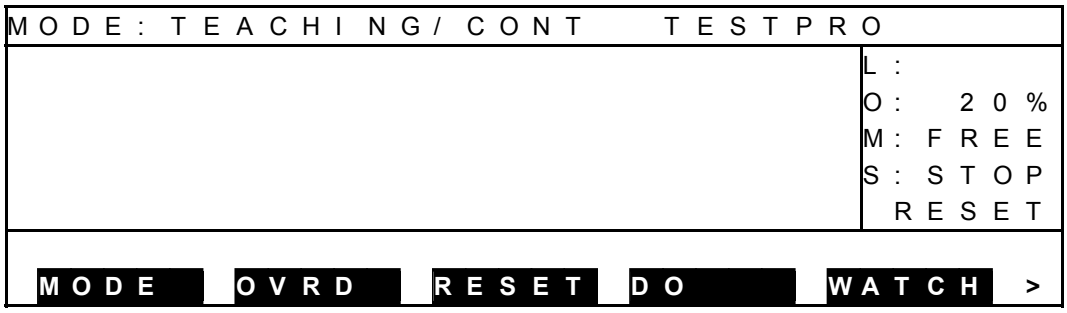
(2) Procedures

(a) Step 1: Setting of running conditions.

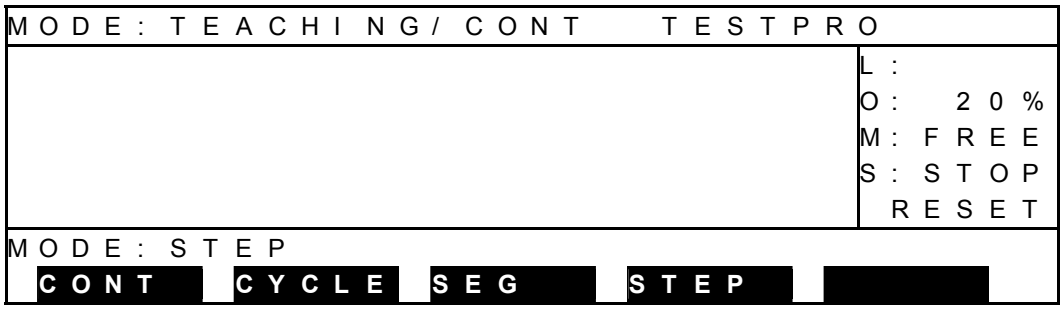
Set the conditions for executing the file selection and speed override setting.

(b) Step 2: Command selection.

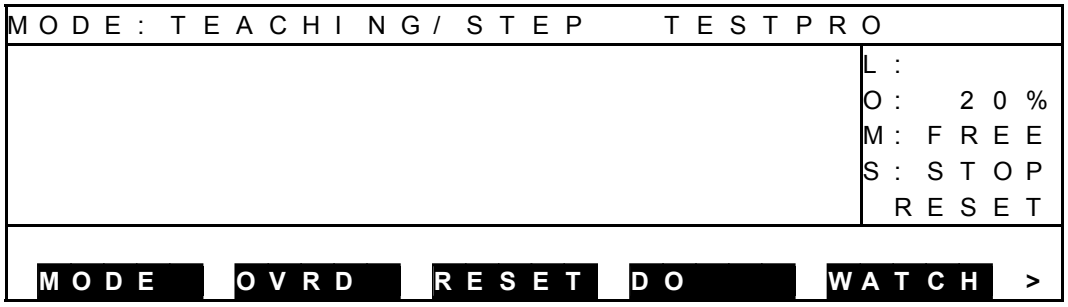
Press the NEXT key repeatedly until [MODE] is displayed in the soft key menu. Press the corresponding function key then.



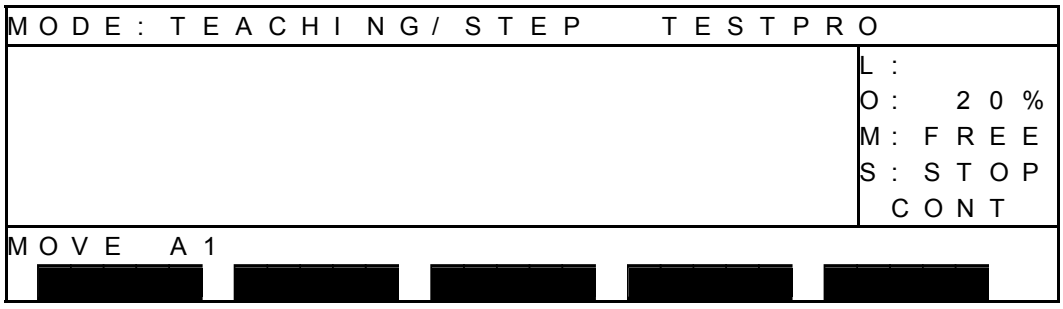
(c) Step 3: Mode selection.
Select [STEP] from the menu.



(d) Step 4: Input of EXE key.
Make sure that your entry is correct, and press the EXE key. The step operation mode is selected then.

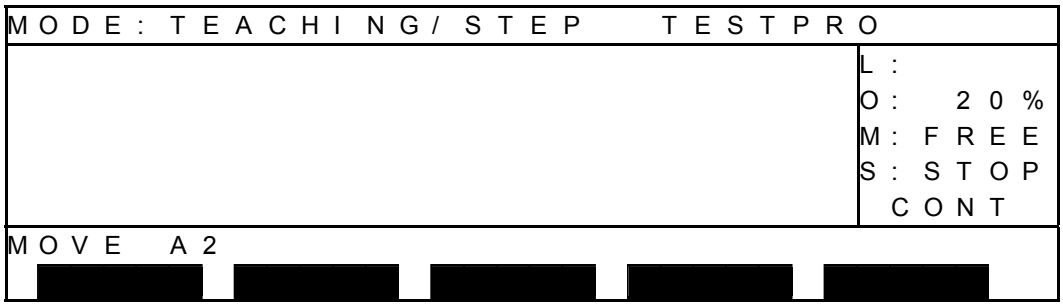


(e) Step 5: Execution of RUN.
Select [RUN] from the menu by means of the NEXT key, then press the EXE key.



(f) Step 6: Execution.

When the program step you wish to execute is displayed in the command input field, press the EXE key. After executing the displayed step, the system waits for the next input of the EXE key, displaying the next program step. Press the EXE key, and the next step is executed.



(3) Caution

To stop the step operation, press the ESC key to quit. To execute a step in another run mode, select the run mode again. During step execution, a step to be executed is displayed in the command filed of the teach pendant.

If a step operation is executed, using a multitask program, tasks are not changed over. The TASK command, SWITCH command or any other command relating the multitask is not executed, either.

7.6 Startup [RUN]

(1) Function

Starts a test operation under the predetermined conditions. This function is used to verify a newly created program and to restart an interrupted program.

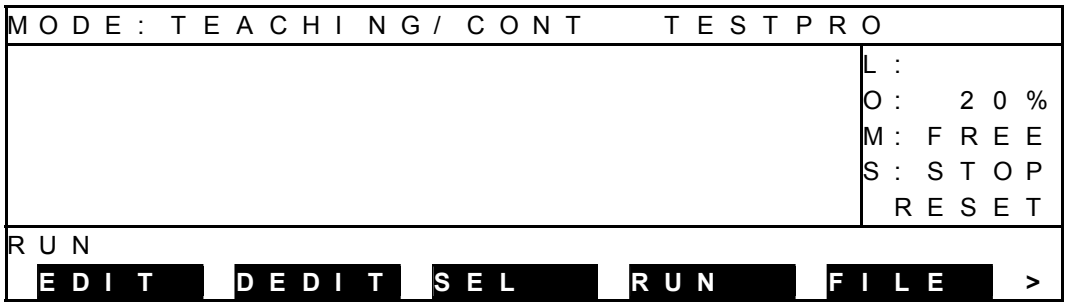
(2) Procedures

(a) Step 1: Setting of running conditions.

Set the conditions for executing the file selection, run mode selection speed override setting.

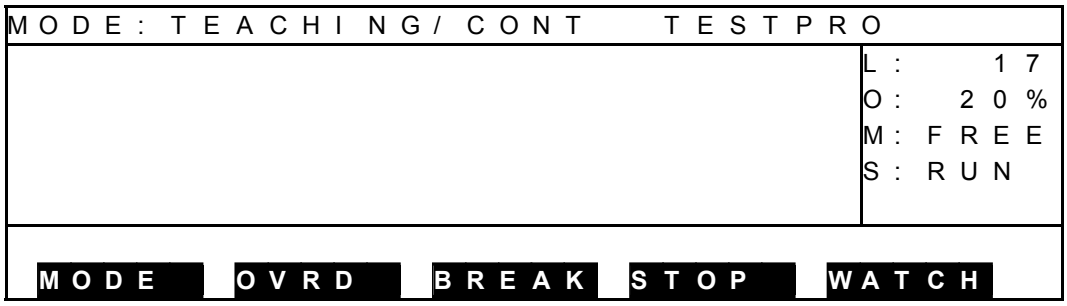
(b) Step 2: Command selection.

Press the NEXT key repeatedly until [RUN] is displayed in the soft key menu. Press the corresponding function key then.



(c) Step 3: Input of EXE key.

Press the EXE key. When program execution has started, the STOP switch LED on the control panel goes out with the RUN switch LED illuminated. During the program execution, the display of the teach pendant is as shown below with "RUN" indicated under "S" (execution status).



(3) Caution

The RUN switch equipped on the control panel and "startup command" external control signal are inoperative in the test operation mode.

To execute a program from the top, execute [RESET] and [PROG] followed by [RUN].

7.7 Stop

7.7.1 CYCLE Stop

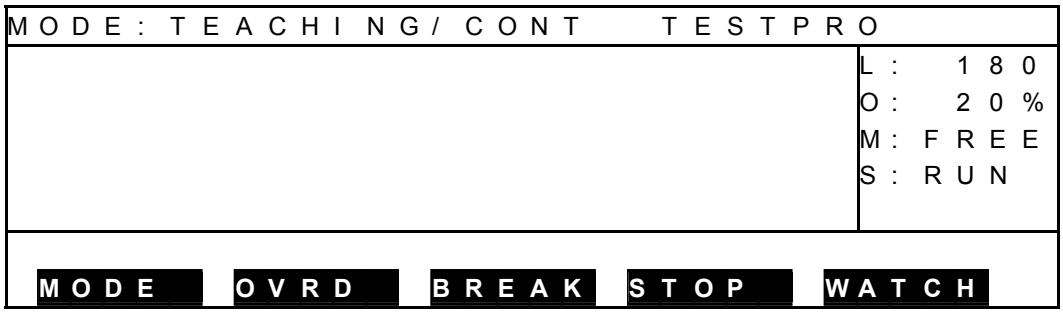
(1) Function

Selects "CYCLE" for the run mode and stops an automatic operation at the program end. This function is used to stop the robot after the end of an operation.

(2) Procedures

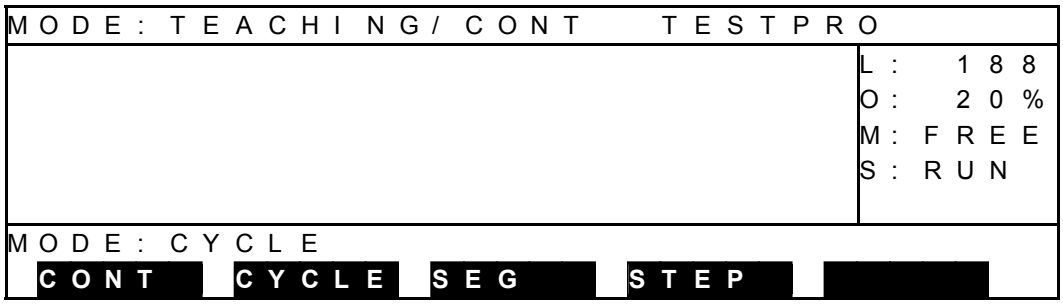
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [MODE] is displayed in the soft key menu, and press the corresponding function key.



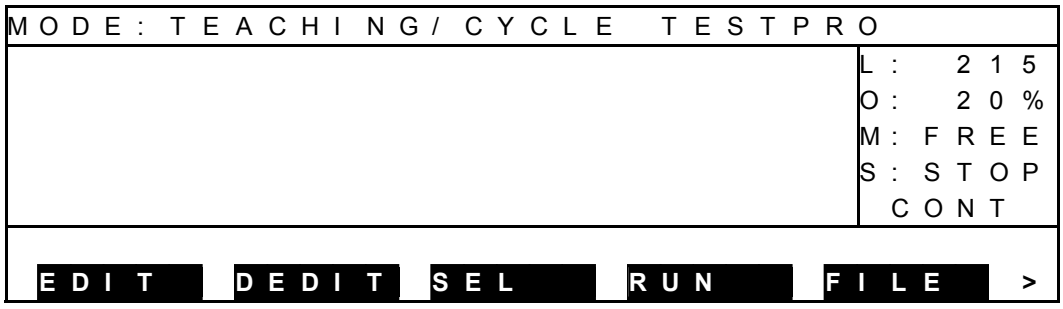
(b) Step 2: Mode selection.

Select "CYCLE" from the menu.



(c) Step 3: Input of EXE key.

Make sure that your entry is correct, and press the EXE key. The CYCLE switch LED equipped on the control panel is illuminated with "CYCLE" shown under "S" (execution status) on the teach pendant display. When the program has been executed to the end, the automatic operation stops at the same time that the RUN switch LED on the control panel goes out with the STOP switch LED illuminated. Also, "STOP CONT" is indicated for "S" (execution status) on the teach pendant display.



(3) Caution

The CYCLE switch equipped on the control panel and "cycle stop" external control signal are inoperative in the test operation mode.

The SCOL program execution unit pre-reads up to four (4) motion commands. Therefore, the program may be executed successively by up to three (3) cycles according to the cycle stop timing or the active program contents.

7.7.2 STOP

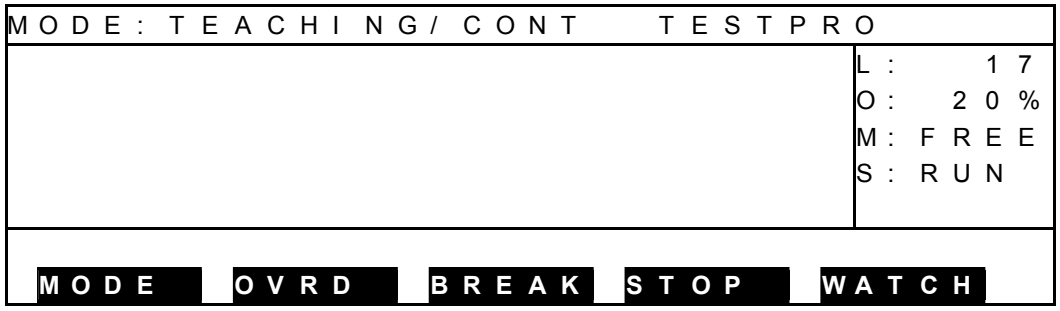
(1) Function

Stops an automatic operation when the active command has been executed. This function is used for normal stop operation.

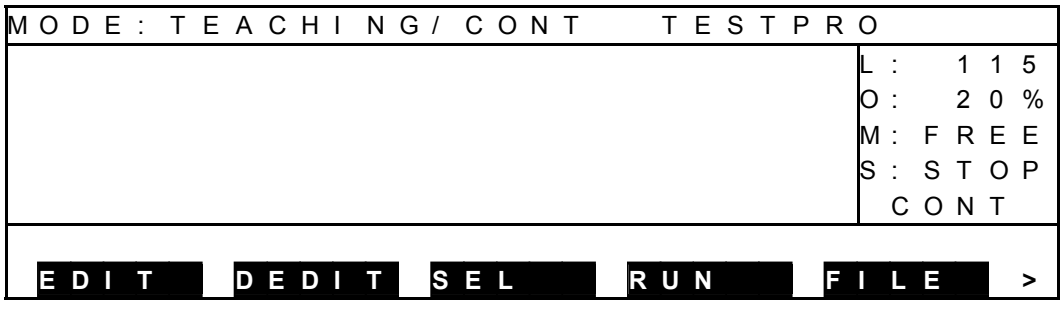
(2) Procedures

(a) Step 1: Input of [STOP].

- Control panel: Press the STOP switch.
- Teach pendant: Press the function key corresponding to the [STOP] menu.



The STOP switch LED on the control panel is illuminated. When the currently executed command has completed, the automatic operation stops at the same time that the RUN switch LED on the control panel turns off with "STOP CONT" shown under "S" (execution status) on the teach pendant display.



(3) Caution

In the test operation mode, the "stop" external control signal is inoperative. To restart, press [RUN] followed by the EXE key.

7.7.3 BREAK

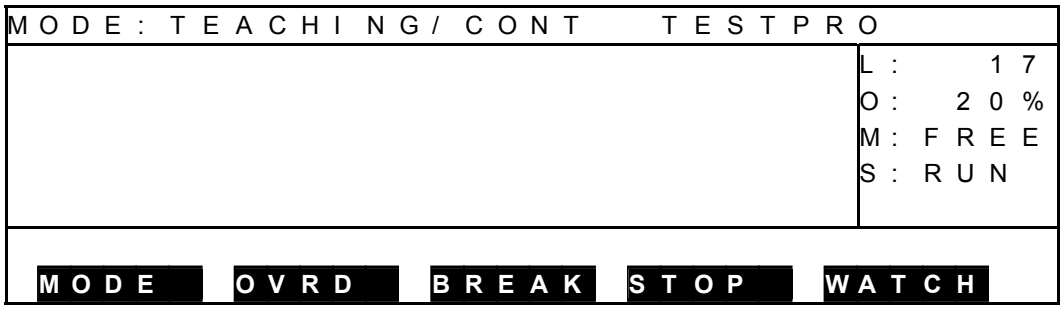
(1) Function

Immediately decelerates and stops the robot, and stops an automatic operation. This function is used to interrupt the automatic operation.

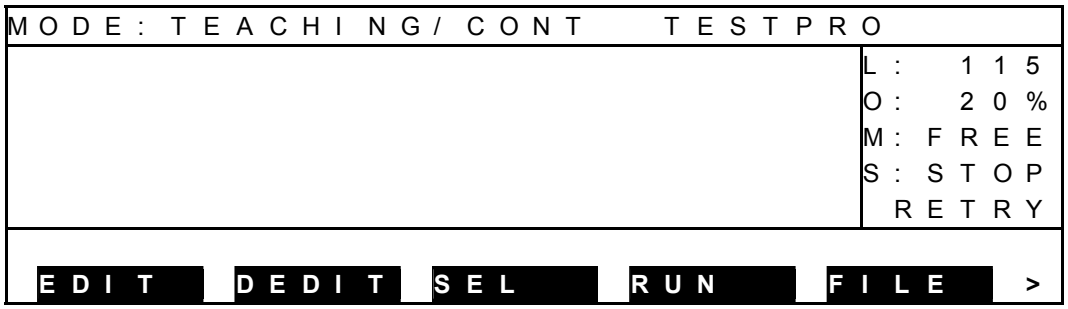
(2) Procedures

(a) Step 1: Input of [BREAK].

Press the function key corresponding to the [BREAK] menu. The robot immediately slows down and stops to stop the automatic operation.



The RUN switch LED on the control panel turns off with the STOP switch LED illuminated. "STOP RETRY" is shown under "S" (execution status) on the teach pendant display.



(3) Caution

During program execution in the step mode, the BREAK function is inoperative. To stop the step operation, press the EMERGENCY stop key or FEED HOLD key. To restart the operation after the break, press [RUN] followed by the EXE key.

7.7.4 FEED HOLD

(1) Function

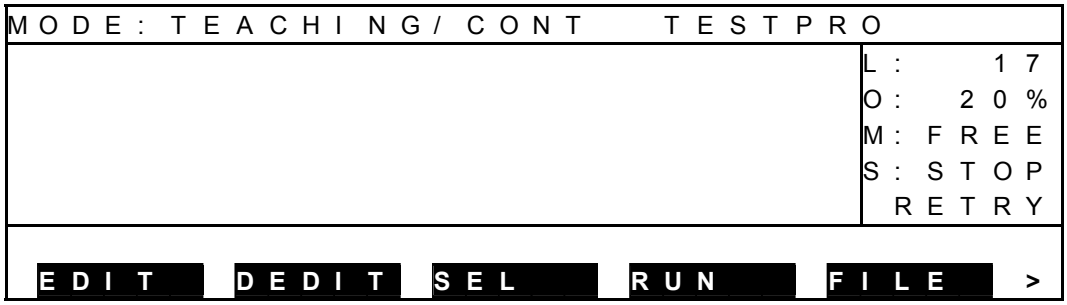
Holds robot motion and program execution. When this function is specified during robot operation, the robot immediately slows down and stop. The feed hold function is used to stop the robot temporarily.

(2) Procedures

(a) Step 1: Input of feed hold.

Press the FEED HOLD pushbutton switch equipped on the teach pendant, and the FEED HOLD switch LED is illuminated. If the robot is working, it decelerates and stops.

When the robot has stopped, the RUN switch LED on the control panel goes out with the STOP switch LED illuminated. The display of the teach pendant is as shown below with "STOP RETRY" indicated under "S" (execution status).



(b) Step 2: Cancel of feed hold.

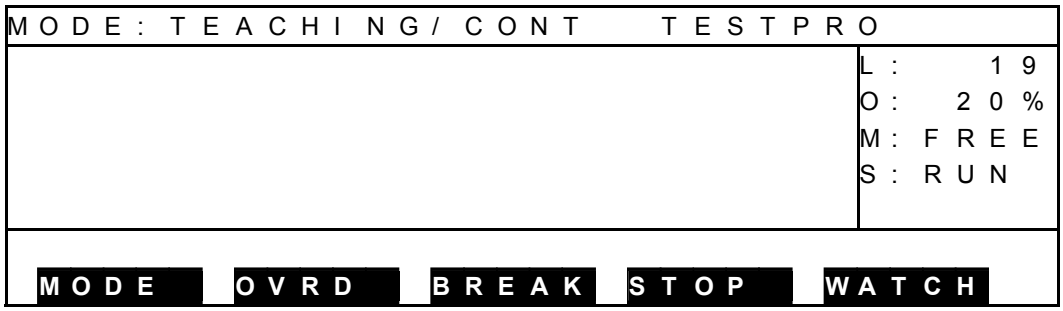
While the FEED HOLD switch LED is turned on, press the FEED HOLD switch again. The [RUN] menu cannot be cancelled.

The FEED HOLD switch LED turns off and automatic operation, step execution, direct execution and manual guidance become operative.

If feed hold was commanded during program execution, program execution can be restarted by cancelling the feed hold.

When the program has restarted, the STOP switch LED on the control panel turned off with the RUN switch LED illuminated.

"RUN" is shown under "S" (execution status) on the teach pendant display.



(3) Caution

In the feed hold mode, automatic operation, step execution, direct execution, home return and manual guidance are invalid.

When the FEED HOLD switch is pressed again, feed hold is cancelled. If the feed hold was commanded during program execution, program execution can be restarted by cancelling the feed hold. When the test operation has finished, program execution will not restart.

7.7.5 EMERGENCY Stop

(1) Function

Turns off the servo power and stops an automatic operation. This function is useful to stop the robot system at the time of an emergency.

(2) Procedures

Press the EMERGENCY stop pushbutton switch equipped on the control panel or teach pendant, or use the "emergency stop" external control signal.
The servo power is turned off and the SERVO ON switch LED on the control panel goes out with the SERVO OFF switch LED illuminated.
The servo ON switch LED on the teach pendant goes out.

(3) Caution

All of the EMERGENCY stop switches on the control panel and teach pendant, and "emergency stop" external control signal are operative. At the time of an emergency, use whichever key you can get at quickest.
The EMERGENCY stop switches on the control panel and teach pendant are of a lock type. When pressed, they remain depressed and locked. To reset, turn and unlock the switch.
While the EMERGENCY stop switch is ON, the servo power cannot be turned on.

7.8 Reset [RESET]

The reset operation comes in the following five (5) modes. Any mode other than (4) cannot be reset unless a file for execution is selected.

(1) Program reset
(2) Step reset
(3) Cycle reset
(4) Output signal reset
(5) Execution file reset

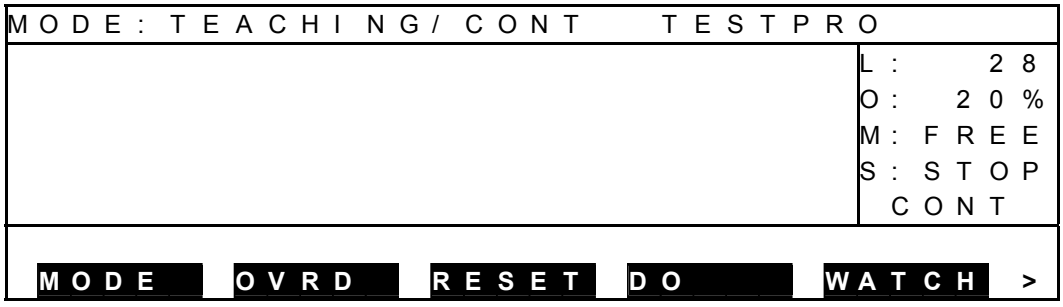
7.8.1 Program Reset

(1) Function

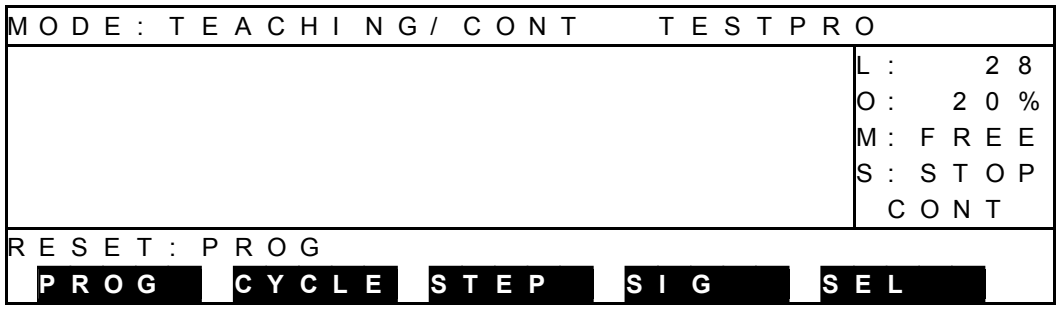
Resets all execution conditions (execution steps, variables) for a current file and allows execution of that program from the top.
This function is useful to execute a current file from the top of the program.

(2) Procedures

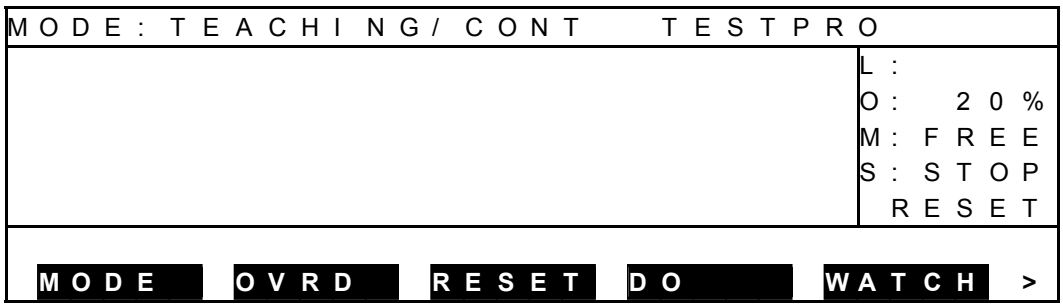
(a) Step 1: Command selection.
Press the NEXT key repeatedly until [RESET] is displayed in the soft key menu, and press the corresponding function key.



(b) Step 2: Program selection.
Select [PROG].



(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key. "STOP RESET" is shown under "S" (execution status) on the teach pendant display.



(3) Caution
All variables used in the program are reset. If you do not wish to the variables, use the step reset function instead.

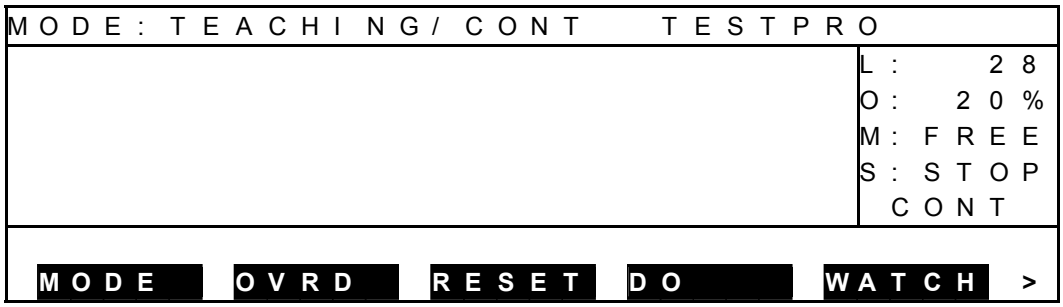
7.8.2 Step Reset

(1) Function

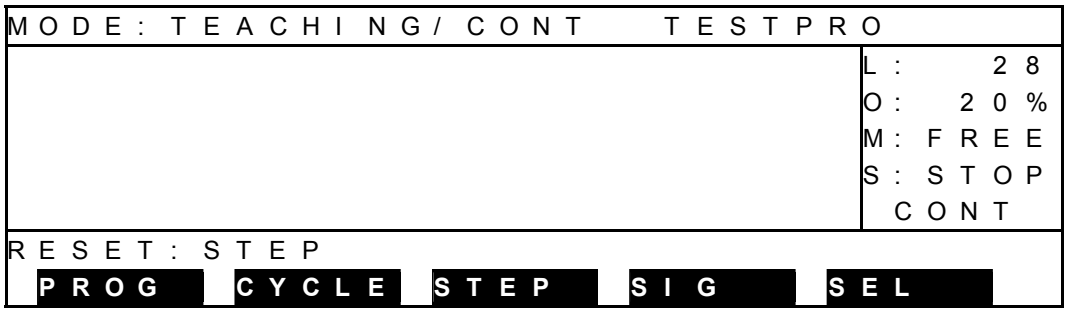
Resets all execution conditions (excluding variables) for a current file and allows execution of that program from the top.
This function is useful to set the repetitive counts in a variable and execute the remaining counts.

(2) Procedures

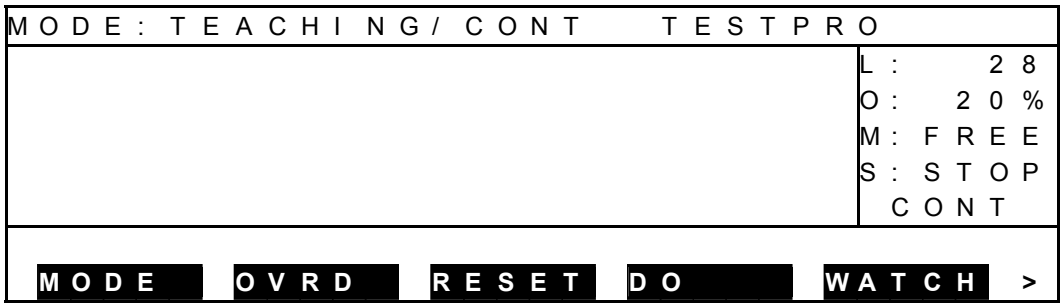
(a) Step 1: Command selection.
Press the NEXT key repeatedly until [RESET] is displayed in the soft key menu, and press the corresponding function key.



(b) Step 2: Step selection.
Select [STEP].



(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key.



(3) Caution
When you do not wish to execute the setting of initial conditions at the head of the program, use the cycle reset instead.

7.8.3 Cycle Reset

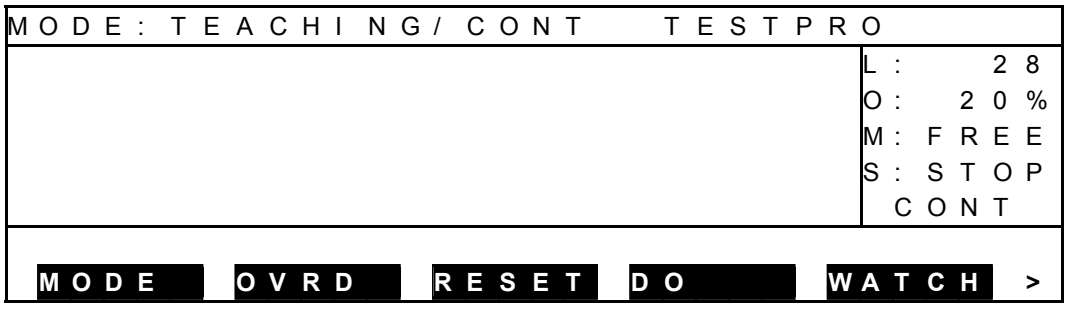
(1) Function

Resets all execution conditions (excluding variables) for a current file and allows execution of the main program from the step indicated by label RCYCLE. This function is useful when you do not wish to execute the setting of the initial condition.

(2) Procedures

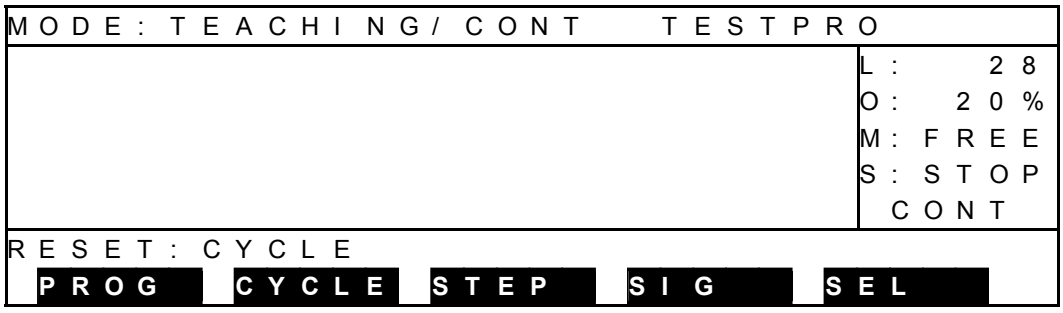
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [RESET] is displayed in the soft key menu, and press the corresponding function key.

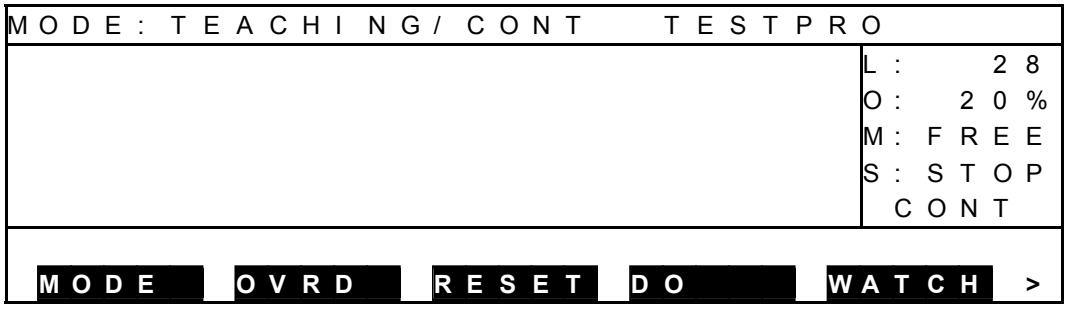


(b) Step 2: Cycle selection.

Select [CYCLE].



(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key. Unless label RCYCLE exists in the main program, the execution conditions, etc., remain unchanged.



(3) Caution
Use label RCYCLE in a desired step of the main program. Also, specify the GOTO RCYCLE command just before the END command not to execute the END command of the main program.

7.8.4 Output Signal Reset

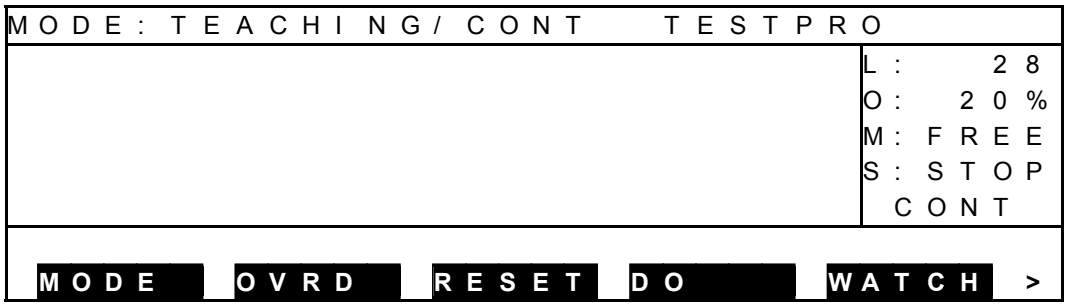
(1) Function

Sets off all user external output signals (DOUT (1) ~ DOUT (164)). The hand signals (201 ~ 204) will not turn off.

(2) Procedures

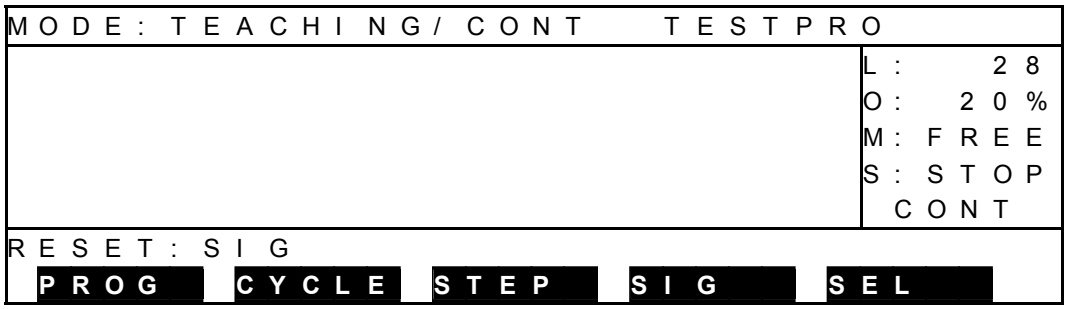
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [RESET] is displayed in the soft key menu, and press the corresponding function key.

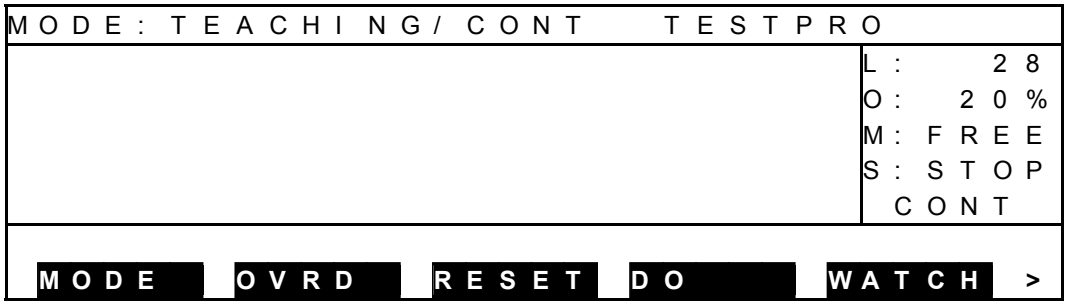


(b) Step 2: SIG selection.

Select [SIG].



(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key.
The system sets off all user output signals and waits for entry of another command.



7.8.5 Execution File Reset

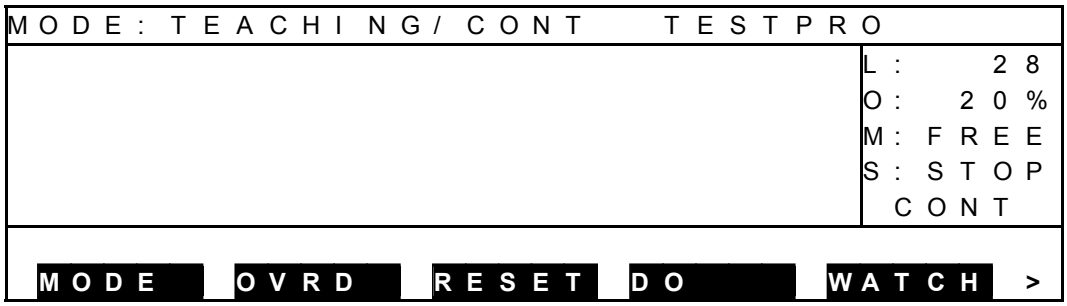
(1) Function

Resets an execution file. This function is used to re-select an execution file and to terminate the test operation mode.

(2) Procedures

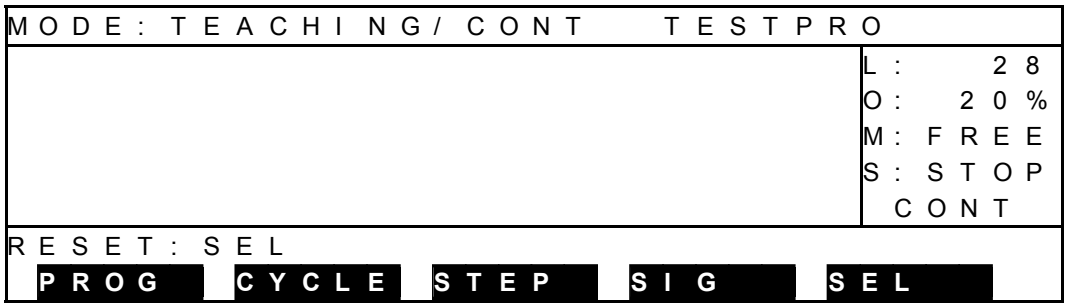
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [RESET] is displayed in the soft key menu, and press the corresponding function key.

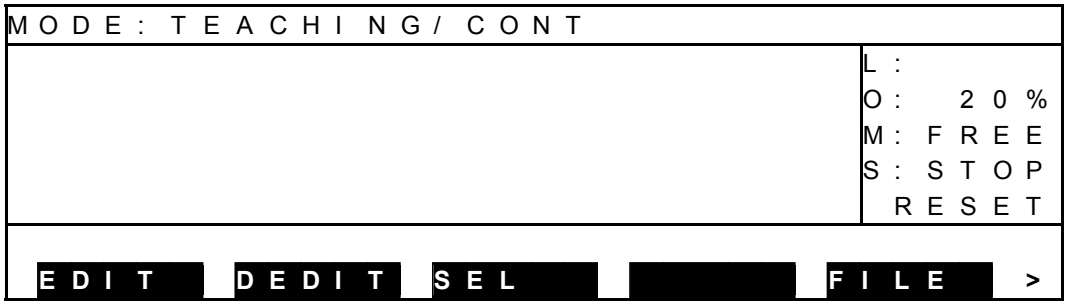


(b) Step 2: SEL selection.

Select [SEL].



(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key.
The execution file name is cleared and "STOP RESET" is indicated under "S" (execution status) on the display.



(3) Caution
All variables used in the program are reset. To start an operation, select the execution file again.

7.9 Direct Execution [DO]

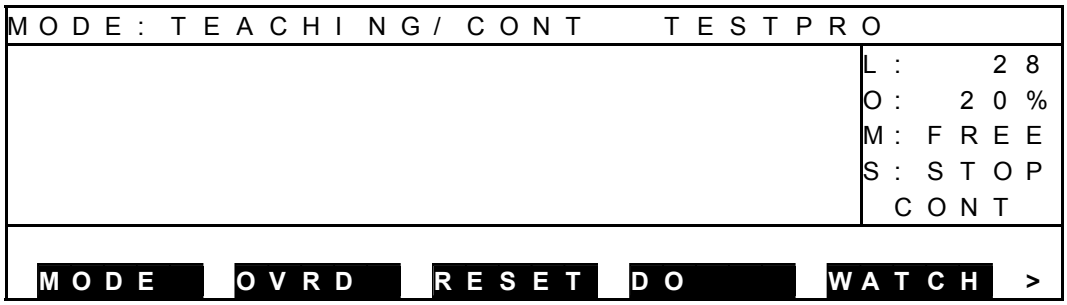
(1) Function

Directly executes a SCOL command by entering it through the teach pendant. This function is used to establish the conditions leading to a specified step when you wish to execute a program from midway. The global variables in the program cannot be used, however.

(2) Procedures

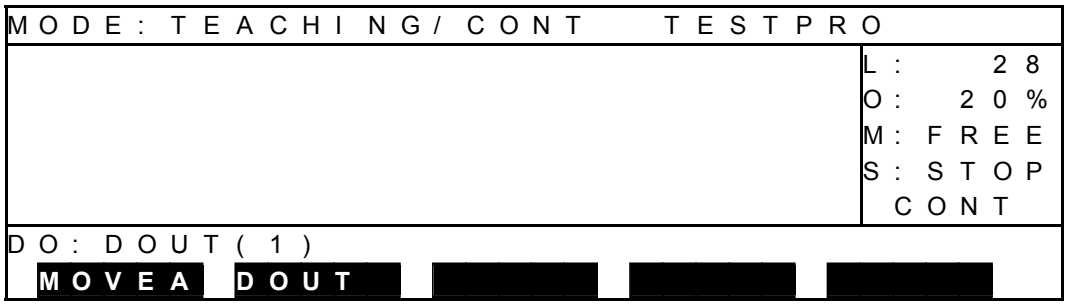
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [DO] is displayed in the soft key menu, and press the corresponding function key.



(b) Step 2: SCOL selection.

Enter the SCOL command you wish to execute through the keyboard. For details of the SCOL command, see the Robot Language Manual.

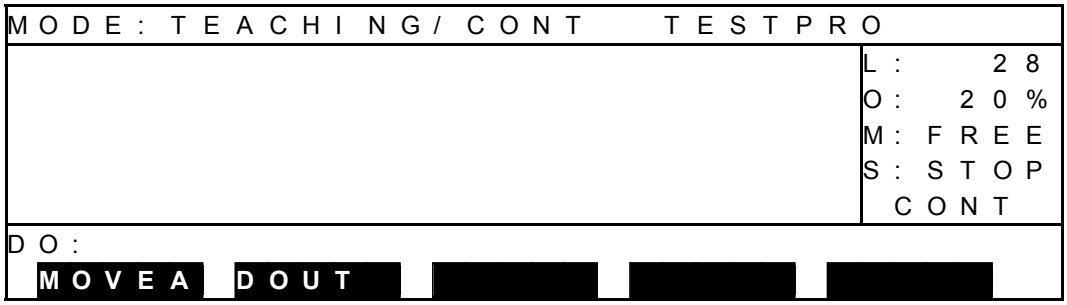


(c) Step 3: Input of EXE key.

Make sure that your entry is correct, and press the EXE key.

During the execution, the menu as shown in Para. 7.6 (2)–(c) is shown.

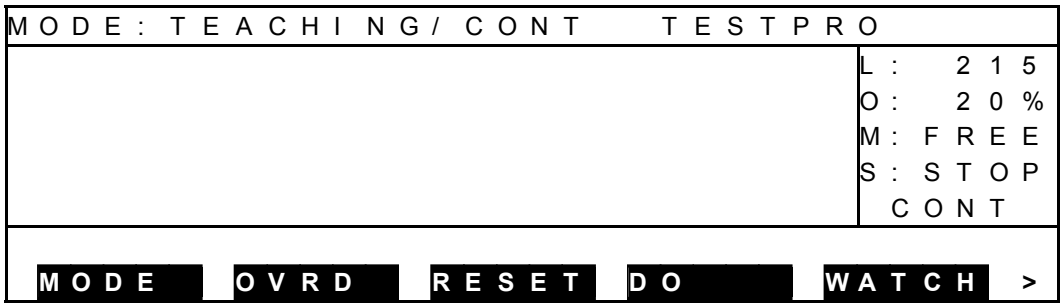
When the operation has finished, the display changes as shown below.



To execute the operation continuously, enter a command again and press the EXE key.

(d) Step 3: Finish.

To terminate the direct operation, press the ESC key.



(3) Caution

This function is inoperative in the feed hold or servo power OFF status.

7.10 Variable Monitor

(1) Function

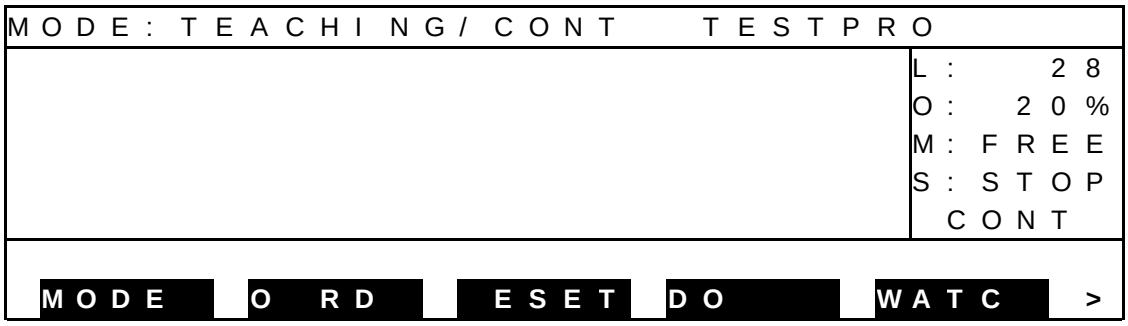
Displays the values of a desired global variable.

(2) Procedures

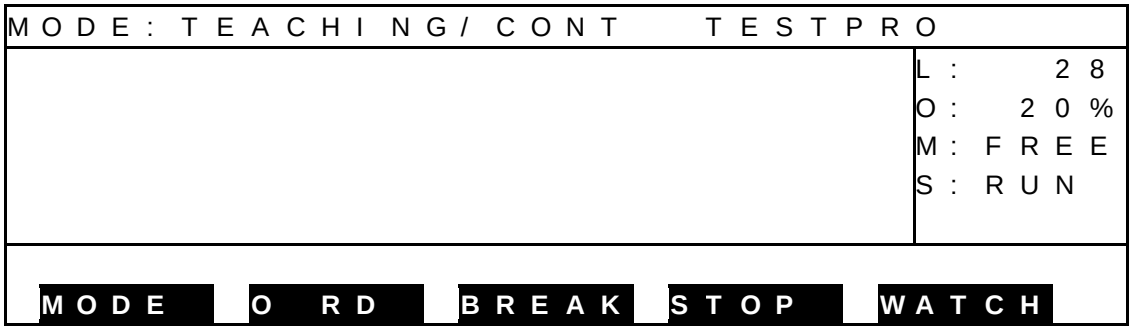
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [WATCH] is displayed in the soft key menu, and press the corresponding function key.

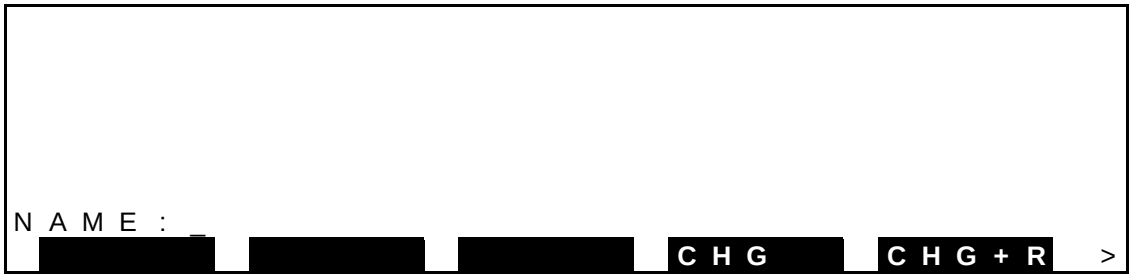
- During stop



- During run

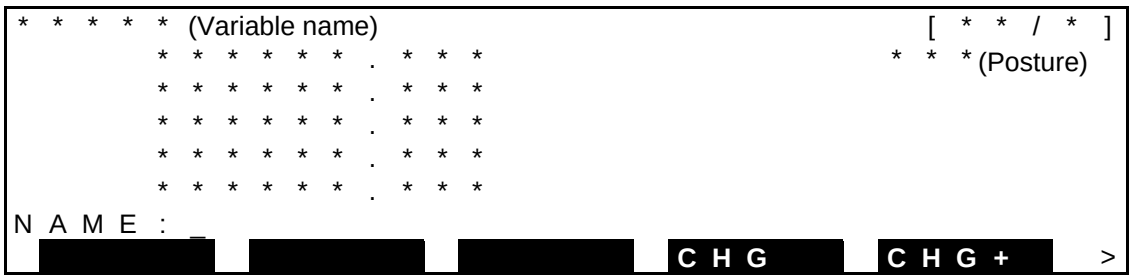


Press the F5 [WATCH] function key, and the variable monitor display appears.



* The global variable signifies the variable (real number type, integer type, position type, coordinate type or load type) as defined in the area flanked by GLOBAL and END, or by DATA and END (for the variable which can be edited by the DEDIT function) in the program

(b) Step 2: Input of variable name.
Enter a variable name you wish to monitor, then press the EXE key. The variable name thus input and its values are displayed.



Unless the specified variable exists in the program, "------.---" is displayed. Up to thirty (30) variable names can be registered.

(c) Step 3: Changeover of monitor variable display.
Monitor variable display can be changed over in the following manner.

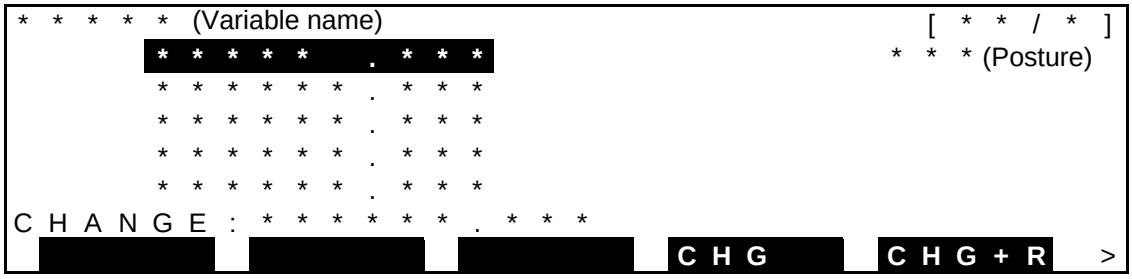
Display of the next variable and after	Press the ⇩ key.
Display of the previous variable and before	Press the ⇧ key.
Display of already registered variable	Enter the already registered variable.

(d) Step 4: Change of variable value [CHG].
It is possible to change a programmed value temporarily. (To rewrite the initial value in the program, execute [CHG+R].)

[1] Press the F4 [CHG] key, and the following display is called.

[2] A reversed indication can be shifted up or down by means of the ⇧ or ⇩ key, respectively.

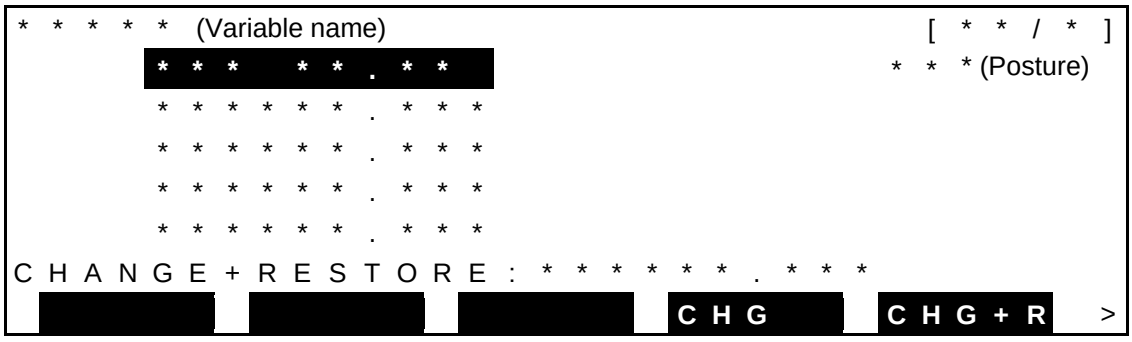
When the cursor is set to an element you wish to change, relevant value of the selected element is displayed as [CHANGE : *******.***■]. Change the value and press the EXE key, then the value is updated.



(e) Step 5: Change of variable [CHG+R].
It is possible to change the initial value in the program. By this operation, a statement itself in the program is rewritten.

[1] Press the F5 [CHG+R] key, and the following display is called.

[2] A reversed indication can be shifted up or down by means of the ↑ or ↓ key, respectively.
When the cursor is set to an element you wish to change, relevant value of the selected element is displayed as [CHANGE+RESTORE : *******.***■]. Change the value and press the EXE key, then the value is updated.



(f) Step 6: Deletion of registered monitor variable.
Change over the monitor variable display until a variable you wish to delete is displayed. Then press the ALT and BS keys (DEL) at the same time.

(g) Step 7: End of variable monitor.
Press the ESC key, and the system returns to the previous display of each mode.

! CAUTION

- Unless any program is selected, displayed value may not be correct.
- The [CHG+R] function cannot be executed for the global variable yet to be initialized (such as position type array variable which is not taught the robot).

  (If you are about to execute this, the error message saying "Can not RESTORE" is displayed.
- The position type data can be changed in the range of ±3 mm or ±3 deg.

7.11 Exiting Test Operation

To quit the test operation mode, execute the execution file reset function as shown in Para. 7.8.5. However, as you can guide the robot even if the execution file is selected, you need not reset the file especially.
You can also quit the test operation mode by selecting a mode other than "TEACHING" by means of the MODE select switch equipped on the control panel.

Section 8 Internal Automatic Operation

! DANGER

NEVER enter the dangerous area of the robot during automatic operation. If you still have to enter the dangerous area, be sure to terminate the automatic operation and turn the servo power off beforehand. Also, select the teaching mode by means of the MODE select switch.

! CAUTION

1. Make sure that no obstacle stands in the robot working range. If any obstacle exists, perform automatic operation at a slow speed and make sure that the robot will not collide with the obstacle at all.
2. If the robot motion becomes abnormal during program execution, press the EMERGENCY stop pushbutton switch immediately to stop the robot. NEVER restart the automatic operation until the cause of the abnormality is identified.
3. The speed of automatic operation is initially set at 100 % of the maximum robot speed.

8.1 Selecting Internal Automatic Operation Mode

(1) Function

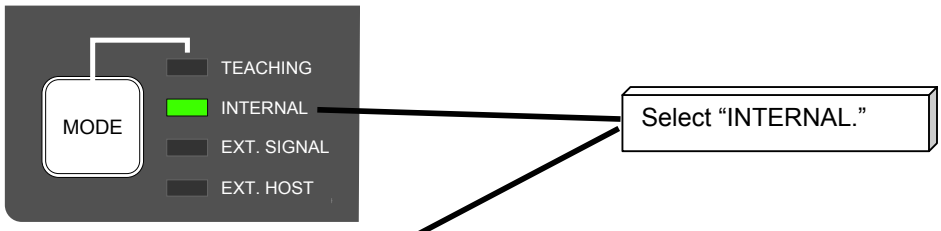
Selects the internal automatic mode where you can execute a program by a start command given through the control panel.

(2) Procedures

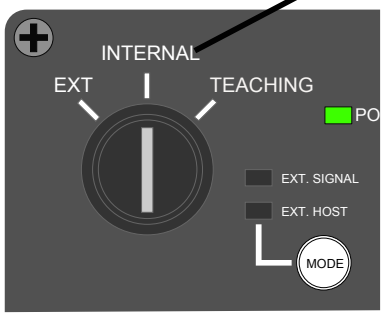
(a) Step 1: Selection of internal automatic mode.

Change over the MODE switch equipped on the control panel to "INTERNAL."

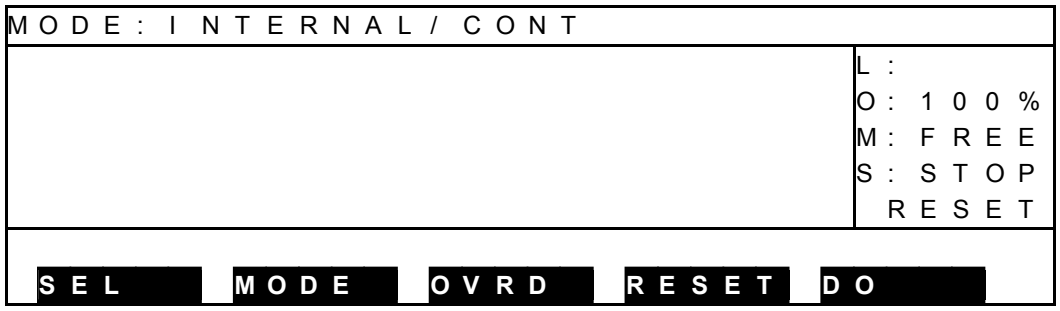
TS1000



TS2000



When the internal automatic mode is selected, the following screen appears.



Descriptions on display

[1] MODE: Indicates the currently selected master mode and run mode.
- CONT : Continuous operation
- CYCLE : Cycle operation
- SEGMNT : Segment operation

[2] Right side on the 1st line: Identifies active file name.

[3] L: No. of execution steps

[4] O: Speed override

[5] M: Machine status
- FREE : Machine lock OFF status

[6] S: Execution status
- RUN : A program is currently executed.
- STOP RESET : A program is stopped in the initialize mode. This is the same as the program reset.
- STOP RETRY : A program is stopped in the restart mode. It is restarted from the interrupted motion.
- STOP CONT : A program is stopped in the continue mode. It is restarted from the current program step.

[7] Bottom line: Function key menu display field
- SEL : Selection of program to be executed
- MODE : Run mode selection
- OVRD : Override setting
- RESET : Reset
- DO : Direct execution
- WATCH : Variable monitor
- STOP : Stop
- BREAK : Interruption

8.2 Selecting File (or Program) [SEL]

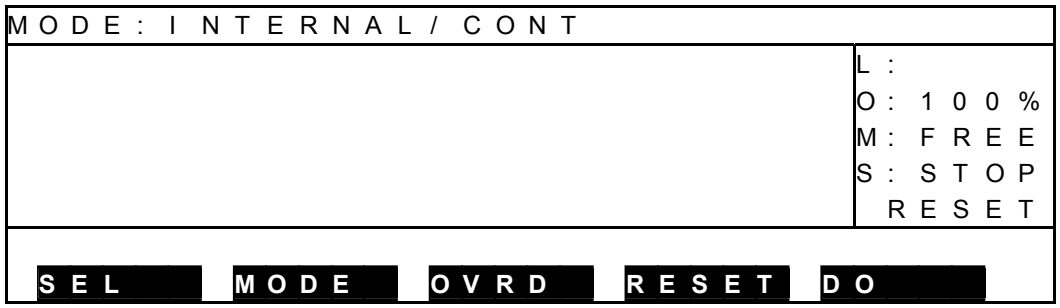
(1) Function

Selects a file (or program) to be executed.

(2) Procedures

(a) Step 1: Command selection.

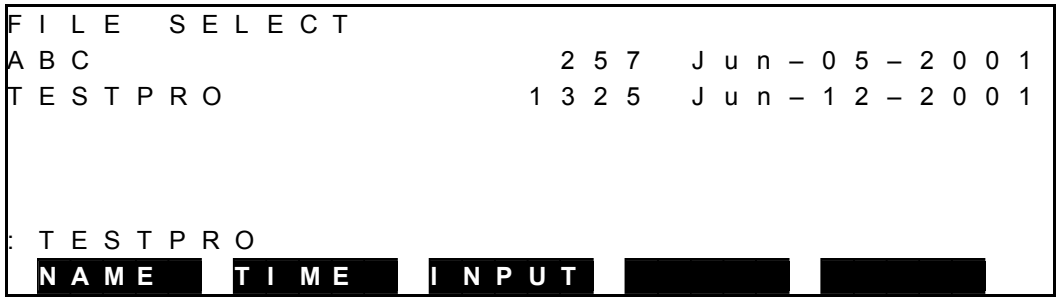
Press the function key corresponding to the [SEL] menu. Unless the SEL menu is available, press the NEXT key repeatedly until [SEL] is displayed in the soft key menu.



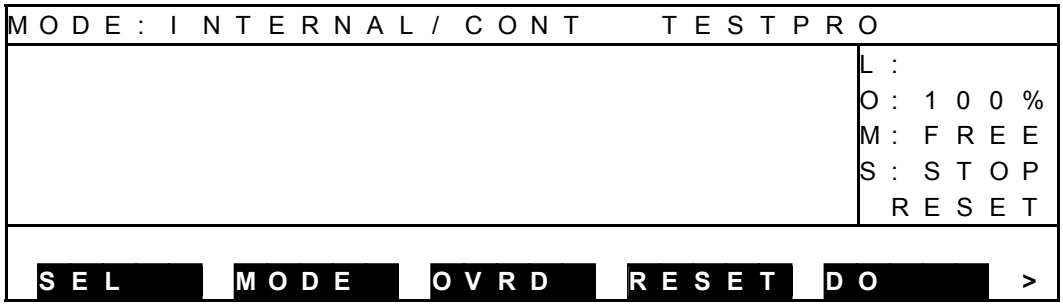
(b) Step 2: Input of file name.

When the file directory is displayed, inversely display and select a desired file name, using the up and down cursor keys, then press the EXE key.

NAME and TIME in the menu are used to sort the file directory by name and time, respectively. To directly enter the file name through the keyboard, press the INPUT function key and input the file name you wish to execute. If the file name contains an extension, be sure to input the extension also.



(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key. Then the specified file name is displayed in the execution file display field on the 1st line.



(3) Caution
Once the execution file has been selected, it remains effective until another file is selected. If a compile error has occurred, however, selection of the execution file is cancelled.
When another file has been selected, the execution environment of automatic operation is reset. To reset the execution environment without changing the execution file, execute [RESET] and [PROG].

8.3 Selecting Run Mode [MODE]

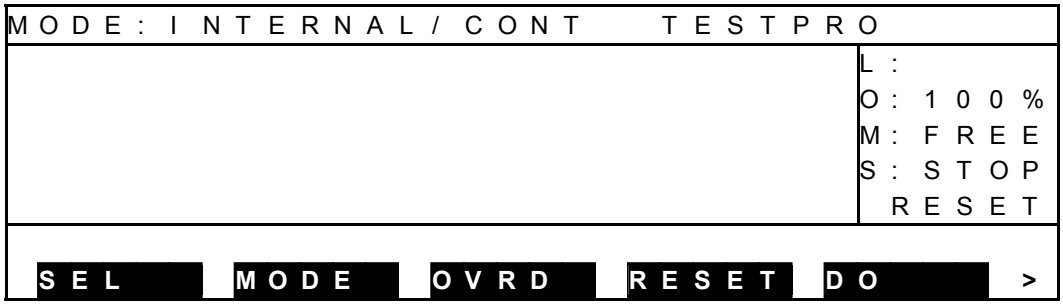
(1) Function

Changes over the run mode of automatic operation.

(2) Procedures

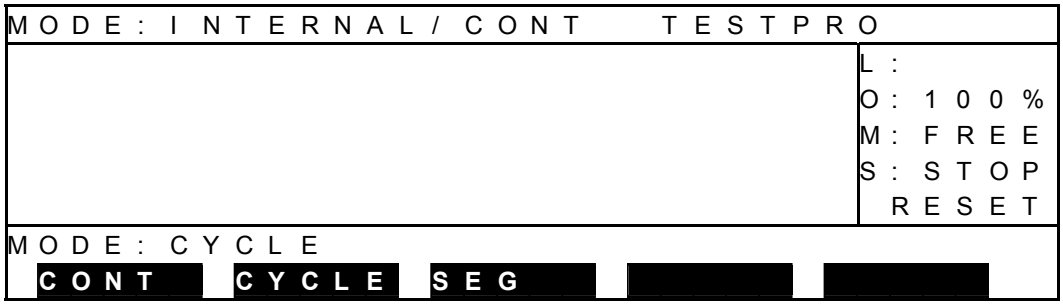
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [MODE] is displayed in the soft key menu. Press the corresponding function key then.



(b) Step 2: Mode selection.

Select a desired run mode from the menu.

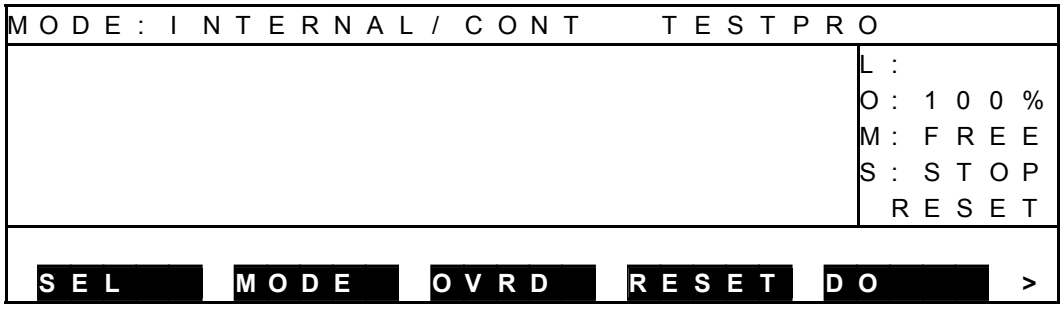


CONT : Repeats program execution continuously.
CYCLE : Stops program execution at the end of the program.
SEG : Stops program execution just before the next motion command.

(c) Step 3: Input of EXE key.

Make sure that your entry is correct, and press the EXE key. If you have entered a wrong data, press the ESC key to cancel the entry and select again. Then the specified run mode is shown on the display.

When the CYCLE mode is selected, the CYCLE switch LED on the control panel is illuminated also.



(3) Caution

The run mode can be selected even while the program is being executed. When the CYCLE switch on the control panel is pressed, "CYCLE" is selected for the run mode. When the same switch is pressed again, the previous run mode is selected.

The execution mode selected is also operative in both the test operation mode and internal automatic mode. The selected run mode remains valid until another run mode is selected.

If the stop command is executed in the program, program execution stops, irrespective of the run mode currently selected.

8.4 Speed Override [OVRD]

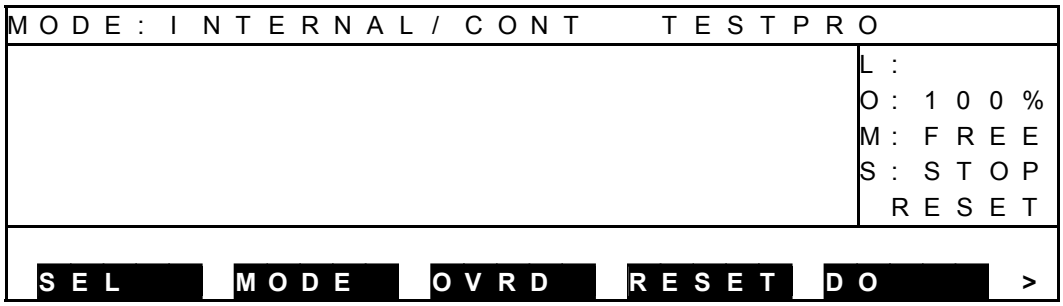
(1) Function

Sets the speed override for automatic operation. This function is useful to effect an additional override, taking the programmed speed as 100 %.

(2) Procedures

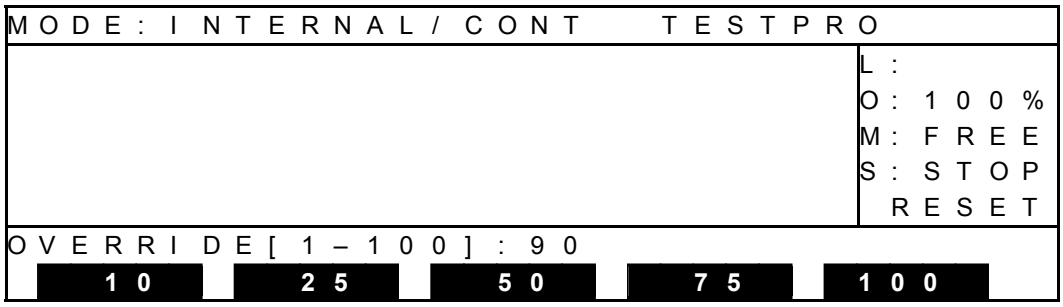
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [OVRD] is displayed in the soft key menu. Press the corresponding function key then.

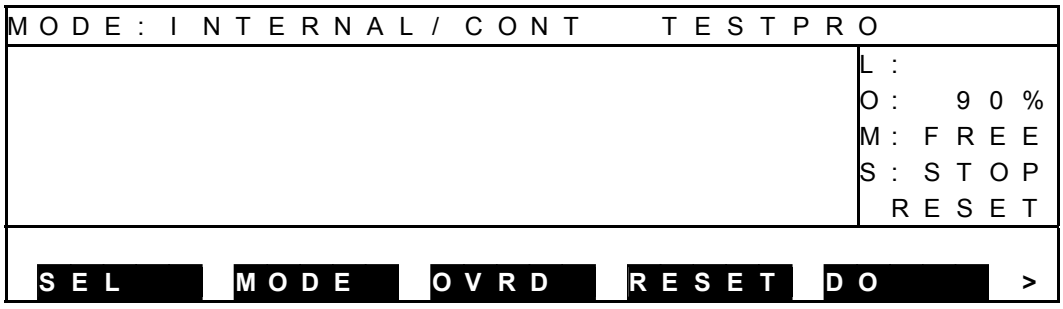


(b) Step 2: Input of speed override.

Set an override of 1 ~ 100 on the speed by means of appropriate function key.



(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key. Then the specified override value is shown under "O" on the display.



If zero (0) is specified for the speed override, a relevant message is displayed and the entry becomes invalid.

(3) Caution
The specified speed override is also operative in both the test operation mode and internal automatic mode. The selected override remains valid until another override is specified.

8.5 Startup [RUN]

(1) Function

Starts an automatic operation under the predetermined conditions.

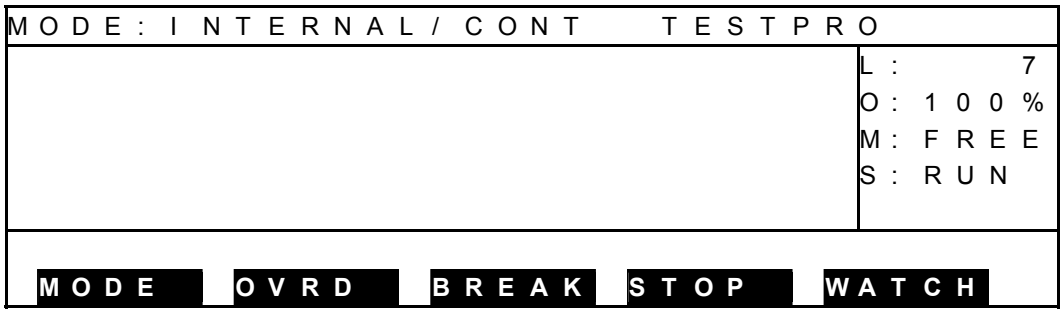
(2) Procedures

(a) Step 1: Setting of running conditions.

Set the conditions for executing the file selection, run mode selection speed override setting as necessary.

When program execution has started, the STOP switch LED on the control panel goes out with the RUN switch LED illuminated.

"RUN" is selected for "S" (execution status) on the teach pendant display.



(3) Caution

The "startup command" external control signal is inoperative in the internal automatic operation mode. Also, the run function cannot be executed through the teach pendant.

8.6 Stop

8.6.1 CYCLE Stop

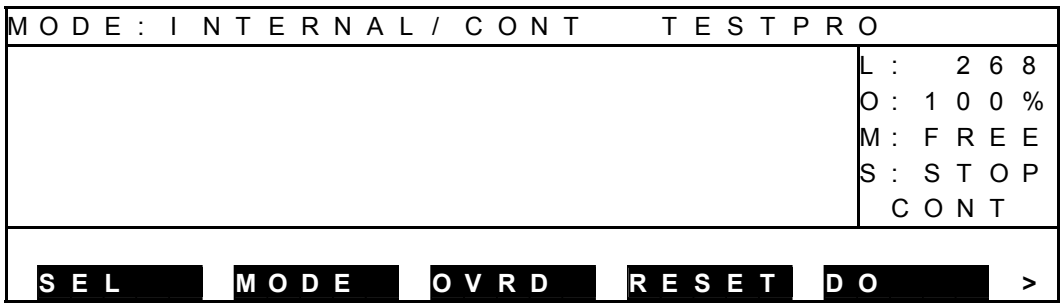
(1) Function

Selects "CYCLE" for the run mode and stops an automatic operation at the program end. This function is used to stop the robot after the end of an operation.

(2) Procedures

(a) Step 1: Input of cycle stop.

Press the CYCLE switch equipped on the control panel. Then the CYCLE switch LED is illuminated with "CYCLE" shown under "S" (execution status) on the teach pendant display. When the program has been executed to the end, the automatic operation stops at the same time that the RUN switch LED on the control panel goes out with the STOP switch LED illuminated.



(b) Step 2: Cancel of cycle stop.

While the CYCLE switch LED is illuminated, press the CYCLE switch again. The CYCLE switch LED goes out with the previous run mode shown on the display. To resume the operation, press the RUN switch.

(3) Caution

The "cycle stop" external control signal is inoperative in the internal automatic operation mode. When the cycle stop is commanded again, the system returns to the previous run mode.

The SCOL program execution unit pre-reads up to four (4) motion commands. Therefore, the program may be executed successively by up to three (3) cycles according to the cycle stop timing or the active program contents.

8.6.2 STOP

(1) Function

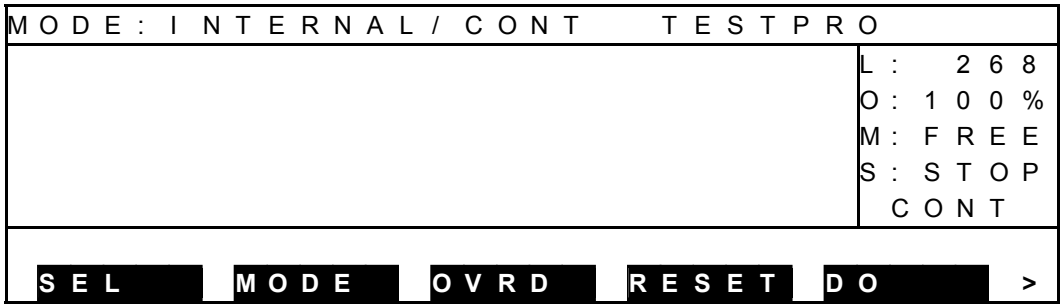
Stops an automatic operation when the active command has been executed. This function is used for normal stop operation.

(2) Procedures

(a) Step 1: Input of [STOP].

- Control panel: Press the STOP switch.
- Teach pendant: Press the function key corresponding to the [STOP] menu.

The STOP switch LED on the control panel is illuminated. When the currently executed command has completed, the automatic operation stops at the same time that the RUN switch LED on the control panel turns off with "STOP CONT" shown under "S" (execution status) on the teach pendant display.



(3) Caution

In the internal automatic operation mode, the "stop" external control signal is inoperative. To restart the operation, press the RUN key.

8.6.3 BREAK

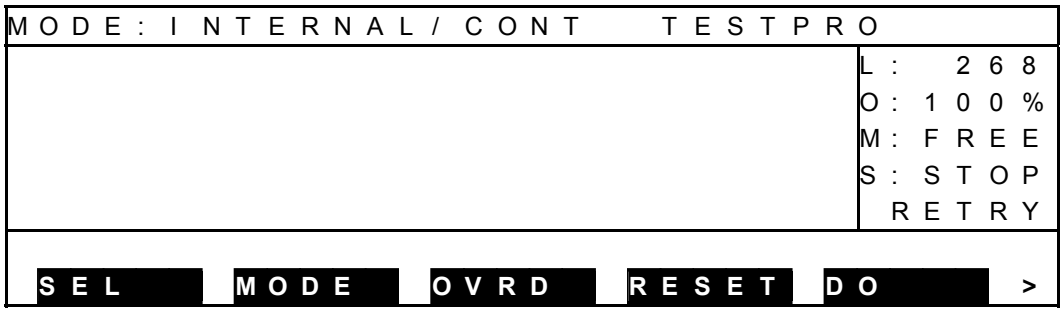
(1) Function

Immediately decelerates and stops the robot, and stops an automatic operation. This function is used to interrupt the automatic operation.

(2) Procedures

(a) Step 1: Input of [BREAK].

Press the function key corresponding to the [BREAK] menu. The robot immediately slows down and stops to stop the automatic operation.
The RUN switch LED on the control panel turns off with the STOP switch LED illuminated. "STOP RETRY" is shown under "S" (execution status) on the teach pendant display.



To restart the operation, press the RUN key.

8.6.4 FEED HOLD

(1) Function

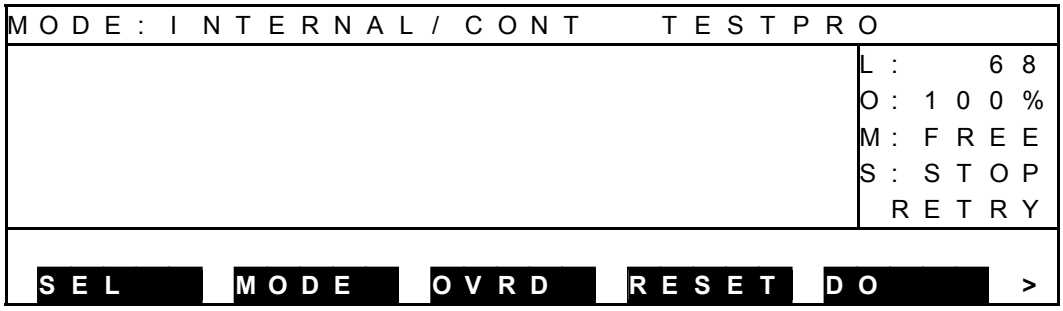
Holds robot motion and program execution. When this function is specified during robot operation, the robot immediately slows down and stop. The feed hold function is used to stop the robot temporarily.

(2) Procedures

(a) Step 1: Input of feed hold.

Press the FEED HOLD pushbutton switch equipped on the teach pendant, and the FEED HOLD switch LED is illuminated. If the robot is working, it decelerates and stops.

When the robot has stopped, the RUN switch LED on the control panel goes out with the STOP switch LED illuminated. The display of the teach pendant is as shown below with "STOP RETRY" indicated under "S" (execution status).



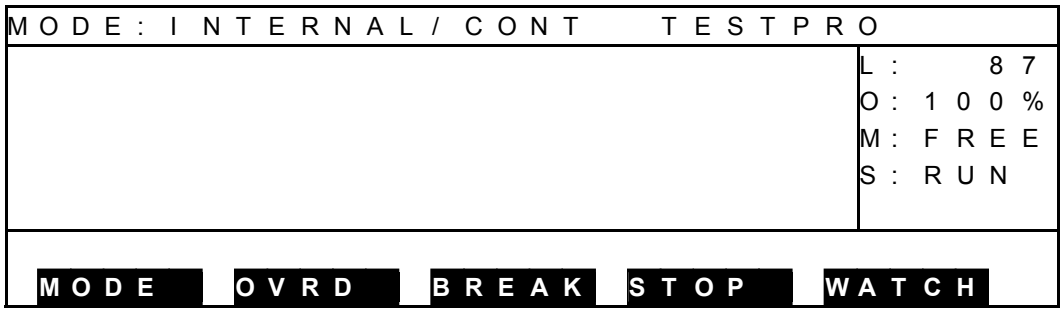
(b) Step 2: Cancel of feed hold.

While the FEED HOLD switch LED is turned on, press the FEED HOLD switch again. The FEED HOLD switch LED turns off and the automatic operation restarts.

If feed hold was commanded during program execution, program execution can be restarted by cancelling the feed hold.

When the program has restarted, the STOP switch LED on the control panel turned off with the RUN switch LED illuminated.

"RUN" is shown under "S" (execution status) on the teach pendant display.



(3) Caution

In the feed hold mode, automatic operation, step execution and direct execution are invalid.

When the FEED HOLD switch is pressed again, feed hold is cancelled. If the feed hold was commanded during program execution, program execution can be restarted by cancelling the feed hold. When the test operation has finished, program execution will not restart.

If you should change the master mode, select a file or reset the execution conditions, the program will not restart.

8.6.5 EMERGENCY Stop

(1) Function

Turns off the servo power and stops an automatic operation. This function is useful to stop the robot system at the time of an emergency.

(2) Procedures

Press the EMERGENCY stop pushbutton switch equipped on the control panel or teach pendant, or use the "emergency stop" external control signal.
The servo power is turned off and the SERVO ON switch LED on the control panel goes out with the SERVO OFF switch LED illuminated.
The servo ON switch LED on the teach pendant goes out.

(3) Caution

All of the EMERGENCY stop switches on the control panel and teach pendant, and "emergency stop" external control signal are operative. At the time of an emergency, use whichever key you can get at quickest.
The EMERGENCY stop switches on the control panel and teach pendant are of a lock type. When pressed, they remain depressed and locked. To reset, turn and unlock the switch.
While the EMERGENCY stop switch is ON, the servo power cannot be turned on.

8.7 Reset [RESET]

The reset operation comes in the following five (5) modes. Any mode other than (4) cannot be reset unless a file for execution is selected.

(1) Program reset
(2) Step reset
(3) Cycle reset
(4) Output signal reset
(5) Execution file reset

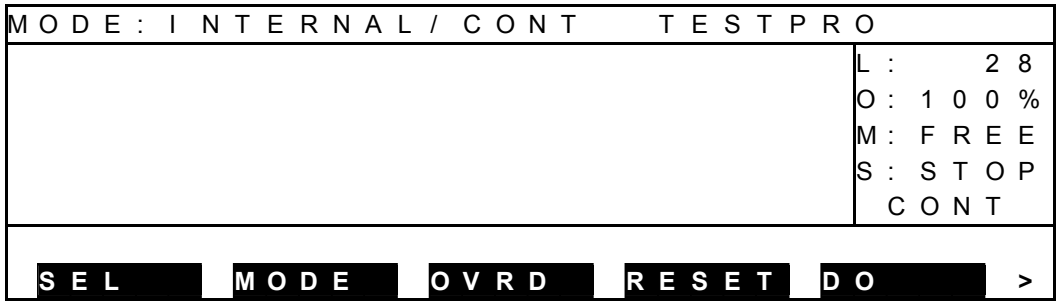
8.7.1 Program Reset

(1) Function

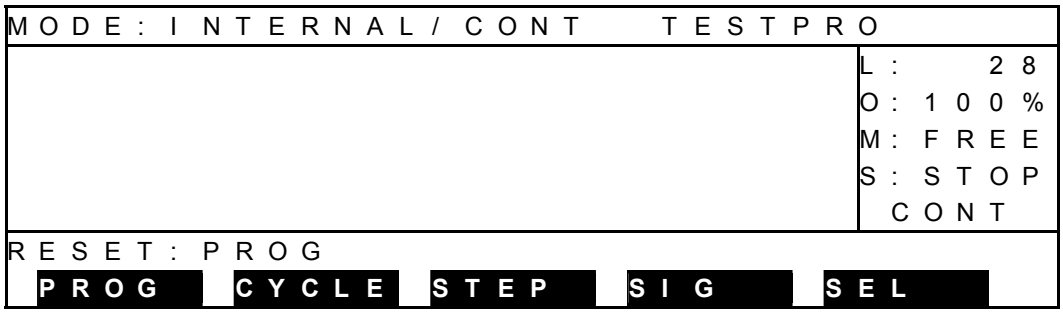
Resets all execution conditions (execution steps, variables) for a current file and allows execution of that program from the top.
This function is useful to execute a current file from the top of the program.

(2) Procedures

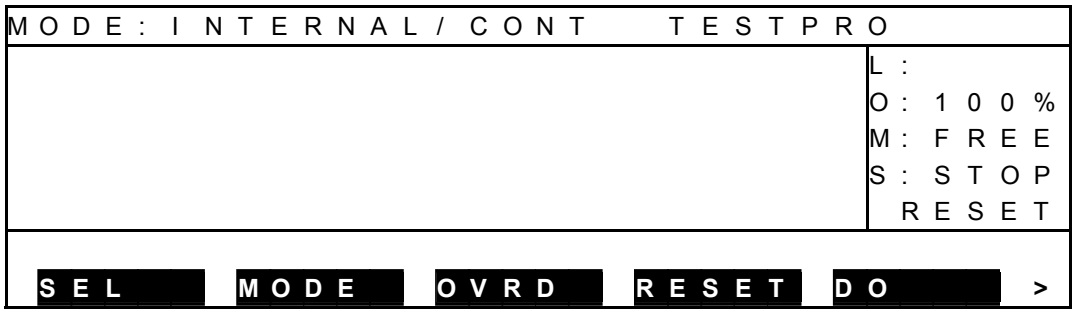
(a) Step 1: Command selection.
Press the NEXT key repeatedly until [RESET] is displayed in the soft key menu, and press the corresponding function key.



(b) Step 2: Program selection.
Select [PROG].



(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key. "STOP RESET" is shown under "S" (execution status) on the teach pendant display.



(3) Caution
All variables used in the program are reset. If you do not wish to the variables, use the step reset function instead.

8.7.2 Step Reset

(1) Function

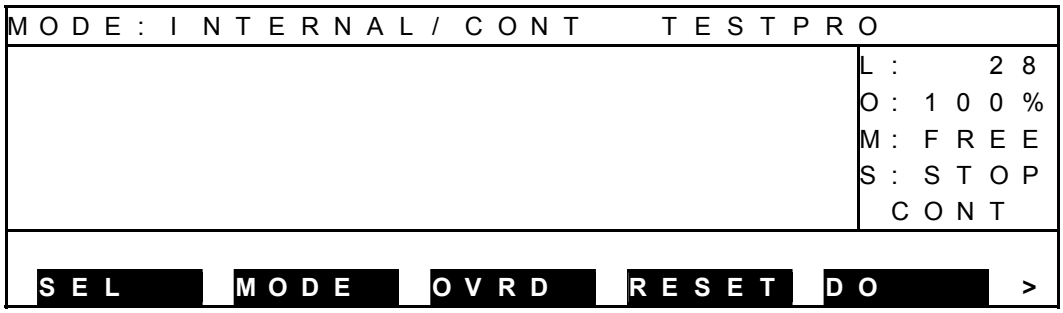
Resets all execution conditions (excluding variables) for a current file and allows execution of that program from the top.

This function is useful to set the repetitive counts in a variable and execute the remaining counts.

(2) Procedures

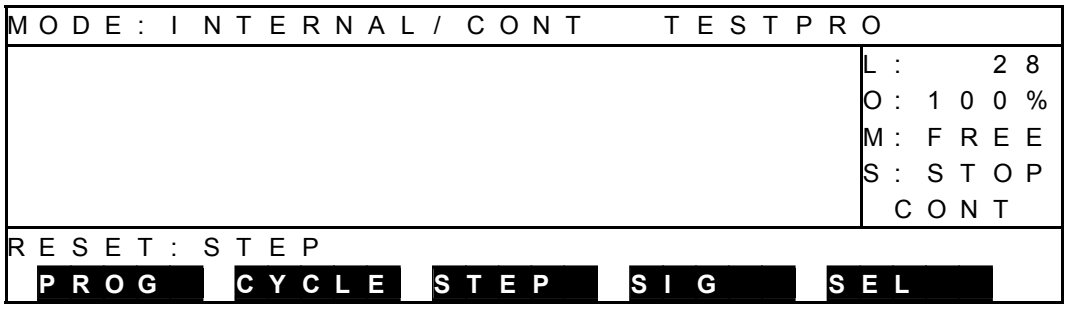
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [RESET] is displayed in the soft key menu, and press the corresponding function key.

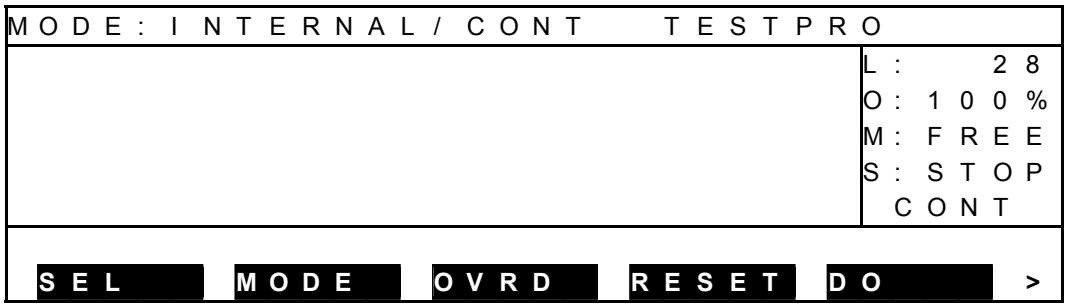


(b) Step 2: Step selection.

Select [STEP].



(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key.



(3) Caution
When you do not wish to execute the setting of initial conditions at the head of the program, use the cycle reset instead.

8.7.3 Cycle Reset

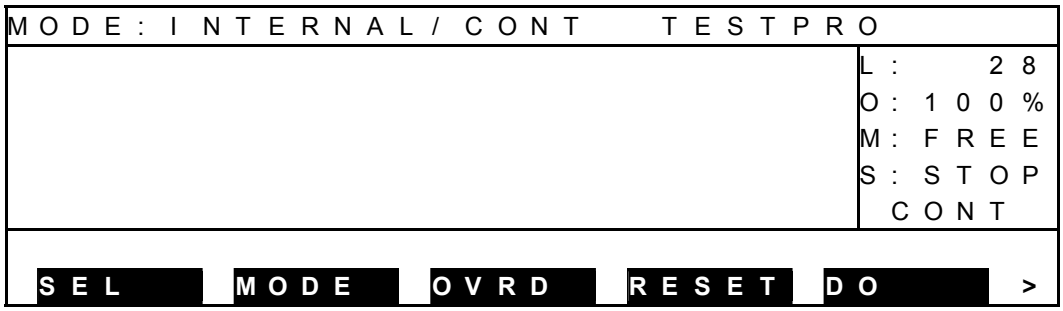
(1) Function

Resets all execution conditions (excluding global variables) for a current file and allows execution of the main program from the step indicated by label RCYCLE. This function is useful when you do not wish to execute the setting of the initial condition.

(2) Procedures

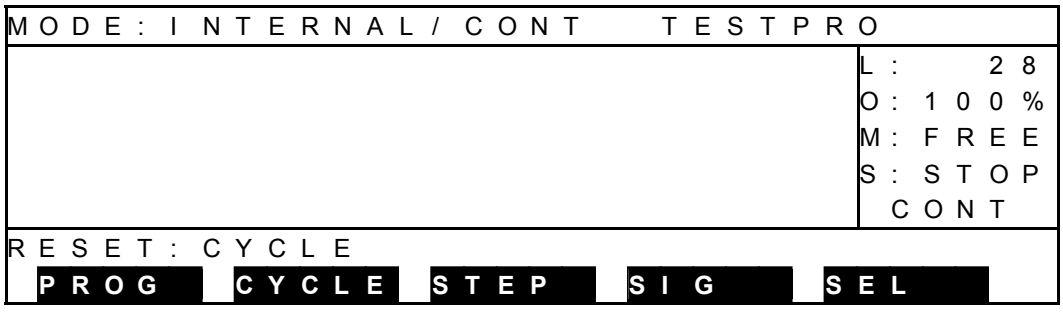
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [RESET] is displayed in the soft key menu, and press the corresponding function key.

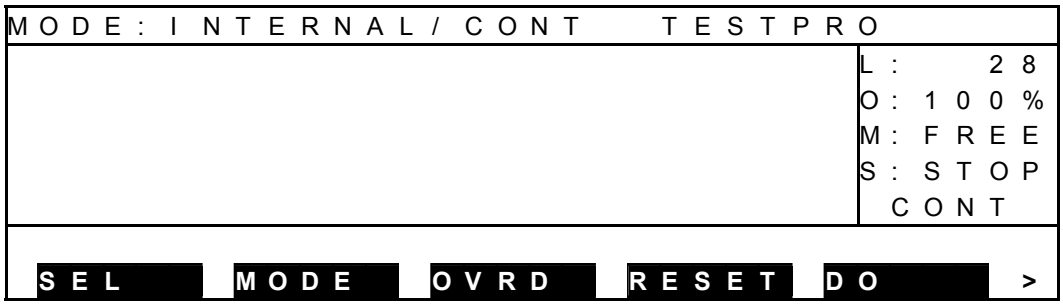


(b) Step 2: Cycle selection.

Select [CYCLE].



(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key.



(3) Caution
Use label RCYCLE in a desired step of the main program. Also, specify the GOTO RCYCLE command just before the END command not to execute the END command of the main program.

8.7.4 Output Signal Reset

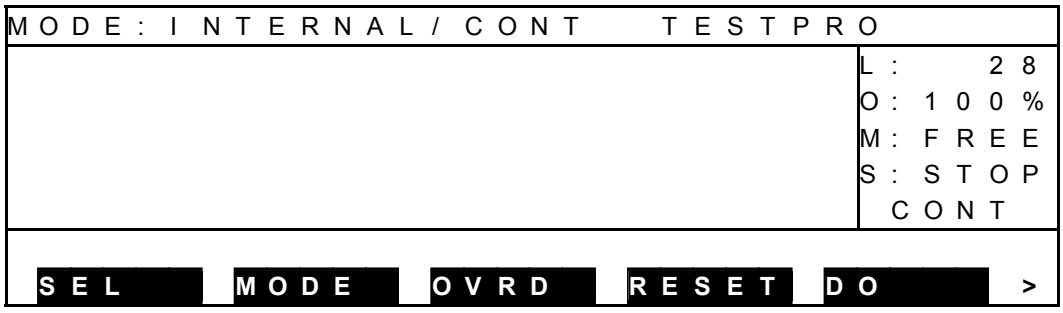
(1) Function

Sets off all user external output signals (DOUT (1) ~ DOUT (164)). The hand signals (201 ~ 204) will not turn off.

(2) Procedures

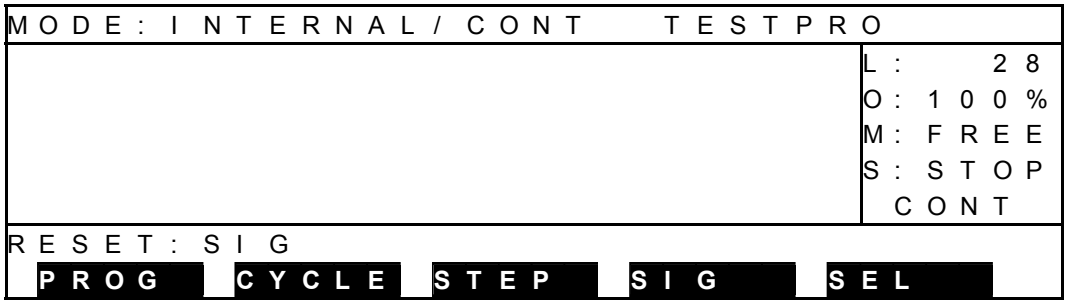
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [RESET] is displayed in the soft key menu, and press the corresponding function key.

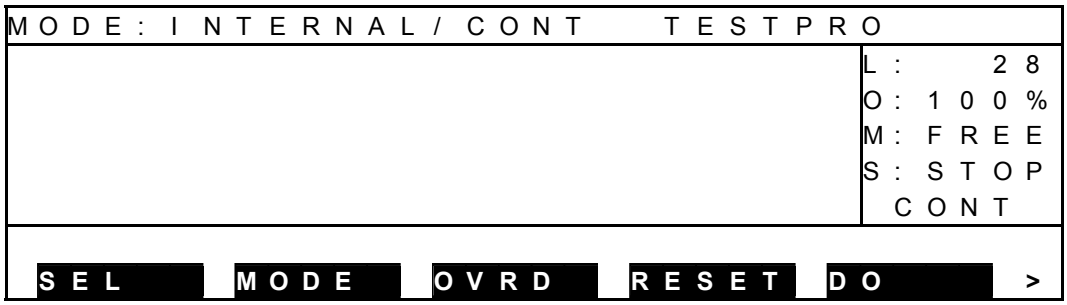


(b) Step 2: SIG selection.

Select [SIG].



(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key.



8.7.5 Execution File Reset

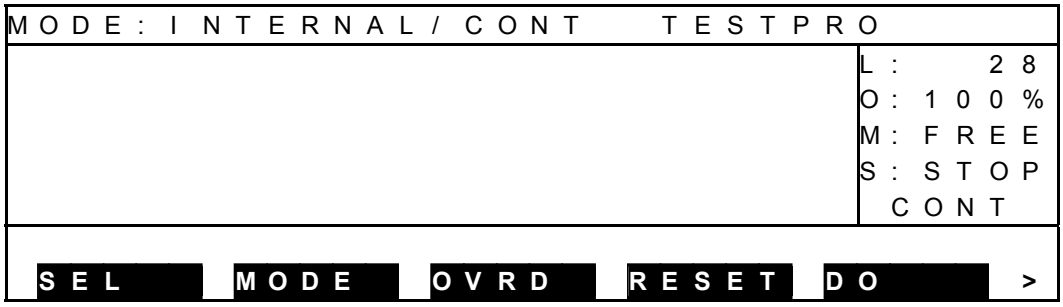
(1) Function

Resets an execution file. This function is used to re-select an execution file.

(2) Procedures

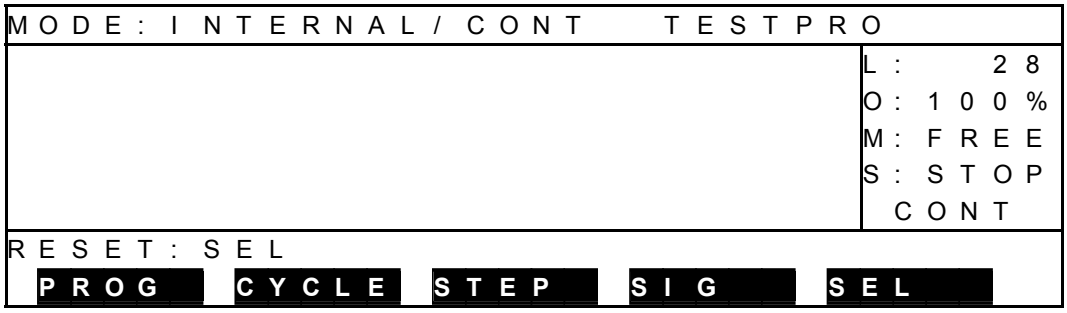
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [RESET] is displayed in the soft key menu, and press the corresponding function key.

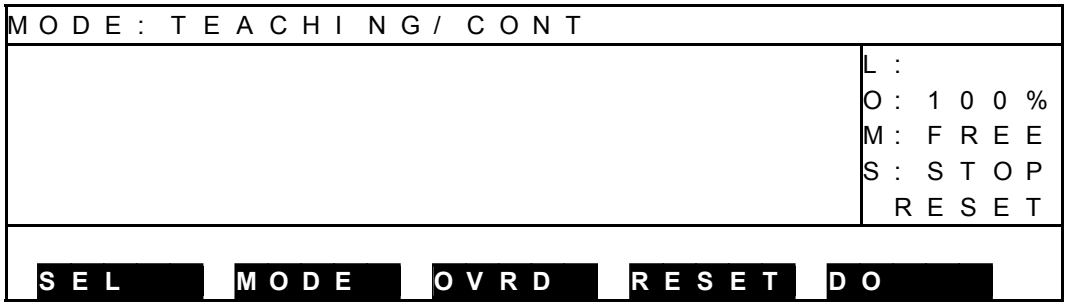


(b) Step 2: SEL selection.

Select [SEL].



(c) Step 3: Input of EXE key.
Make sure that your entry is correct, and press the EXE key.
The execution file name is cleared and "STOP RESET" is indicated under "S" (execution status) on the display.



(3) Caution
All variables used in the program are reset. To start an operation, select the execution file again.

8.8 Direct Execution [DO]

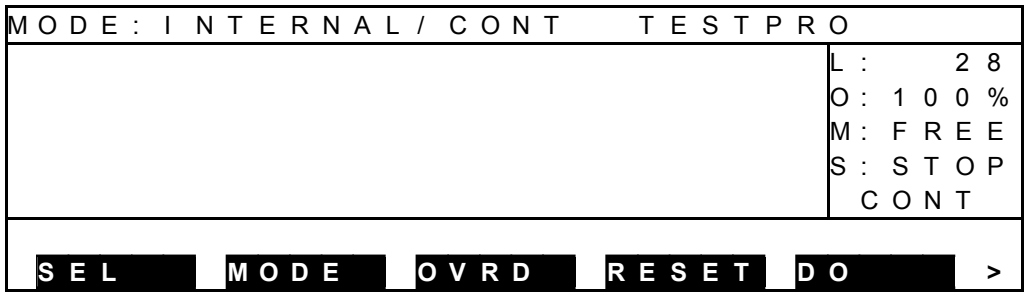
(1) Function

Directly executes a SCOL command by entering it through the teach pendant. This function is used to establish the conditions leading to a specified step when you wish to execute a program from midway. The global variables in the program cannot be used, however.

(2) Procedures

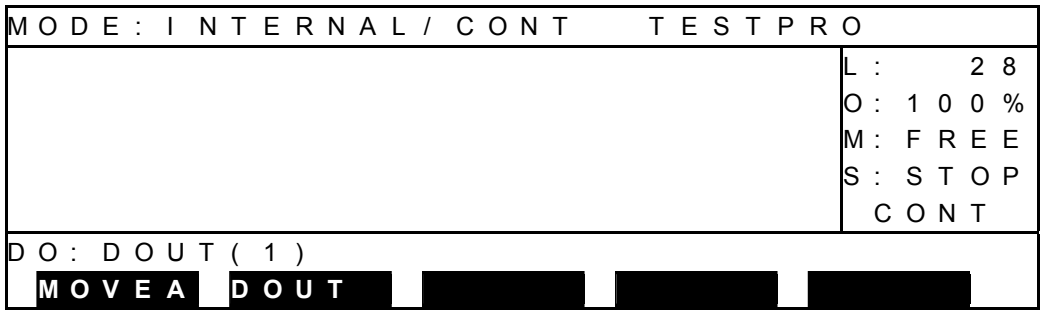
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [DO] is displayed in the soft key menu, and press the corresponding function key.



(b) Step 2: SCOL selection.

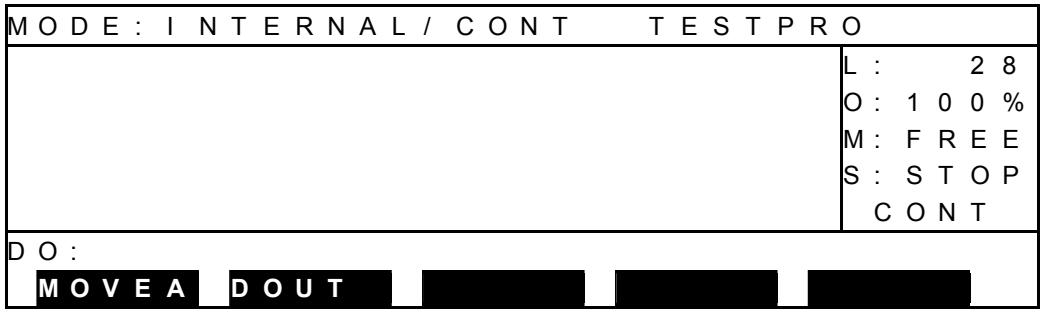
Enter the SCOL command you wish to execute through the keyboard. For details of the SCOL command, see the Robot Language Manual.



(c) Step 3: Input of EXE key.

Make sure that your entry is correct, and press the EXE key.

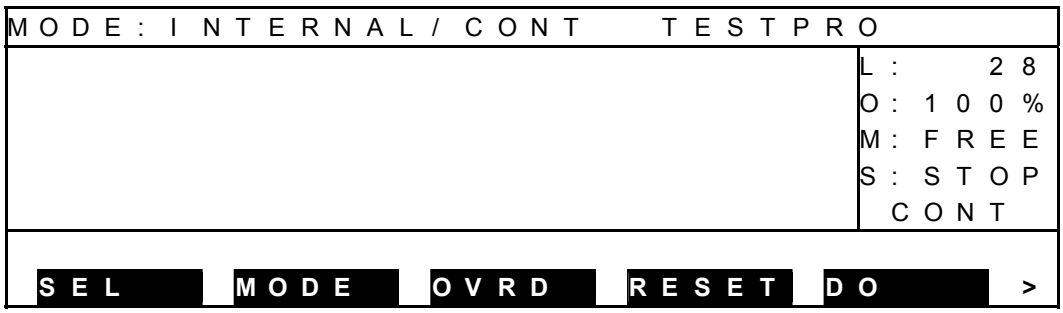
During the execution, the menu as shown in Para. 8.5 is shown. When the operation has finished, the display changes as shown below.



To execute the operation continuously, enter a command again and press the EXE key.

(d) Step 3: Finish.

To terminate the direct operation, press the ESC key.



(3) Caution

This function is inoperative in the feed hold or servo power OFF status.

8.9 Variable Monitor

(1) Function

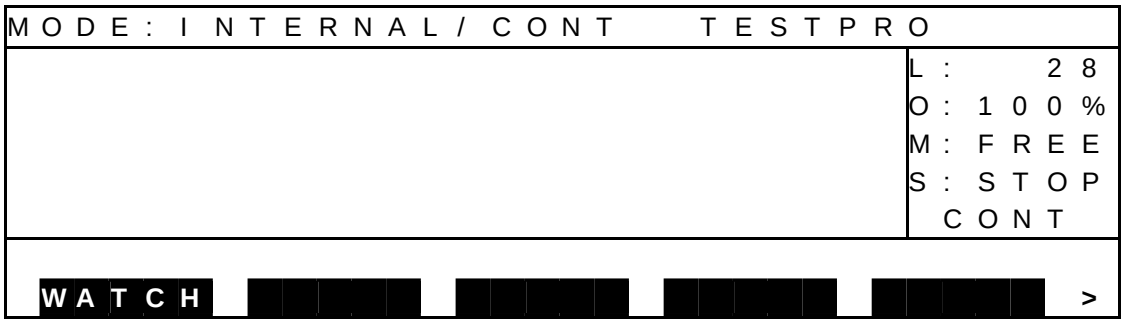
Displays the values of a desired global variable.

(2) Procedures

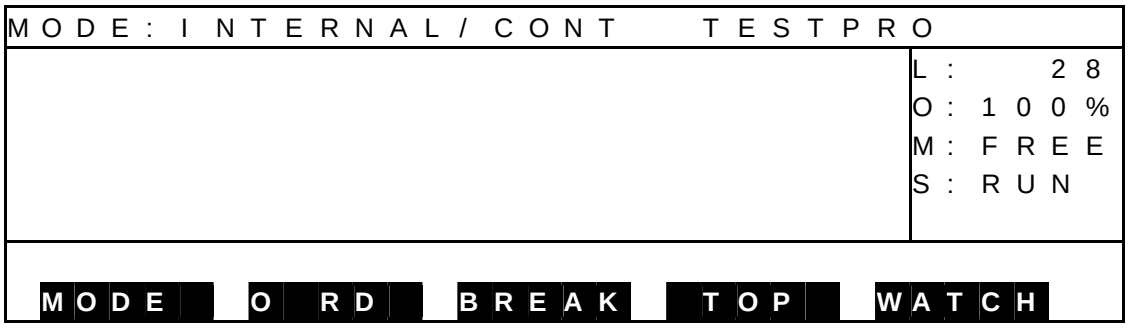
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [WATCH] is displayed in the soft key menu, and press the corresponding function key.

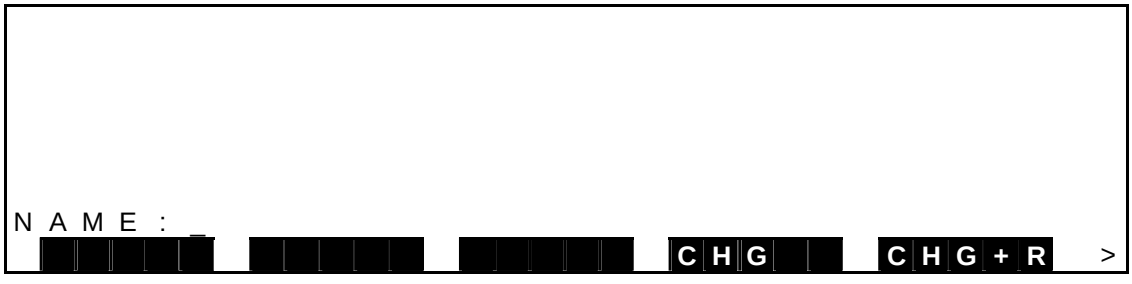
- During stop



- During run

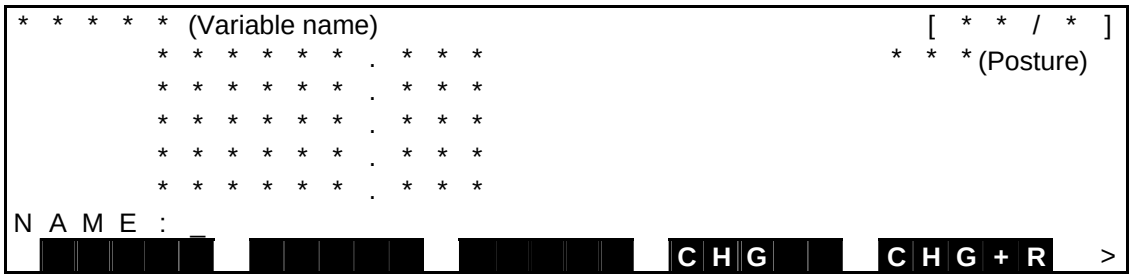


Press the F5 [WATCH] function key, and the variable monitor display appears.



* The global variable signifies the variable (real number type, integer type, position type, coordinate type or load type) as defined in the area flanked by GLOBAL and END, or by DATA and END (for the variable which can be edited by the DEDIT function) in the program.

(b) Step 2: Input of variable name.
Enter a variable name you wish to monitor, then press the EXE key. The variable name thus input and its values are displayed.



Unless the specified variable exists in the program, "------.---" is displayed.
Up to thirty (30) variable names can be registered.

(c) Step 3: Changeover of monitor variable display.
Monitor variable display can be changed over in the following manner.

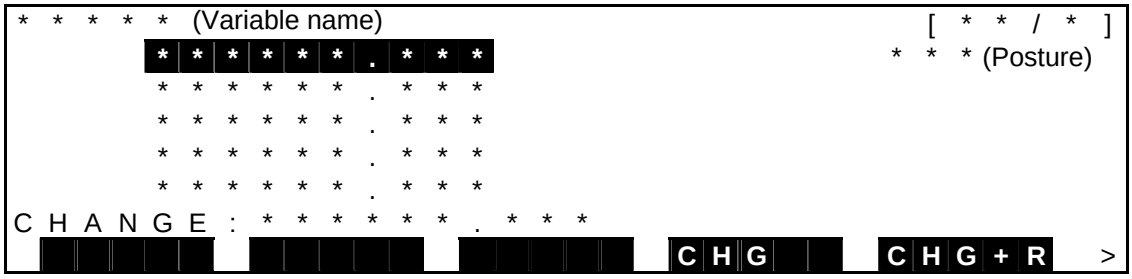
Display of the next variable and after	Press the ⇓ key.
Display of the previous variable and before	Press the ⇑ key.
Display of already registered variable	Enter the already registered variable.

(d) Step 4: Change of variable value [CHG].
It is possible to change a programmed value temporarily. (To rewrite the initial value in the program, execute [CHG+R].)

[1] Press the F4 [CHG] key, and the following display is called.

[2] A reversed indication can be shifted up or down by means of the ⇑ or ⇓ key, respectively.

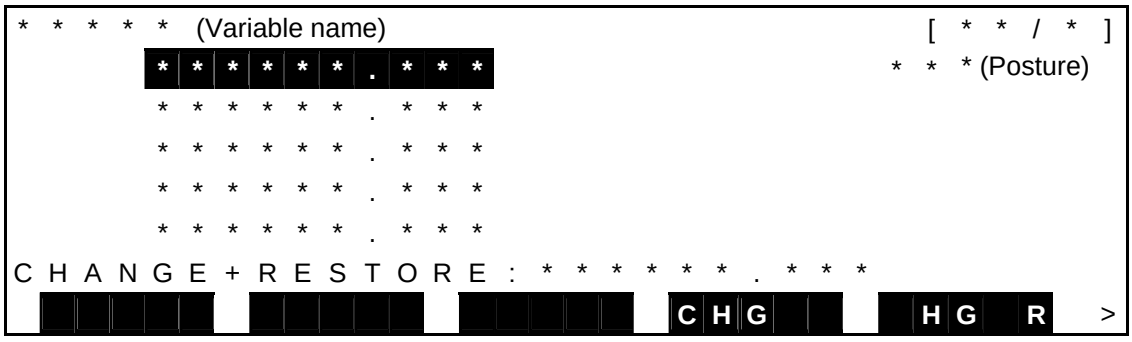
When the cursor is set to an element you wish to change, relevant value of the selected element is displayed as [CHANGE : *******.***■]. Change the value and press the EXE key, then the value is updated.



(e) Step 5: Change of variable [CHG+R].
It is possible to change the initial value in the program. By this operation, a statement itself in the program is rewritten.

[1] Press the F5 [CHG+R] key, and the following display is called.

[2] A reversed indication can be shifted up or down by means of the ↑ or ↓ key, respectively.
When the cursor is set to an element you wish to change, relevant value of the selected element is displayed as [CHANGE+RESTORE : *******.***■].
Change the value and press the EXE key, then the value is updated.



(f) Step 6: Deletion of registered monitor variable.
Change over the monitor variable display until a variable you wish to delete is displayed. Then press the ALT and BS keys (DEL) at the same time.

(g) Step 7: End of variable monitor.
Press the ESC key, and the system returns to the previous display of each mode.

! CAUTION

- Unless any program is selected, displayed value may not be correct.
- The [CHG+R] function cannot be executed for the global variable yet to be initialized (such as position type array variable which is not taught the robot).

  (If you are about to execute this, the error message saying "Can not RESTORE" is displayed.
- The position type data can be changed in the range of ±3 mm or ±3 deg.

Section 9 External Automatic Operation

! DANGER

NEVER enter the dangerous area of the robot during automatic operation. If you still have to enter the dangerous area, be sure to terminate the automatic operation and turn the servo power off beforehand. Also, select the teaching mode by means of the MODE select switch.

! CAUTION

1. Make sure that no obstacle stands in the robot working range. If any obstacle exists, perform automatic operation at a slow speed and make sure that the robot will not collide with the obstacle at all.
2. If the robot motion becomes abnormal during program execution, press the EMERGENCY stop pushbutton switch immediately to stop the robot. NEVER restart the automatic operation until the cause of the abnormality is identified.
3. The speed of automatic operation is initially set at 100 % of the maximum robot speed.

9.1 Selecting External Automatic Operation Mode

(1) Function

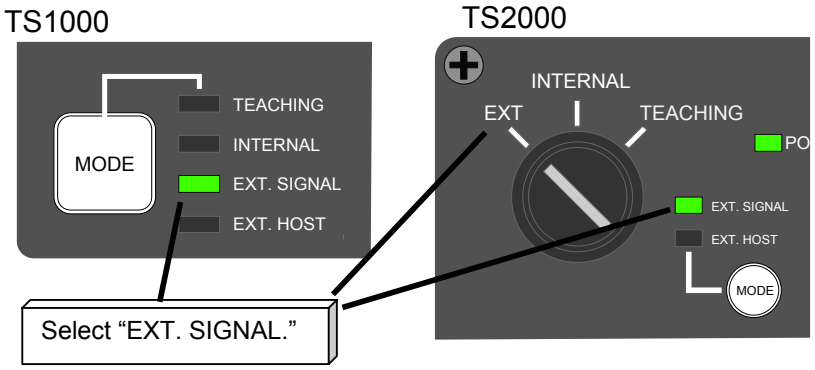
Selects the external automatic mode where you can execute a program by external control signals or commands given from the host computer.

(2) Procedures

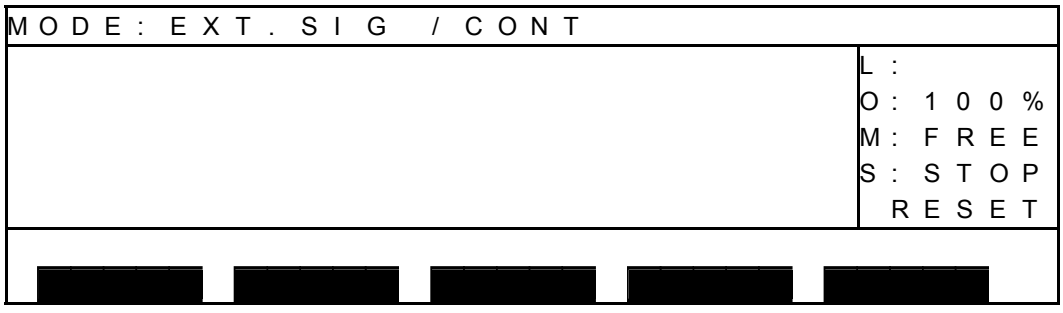
(a) Step 1: Selection of external automatic mode.

- External signal mode

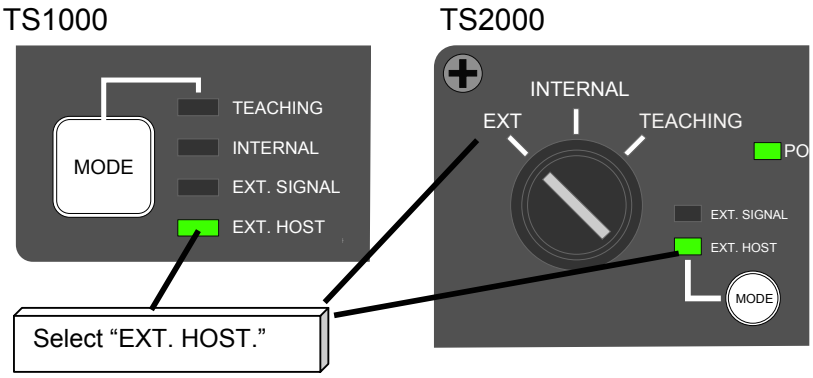
  Press the MODE switch equipped on the control panel to select "EXT. SIGNAL."



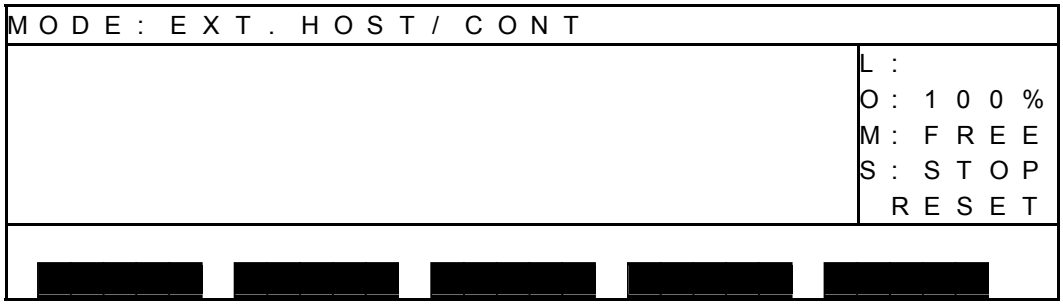
When the external automatic mode is selected, the following screen appears.



- Host mode
  Press the MODE switch equipped on the control panel to select "EXT. HOST."



When the external automatic mode is selected, the following screen appears.



Descriptions on display

[1] MODE: Indicates the currently selected master mode and run mode.
- EXT. SIG : External control signal mode
- EXT. HOST : Host mode (Control by host computer)
- CONT : Continuous operation
- CYCLE : Cycle operation

[2] Right side on the 1st line: Identifies active file name.

[3] L: No. of execution steps

[4] O: Speed override

[5] M: Machine status
- FREE : Machine lock OFF status

[6] S: Execution status
- RUN : A program is currently executed.
- STOP RESET : A program is stopped in the initialize mode. This is the same as the program reset.
- STOP RETRY : A program is stopped in the restart mode. It is restarted from the interrupted motion.
- STOP CONT : A program is stopped in the continue mode. It is restarted from the current program step.

[7] Bottom line: Function key menu display field
- WATCH : Variable monitor
- STOP : Stop
- BREAK : Interruption

(3) Caution

For the functions and use of the external control signals, see the Interface Manual.

For the data communication function with the host computer, see the Communication Manual.

In the external automatic mode, only the MODE select switch, STOP switch, SERVO ON switch, SERVO OFF switch and EMERGENCY stop pushbutton switch provided on the control panel are operative.

9.2 Selecting File (or Program)

(1) Function

Selects or resets an execution file (or program).

(2) Procedures

Observe one of the following manners to select a file.

(a) File selection by automatic execution file.

Create in the drive an automatic execution file which specifies a selected file name by the SELECT command, then turn on the controller power.

For the automatic execution file, see Section 10.

(b) File selection by external signal.

Specify a file to be selected externally in the user parameter beforehand. While the external signal mode is selected, an execution file can be selected by the external signal.

For the externally selected file and external signals, see the User Parameter Manual and Interface Manual, respectively.

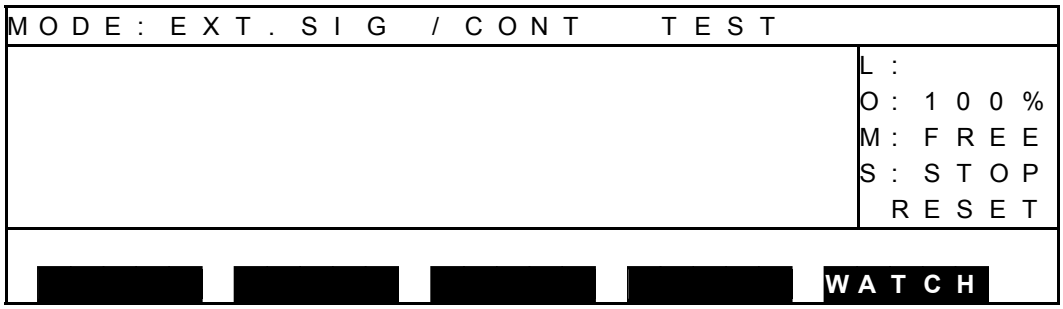
(c) File selection by internal automatic mode or test operation mode.

Select in advance a file you wish to execute in the internal automatic mode or test operation mode. Select the either mode by means of the MODE select switch on the control panel.

(d) File selection via data transmission.

While the host mode is selected, a file can be selected by data transmission. For details, see the Communication Manual.

When the file is selected, the following display appears.



(3) Caution

In the external control mode, an execution file cannot be selected through the teach pendant, using a command.

9.3 Selecting Run Mode [MODE]

(1) Function

Changes over the run mode of automatic operation.

(2) Procedures

(a) External control signal mode:

When the cycle operation mode signal is turned on, the cycle operation mode is selected. When the same signal is turned off, the continuous operation mode takes effect.

For the cycle operation mode signal, see the Interface Manual.

(b) Host mode:

Specify the run mode by sending the MODE command as an internal command for data communication (EC).

For details of sending the internal command, see the Communication Manual.

(3) Caution

In the external control signal mode, the run mode is selected by the cycle operation mode signal.

In the host mode, the cycle operation mode signal is ineffective.

If the external automatic mode has replaced the internal automatic mode in the host mode, the operation mode effective then is maintained. To specify the run mode in the host mode, send the MODE command as an internal command through data communication.

In the external automatic mode, the CYCLE switch equipped on the control panel is inoperative.

In the external automatic mode, you cannot specify the run mode, using the command through the teach pendant.

9.4 Speed Override

(1) Function

Sets the speed override for automatic operation. This function is useful to effect an additional override, taking the programmed speed as 100 %.

(2) Procedures

(a) When specifying speed override in the internal automatic mode:
Specify a desired speed override beforehand in the internal automatic mode. Then change over the MODE select switch.

(b) When using an automatic execution file:
Create in the RAM drive an automatic execution file which designates the speed override by the OVRD command, then turn on the controller power. For the automatic execution file, see Section 10.

(3) Caution

A speed override which was set in the internal automatic mode or test operation mode remains effective as it is. Just after the controller power is turned on, the override is 100 %.

You can specify the speed override valid at power ON by using the OVRD command in the automatic execution file.

In the external automatic mode, you cannot specify the speed override, using the command through the teach pendant.

9.5 Startup

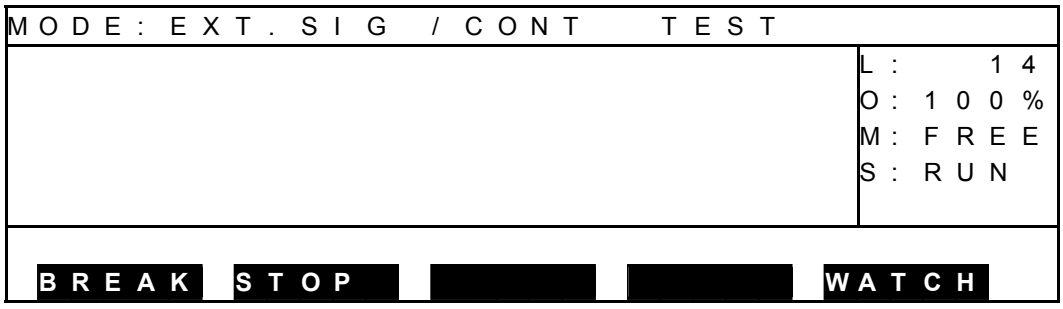
(1) Function

Starts an automatic operation under the predetermined conditions.

(2) Procedures

(a) Startup by using the start signal in the external signal mode:
Input the start signal.
For the start signal, see the Interface Manual.

(b) Startup by data communication in the host mode:
Start up the program through data communication.
For the data communication, see the Communication Manual.



(3) Caution

Startup can be performed, using the "startup" signal or data communication. In the external control signal mode, only the "startup" signal is effective. In the host mode, only the startup by data communication takes effect. The START switch on the control panel is inoperative.

9.6 Stop

9.6.1 CYCLE Stop

(1) Function

Selects "CYCLE" for the run mode and stops an automatic operation at the program end.

(2) Procedures

(a) Step 1: Input of "cycle operation mode" signal.

Input the "cycle operation mode" signal, and the cycle operation mode is selected, where an automatic operation stops at the end of the program. With the stop of the automatic operation, the "cycle end" signal is output. In the external automatic mode, the CYCLE switch equipped on the control panel is inoperative.

For the "cycle operation mode" signal and "cycle end" signal, see the Interface Manual.

(3) Caution

In the external control signal mode, the cycle operation mode is selected by the input of "cycle operation mode" signal. When the program has been executed to the end in the cycle operation mode, the automatic operation stops with the "cycle end" signal output.

In the host mode, the operation mode cannot be changed over by the external control signal. If the program is started in the continuous operation mode, cycle stop cannot be effected.

The SCOL program execution unit pre-reads up to four (4) motion commands. Therefore, the program may be executed successively by up to three (3) cycles according to the cycle stop timing or the active program contents.

9.6.2 STOP

(1) Function

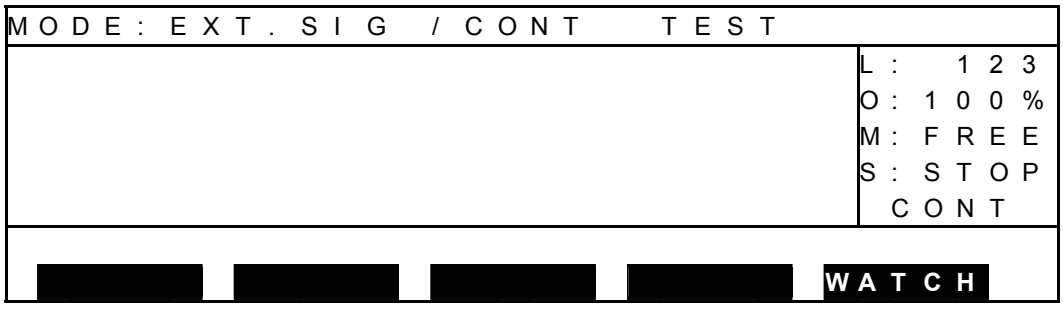
Stops an automatic operation when the active command has been executed. This function is used for normal stop operation.

(2) Procedures

(a) Step 1: Input of [STOP].

- Control panel: Press the STOP switch.
- Teach pendant: Press the function key corresponding to the [STOP] menu.
- "Stop" signal (only in the external control signal mode): Input the "stop" signal. (For details, see the Interface Manual.)
- Data communication (only in the host mode): Send the stop command. (For details, see the Communication Manual.)

The STOP switch LED on the control panel is illuminated. When the currently executed command has completed, the automatic operation stops at the same time that the RUN switch LED on the control panel turns off with "STOP CONT" shown under "S" (execution status) on the teach pendant display.



(3) Caution

In the host mode, the "stop" signal is invalid.
In the external control signal mode, stop by data communication is ineffective.
For the "stop" signal and data communication, see the Interface Manual and Communication Manual, respectively.

9.6.3 BREAK

(1) Function

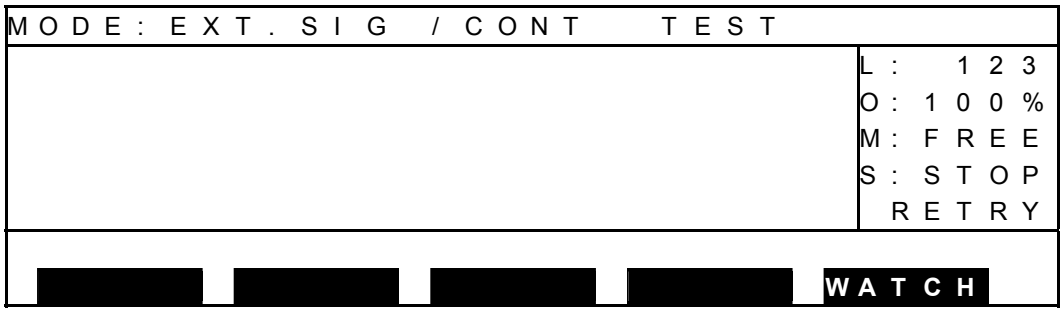
Immediately decelerates and stops the robot, and stops an automatic operation. This function is used to interrupt the automatic operation.

(2) Procedures

(a) Step 1: Input of [BREAK].

- Teach pendant: Press the function key corresponding to the [BREAK] menu.
- "Stop" signal (only in the external control signal mode):
  Input the "break" signal. (For details, see the Interface Manual.)

The robot immediately slows down and stops to stop the automatic operation. The RUN switch LED on the control panel turns off with the STOP switch LED illuminated. "STOP RETRY" is shown under "S" (execution status) on the teach pendant display.



(3) Caution

To restart the operation after the stop by the break command, perform the startup operation.

9.6.4 FEED HOLD

(1) Function

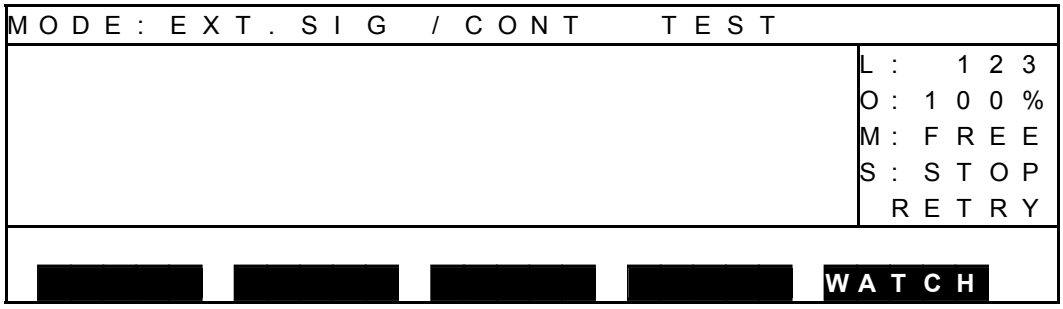
Holds robot motion and program execution. When this function is specified during robot operation, the robot immediately slows down and stop. The feed hold function is used to stop the robot temporarily.

(2) Procedures

(a) Step 1: Input of feed hold.

Press the FEED HOLD pushbutton switch equipped on the teach pendant, and the FEED HOLD switch LED is illuminated. If the robot is working, it decelerates and stops.

When the robot has stopped, the RUN switch LED on the control panel goes out with the STOP switch LED illuminated. The display of the teach pendant is as shown below with "STOP RETRY" indicated under "S" (execution status).



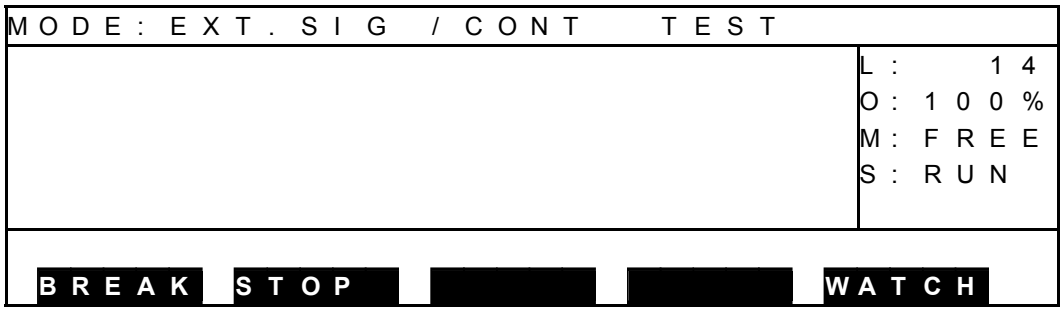
(b) Step 2: Cancel of feed hold.

While the FEED HOLD switch LED is turned on, press the FEED HOLD switch again. The FEED HOLD switch LED turns off and the automatic operation restarts.

If feed hold was commanded during program execution, program execution can be restarted by cancelling the feed hold.

When the program has restarted, the STOP switch LED on the control panel turned off with the RUN switch LED illuminated.

"RUN" is shown under "S" (execution status) on the teach pendant display.



(3) Caution

If the feed hold was commanded during program execution, program execution can be restarted by cancelling the feed hold.

If you should change the master mode, select a file or reset the execution conditions, the program will not restart.

9.6.5 EMERGENCY Stop

(1) Function

Turns off the servo power and stops an automatic operation. This function is useful to stop the robot system at the time of an emergency.

(2) Procedures

Press the EMERGENCY stop pushbutton switch equipped on the control panel or teach pendant, or use the "emergency stop" external control signal.
The servo power is turned off and the SERVO ON switch LED on the control panel goes out with the SERVO OFF switch LED illuminated.
The servo ON switch LED on the teach pendant goes out.

(3) Caution

All of the EMERGENCY stop switches on the control panel and teach pendant, and "emergency stop" external control signal are operative. At the time of an emergency, use whichever key you can get at quickest.
The EMERGENCY stop switches on the control panel and teach pendant are of a lock type. When pressed, they remain depressed and locked. To reset, turn and unlock the switch.
While the EMERGENCY stop switch is ON, the servo power cannot be turned on.

9.7 Reset

(1) Function

(a) Program reset
Resets all execution conditions (execution steps, variables) for a current file and allows execution of that program from the top.
This function is useful to execute a current file from the top of the program.

(b) Step reset
Resets all execution conditions (excluding global variables) for a current file and allows execution of that program from the top.

(c) Cycle reset
Resets all execution conditions (excluding global variables) for a current file and allows execution of the main program from the step indicated by label RCYCLE. This function is useful when you do not wish to execute the setting of the initial condition. Also, specify the GOTO RCYCLE command just before the END command not to execute the END command of the main program.

(d) Output signal reset
Sets off all user external output signals (DOUT (1) ~ DOUT (164)). The hand signals (201 ~ 204) will not turn off.

(2) Procedures

- External control signal mode:
  Turn on the input signal corresponding to either the program reset, step reset, cycle reset or output signal reset.
  In the external control signal mode, reset by data communication is not possible. For details, see the Interface Manual.

- Host mode:
  Send the command corresponding to the reset operation you wish to perform through data communication. If the command is accepted by the controller, an "acknowledge" signal is sent back from the controller.
  In the host mode, reset by external control signal is not possible. For details, see the Communication Manual.

(3) Caution

In the program reset mode, all variables used in the program are reset. If you do not wish to the variables, use the step reset function instead.

If you do not wish to execute the setting of initial conditions at the head of the program, use the cycle reset instead.

In the cycle reset mode, use label RCYCLE in a desired step of the main program.

9.8 Variable Monitor

(1) Function

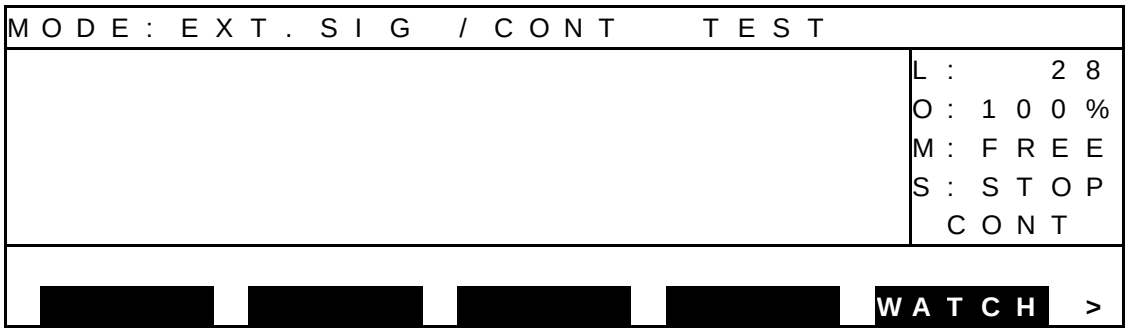
Displays the values of a desired global variable.

(2) Procedures

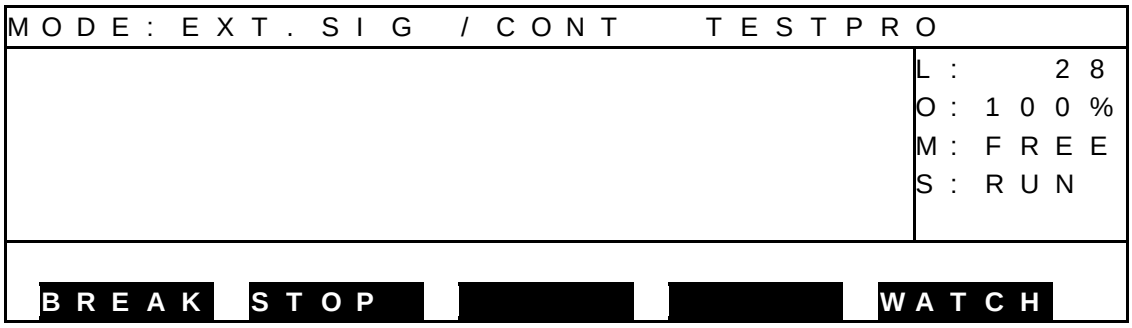
(a) Step 1: Command selection.

Press the NEXT key repeatedly until [WATCH] is displayed in the soft key menu, and press the corresponding function key.

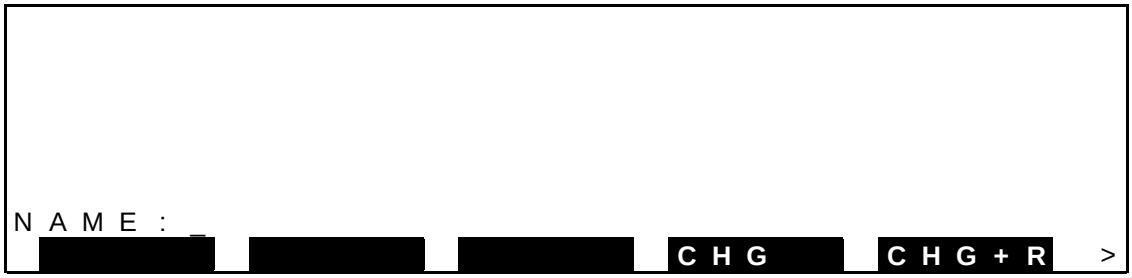
- During stop



- During run

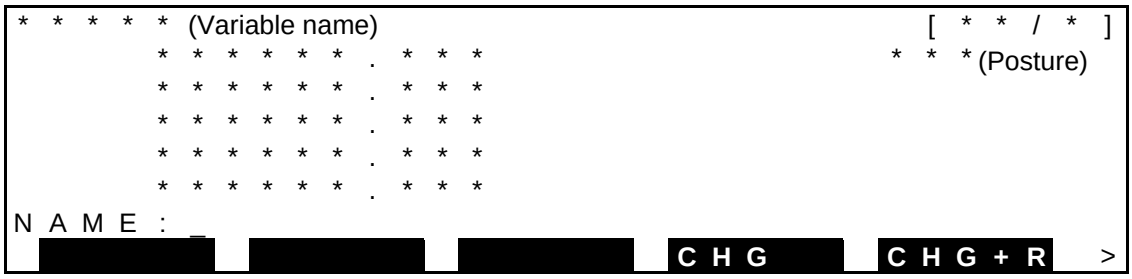


Press the F5 [WATCH] function key, and the variable monitor display appears.



* The global variable signifies the variable (real number type, integer type, position type, coordinate type or load type) as defined in the area flanked by GLOBAL and END, or by DATA and END (for the variable which can be edited by the DEDIT function) in the program.

(b) Step 2: Input of variable name.
Enter a variable name you wish to monitor, then press the EXE key. The variable name thus input and its values are displayed.



Unless the specified variable exists in the program, "------.---" is displayed. Up to thirty (30) variable names can be registered.

(c) Step 3: Changeover of monitor variable display.
Monitor variable display can be changed over in the following manner.

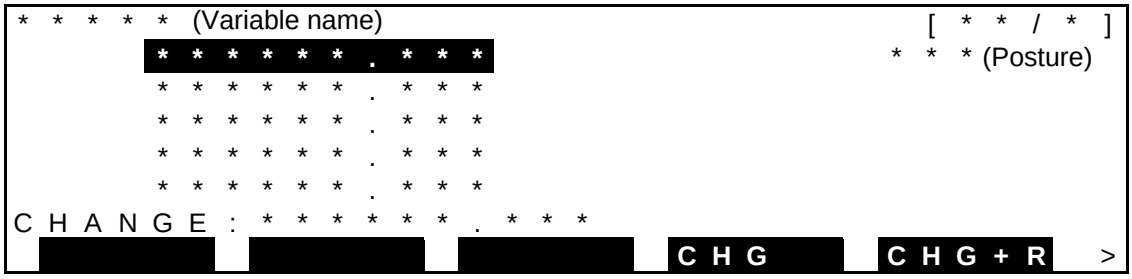
Display of the next variable and after	Press the ⇓ key.
Display of the previous variable and before	Press the ⇑ key.
Display of already registered variable	Enter the already registered variable.

(d) Step 4: Change of variable value [CHG].
It is possible to change a programmed value temporarily. (To rewrite the initial value in the program, execute [CHG+R].)

[1] Press the F4 [CHG] key, and the following display is called.

[2] A reversed indication can be shifted up or down by means of the ⇑ or ⇓ key, respectively.

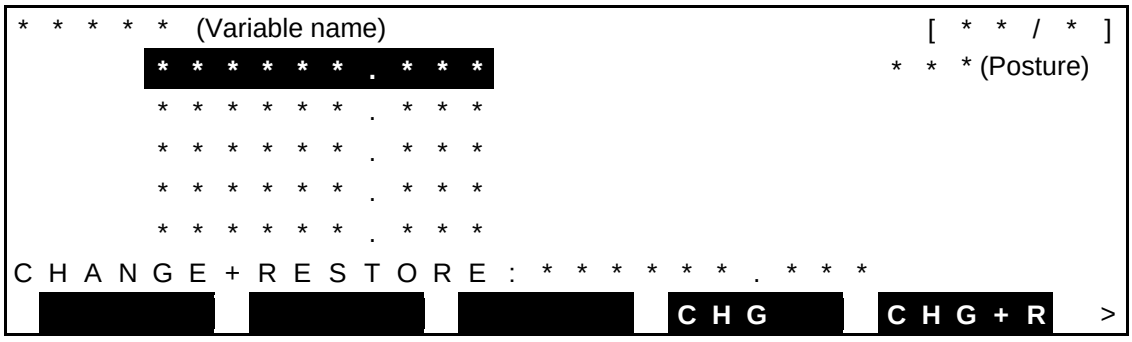
When the cursor is set to an element you wish to change, relevant value of the selected element is displayed as [CHANGE : *******.***■]. Change the value and press the EXE key, then the value is updated.



(e) Step 5: Change of variable [CHG+R].
It is possible to change the initial value in the program. By this operation, a statement itself in the program is rewritten.

[1] Press the F5 [CHG+R] key, and the following display is called.

[2] A reversed indication can be shifted up or down by means of the ↑ or ↓ key, respectively.
When the cursor is set to an element you wish to change, relevant value of the selected element is displayed as [CHANGE+RESTORE : *******.***■]. Change the value and press the EXE key, then the value is updated.



(f) Step 6: Deletion of registered monitor variable.
Change over the monitor variable display until a variable you wish to delete is displayed. Then press the ALT and BS keys (DEL) at the same time.

(g) Step 7: End of variable monitor.
Press the ESC key, and the system returns to the previous display of each mode.

! CAUTION

- Unless any program is selected, displayed value may not be correct.
- The [CHG+R] function cannot be executed for the global variable yet to be initialized (such as position type array variable which is not taught the robot).

  (If you are about to execute this, the error message saying "Can not RESTORE" is displayed.
- The position type data can be changed in the range of ±3 mm or ±3 deg.

Section 10 System Control

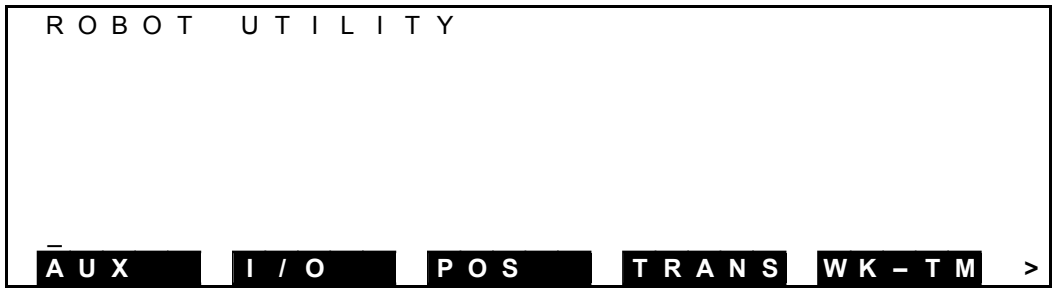
10.1 Setting the Date and Time

(1) Function

Sets and displays the date and time used by the controller. Specify the date or time if it is found incorrect after the controller has started up.
This function is used for directory management and error history.

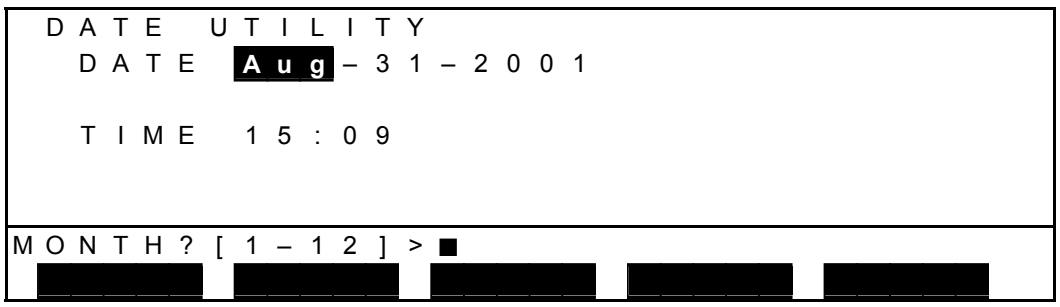
(2) Procedures

(a) Step 1: Selection of utility mode.
Press the UTILITY key equipped on the teach pendant. When the utility mode is selected, the following display appears.



(b) Step 2: Command selection.
Press the NEXT key repeatedly until [DATE] is displayed in the soft key menu. Press the corresponding function key then.



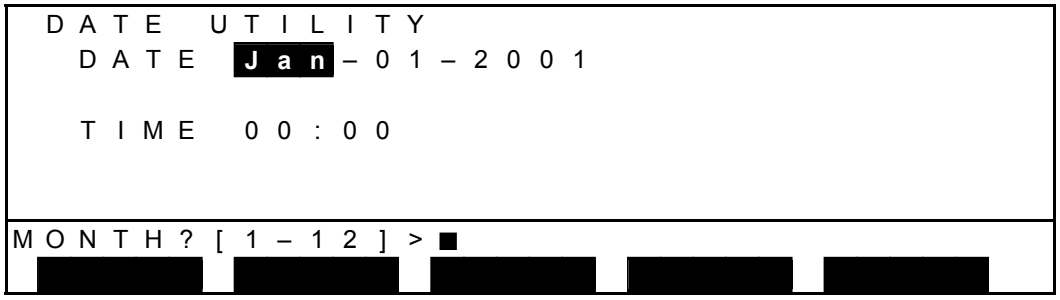


(c) Step 3: Enter the year, month, day, hours and minutes.
Move the cursor to each data item, then input the data.



(d) Step 4: Input of EXE key.
Make sure that your entry is correct, and press the EXE key. If you have entered a wrong data, use the BS key to correct the data, or cancel your entry (press the ESC key), enter a legal data again and press the EXE key.

If you have entered a value outside the specified range, you cannot change the data. Or if you have entered only "*", the calendar is initialized as shown below.



(3) Caution
As the controller has a built-in calendar which is backed up by battery, setting is usually unnecessary. Set the data again only when the date or time is incorrect.

10.2 Automatic Execution File

(1) Function

If you create a batch file named "AUTOSTR. BAT" and specify the command in it beforehand, it is executed automatically at power ON.

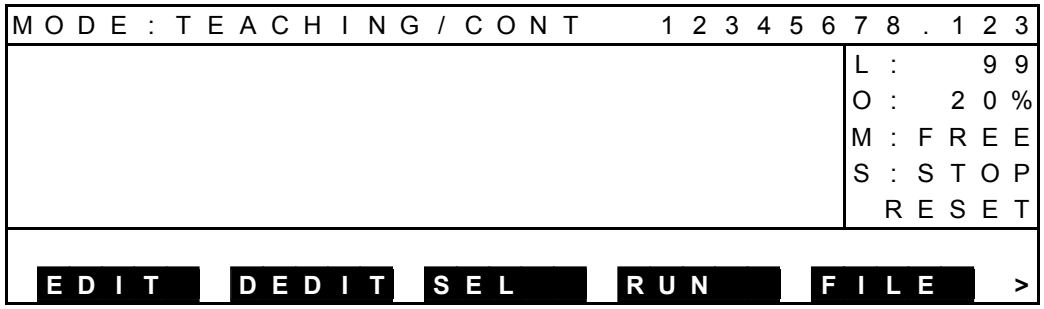
(2) Procedures for file creation

(a) Step 1: Mode selection.

Press the MODE select switch equipped on the control panel to select "TEACHING."

(b) Step 2: Menu selection.

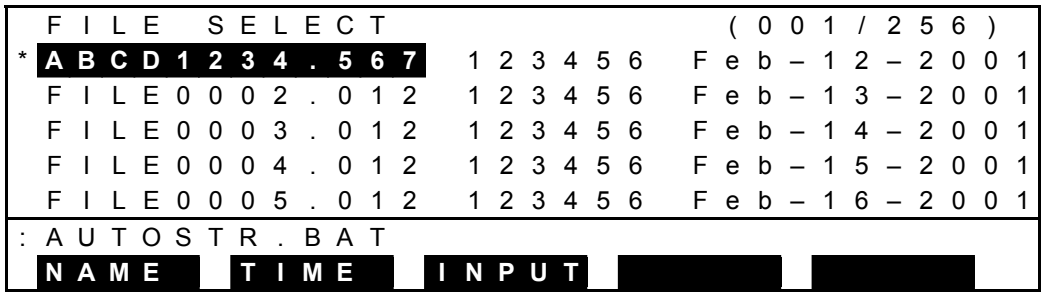
Press the NEXT key repeatedly until [EDIT] is displayed in the soft key menu. Press the corresponding function key then.



(c) Step 3: Designation of file name.

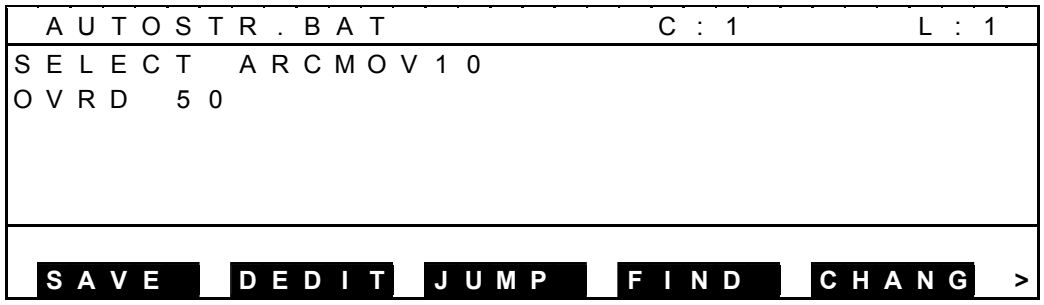
Specify "AUTOSTR. BAT" for the file name. The extension is BAT.

To create a new file, press [INPUT] and enter the name.



(d) Step 4: Check of file name you wish to edit.
Make sure that your entry is correct, and press the EXE key. If you wish to change the file name, return the cursor, using the BS key, enter another file name through the keyboard, then press the EXE key.

(e) Step 5: Creation of batch file.
Enter a command which becomes valid in batch processing, and a file name to execute that command line by line.



(f) Step 6: Quit of the editor.
Press [SAVE] to quit the editor.

Now, the batch file for batch processing has been created.

(3) Execution of batch file
If the AUTOSTR. BAT file exists in the RAM drive at power ON, it is automatically executed after the controller has started up.

(4) Caution
The following commands become valid in the batch processing.
SELECT
OVRD
DATE
TIME

Section 11 File Operation

11.1 Directory Display [DIR]

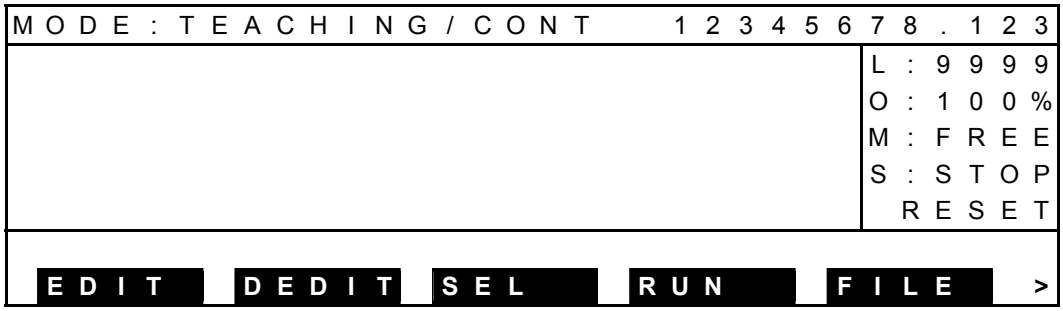
(1) Function

Displays a file name directory registered in the RAM drive.

(2) Procedures

(a) Step 1: Selection of file mode.

Press the NEXT key repeatedly until [FILE] is displayed in the soft key menu. Press the corresponding function key then.

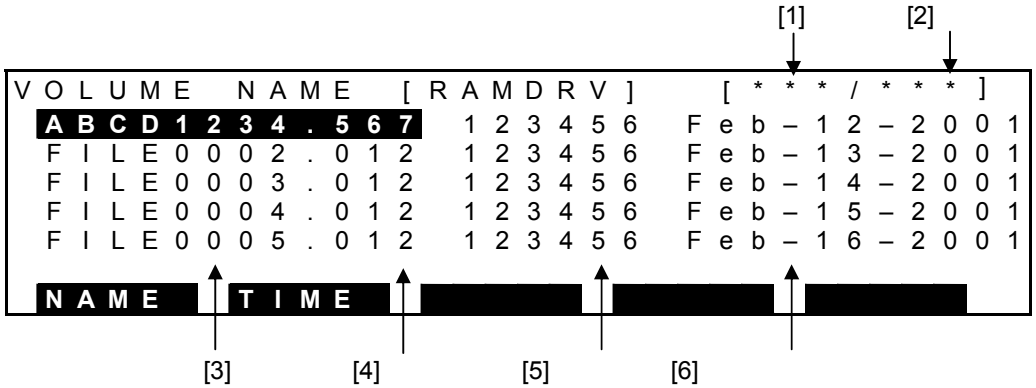


↓ Press [FILE].



(b) Step 2: Command selection.

Press [DIR], and a directory of all files registered is displayed.



Display contents:

[1] Order of a file shown on the 1st line.
[2] Total number of files
[3] File name
[4] File name extension
[5] File size in bytes
[6] Date of file creation (or alteration)

Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[↑]	Used to display the contents which are just one (1) line before the cursor line.
[↓]	Used to display the contents which are just one (1) line after the cursor line.
[ALT] + [↑]	Used to display the contents of the just preceding page.
[ALT] + [↓]	Used to display the contents of the just succeeding page.
[F1]	Used to sort the data by file name.
[F2]	Used to sort the data by date.
[ESC]	Used to return the system to the previous display.

11.2 File Copy [COPY]

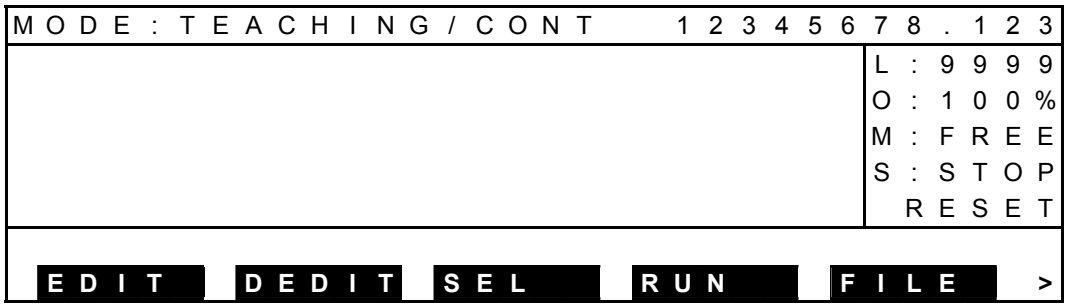
(1) Function

Copies one file to another file. Both program section and data section are copied. This function is used to back up the editing and create a similar file.

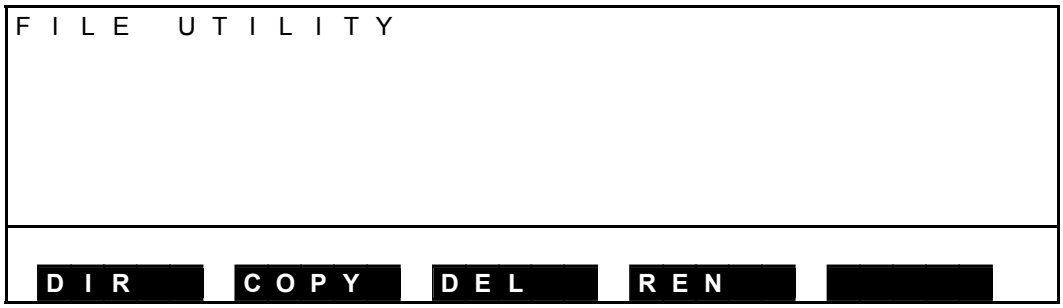
(2) Procedures

(a) Step 1: Selection of file mode.

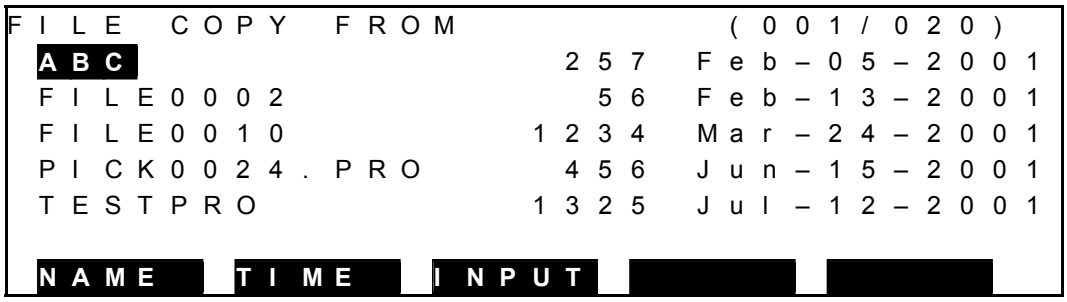
Press the NEXT key repeatedly until [FILE] is displayed in the soft key menu. Press the corresponding function key then.



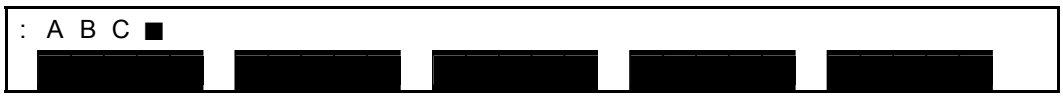
↓ Press [FILE].



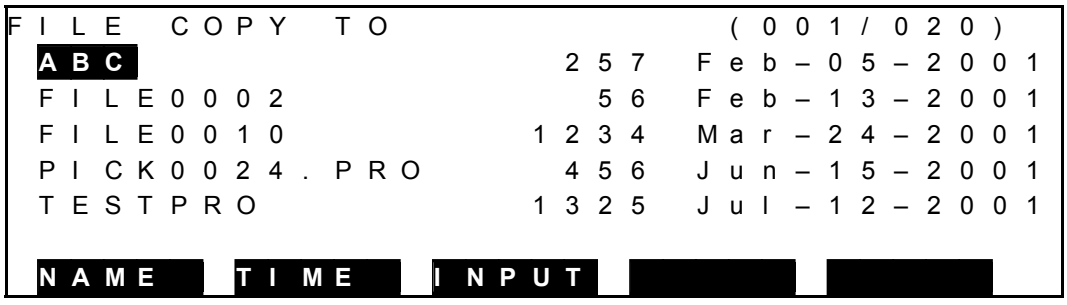
(b) Step 2: Command selection.
Press [COPY].
If you press the ESC key without any character input in the step below, you can exit this mode.
For the operation of each function key, see Para. 7.2.



(c) Step 3: Input of source file name.
Select a desired source file name, using the cursor, then press the EXE key. The file name thus selected is displayed on the command line. If it is OK, press the EXE key. It is also possible to change the file name by deleting one (1) character by means of the BS key or all characters by means of the ESC key.



(d) Step 4: Input of copied file name.
Select a file name where you wish to copy the source file.
If you press the ESC key without any character input in the following step, the system returns to the source file name input mode.



After selecting the copied file name, press the EXE key. When you wish to create a new file, press [INPUT] and enter the file name. The selected file name can also be changed like the source file name. If you press the ESC key, you can delete a file name. If you further press the same key, the system returns to the source file name input mode.



Make sure that the input file name is correct, then press the EXE key.

(e) Step 5: Input of EXE key.
When the confirmation message appears, make sure of the input file name again. Then press the EXE key.



(f) Step 6: Input of any key.
When the file has been copied, the following message appears.
When you press any key on the teach pendant, the system returns to Step 2 and you can continue copying.



If the copied file name specified already exists, the system waits for your answer, displaying a message saying "Already exist. Delete?".



If you wish to delete the already registered file, press the EXE key. If you do not wish to delete it, press the ESC key.

(3) Caution

If the size of the copying data exceeds the memory capacity as a result of the copy operation, an error message of "RAM file I/O error" is displayed.

If the number of files registered in the RAM drive exceeds the limit by the copy operation, the system waits for key input, displaying an error message of "MTX file I/O error" or "RAM file I/O error". Delete any unnecessary file or files and try again

Unless the source file exists, an error message saying "File name not found" is displayed, and this operation becomes invalid.

If the source file name is identical with the copied file name, an error message of "Illegal operand" is displayed, and this operation becomes invalid.

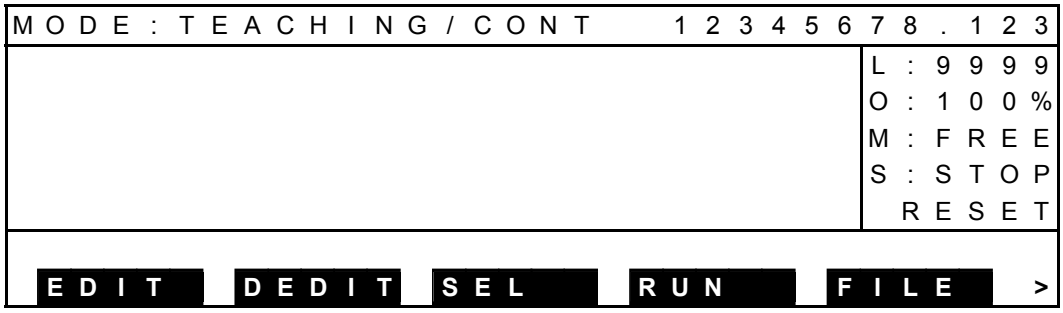
You cannot copy a file to the file selected as the execution file. When this happens, an error message saying "File name file already selected" is displayed.

11.3 File Rename [REN]

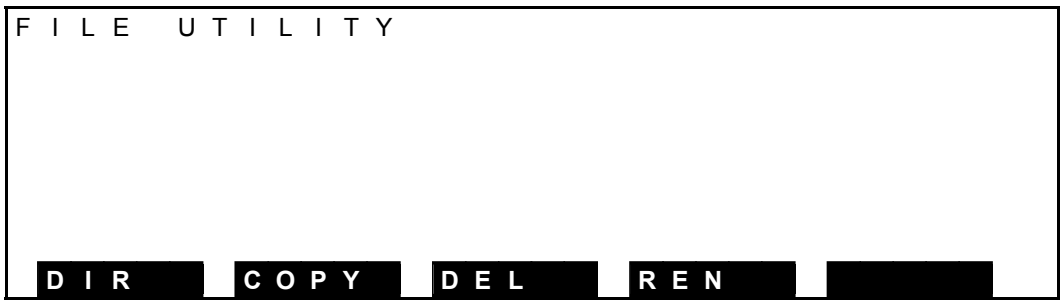
(1) Function
Renames a file.

(2) Procedures

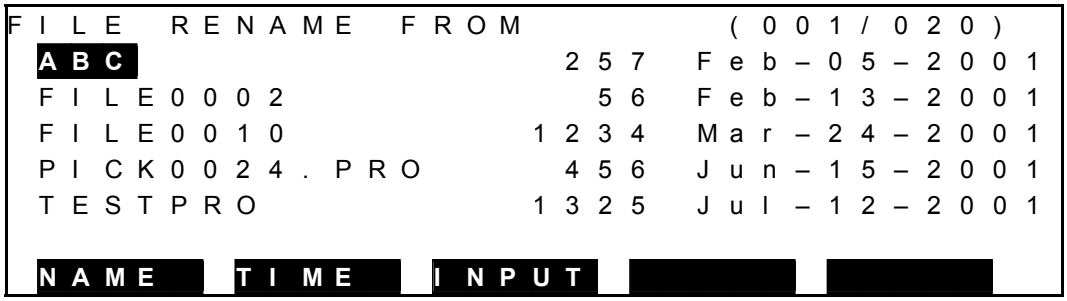
(a) Step 1: Selection of file name.
Press the NEXT key repeatedly until [FILE] is displayed in the soft key menu.
Press the corresponding function key then.



↓ Press [FILE].



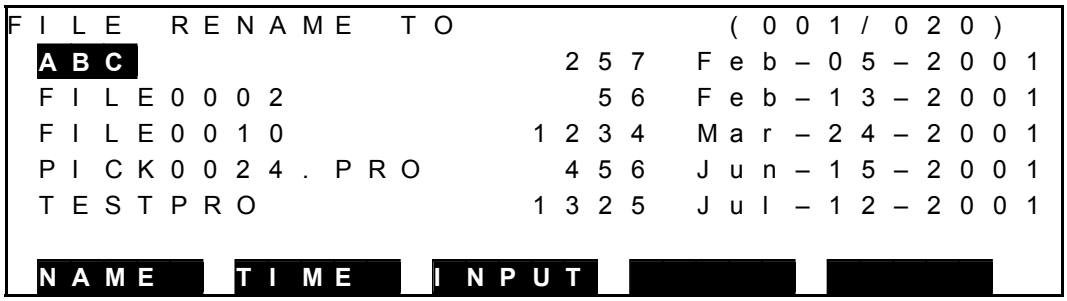
(b) Step 2: Command selection.
Press [REN].
If you press the ESC key without any character input in the step below, you can exit this mode.



(c) Step 3: Input of file name before change.
Select a file name before change, using the cursor, then press the EXE key. The file name thus selected is displayed on the command line. If it is OK, press the EXE key. It is also possible to change the file name by deleting one (1) character by means of the BS key or all characters by means of the ESC key.



(d) Step 4: Input of file name after change.
Select a file name after change.
If you press the ESC key without any character input in the following step, the system returns to the "input of file name before change" mode.



After selecting the file name after change, press the EXE key. When you wish to create a new file, press [INPUT] and enter the file name. The selected file name can also be changed like the file name before change. If you press the ESC key, you can delete a file name. If you further press the same key, the system returns to the "input of file name before change" mode.



Make sure that the input file name is correct, then press the EXE key.

(e) Step 5: Input of EXE key.
When the confirmation message appears, make sure of the input file name again. Then press the EXE key.



(f) Step 6: Input of any key.
When the file has been renamed, the following message appears.
When you press any key on the teach pendant, the system returns to Step 2 and you can continue renaming.



If the specified file name before change already exists, the system waits for your answer, displaying a message saying "Already exist. Delete?".



If you wish to delete the already registered file, press the EXE key. If you do not wish to delete it, press the ESC key.

(3) Caution

Unless the file name before change exists, an error message of "File name not found" is displayed, and this operation becomes invalid.

If the file names before and after change are identical, an error message saying "Illegal operand" is displayed, and this operation becomes invalid.

If you try to rename a file selected as the execution file, an error message saying "File name file already selected" is displayed, and this operation is ineffective.

If the file name after change is identical with the file selected as the execution file, an error message saying "File name file already selected" is displayed, and this operation is ineffective.

11.4 File Delete [DEL]

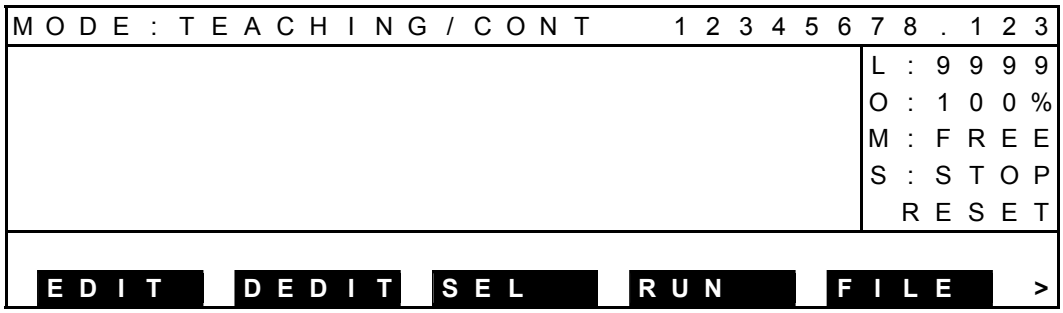
(1) Function

Deletes a specified file. This function is used to delete an unnecessary file or files when no free area is left in the RAM drive.

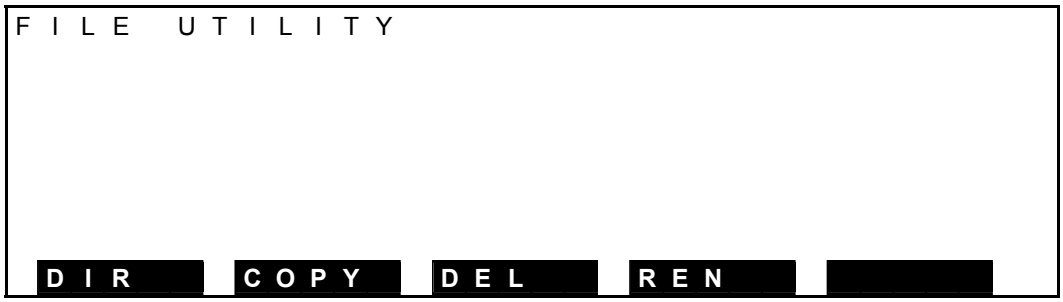
(2) Procedures

(a) Step 1: Selection of file mode.

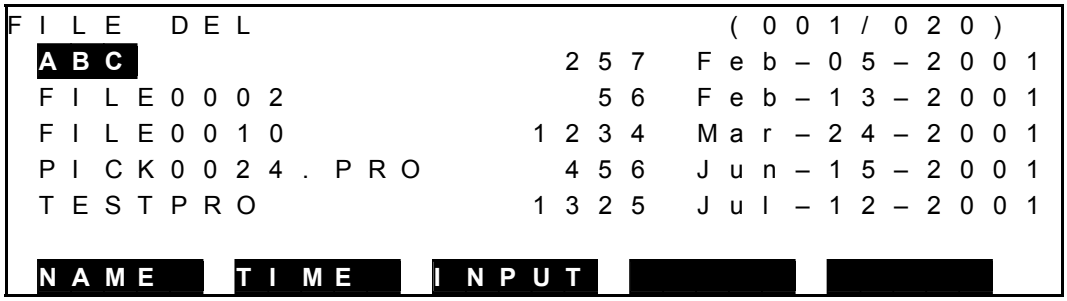
Press the NEXT key repeatedly until [FILE] is displayed in the soft key menu. Press the corresponding function key then.



↓ Press [FILE].



(b) Step 2: Command selection.
Press [DEL].
If you press the ESC key without any character input in the step below, you can exit this mode.



(c) Step 3: Input of file name to be deleted.
Select a file name you wish to delete and press the EXE key. If there are many files to be deleted, use wild cards or execute this function over a few times. The file name thus selected is displayed on the command line. When you press the ESC key, a file name can be deleted. If you further press the same key, you can select a file name again.



(d) Step 4: Input of EXE key.
After the confirmation message appears, make sure that your entry is correct, and press the EXE key.



(e) Step 5: Input of any key.
When the file has been deleted, the following message appears.
When you press any key on the teach pendant, the system returns to Step 2 and you can continue deleting.



When you use the wild card (*.*), the system waits for your answer, displaying a message of "Are you sure? (Y/N)".



When you wish to delete all files, press the EXE key. When you do not wish to delete them, press the ESC key.

(3) Caution
If you use the wild card (*.*) for the RAM drive, you will clear the reserved files, too. A file selected as the execution file cannot be deleted. When this happens, an error message saying "File name file already selected" is displayed.

Section 12 Utility

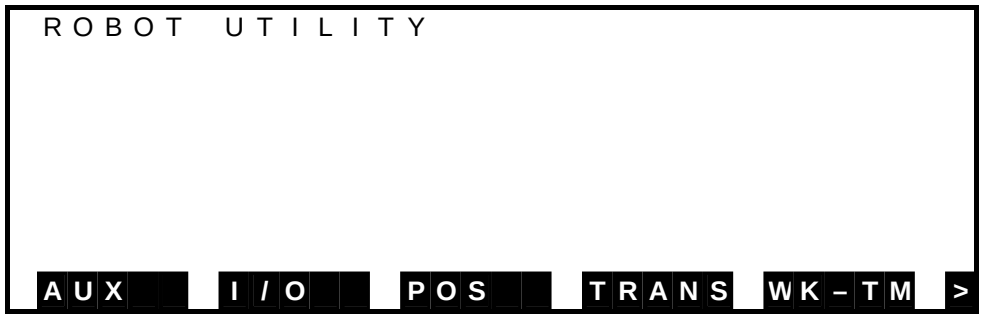
The utility mode allows external input/output signal display, current position display, selection of work/tool coordinate system, etc. It can be used in the teaching, internal automatic and external automatic modes.
When you press the ESC key, the system returns to the previous display, quitting the utility mode.

12.1 Selecting Utility Mode

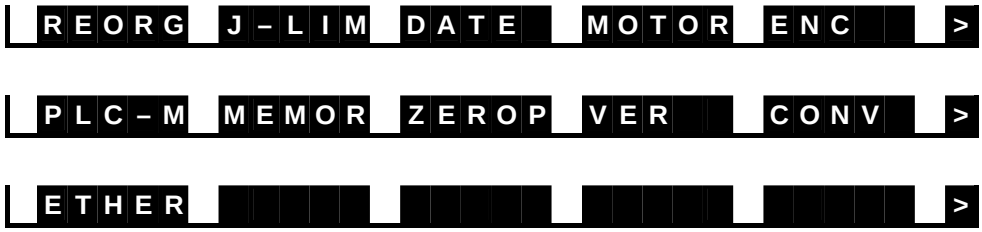
(1) Function
Selects the utility mode.

(2) Procedures

(a) Step 1:
Press the UTILITY key equipped on the teach pendant, and the following display appears.



Each time the NEXT key is pressed, the menu is changed over as follows:



The key operative in this mode and its operation are tabled below.

Key	Descriptions
ESC	Used to quit the utility mode and return the system to the previous display.

Operations of respective soft keys are shown below.

Operations of respective soft keys are shown below.

Soft key	Descriptions
[AUX]	Used to display the auxiliary signals.
[I/O]	Used to display the external input/output signals.
[POS]	Used to display the robot current position in the joint, world or work coordinate system.
[TRANS]	Used to select the work coordinate system or tool coordinate system.
[WK–TM]	Used to display and set the robot working time.
[REORG]	Used to reproduce the origin (or home point). Note: This function is intended for the maintenance. For details, see the maintenance manual provided separately.
[J–LIM]	Used to display the joint limit set values and set a joint limit value as necessary.
[DATE]	Used to display and set the current date and time.
[MOTOR]	Used to display the servo motor torque, electronic thermal total value and load factor. Note: This is the extended function of TS2000.

Soft key	Descriptions
[ENC]	Used to display the servo encoder data. Note: This function is intended for the maintenance. Operate this key according to the instructions of our service engineers.
[PLC–M]	Used to display the PLC data.
[MEMOR]	Used to display the system internal memory. Note: This function is intended for the maintenance. Operate this key according to the instructions of our service engineers.
[ZEROP]	Used to set the origin (or home point). Note: This function is intended for the maintenance. Operate this key according to the instructions of our service engineers.
[VER]	Used to display the system version.
[CONV]	Used to display each status of the conveyor synchronization function. Note: This function is an option of TS2000. For details, see the conveyor synchronous function instruction manual provided separately.
[ETHER]	Used to display the Ethernet status. Note: This function is an option of TS2000. For details, see the Ethernet interface instruction manual provided separately.

(3) Caution

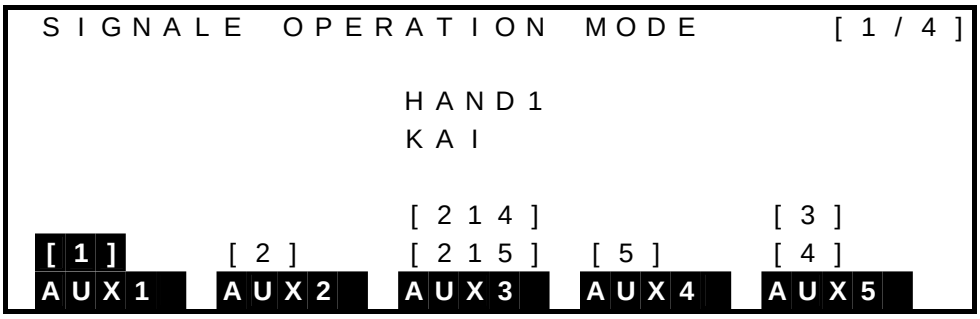
- This function can be used in the teaching, internal automatic and external automatic modes.

12.2 Auxiliary Signal Display [AUX]

(1) Function
Displays the status of auxiliary signals.

(2) Procedures

1. Press the NEXT key repeatedly until [AUX] is displayed in the soft key menu. Press the corresponding function key, and the following auxiliary signal display appears.



- The numbers on the display correspond to the output signal ports.
- Auxiliary signals having two (2) output signal ports are set for the double solenoid, and those having one (1) output signal port are intended for the single solenoid. Those having no output signal ports are not intended for anything.
- Ports inversely displayed are ON and ports not inversely displayed are OFF. (On the above display, only [1] is ON.)
- It is possible to display messages for respective signals on the 3rd to 4th lines of the display only when they are defined in the user parameter file beforehand. (For further information, see the Parameter Manual.)

Keys operative in this mode and their operations are tabled below.

Key	Descriptions
ALT + ↓	Used to display the next page. If the cursor is located at the final page, the page will not change over.
ALT + ↑	Used to display the previous page. If the cursor is located at the top page, the page will not change over.
ESC	Used to return the system to the utility display.

Operation of the soft key is shown below.

Key	Descriptions
[AUXn]	Each time this key is pressed, the solenoid state changes from ON to OFF or vice versa. For the double solenoid, the state of two (2) ports are changed over alternately.

(3) Caution

- The output signal port and solenoid type corresponding to each soft key should be defined in the user parameter file in advance. For more information, see the User Parameter Manual.
- During emergency stop, auxiliary signals can be turned on and off.
  The initial state at power ON, including signals which are set for the double solenoid, is OFF.

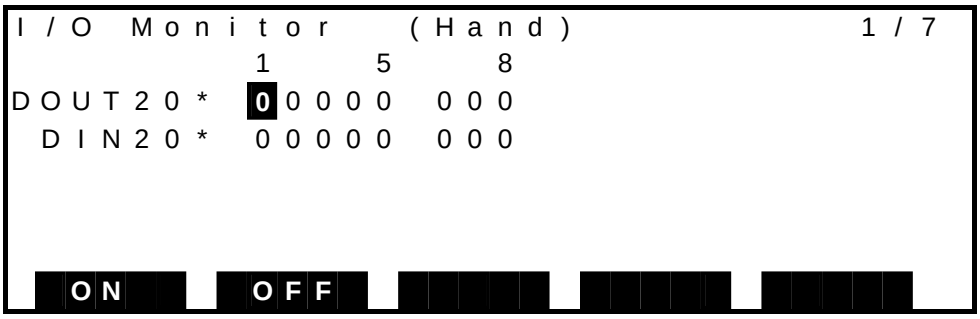
12.3 External Input/Output Signal Display [I/O]

(1) Function

Displays the status of external input/output signals.

(2) Procedures

1. Press the NEXT key repeatedly until [I/O] is displayed in the soft key menu. Press the corresponding function key, and the following hand input/output signal status display appears.



- For the display contents, see Para. 2 below.

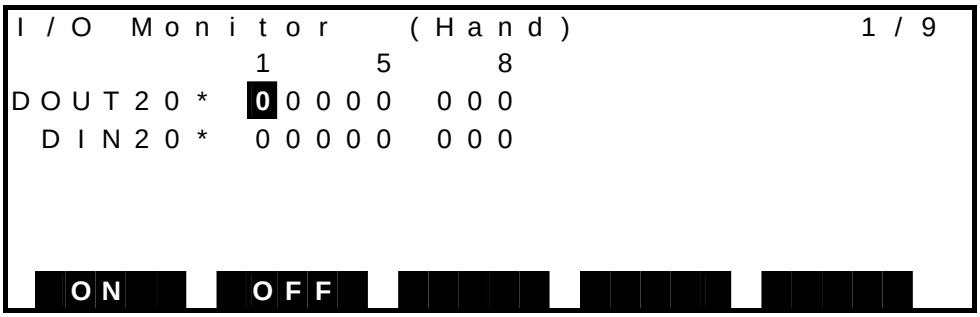
Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[←]	Used to move the cursor to the left. When the cursor is located on the left end, it will not move at all.
[→]	Used to move the cursor to the right. When the cursor is located on the right end, it will not move at all.
[↑]	Used to move up the cursor. When the cursor is located at the top, it will not move at all.
[↓]	Used to move down the cursor. When the cursor is located at the bottom, it will not move at all.
[ALT]+[↓]	Used to display the next page. If the cursor is located at the final page, the page will not change over.
[ALT]+[↑]	Used to display the previous page. If the cursor is located at the top page, the page will not change over.
[ESC]	Used to return the system to the utility display.

Operations of respective soft keys are shown below.

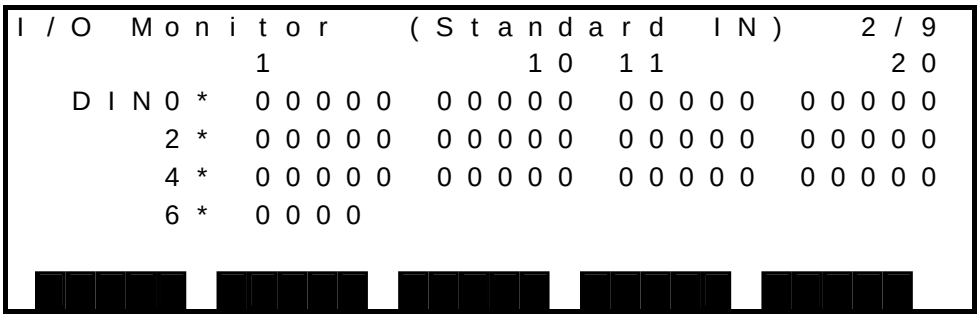
Key	Descriptions
[ON]	Used to set ON the bit indicated by the cursor.
[OFF]	Used to set OFF the bit indicated by the cursor.

2. Press the ALT and ↓ keys at the same time, and the display items are changed over. Display contents of each page are shown below.



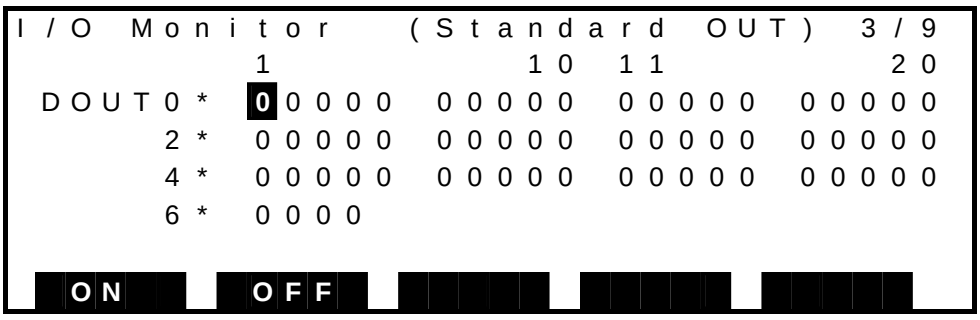
- The status of input and output signals are as follows:
  When the contact is open, the I/O signal status is "1". When closed, the signal status is "0".
- The number of input and output signals is eight (8) each. The number of the leading signal is indicated on the left end.
- For the output signals, data indicated by the cursor (i.e., data inversely displayed) can be set ON and OFF by means of respective soft keys.

2nd page (general input)



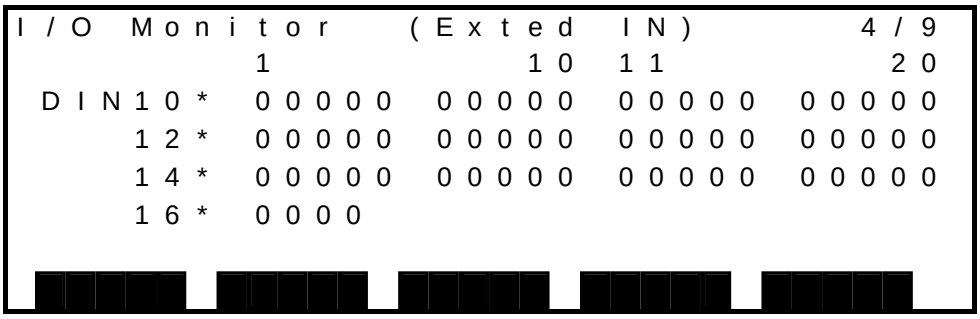
- The status of input and output signals are as follows:
  When the contact is open, the I/O signal status is "1". When closed, the signal status is "0".
- A total of twenty (20) signals are displayed on the same line by grouping five (5) signals each. The number of the leading signal is indicated on the left end.
- The number of signals is 64.

3rd page (general output)



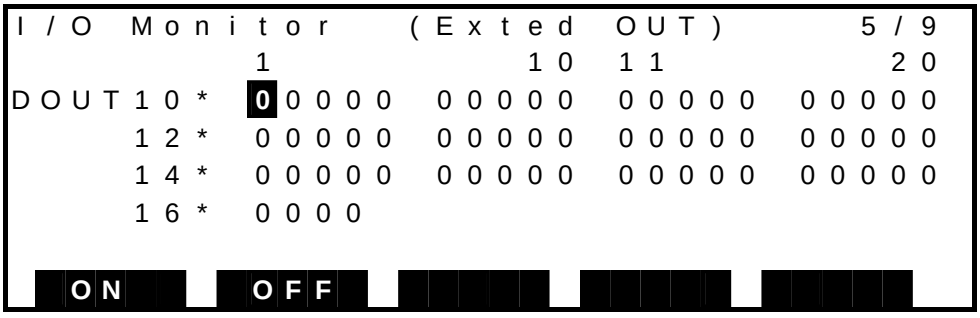
- The status of input and output signals are as follows:
  When the contact is open, the I/O signal status is "1". When closed, the signal status is "0".
- A total of twenty (20) signals are displayed on the same line by grouping five (5) signals each. The number of the leading signal is indicated on the left end.
- The number of signals is 64.
- Data indicated by the cursor (i.e., data inversely displayed) can be set ON and OFF by means of respective soft keys.

4th page (extension input)



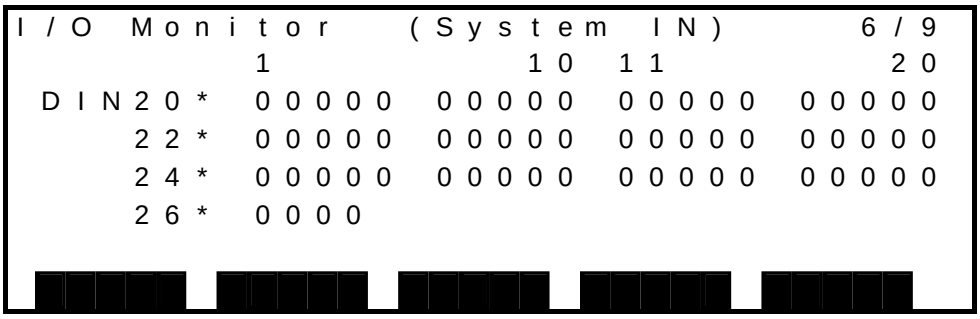
- The status of input and output signals are as follows:
  When the contact is open, the I/O signal status is "1". When closed, the signal status is "0".
- A total of twenty (20) signals are displayed on the same line by grouping five (5) signals each. The number of the leading signal is indicated on the left end.
- The number of signals is 64.

5th page (extension output)

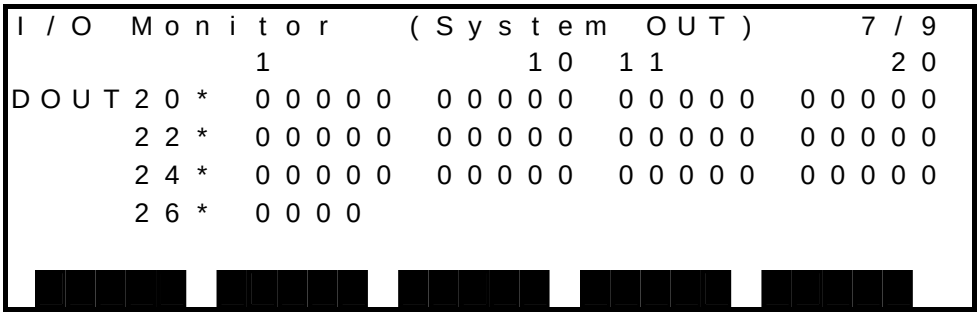


- The status of input and output signals are as follows:
  When the contact is open, the I/O signal status is "1". When closed, the signal status is "0".
- A total of twenty (20) signals are displayed on the same line by grouping five (5) signals each. The number of the leading signal is indicated on the left end.
- The number of signals is 64.
- Data indicated by the cursor (i.e., data inversely displayed) can be set ON and OFF by means of respective soft keys.

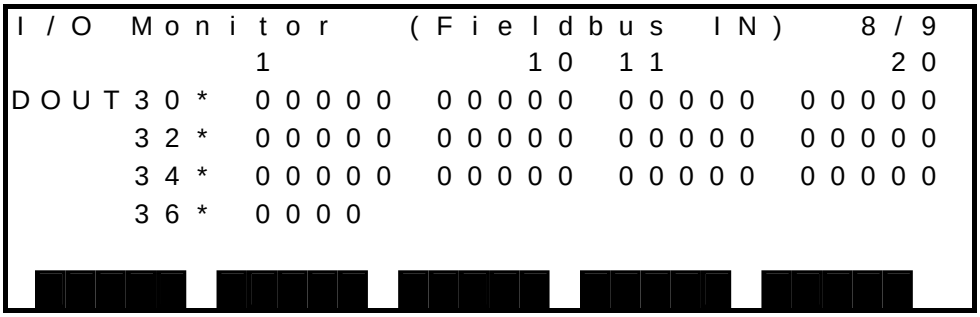
6th page (system input)



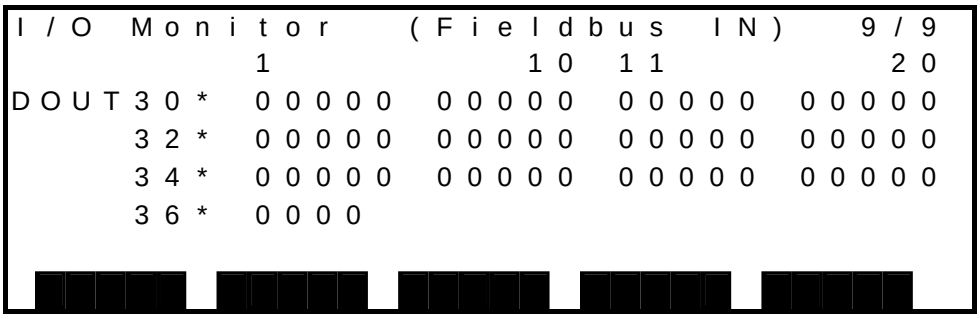
7th page (system output)



8th page (field bus input)



9th page (field bus output)



- The status of input and output signals are as follows:
  When the contact is open, the I/O signal status is "1". When closed, the signal status is "0".
- A total of twenty (20) signals are displayed on the same line by grouping five (5) signals each. The number of the leading signal is indicated on the left end.
- The number of signals is 64.

(3) Caution

- Signal display is also possible even during the automatic operation.

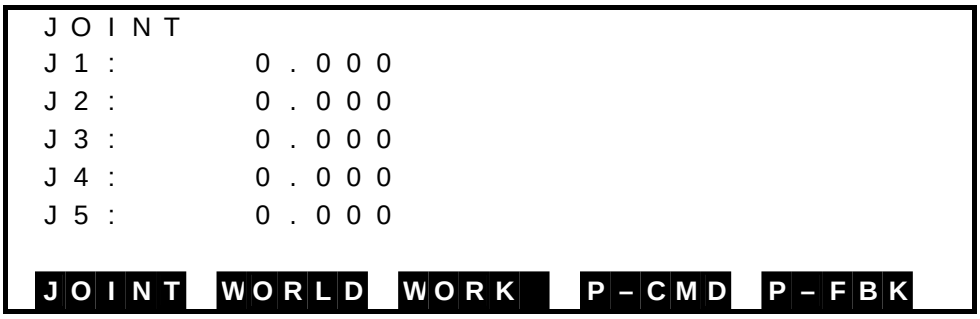
12.4 Current Position Display [POS]

(1) Function

Displays the robot position in the joint, world or work coordinate system.

(2) Procedures

1. Press the NEXT key repeatedly until [POS] is displayed in the soft key menu. Press the corresponding function key, and the following joint coordinate system position display appears.



- Coordinate value of respective joints are displayed.

The key operative in this mode and its operation are tabled below.

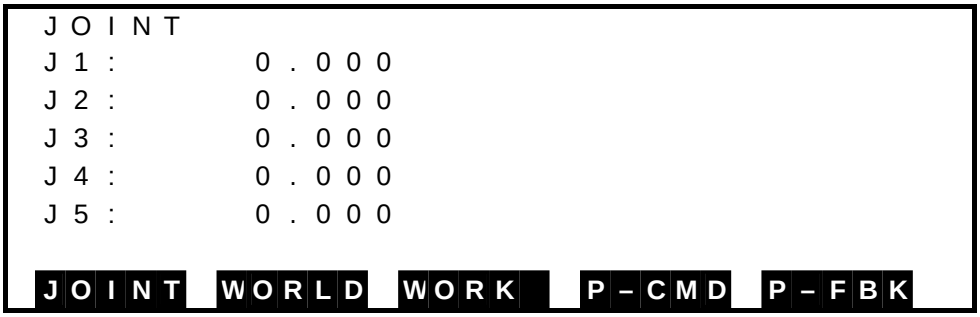
Key	Descriptions
ESC	Used to return the system to the utility display.

Operations of respective soft keys are shown below.

Key	Descriptions
[JOINT]	Used to call the position display in the joint coordinate system.
[WORLD]	Used to call the position display in the world coordinate system.
[WORK]	Used to call the position display in the work coordinate system.
[P–CMD]	Used to call the position command display.
[P–FBK]	Used to call the position feedback display.

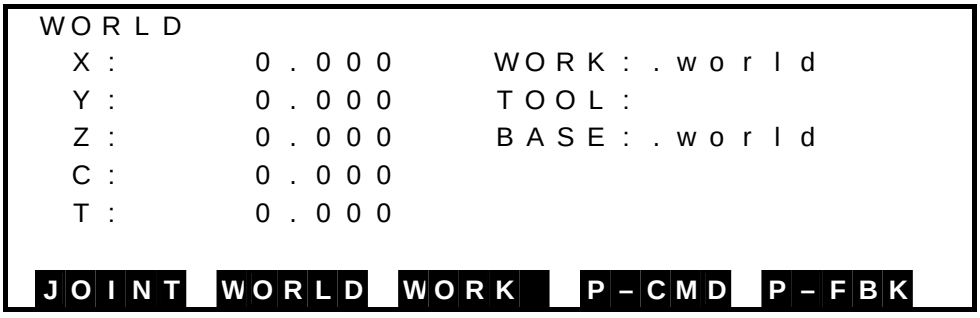
2. Press the soft key corresponding to the coordinate system you wish to display, and the position display in the selected coordinate system appears as shown below.

Position display in joint coordinate system



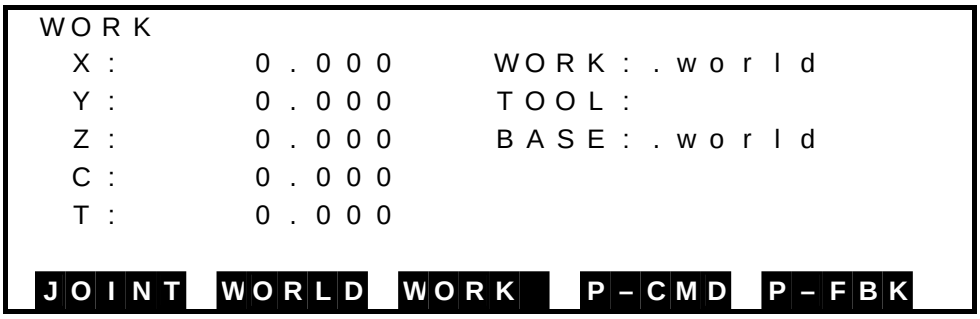
- Coordinates of respective joints are displayed.

Position display in world coordinate system



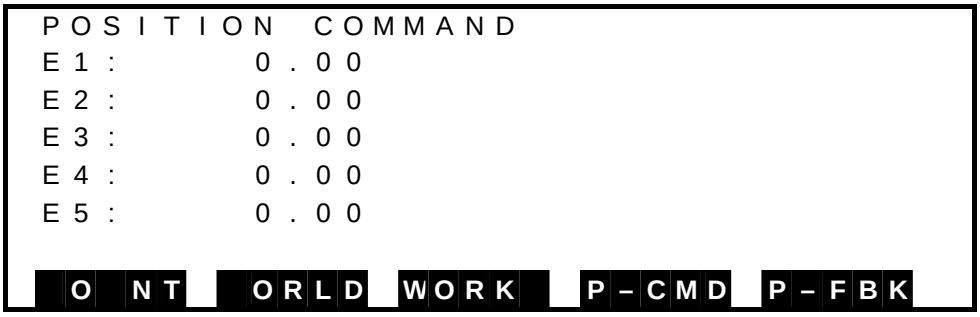
- A world coordinate of each axis is displayed.
- "WORK" signifies the coordinate name selected as the work coordinate system.
- "TOOL" represents the coordinate name selected as the tool coordinate system.
- "BASE" indicates the coordinate name selected as the base coordinate system.

Position display in work coordinate system



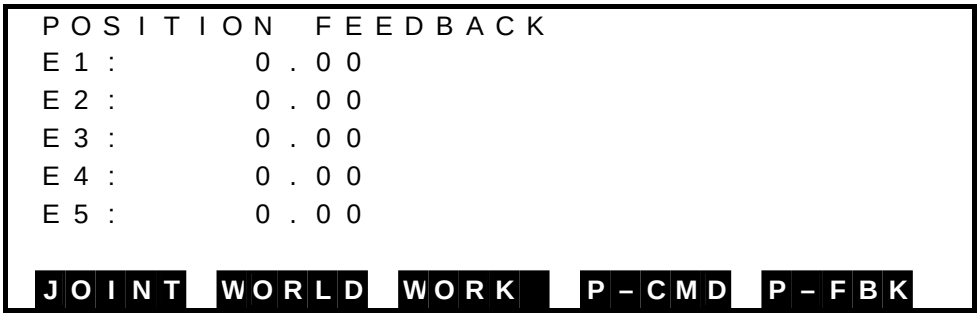
- A work coordinate of each axis is displayed.
- Display of "WORK", "TOOL" and "BASE" is the same as in the position display in the world coordinate system.

Position command display



- A position command value to each axis servo is displayed.

Position feedback display



- A position feedback from each axis servo is displayed.

(3) Caution

- Current position display is also possible even during the automatic operation.

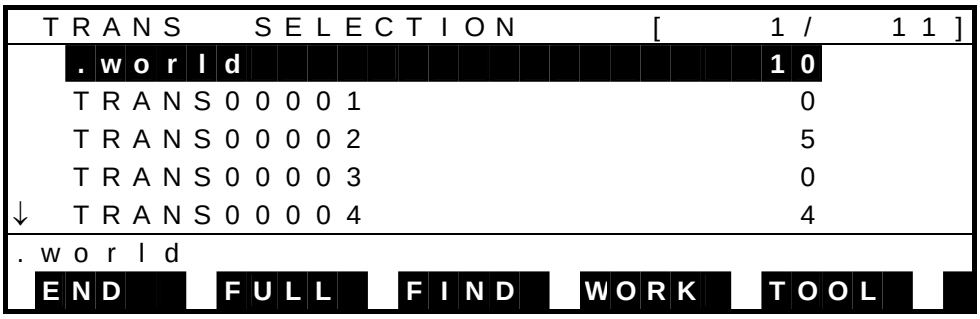
12.5 Selecting Work/Tool Coordinate System [TRANS]

(1) Function

Selects the work coordinate system or tool coordinate system.

(2) Procedures

1. Press the NEXT key repeatedly until [TRANS] is displayed in the soft key menu. Press the corresponding function key, and the following coordinate data directory display appears.



- Unless a program is selected, this key is inoperative.
- Coordinates displayed in the directory are those defined in the selected program.
- Defined coordinate data are displayed at a rate of one (1) data per line.
- An inversely displayed section in the data is called the "main cursor" which indicates that this data is selected for editing.
- Symbol "↑" shown in the 1st column signifies that there is coordinate data before the top data.
- Symbol "↓" shown in the 1st column signifies that there is coordinate data after the final data.
- Alphabet "t" displayed in the 1st column represents that the coordinate data on the same line is based on the tool coordinate system.
- Alphabet "w" displayed in the 1st column represents that the coordinate data on the same line is based on the work coordinate system.
- On the right side, a coordinate name is displayed. On the left side, the number of position data belonging to that coordinate is displayed.
- Regarding the data of [nnnn/mmmm] on the 1st line, "nnnn" identifies the selected coordinate index, and "mmmm" the number of coordinate data.

Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[↑]	Used to move up the main cursor. If the main cursor is located on the top line of screen, the screen scrolls down.
[↓]	Used to move down the main cursor. If the main cursor is located on the last line of screen, the screen scrolls up.
[ALT]+[↑]	Used to move up the main cursor on the screen basis.
[ALT]+[↓]	Used to move down the main cursor on the screen basis.
[ESC]	Used to quit the coordinate selection and return the system to the utility display.

Operations of respective soft keys are shown below.

Key	Descriptions
[END]	Used to quit the coordinate selection and return the system to the utility display.
[FULL]	Used to call the detailed display of a coordinate indicated by the main cursor.
[WORK]	Used to specify the work coordinate system for a coordinate system indicated by the main cursor, and cancel the designation.
[TOOL]	Used to specify the tool coordinate system for a coordinate system indicated by the main cursor, and cancel the designation.
[FIND]	Used to search a coordinate name.

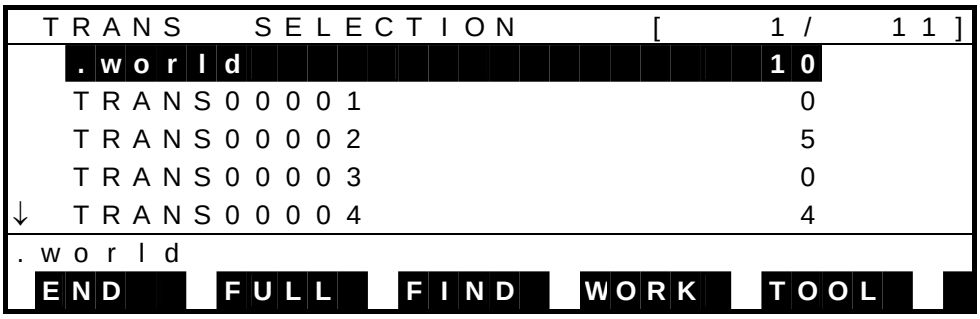
12.5.1 End of Coordinate Selection [END]

(1) Function

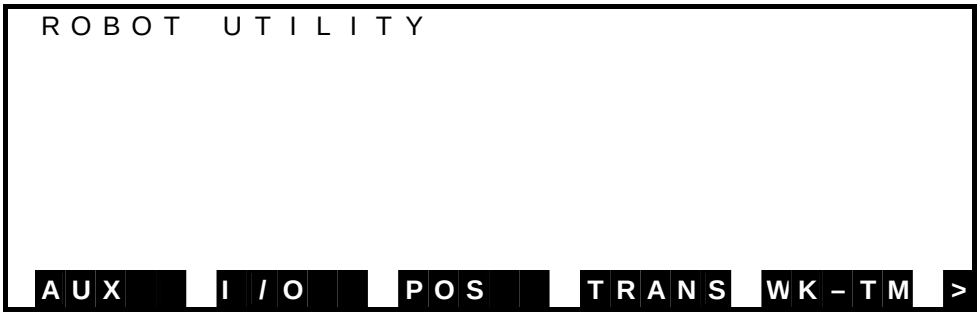
Quits the coordinate selection and return the system to the utility display.

(2) Procedures

Assume that the following coordinate data directory display appears now.



1. Press [END], and the following utility display is called.



(3) Caution

- You can press the ESC key in lieu of the [END] key. The result is the same.

12.5.2 Coordinate Data Detailed Display [FULL]

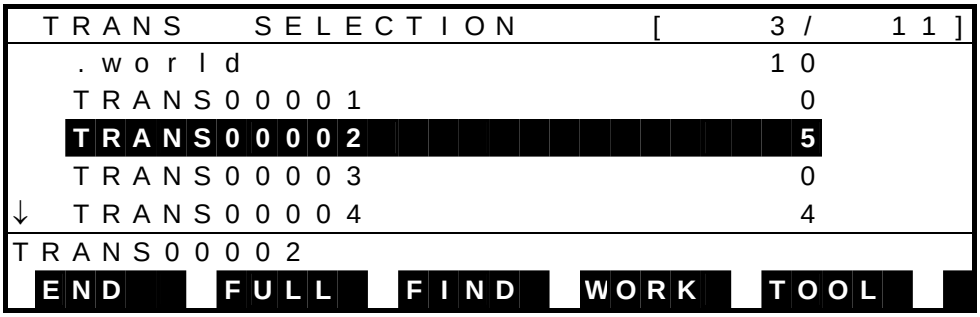
(1) Function

Calls the coordinate data detailed display.

(2) Procedures

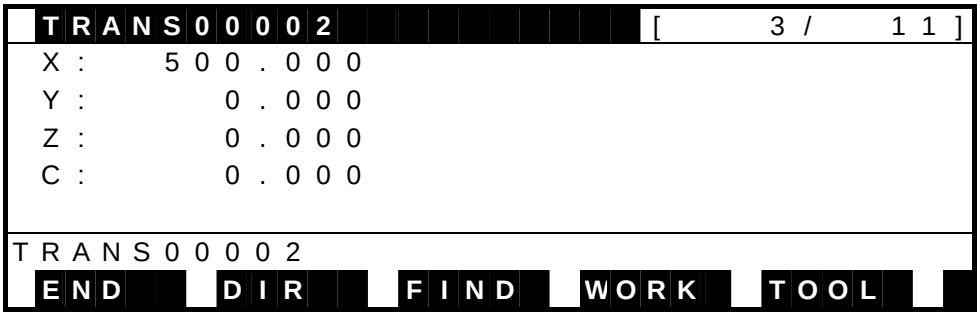
Assume that you are going to display "TRANS00002" in detail.

1. Move the main cursor to "TRANS00002", and the following display appears.



- For the data of [nnnn/mmmm] displayed on the 1st line, "nnnn" signifies the number of positions contained in the selected coordinate system. "mmmm" represents the total number of position data in all coordinate systems.

2. Press [FULL], and the following detailed display is called.



Operations of respective soft keys are shown below.

Key	Descriptions
[END]	Used to quit the coordinate selection and return the system to the utility display.
[DIR]	Used to call the coordinate data directory display.
[WORK]	Used to specify the work coordinate system for a coordinate system indicated by the main cursor, and cancel the designation.
[TOOL]	Used to specify the tool coordinate system for a coordinate system indicated by the main cursor, and cancel the designation.
[FIND]	Used to search a coordinate name.

(3) Caution

- Detailed display of world coordinates is not possible.
- Processing of the [END], [WORK], [TOOL] and [FIND] soft keys is the same as in the coordinate data directory display. For details, see the descriptions on the coordinate data directory display.

12.5.3 Setting of Work Coordinate System [WORK]

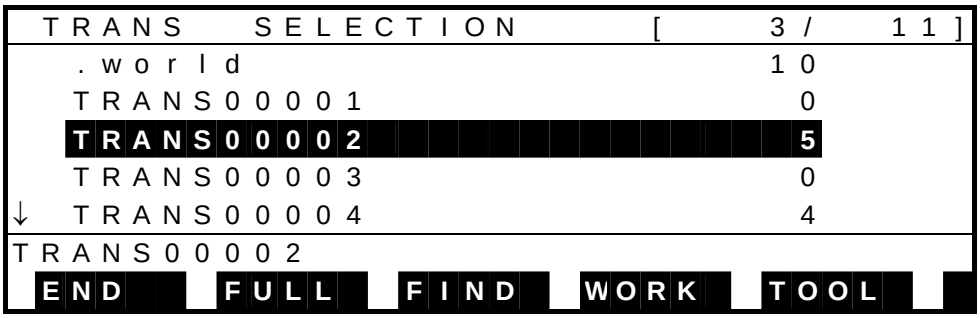
(1) Function

Sets a specified coordinate in the work coordinate system.
If this function is executed for a coordinate already set in the work coordinate system, the work coordinate system is cancelled.

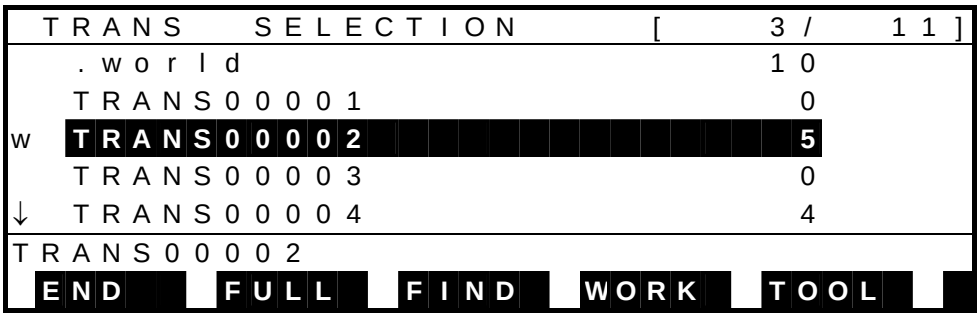
(2) Procedures

Assume that you are going to set "TRANS00002" in the work coordinate system.

1. Move the main cursor to "TRANS00002", and the following display appears.



2. Press [WORK], and the data of "TRANS00002" is set in the work coordinate system. Alphabet "w" is indicated in the 1st column, which signifies that the coordinate is set in the work coordinate system.



(3) Caution

- If this operation is done for the world coordinate system, setting of the work coordinate system is cancelled.
- If this operation is done for a coordinate already set in the work coordinate system, the work coordinate is cancelled.
- If another coordinate was already set as the work coordinate, it is cancelled by this operation.
- A coordinate set as the work coordinate is identified by "w" in the 1st column on the coordinate data directory display.

12.5.4 Setting and Cancel of Tool Coordinate System [TOOL]

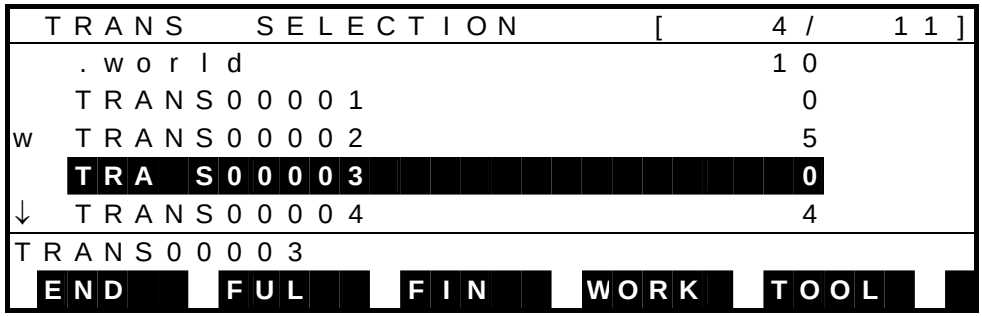
(1) Function

Sets a specified coordinate in the tool coordinate system. If this operation is done for a coordinate already set as the tool coordinate, setting of the tool coordinate system is cancelled.

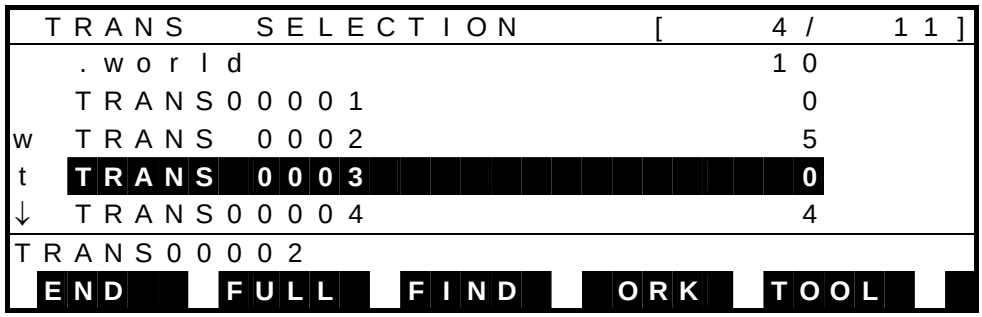
(2) Procedures

Assume that you are going to set "TRANS00003" in the tool coordinate system.

1. Move the main cursor to "TRANS00003", and the following display appears.



2. Press [TOOL], and the data of "TRANS00003" is set in the tool coordinate system and the following display appears.



(3) Caution

- If this operation is done for the world coordinate system, setting of the tool coordinate system is cancelled.
- If this operation is done for a coordinate already set in the tool coordinate system, the tool coordinate is cancelled.
- If another coordinate was already set as the tool coordinate, it is cancelled by this function.
- A coordinate set as the tool coordinate is identified by "t" in the 1st column of the display.

12.5.5 Search of Coordinate Data [FIND]

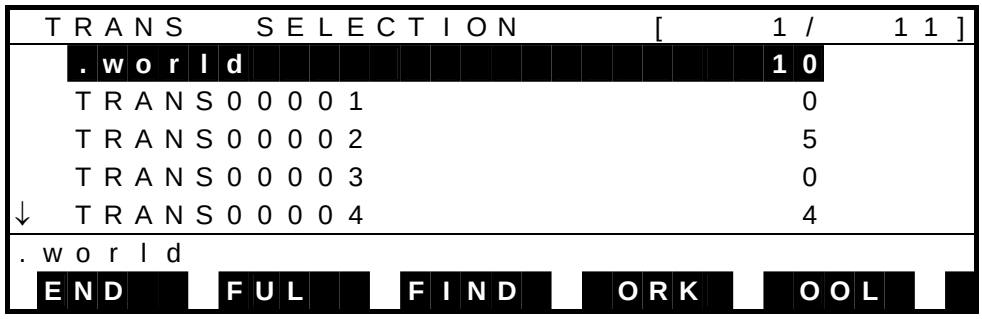
(1) Function

Searches the data bearing the specified coordinate name.

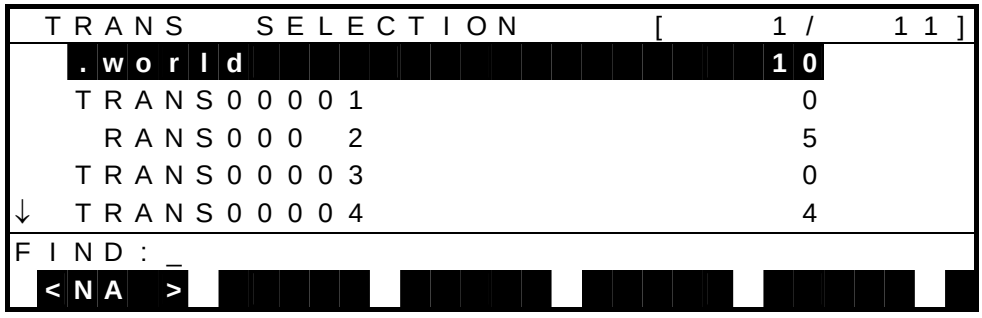
(2) Procedures

Assume that you are going to search the coordinate data named "TRANS00008."

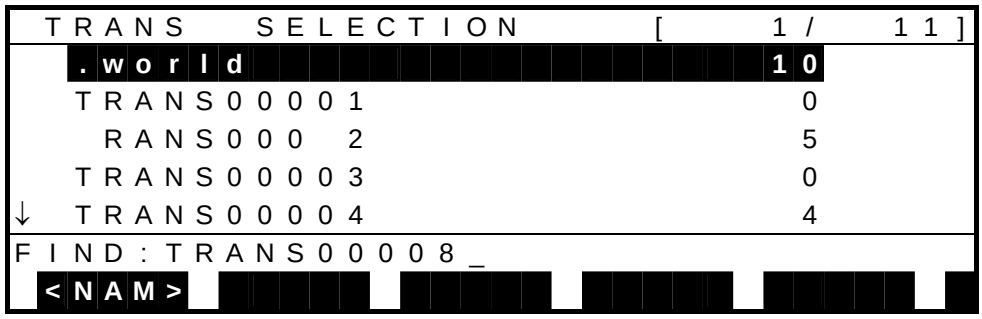
1. Assume that the current display is as shown below.



2. Press [FIND], and the following display appears.



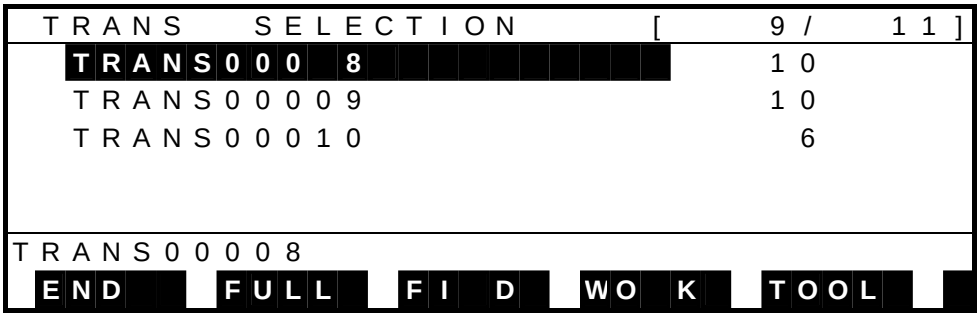
3. Enter "TRANS 00008", and the display is changed over as shown below.



Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[←]	Used to move left the cursor on the command line.
[→]	Used to move right the cursor on the command line.
[Numeral] [Alphabets]	Used to input characters to the command line.
[EXE]	Used to search the specified coordinate name and return the system to the coordinate data directory display. Unless the specified data is present, a message saying "Not found" is displayed on the command line.
[INS]	Used to select the insert or replace mode for the command line.
[BS]	Used to delete one (1) character just before the cursor position on the command line.
[ALT]+[BS]	Used to delete one (1) character at the cursor position on the command line.
[ESC]	This function works in a different manner according to the presence or absence of input character(s). Presence of input character(s): All input character(s) on the command line are cleared. Absence of input character(s): The system returns to the coordinate data directory display.

4. Press the EXE key, and search of the specified coordinate name starts. If found, the coordinate data directory is displayed with the specified data at the head of the screen and the main cursor moved there. The display is as shown below.



(3) Caution

- If a data you wish to search exists on the current display, the main cursor moves there. Under this condition, the main cursor will not move in such a manner that the coordinate data directory is displayed with the specified data at the head of the screen.
- Unless the specified name exists, the system returns to the previous directory display with a message of "Not found" shown on the command line.
- As the search is performed throughout the file, the search start position is unrelated to the search operation.

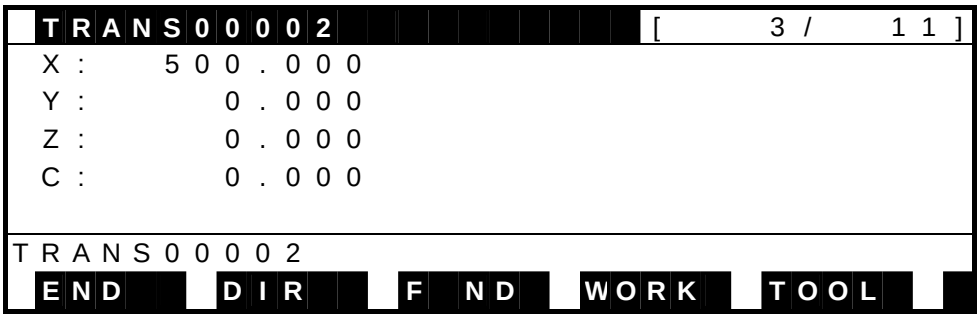
12.5.6 Coordinate Data Directory Display [DIR]

(1) Function

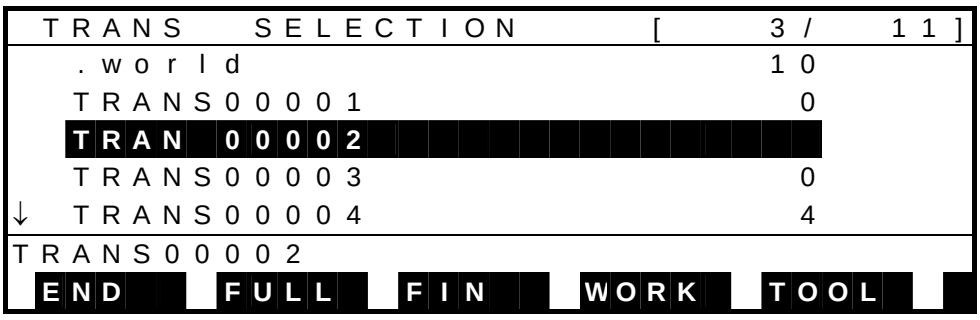
Calls the coordinate data directory display.

(2) Procedures

Assume that the data of "TRANS00002" is displayed in detail.



1. Press [DIR], and the following directory display appears.



(3) Caution

- The main cursor position on the directory display is the position of the coordinate which was displayed in detail.

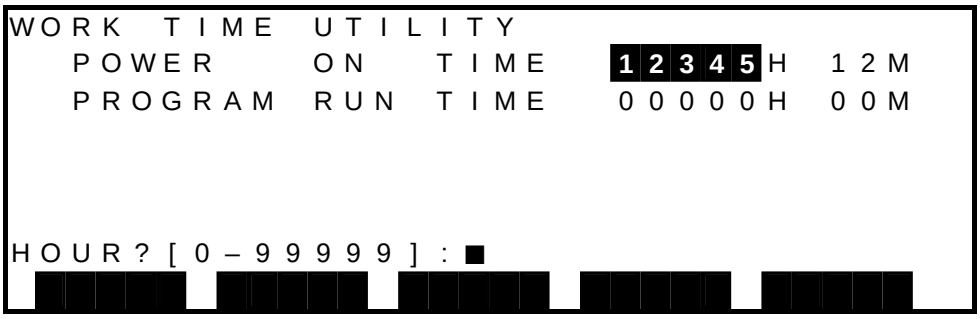
12.6 Working Time Display [WK–TM]

(1) Function

Displays the robot working time.

(2) Procedures

1. Press the NEXT key repeatedly until [WK–TM] is displayed in the soft key menu. Press the corresponding function key, and the following working time display appears.

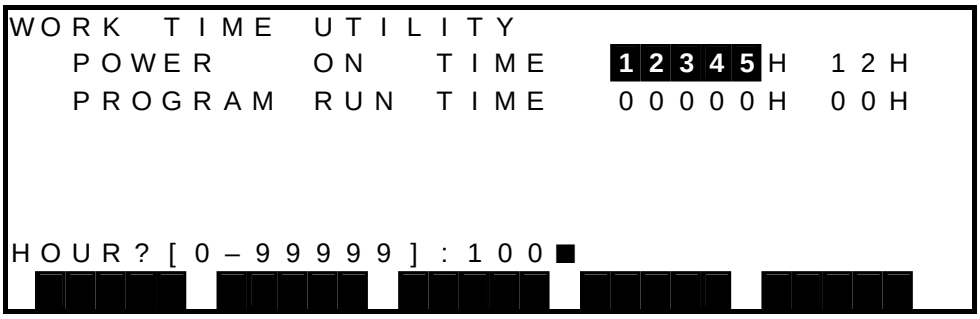


- "POWER ON TIME" displays the total power ON time in hours and minutes (H M).
- "PROGRAM RUN TIME" indicates the total program run time in hours and minutes (H M).
- Each time can be changed in this mode.

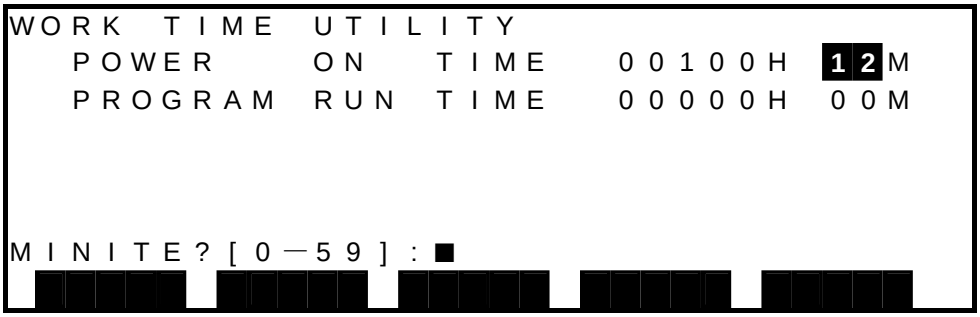
Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[↑]	Used to move up the cursor. The command line is cleared.
[↓]	Used to move down the cursor. The command line is cleared.
[←]	Used to move the cursor to the left. The command line is cleared.
[→]	Used to move the cursor to the right. The command line is cleared.
[Numeral]	Used to input characters to the command line.
[EXE]	Used to change the data indicated by the cursor to the input value.
[BS]	Used to delete one (1) character just before the cursor position on the command line.
[ALT]+[BS]	Used to delete one (1) character at the cursor position on the command line.
[ESC]	Used to return the system to the utility display.

2. Move the main cursor to a data you wish to change, and enter a new data. Character input to the command line is always effective.



3. Press the EXE key, and the value indicated by the main cursor is replaced with the new value you entered. The display is as shown below.



(3) Caution

- The input range of each data is as follows:
  Hour : 0 ~ 99999
  Minute : 0 ~ 59

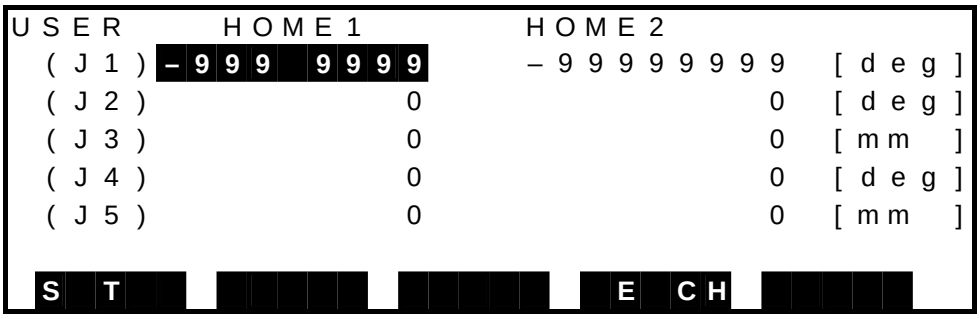
12.7 Reproducing the Origin [REORG]

(1) Function
Reproduces the origin.

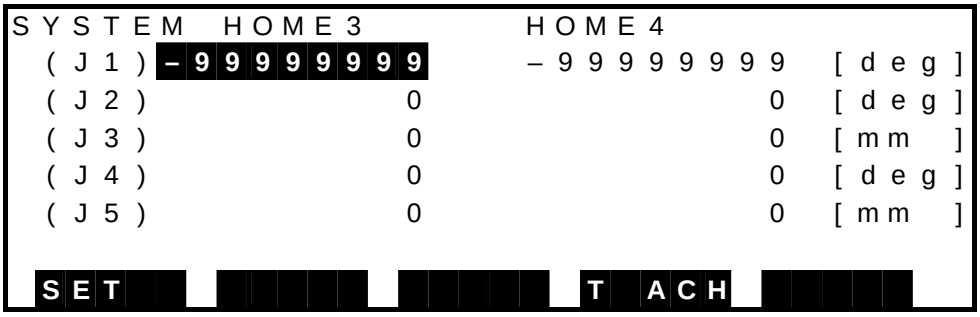
(2) Procedures

1. Set the servo OFF.
Press the NEXT key repeatedly until [REORG] is displayed in the soft key menu. Press the corresponding function key, and the following origin reproduction display appears.

[Page 1]



[Page 2]



- The origin coordinate of each axis is displayed.
- The origin coordinate as specified by the ZEROP function can be set on the display, using the TEACH function.
- The SET function works in the same manner as the ZEROP function.

- This function is intended for reproducing the machine origin. DO NOT use it at other than origin reproduction.

(3) Caution

- For details of the display contents and setting, see the separate manual.

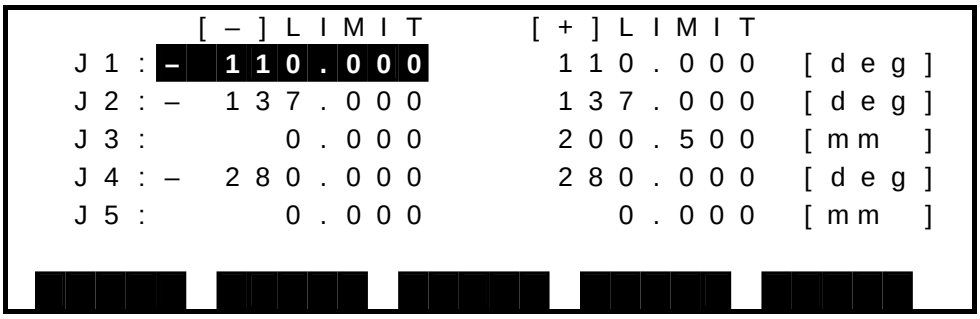
12.8 Joint Limit Setting [J–LIM]

(1) Function

Displays joint limit set values and sets a joint limit value as necessary.

(2) Procedures

1. Press the NEXT key repeatedly until [J–LMT] is displayed in the soft key menu. Press the corresponding function key, and the following joint limit set value display appears.

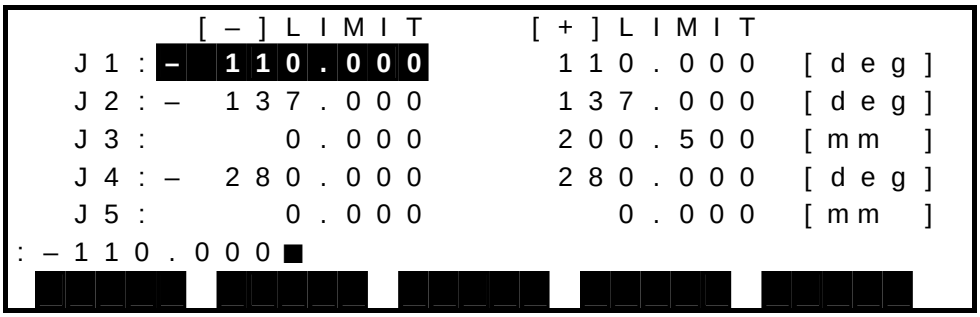


- Both minus (–) and plus (+) limit coordinates of each joint are displayed.

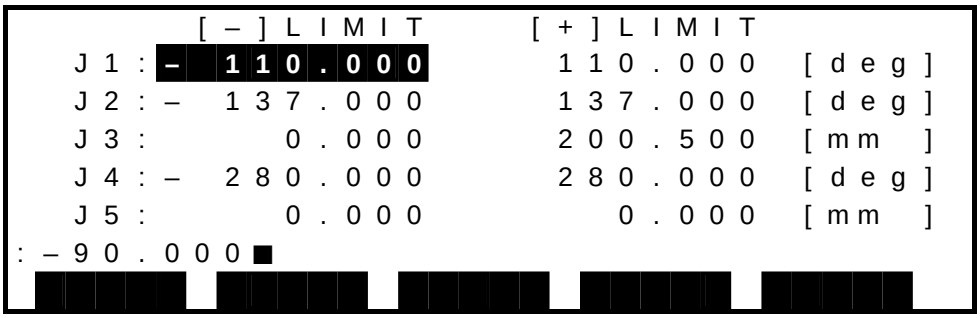
Keys operative in this mode and their operations are tabled below.

Key	Descriptions
↑	Used to move up the main cursor. If it is located at the top, the cursor will not move at all.
↓	Used to move down the main cursor. If it is located at the bottom, the cursor will not move at all.
←	Used to move the main cursor to the left. If it is located on the right end, the cursor will not move at all.
→	Used to move the cursor to the right. The command line is cleared.
EXE	Used to change the data indicated by the main cursor.
ESC	Used to return the system to the utility display.

2. Move the main cursor to a data you with to change, using the cursor keys (→, ←, ↑, ↓), and press the EXE key. Now, the current data is displayed on the command line. The display is changed over as shown below.



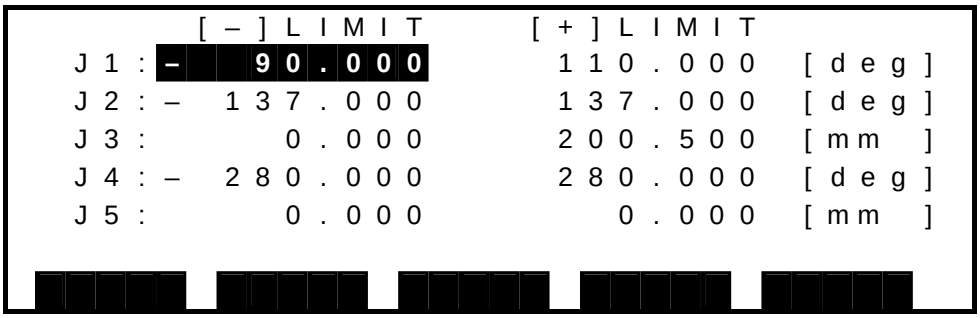
3. Enter the data to be changed, and the following display appears.



Keys operative in this mode and their operations are tabled below.

Key	Descriptions
Numeral . + –	Used to input characters to the command line.
EXE	Used to change a data indicated by the cursor to an entered one. The system returns to the joint limit display.
INS	Used to select the insert or replace mode for the command line.
BS	Used to delete one (1) character just before the cursor position on the command line.
ALT + BS	Used to delete one (1) character at the cursor position on the command line.
ESC	This function works in a different manner according to the presence or absence of input character(s). Presence of input character(s): All input character(s) on the command line are cleared. Absence of input character(s): The system returns to the joint limit display.

4. Press the EXE key, and the data at the cursor position is changed. The display is changed over as shown below.



(3) Caution

- The rotary joint is input in units of degree (°) and the linear joint in units of mm.
- These limit values are used to limit the automatic operation range and teaching range of the robot for each joint. They are factory-set to the maximum working range for each axis.
- Each joint set values are registered in the user parameter file (file name: USER. PAR). If any set value has been changed, relevant data in the user parameter file in the RAM drive is also updated automatically.

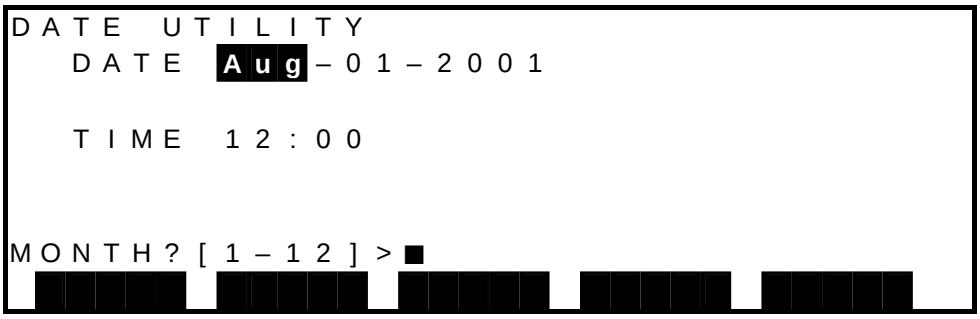
12.9 Date and Time Setting [DATE]

(1) Function

Displays and sets current date and time.

(2) Procedures

1. Press the NEXT key repeatedly until [DATE] is displayed in the soft key menu. Press the corresponding function key, and the following date and time setting display appears.

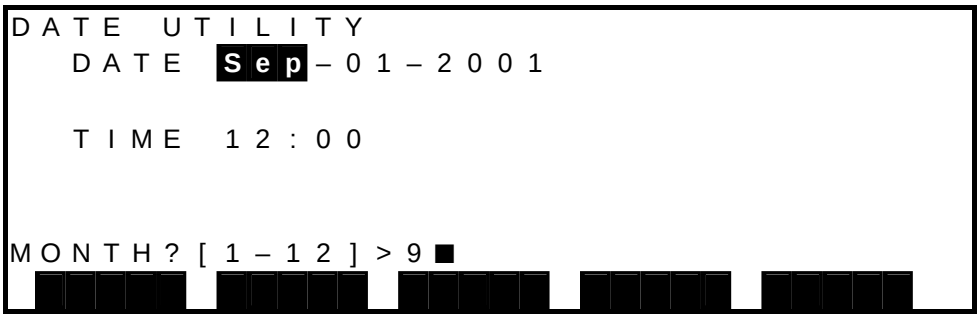


- "DATE" displays the current date by "month–day–year."
- "TIME" indicates the current time in hours and minutes.

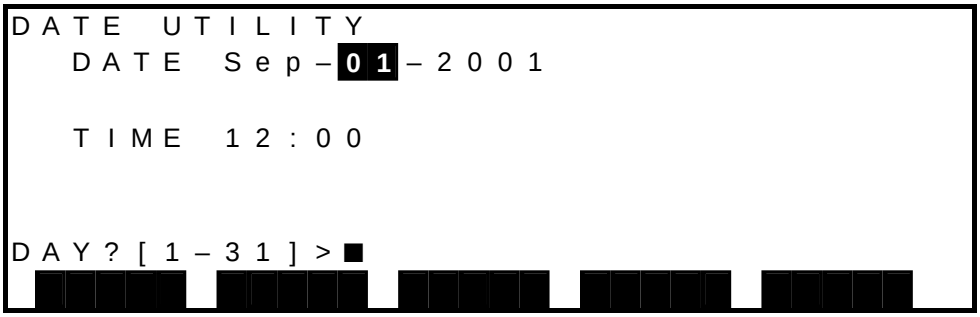
Keys operative in this mode and their operations are tabled below.

Key	Descriptions
[↑]	Used to move up the main cursor. The command line is cleared.
[↓]	Used to move down the main cursor. The command line is cleared.
[←]	Used to move the main cursor to the left. The command line is cleared.
[→]	Used to move the main cursor to the right. The command line is cleared.
[Numeral]	Used to input characters to the command line.
[EXE]	Used to change the data indicated by the main cursor to the input value.
[BS]	Used to delete one (1) character just before the main cursor position on the command line.
[ALT]+[BS]	Used to delete one (1) character at the main cursor position on the command line.
[ESC]	Used to return the system to the utility display.

2. Move the main cursor to a data you wish to change, and enter a new data. Character input to the command line is always effective.



3. Press the EXE key, and the value indicated by the main cursor is replaced with the new value you entered. The display is as shown below.



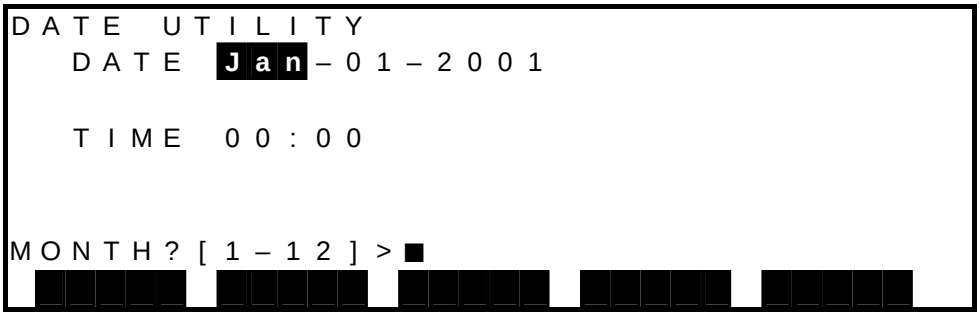
(3) Caution

- As the controller has a built-in calendar which is backed up by battery, setting is usually unnecessary. Set the data again only when the date or time is incorrect.

- The input range of each data is as follows:

  Year : 0 ~ 99 (2000 ~ 2099)
  Month : 1 ~ 12
  Day : 1 ~ 31
  Hour : 0 ~ 23
  Minute : 0 ~ 59

- If only an asterisk (*) mark is input as the data, the display is initialized as shown below. Use this mark if the calendar display is illegal in the initial state.



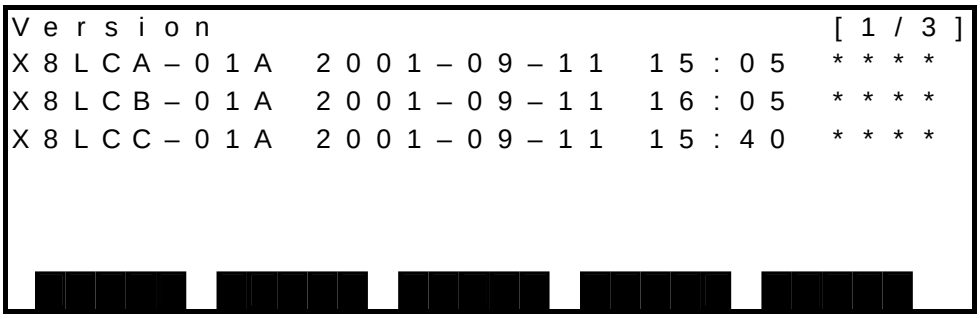
12.10 Version Display [VER]

(1) Function

Displays the system version.

(2) Procedures

1. Press the NEXT key repeatedly until [VER] is displayed in the soft key menu. Press the corresponding function key, and the following version display appears.



- The data is displayed in the order of the system version name, date of creation and checksum value from the left.
- The date of creation is shown in the order of "year, month, day, hour and minute."
- A checksum value is expressed in the hexadecimal notation.
- Pages 1 and 2 display the system information, and page 3 includes other information.

Keys operative in this mode and their operations are tabled below.

Key	Descriptions
ALT + ↓	Used to display the next page. If the cursor is located at the final page, the page will not change over.
ALT + ↑	Used to display the previous page. If the cursor is located at the top page, the page will not change over.
ESC	Used to return the system to the utility display.

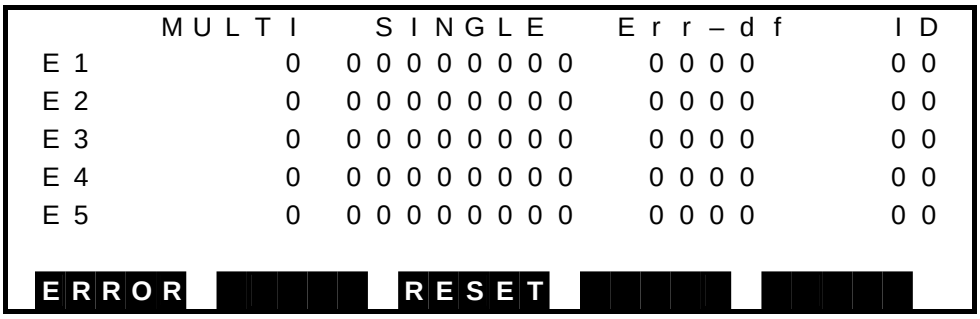
12.11 Servo Encoder Data Display [END]

(1) Function

Displays the servo encoder data.

(2) Procedures

1. Press the NEXT key repeatedly until [ENC] is displayed in the soft key menu. Press the corresponding function key, and the following encoder data monitor display appears.



- Each axis encoder data is displayed.
- This function is intended for the maintenance only. To make valid any change in the data, therefore, subsequent input of combined keys is necessary.

The key operative in this mode and its operation are tabled below.

Key	Descriptions
ESC	Used to return the system to the utility display.

(3) Caution

- For details of the display contents, see the separate manual.

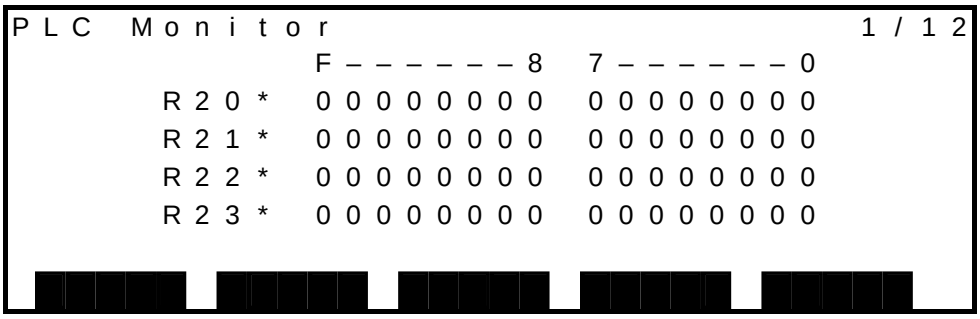
12.12 PLC Monitor [PLC–M]

(1) Function

Displays the PLC data.

(2) Procedures

1. Press the NEXT key repeatedly until [PLC–M] is displayed in the soft key menu. Press the corresponding function key, and the following PLC monitor display appears.



- Each contact data of the PLC is displayed.

Keys operative in this mode and their operations are tabled below.

Key	Descriptions
ALT + ↓	Used to display the next page. If the cursor is located at the final page, the page will not change over.
ALT + ↑	Used to display the previous page. If the cursor is located at the top page, the page will not change over.
ESC	Used to return the system to the utility display.

(3) Caution

- For details of the display contents, see the separate manual.

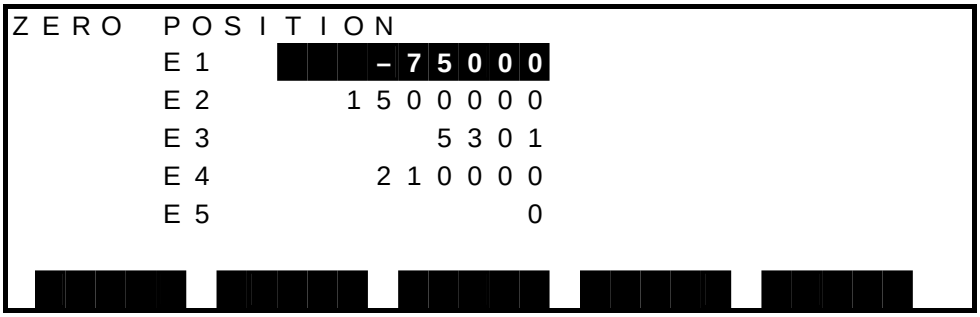
12.13 Zero Position Setting [ZEROP]

(1) Function

Sets the zero position (or origin).

(2) Procedures

1. Set the servo OFF.
   Press the NEXT key repeatedly until [ZEROP] is displayed in the soft key menu. Press the corresponding function key, and the following zero position setting display appears.



- Zero position of each axis encoder is displayed.
- This function is intended for the maintenance only. To make valid any change in the data, therefore, subsequent input of combined keys is necessary.
- This function is used to establish the machine zero position (or origin). If the operation is done incorrectly, the zero position will shift, causing a very dangerous situation. To avoid this, take careful precautions on the use of this function.

(3) Caution

- For details of the display contents, see the separate manual.

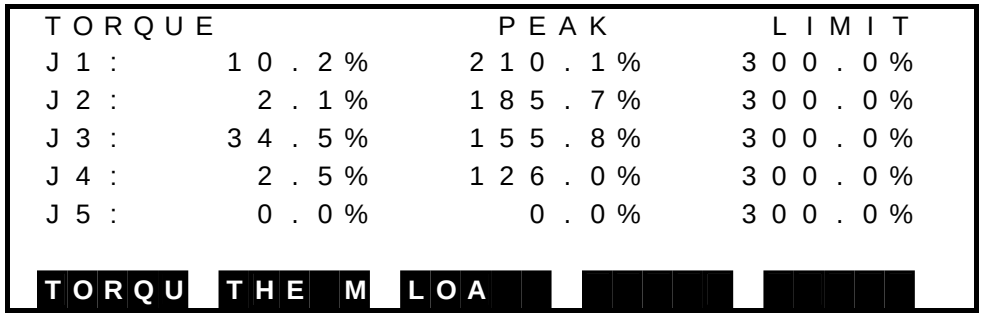
12.14 Motor Load Display [MOTOR]

(1) Function

Displays the torque, electronic thermal total value and load factor of each servo motor.

(2) Procedures

1. Press the NEXT key repeatedly until [MOTOR] is displayed in the soft key menu. Press the corresponding function key, and the following PLC monitor display appears.



- Each contact data of the PLC is displayed.

The key operative in this mode and its operation are tabled below.

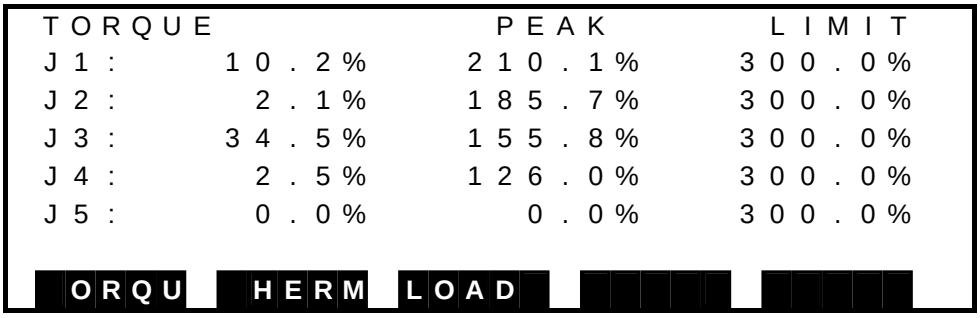
Key	Descriptions
ESC	Used to return the system to the utility display.

Operations of respective soft keys are shown below.

Soft key	Descriptions
[TORQU]	Used to display the torque value of each motor.
[THERM]	Used to display the electronic thermal total value of each motor.
[LOAD]	Used to display the load factor of each motor.

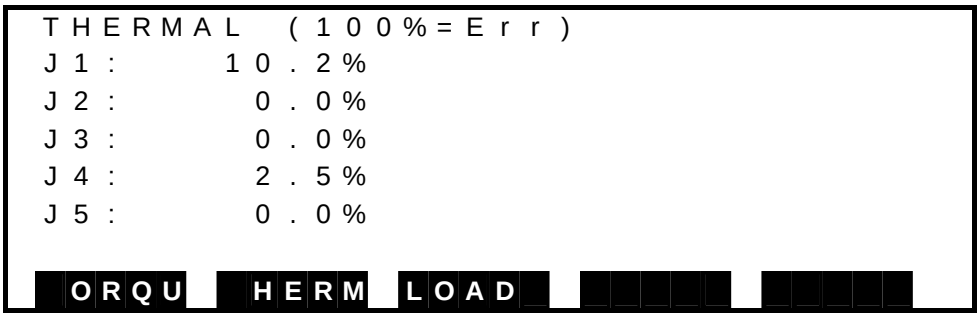
2. Press the soft key for the coordinates you wish to display. Then the following position display of each coordinate system appears.

Torque display



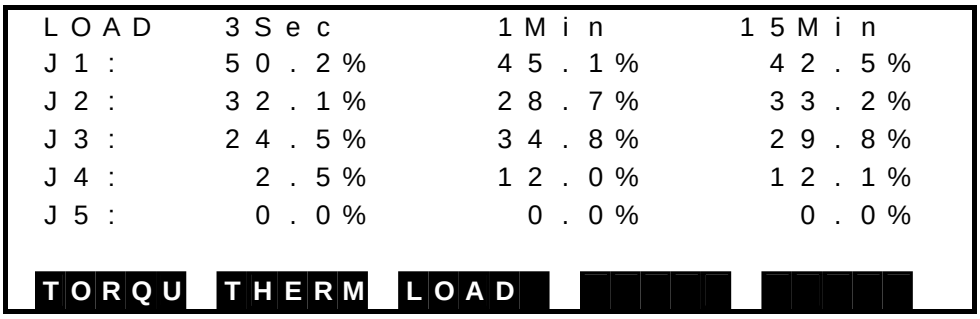
- The current motor torque of each axis is displayed. (100 % = Rated torque)
- In the PEAK column, the maximum value from the power ON to the current time is displayed.
- Under the LIMIT column, the ratio of the maximum torque to the rated torque is displayed.

Electronic thermal total value display



- The current total value of the electronic thermal is displayed.
- When the value reaches 100 %, the alarm of "Axis* Electric Thermal" occurs.

Load factor display



- The current motor factor of each axis is displayed.
  (100 % = Rated load)
- The displayed values are the running averages for the duration of the past three (3) seconds, one (1) minute and fifteen (15) minutes.
- When the load factor under the column of "1Min" or "15Min" reaches the specified value, the alarm saying "Axis* Motor overload" generates.

Section 13 Self-Diagnosis

13.1 Classification of Errors

Errors detected by the self-diagnosis function are grouped and processed as described in Para. 13.8.

13.2 Display at Error Generation

If an error has occurred, the buzzer sounds intermittently and the relevant error number flickers on the display when "ALARM" is selected beforehand by means of the SELECT switch on the control panel.
If the error level is 2, 4 or 8, the "fault" external control signal turns on.
At the time of emergency stop, the buzzer will not sound and the "fault" signal will not turn on.
To reset the alarm buzzer, press the STOP pushbutton switch on the control panel, or execute the reset operation stated in Para. 13.4.

13.3 Error Display

(1) Function

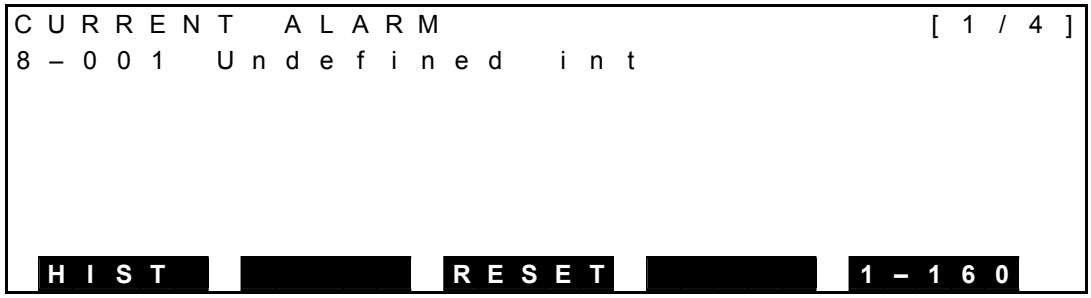
Displays the code number and description of an error which is detected currently or which is yet to be reset.

(2) Procedures

(a) Step 1: Input of error display.

Press the ERROR key equipped on the teach pendant, and the following display appears.

[Error message display]

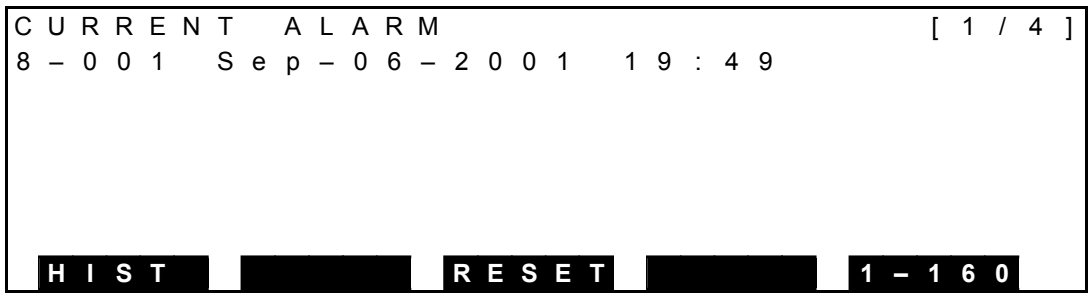


(b) Step 2: Error display.

Keys operative in this mode and their operations are tabled below.

Key	Descriptions
ALT + ↑	Used to display the previous page. (If errors which have not yet been reset are covered on two or more pages.)
ALT + ↓	Used to display the next page. (If errors which have not yet been reset are covered on two or more pages.)
←	Used t display an error message.
→	Used to display the time of error generation.
ESC	Used to return the system to the previous display.

[Error time display]



(3) Caution

- When the ESC key is pressed, the system quits the error display and returns to the previous display. Error display is possible, irrespective of the mode currently selected.

13.4 Error Reset

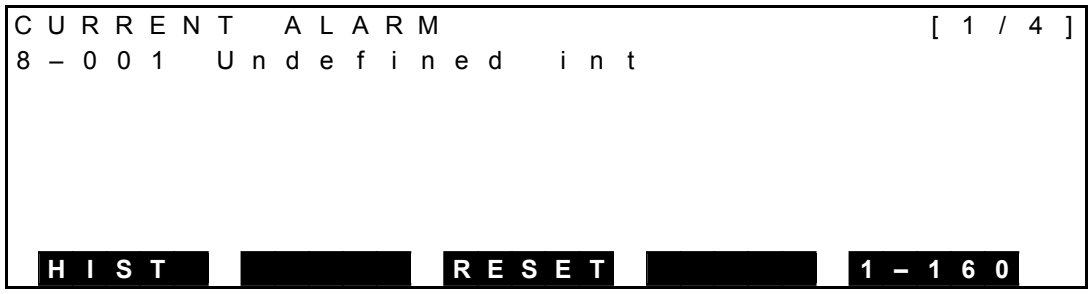
(1) Function

Resets the error display, "fault" signal and buzzer after the error has been corrected.

(2) Procedures

(a) Step 1: Input of error display.

Press the ERROR key equipped on the teach pendant, and following display appears.



(b) Step 2: Input of reset.

Press the function key corresponding to the [RESET] menu. After the error has been reset, another error, if any, is displayed in its place.

(3) Caution

- The soft limit and emergency stop are automatically reset after a relevant error has been corrected. Unless the error cause is remedied, the error is not reset.

13.5 All Errors Reset

(1) Function
Resets all errors whose causes have been cleared.

(2) Procedures

(a) Select "ALARM" by pressing the SELECT switch provided on the control panel.

(b) Press the ALARM RESET switch on the control panel, and the buzzer sounding to inform an operator of error generation stops, and all errors whose causes are already cleared are reset.

(c) When the "ALARM_RST" signal is set ON in the external signal mode, the same function as (b) above can be realized.

(3) Caution
This function can be executed, irrespective of presence or absence of the teach pendant.

13.6 Error History Display

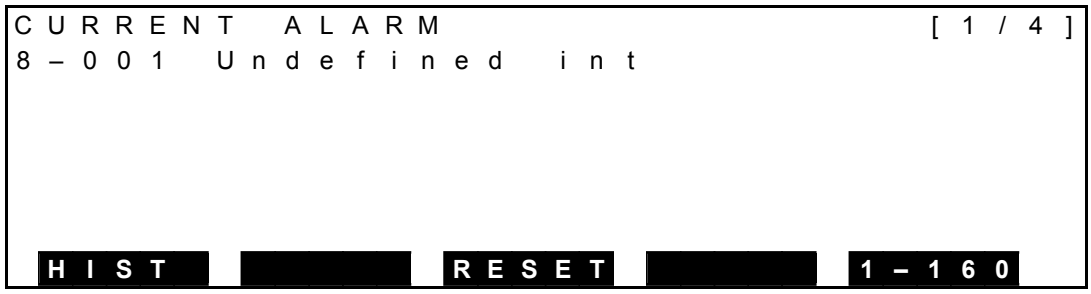
(1) Function

Displays the record number, code number and time of error generation in the order in which errors occurred.

(2) Procedures

(a) Step 1: Input of error display.

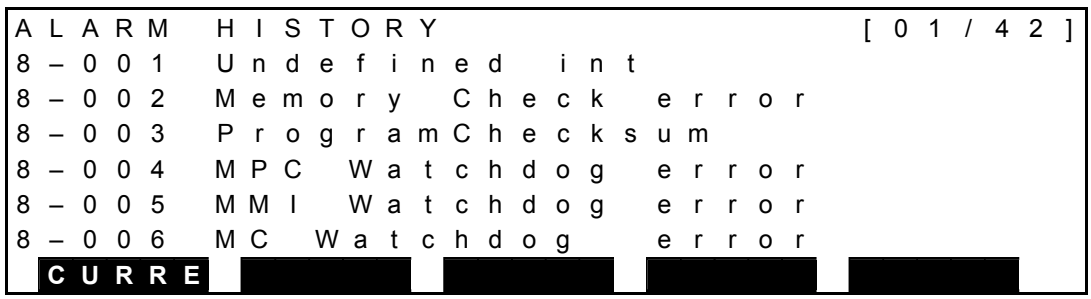
Press the ERROR key on the teach pendant, and the following display appears.



(b) Step 2: Input of history.

Press the function key corresponding to the [HIST] menu, and the following error history display appears.

[Error message display]

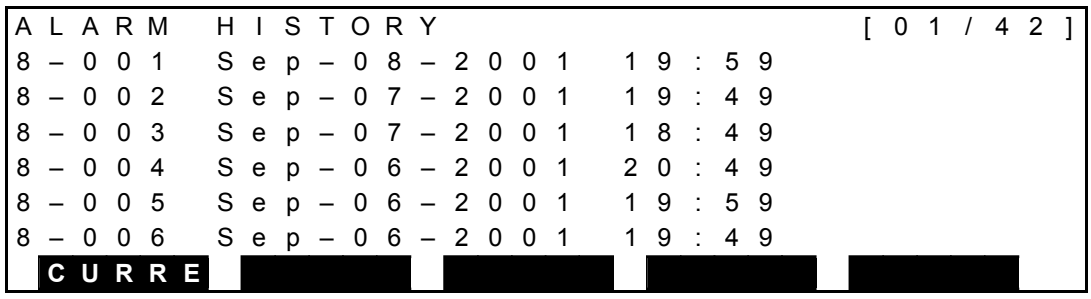


(c) Step 3: Display of error history.

Keys operative in this mode and their operations are tabled below.

Key	Descriptions
ALT + ↑	Used to display the previous page. (If errors which have not yet been reset are covered on two or more pages.)
ALT + ↓	Used to display the next page. (If errors which have not yet been reset are covered on two or more pages.)
←	Used t display an error message.
→	Used to display the time of error generation.
ESC	Used to return the system to the previous display.

[Error time display]



(d) Step 4: Error display.
Press the function key corresponding to the [CURRE] menu, and the display of error or errors currently generated is shown.

(3) Caution

- A maximum of 128 errors can be recorded. If the number of errors saved exceeds 128, old errors are overwritten in order by the new ones.

13.7 Compile Error Display

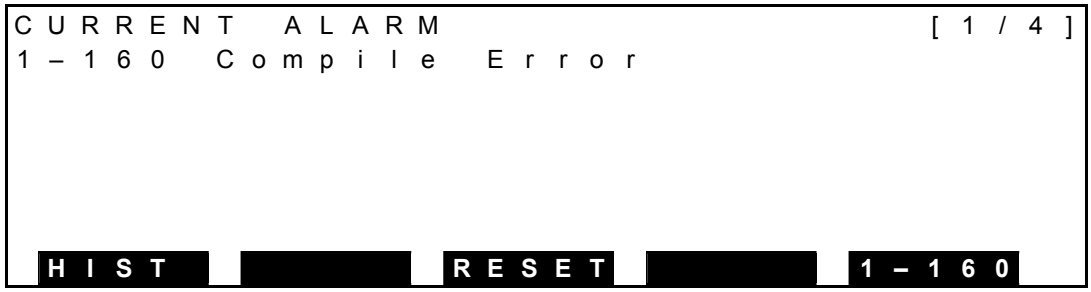
(1) Function

Displays the details of a compile error and the line in which the error occurred.

(2) Procedures

(a) Step 1: Input of error display.

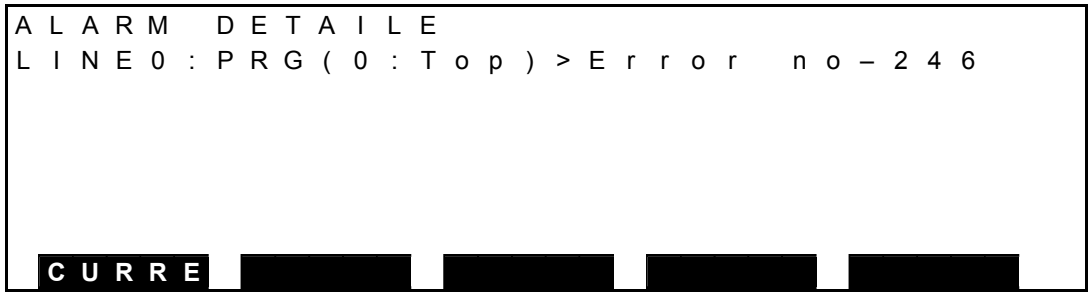
Press the ERROR key on the teach pendant, and the following display appears.



(b) Step 2: Input of compile error detailed display.

Press the function key corresponding to the [1–160] menu, compile error detailed display menu, and the compiled error message detailed display is called, as shown below.

[Compiled error message detailed display]



(c) Step 3: Error display.

Press the function key corresponding to the [CURRE] menu, and the display of error or errors currently generated is shown.

13.8 Error Code Table

Alarms of level 8:

An alarm whose number starts with 8. If this alarm has been detected, the robot emergency-stops.

Error code	Descriptions	
8–001	Undefined int	
	Possible cause:	Abnormal interruption has been done in the main control board.
	Remedy:	Call us at the after-sale service department or our after-sale service agent in your territory.
8–002	Memory Check error	
	Possible cause:	Abnormal interruption has been done in the main control board.
	Remedy:	Call us at the after-sale service department or our after-sale service agent in your territory.
8–004	Watchdog error	
	Possible cause:	An error has been detected in the CPU regular monitor circuit of the main circuit board.
	Remedy:	Call us at the after-sale service department or our after-sale service agent in your territory.
8–005	PLC CPU error	
	Possible cause:	A CPU error has been detected in the PLC unit of the main control circuit.
	Remedy:	Call us at the after-sale service department or our after-sale service agent in your territory.
8–006	COM CPU error	
	Possible cause:	A CPU error has been detected in the communication unit of the main control circuit.
	Remedy:	Call us at the after-sale service department or our after-sale service agent in your territory.

Error code	Descriptions
8–007	Axis 1 Servo CPU error
8–008	Axis 2 Servo CPU error
8–009	Axis 3 Servo CPU error
8–010	Axis 4 Servo CPU error
8–011	Axis 5 Servo CPU error
8–012	Axis 6 Servo CPU error Possible cause: The main control board has detected a CPU error in the servo printed board at mutual monitoring of the main control board and servo printed board (i.e., monitoring by software). Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
8–013	AC Alarm Possible cause: A power error has been detected in the AC power monitor circuit. The following conditions are monitored in the servo power unit. • When the input power has been cut off for more than 47 msec. • When the input power voltage has dropped to 120 V or less. Remedy: Check the controller power supply for presence or absence of instantaneous power failure and voltage, then turn the power off and on again.
8–014	Emergency Stop SW ON Possible cause: The EMERGENCY stop pushbutton switch has been pressed. Remedy: Check for the EMERGENCY stop switch. This switch is provided on the controller and teach pendant, and the "emergency stop" external control signal is also provided. Unless the switch has been pressed, make sure that the EMERGENCY stop switch is not defective and that the wiring is correct.

Error code	Descriptions
8–015	Parameter error Possible cause: The set data of the parameter file is illegal. Remedy: Re-load the data from the parameter disk attached to the controller. If the error still occurs, call us at the after-sale service department or our after-sale service agent in your territory.
8–016	Servo Type error Possible cause: The type of the servo (digital or analog) set in the parameter differs from the type of the servo printed board actually incorporated. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
8–018 8–019 8–020 8–021 8–022 8–023	Axis 1 Servo CPU error Axis 2 Servo CPU error Axis 3 Servo CPU error Axis 4 Servo CPU error Axis 5 Servo CPU error Axis 6 Servo CPU error Possible cause: The main control board has detected a CPU error of the servo printed board. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
8–026	Servo Gate OFF error Possible cause: The servo has turned off unexpectedly. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.

Error code	Descriptions
8–027	<u>Slow Charge error</u> Possible cause: The servo could not turn on due to abnormal operation of the condenser charge at servo ON. Contents: This error occurs unless the servo power voltage has reached the predetermined voltage after lapse of the specified time at servo ON. Remedy: • Make sure that the supply power voltage satisfies the controller input power specified and is stabilized. The specified power voltage are follwings. TS1000: AC170-250V , single phase 50/60 Hz ±1% TS2000: AC180-250V , single phase 50/60 Hz ±1% • If the voltage at the power connector is low, compared with the power voltage of the facility, there is a possibility that the power cable is too thin or too long. (Thickness of power cable conductor: 3.5 ~ 5.5 mm^2) • The servo power unit has possibly malfunctioned. Call us at the after-sale service department or our after-sale service agent in your territory.
8–028	<u>IPM error</u> Possible cause: The motor drive transistor is abnormal. Remedy: This error occurs when error 8–073 (Servo Amp error) has generated. For details, see the descriptions on Error 8–073.
8–029	<u>DC24 Regulator error</u> (only for TS2000) Possible cause: The DC24 V power voltage in the controller has dropped to 21 V or less. Remedy: Make sure that the DC24 V circuit protector on the controller front side has not been cut off. The 24 V power is also used for the external I/O. Short-circuited external wiring will cause voltage drop. Make sure that the external wiring is not short-circuited. Also, output failure of the switching power may be the cause.

Error code	Descriptions
8–030	Servo Power Volt Low Possible cause: The servo power in the controller (approx. DC300 V) has dropped to 130 V or less. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
8–031	Volt Overheat (only for TS2000) Possible cause: The temperature of the cement resistor for overvoltage absorption exceeds 150°C. Remedy: Turn the power off, and leave the cement resistor as it is until its temperature drops. The following factors are considered for overheat of the resistor. • The robot is overloaded. → Confirm the weight of "hand + workpiece". • Sharp acceleration/deceleration. → Regulate the acceleration/deceleration by the PAYLOAD, ACCEL or DECEL command. • The robot has collided with an obstacle. → Remove the cause of the collision. • The robot drive system has caused the trouble. → Move the robot by hand and check to see if there is a problem.
8–032	PW Over Volt Possible cause: The servo power in the controller (approx. DC300 V) exceeds 440 V. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
8–033	Distribution error Possible cause: As a result of the arithmetic operation of the motion command (such as MOVE), the robot cannot be moved any further. This alarm will not occur due to an error in the SCOL program. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.

Error code	Descriptions
8–034	Unable to Trans Possible cause: As a result of coordinate conversion of the internal operation, the robot cannot be moved any further. This alarm will not occur due to an error in the SCOL program. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
8–035	Point Invalid of Range Possible cause: The robot was about to move to a position where it cannot be moved due to the arm length, by the motion command (such as MOVE). Remedy: Make sure that the locus of MOVES or MOVEC is within the robot working envelope.
8–041 8–042 8–043 8–044 8–045 8–046	Axis1 Exceed Speed Axis2 Exceed Speed Axis3 Exceed Speed Axis4 Exceed Speed Axis5 Exceed Speed Axis6 Exceed Speed Possible cause: As a result of internal operation of the move command, the command value exceeds the limit speed of the joint axis. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.

Error code	Descriptions
8–065 8–097 8–129 8–161 8–193 8–225	Axis1 Encoder abnormal Axis2 Encoder abnormal Axis3 Encoder abnormal Axis4 Encoder abnormal Axis5 Encoder abnormal Axis6 Encoder abnormal Possible cause: [1] Communication with the feedback detector is not possible. [2] An alarm has been detected in the feedback detector. For the alarms detected in the feedback detector, the following conditions are predetermined. • Overspeed The motor has rotated in excess of the specified value during power OFF. • Data error The encoder internal data is abnormal. • Battery error The battery voltage has dropped to the specified value or less. • Thermal error The temperature in the feedback detector has reached 90°C. Remedy: [1] Make sure that the cable and encoder are not damaged and are in the good condition. [2] The thermal error can be cancelled if the encoder internal temperature has dropped. For the overspeed error, data error and battery error, position data saved are not reliable. To cancel, the feedback detector should be reset. There is a possibility that the robot coordinates are shifted. Call us at the after-sale service department or our after-sale service agent in your territory.

Error code	Descriptions
8–065 8–097 8–129 8–161 8–193 8–225	To prevent an error of the feedback detector, observe the following precautions. • While the robot power is turned off, DO NOT move the arm suddenly. • Replace the battery for the feedback detector on a regular basis. • When the robot is stopped and stored for a long term, call us at the after-sale service department or our after-sale service agent in your territory.
8–067 8–099 8–131 8–163 8–195 8–227	Axis1 Position error Axis2 Position error Axis3 Position error Axis4 Position error Axis5 Position error Axis6 Position error Possible cause: The difference between the motor command position and its current position exceeds the predetermined value. Remedy: If this alarm has occurred because the robot could not move due to the external factor such as collision (or interference), remove the cause. If the axis 3 or 4 has collided, carry out the servo-free operation just after the servo ON to retract the axis. If no external factor exists, the motor or servo amplifier may be the cause. When this happens, call us at the after-sale service department or our after-sale service agent in your territory.

Error code	Descriptions
8–070	Axis1 Motor Locked
8–102	Axis2 Motor Locked
8–134	Axis3 Motor Locked
8–166	Axis4 Motor Locked
8–198	Axis5 Motor Locked
8–230	Axis6 Motor Locked
	Possible cause: As the robot collided with an obstacle, the motor will not rotate even if the maximum torque is output. [Conditions for detecting motor lock] The following status has continued for 100 msec or over. • Torque is maximum (300 %). • Speed command is more than 1 % of the maximum speed. • Motor speed is under 1 % of the maximum speed If 299 % or less is specified for the torque, using the TORQUE command, this alarm will not occur.
	Remedy: Investigate into the cause of the collision and correct. If the axis 3 or 4 has collided, carry out the servo-free operation just after the servo ON to retract the axis. If torque control is necessary, use the TORQUE command.

Error code	Descriptions
8–071	Axis1 Motor Speed Over
8–103	Axis2 Motor Speed Over
8–135	Axis3 Motor Speed Over
8–167	Axis4 Motor Speed Over
8–199	Axis5 Motor Speed Over
8–231	Axis6 Motor Speed Over
	Possible cause: The motor speed exceeds the predetermined value. If the torque has fluctuated sharply at the robot interference or by the torque command, the motor speed may exceed the specified value.
	Remedy: Remove the cause of sharp torque fluctuation.
8–072	Axis1 Analog Amp error
8–104	Axis2 Analog Amp error
8–136	Axis3 Analog Amp error
8–168	Axis4 Analog Amp error
8–200	Axis5 Analog Amp error
8–232	Axis6 Analog Amp error
	Possible cause: The analog command amplifier has detected an error.
	Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
	This alarm will occur if an amplifier error has been detected in the system using analog command servo amplifier AD, AB or AL series. Check for the servo amplifier LED.
	GTR GTR overheat
	MOT Motor overheat
	OVR Overvoltage absorption resistor overheat
	TPD Instantaneous power failure
	OVC Overcurrent
	SCERR Slow charge error
	SPUV Servo power voltage drop
	SPOV Servo power overvoltage

Error code	Descriptions
8–073	Axis1 Servo Amp error
8–105	Axis2 Servo Amp error
8–137	Axis3 Servo Amp error
8–169	Axis4 Servo Amp error
8–201	Axis5 Servo Amp error
8–233	Axis6 Servo Amp error Possible cause: The motor drive transistor is abnormal. The transistor comes with the following three (3) errors. • Overcurrent • Overtemperature (only for TS2000) Detected when the temperature exceeds 150°C. • Power voltage drop Detected when the voltage has dropped to 13.5 V or less (normal value 15 V). Error 8–028 occurs at the same time. Remedy: Turn the power off and drop the transistor temperature. Correct the program to alleviate load imposed on the motor (transistor).

Error code	Descriptions
8–074	Axis1 Electric Thermal
8–106	Axis2 Electric Thermal
8–138	Axis3 Electric Thermal
8–170	Axis4 Electric Thermal
8–202	Axis5 Electric Thermal
8–234	Axis6 Electric Thermal
	Possible cause: If the load exerted on the motor is large and the current exceeding the motor rated value flows continuously, this alarm occurs to protect the motor. The alarm is monitored by the software. Remedy: Confirm the weight of "hand + workpiece". If the robot has collided with an obstacle, remove the cause of the collision. If torque control is necessary, use the TORQUE command.
8–076	Axis1 Current Feedback
8–108	Axis2 Current Feedback
8–140	Axis3 Current Feedback
8–172	Axis4 Current Feedback
8–204	Axis5 Current Feedback
8–236	Axis6 Current Feedback
	Possible cause: The current feedback value just before servo ON is illegal. If the servo is turned on under this condition, the motor may burn. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.

Error code	Descriptions
8–077	Axis1 Servo Parameter
8–109	Axis2 Servo Parameter
8–141	Axis3 Servo Parameter
8–173	Axis4 Servo Parameter
8–205	Axis5 Servo Parameter
8–237	Axis6 Servo Parameter
	Possible cause: At calculation of the servo valid parameter (at power ON), the calculation is not possible because the parameter set value is outside the specified range.
	Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
8–175	Temperature is abnormal
	Possible cause: The temperature inside the controller is too high.
	Remedy: Check for the controller vent holes to make sure that they are not blocked to prevent air flow.
8–081	Axis1 Communication error
8–113	Axis2 Communication error
8–145	Axis3 Communication error
8–177	Axis4 Communication error
8–209	Axis5 Communication error
8–241	Axis6 Communication error
	Possible cause: Communication with the encoder is not possible.
	Remedy: Check for the encoder wiring. Unless there is a problem in the wiring, the servo printed board or encoder trouble may be the cause. Call us at the after-sale service department or our after-sale service agent in your territory.

Error code	Descriptions
8–082	Axis1 Encoder Type error
8–114	Axis2 Encoder Type error
8–146	Axis3 Encoder Type error
8–178	Axis4 Encoder Type error
8–210	Axis5 Encoder Type error
8–242	Axis6 Encoder Type error
	Possible cause: The type of the encoder cannot be identified.
	Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
8–086	Axis1 Internal RAM error
8–118	Axis2 Internal RAM error
8–150	Axis3 Internal RAM error
8–182	Axis4 Internal RAM error
8–214	Axis5 Internal RAM error
8–246	Axis6 Internal RAM error
	Possible cause: The self-diagnosis check performed at power ON has found an error in the built-in RAM.
	Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
8–087	Axis1 Common RAM error
8–119	Axis2 Common RAM error
8–151	Axis3 Common RAM error
8–183	Axis4 Common RAM error
8–215	Axis5 Common RAM error
8–247	Axis6 Common RAM error
	Possible cause: The self-diagnosis check performed at power ON has found an error in the common RAM.
	Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.

Error code	Descriptions
8–088	Axis1 Hard Watchdog error
8–120	Axis2 Hard Watchdog error
8–152	Axis3 Hard Watchdog error
8–184	Axis4 Hard Watchdog error
8–216	Axis5 Hard Watchdog error
8–248	Axis6 Hard Watchdog error
	Possible cause: The sub printed board has detected an error at regular mutual software monitoring (watchdog) with the main control board or an error signal of the main control board.
	Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
8–089	Axis1 Bus error
8–121	Axis2 Bus error
8–153	Axis3 Bus error
8–185	Axis4 Bus error
8–217	Axis5 Bus error
8–249	Axis6 Bus error
	Possible cause: An execution error relating to CPU, such as CPU address error and general illegal command, has occurred in the servo printed board.
	Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
8–093	Axis1 Background OS error
8–125	Axis2 Background OS error
8–157	Axis3 Background OS error
8–189	Axis4 Background OS error
8–221	Axis5 Background OS error
8–253	Axis6 Background OS error
	Possible cause: An error has occurred in the servo background multitask OS.
	Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.

Error code	Descriptions
8–257	External Watchdog error Possible cause: A watchdog error has occurred in the CPU other than the main control CPU. Other watchdog error occurs also. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
8–258	Backup Memory error Possible cause: The data in the backup memory has been destroyed. Remedy: The memory is formatted automatically. Load the parameter data from the system disk as factory-set
8–269 8–270 8–271 8–272	PLC USER ALARM Possible cause: A user alarm of level 8 has occurred in the PLC. The message can be defined by the user parameter. Remedy: Remove the alarm conditions which were set by the user.
8–273	EI/O Fuse Broken Possible cause: The fuse of the following printed board has brown out. TS1000:X8LC TS2000:X8HN/X8HI Remedy: Replace the fuse with a new one. The maximum external I/O current is 2 A. Make sure that the load is not over the specified value. Also, make sure that the external I/O wiring is not short-circuited. For the fuse replacement, see the Maintenance Manual.

Error code	Descriptions
8–274	Enable SW error Possible cause: The status of the two (2) contacts of the enable switch provided on the teach pendant is not consistent. Remedy: Turn the power on again. If the error still occurs, the switch or wiring of the teach pendant may be the cause. Call us at the after-sale service department or our after-sale service agent in your territory.
8–275	EMG SW error Possible cause: The emergency stop status of the two (2) circuits is not consistent. Remedy: The emergency stop wiring or EMERGENCY stop switch may be the cause. Perform check on them.
8–276	TP Disconnect SW error Possible cause: When connecting or disconnecting the teach pendant, the TP disconnect switch has been pressed for more than 100 seconds(TS1000)/ 30 seconds(TS2000). Remedy: The time of pressing the TP disconnect switch should be less than 100 seconds(TS1000)/30 seconds(TS2000) at connection or disconnection of the teach pendant.
8–277	PLC Stop Possible cause: The sequence of the PLC unit remains stopped. Remedy: For an error which was caused by stopping the PLC unit deliberately, there is no problem at all. Turn the power on, or restart the PLC unit by sending the "startup" command from the PLC programmer.
8–353	Fieldbus Parameter Error Possible cause: The set value of the field bus parameter is illegal. Remedy: Set a legal parameter.

Error code	Descriptions	
8–354	Fieldbus Offline	
	Possible cause:	An error occurred during the field bus communication and the online mode prevails.
	Remedy:	Check for the field bus master device to locate the cause.

Alarms of level 4:

An alarm whose number starts with 4. If this alarm has been detected, the robot stops instantly.

Error code	Descriptions	
4–017	Axis1 +Soft Limit	
4–018	Axis2 +Soft Limit	
4–019	Axis3 +Soft Limit	
4–020	Axis4 +Soft Limit	
4–021	Axis5 +Soft Limit	
4–022	Axis6 +Soft Limit	
4–023	Axis1 –Soft Limit	
4–024	Axis2 –Soft Limit	
4–025	Axis3 –Soft Limit	
4–026	Axis4 –Soft Limit	
4–027	Axis5 –Soft Limit	
4–028	Axis6 –Soft Limit	
	Possible cause:	The axis has overrun the stroke limit set in JLIMIT.
	Remedy:	Select the joint coordinate system for manual guide and guide an appropriate axis to the "–" direction if it is located at the "+" stroke end. If the axis is at the "–" stroke end, guide it to the "+" direction. If the robot enters the working envelope, this error is reset automatically. If this error has occurred during automatic operation, the robot passes the outside of the soft limit. Correct the program.
4–033	Interfere Error	
	Possible cause:	The robot has entered the off-limit area (i.e., interference area).
	Remedy:	Select the joint coordinate system for manual guide, then move the robot to outside of the off-limit area. If the robot goes out of the off-limit area, this error is reset automatically.

Error code	Descriptions
4–049	Interpolation limit area Possible cause: The interpolation command (MOVES or MOVEC) has been specified in the area where it is prohibited. Remedy: Correct the robot program (teach points).
4–065	Transfer function Possible cause: The robot adjustment function has been used. Remedy: Turn the power off once, then on again.
4–077 4–078 4–079 4–080	PLC USER ALARM Possible cause: A user alarm of level 4 has occurred in the PLC. The message can be defined by the user parameter. Remedy: Remove the alarm conditions which were set by the user.

Alarms of level 2:

An alarm whose number starts with 2. If this alarm has been detected, the robot stops just before the error point in the SCOL program.

Error code	Descriptions
2–018	<u>SVM error</u> Possible cause: [1] The latch function has been commanded though the system does not have an latch function option. [2] Illegal processing has occurred in the internal operation at execution of the SCOL program (ON–DO command). Remedy: [1] Check to see according to the specifications manual if the latch function option is selected. [2] Call us at the after-sale service department or our after-sale service agent in your territory.
2–019	<u>Direct Program error</u> Possible cause: Illegal processing has occurred in the internal operation at direct execution or execution of manual guidance. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
2–020	<u>Unable to Start</u> Possible cause: The selected program cannot be executed. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
2–021	<u>Unable to Continue</u> Possible cause: Illegal processing has occurred in the internal operation during restart operation. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.

Error code	Descriptions
2–022	Program Counter error Possible cause: An illegal value has been detected in the internal program counter during execution of the SCOL program. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
2–023	Program Counter error Possible cause: The ON condition command cannot be executed in the subtask. Remedy: Delete the ON condition command from the subtask program of the SCOL program.
2–024	Operand error Possible cause: Illegal processing has occurred in the internal operation of the program for system variable operation. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
2–025	Invalid Command Possible cause: This error occurs if the command invalid currently has been executed. • The conveyor command has been executed though the system is not provided with a conveyor option. • The motion command has been executed in the subtask. • The task command has been executed in the ON–DO statement. • If the auto variable is used during execution of the ON–DO statement, this error may occur due to the timing of establishing the ON condition. Remedy: Correct (or delete) the command causing this error.

Error code	Descriptions
2–029	Invalid Motion Command Possible cause: Illegal processing has occurred in the internal operation during execution of the BREAK or RESUME command. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
2–031	Unable to Resume Possible cause: An unexpected error has occurred during execution of the RESUME command. This alarm will not occur due to an error in the SCOL program. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
2–033	Conveyor Parameter error Possible cause: An error has been found in the argument of the CONV command. Remedy: Modify the argument of the CONV command programmed.
2–034	Ethernet connection Error Possible cause: While the Ethernet connection was not established, the PRINT or INPUT command has been executed. Remedy: Make sure of the Ethernet connection (parameters). Check for the SCOL program.

Error code	Descriptions
2–041	Invalid Variable Possible cause: The value specified as the argument cannot be used. • TAN: A multiple of integer of 90° has been input. • ASIN/ACOS: A value smaller than "–1" or larger than "1" has been input. • LN/LOG10/SQRT: A value smaller than zero (0) has been input. • DIN/BCDIN/DOUT/BCDOUT/PULOUT: A non-existent line number has been specified. • MOVE1/MOVEA: Axis 5 has been specified through the number of controlled axes is four (4). • DIM: An index in excess of the maximum of the value defined by the array command has been specified. • Data other than conveyor 1 and 2 has been commanded by the conveyor command. Remedy: Make sure that the value of each argument is correct and correct, if necessary.

Error code	Descriptions
2–042	<u>Data Type error</u> Possible cause: Illegal processing has occurred in the variable processing at execution of the SCOL program. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
2–043	<u>Invalid ON Command</u> Possible cause: Illegal processing has occurred in the internal operation of the ON condition command. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
2–045	<u>Write to Const</u> Possible cause: In the subprogram where a constant (such as LEFTY, PI, 1, 100) is used for the argument, the value of the constant transferred was about to be changed. Ex. PROGRAM MAIN ..... SUB2 (1) ..... END → PROGRAM SUB2 (X) ..... X=X+1 ..... END ← Not allowed as the constant of X is "1". Remedy: Correct the program.
2–046	<u>Invalid Channel</u> Possible cause: A character data other than numerals has reached during communication via the COM port. Remedy: Check for the data sent or received.
2–047	<u>Task Limits Over</u> Possible cause: A task exceeding the number of the maximum tasks as defined by MAXTASK was about to be created. Remedy: Increase the value of global variable MAXTASK.

Error code	Descriptions
2–049	Task Limits Over Possible cause: A space required for the internal operation has run short at execution of the SCOL program. Remedy: Reduce the size of one (1) program, using subprograms.
2–050	Stack Overflow Possible cause: A space required for the internal operation has run short at execution of the SCOL program. Remedy: Reduce the size of one (1) program, using subprograms.
2–051	ON–DO Overflow Possible cause: A total of ten (10) ON condition commands can be executed simultaneously. This alarm will occur if eleven (11) or more commands have been executed at the same time. Remedy: Reduce the number of the ON–DO commands.
2–053	RESTORE error Possible cause: As the variable specified by the variable name of the RESTORE command was illegal, the RESTORE command could not be executed. Remedy: Check the RESTORE command for the argument (variable name).
2–065	Motion Command error Possible cause: A numeral with which a locus cannot be calculated has been detected in the locus creation of the internal operation. This alarm will not occur due to an error in the SCOL program. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.

Error code	Descriptions
2–066	Trans Code error Possible cause: Due to non-conformity of the results of coordinate conversion in the internal operation, the robot cannot be moved. This alarm will not occur due to an error in the SCOL program. Remedy: Call us at the after-sale service department or our after-sale service agent in your territory.
2–067	NOVEC error Possible cause: A point with which an arc cannot be created has been found in the circular (MOVEC) command. Remedy: Make sure that the coordinates of start point, pass point and end point of the circular command differ with each other and that they are not located on the straight line.
2–068	PAYLOAD error Possible cause: An error has been detected in the argument of the PAYLOAD command. Remedy: Check the PAYLOAD command of the SCOL program for the argument.
2–069	CONFIG error Possible cause: An illegal configuration has been specified in the circular (MOVEC) or straight line (MOVES) interpolation command. Remedy: Make sure that the teach configuration (LEFTY, RIGHTY) of start point, pass point (circular) and end point of the interpolation movement are identical.

Error code	Descriptions
2–073	Axis1 Out of Limit
2–074	Axis2 Out of Limit
2–075	Axis3 Out of Limit
2–076	Axis4 Out of Limit
2–077	Axis5 Out of Limit
2–078	Axis6 Out of Limit Possible cause: As the position passing the stroke limit is specified by the motion command in the program, the axis cannot move. Remedy: Make sure that the teach point data will not pass the stroke limit.
2–079	Point Out of Range Possible cause: As the position where the robot cannot move (such as outside of the arm length) is specified by the motion command in the program, the robot cannot move. Remedy: Make sure that the teach point data is within the robot working envelope.
2–081	Dest Interfere Possible cause: As the off-limit position (i.e., interference area) is specified by the motion command in the program, the robot cannot move. Remedy: Make sure that the teach point data is outside the off-limit area.
2–082	Positioning limit area Possible cause: The commanded position cannot be specified as the target position. Remedy: Correct the robot program (teach points).
2–089	MOVEJ Parameter error Possible cause: The argument of MOVEJ in the SCOL program is illegal. Remedy: Check the MOVEJ command of the SCOL program for the argument.

Error code	Descriptions
2–113	Axis1 Motor overload (15m)
2–114	Axis2 Motor overload (15m)
2–115	Axis3 Motor overload (15m)
2–116	Axis4 Motor overload (15m)
2–117	Axis5 Motor overload (15m) Possible cause: The motor load running average for the duration of fifteen (15) minutes exceeds the predetermined value. Remedy: Reduce the load exerted on the motor. [1] Specify the appropriate load data, using the PAYLOAD command. [2] Make sure that the weight of "workpiece + robot hand" will not exceed the specified range.
2–121	Axis1 Motor overload (1m)
2–122	Axis2 Motor overload (1m)
2–123	Axis3 Motor overload (1m)
2–124	Axis4 Motor overload (1m)
2–125	Axis5 Motor overload (1m) Possible cause: The motor load running average for the duration of one (1) minute exceeds the predetermined value. Remedy: Reduce the load exerted on the motor. [1] Specify the appropriate load data, using the PAYLOAD command. [2] Make sure that the weight of "workpiece + robot hand" will not exceed the specified range.

Alarms of level 1:

A message alarm whose number starts with 1. If this alarm has been detected, the robot will display an appropriate message alone.

Error code	Descriptions
1–009	Unable to RCYCLE Possible cause: During execution of the multi-task program or ON ~ DO command, you were about to stop the DO statement currently executed to use this function. Remedy: The cycle reset command cannot be used while the DO statement is being executed in the multi-task program or ON ~ DO command. Change the place where the cycle reset is commanded, or correct the SCOL program.
1–017	Invalid PASS command Possible cause: The ENABLE PASS command has been specified during execution of the ENABLE SMOOTH command. Remedy: Correct the SCOL program.
1–018	Invalid SMOOTH command Possible cause: The ENABLE SMOOTH command has been specified during execution of the ENABLE PASS command. Remedy: Correct the SCOL program.
1–025	Over Deceleration Possible cause: In the program of SMOOTH connection, the travel distance for the last motion is short and slowdown is not possible. Remedy: Correct the SCOL program (teach points).
1–026	SMOOTH impossible Possible cause: In the program of SMOOTH connection, the travel distance is short and internal calculation cannot be executed in time. Remedy: Correct the SCOL program (teach points).

Error code	Descriptions
1–027	Enable SW error Possible cause: The status of two (2) contacts of the ENABLE switch provided on the teach pendant (TP) disagrees with each other. Remedy: Turn the power off once, then on again. If this error still occurs, the TP switch or wiring may be faulty. Contact our after-sale service department for repair.
1–028	Already Enable SW ON Possible cause: The ENALBE switch equipped on the teach pendant (TP) was already pressed when the power has been turned on. Remedy: Turn the power off once, then on again. If this error still occurs, the TP switch or wiring may be faulty. Contact our after-sale service department for repair.
1–049 1–065 1–081 1–097 1–113 1–129	Axis1 Battery alarm Axis2 Battery alarm Axis3 Battery alarm Axis4 Battery alarm Axis5 Battery alarm Axis6 Battery alarm Possible cause: The feedback detector battery is approaching its life end. Remedy: Replace the battery with a new one.

Error code	Descriptions
1–053	Axis1 Packet Trans error
1–069	Axis2 Packet Trans error
1–085	Axis3 Packet Trans error
1–101	Axis4 Packet Trans error
1–117	Axis5 Packet Trans error
1–133	Axis6 Packet Trans error Possible cause: An unexpected command has been sent from the main control to the servo unit. Remedy: This alarm will not occur due to an error in the SCOL program. The robot system possibly went wrong. Call us at the after-sale service department or our after-sale service agent in your territory.
1–145	MAIN Battery alarm Possible cause: The battery for storing programs, parameter data, etc., is approaching its life end. Remedy: Replace the battery with a new one, referring to the Maintenance Manual.
1–146	CONV Need Reset Possible cause: Program execution has stopped during synchronous conveyor operation. Remedy: Reset the synchronous conveyor operation.
1–147	COM Buffer Over Possible cause: There is no more receiver buffer space left for the serial communication. Remedy: Check for the communication protocol (i.e., flow control) with the partner equipment.
1–148	Brake ON Possible cause: As the brake is effected, the robot cannot move. Remedy: Turn the servo power off, then on again.

Error code	Descriptions
1–149	Extern SELECT error Possible cause: An error has occurred during external file selection. Remedy: Perform check on the following items. • Setting of user parameters 7 and 8. • "File select number" external signal • "Program reset" external signal
1–153	Auto file selection error Possible cause: While the power was turned on, the program could not be selected. Remedy: Contact our after-sale service department for repair.
1–154	Data Block Format error Possible cause: A data format error has occurred at the startup of the data editor. Remedy: Reading of a line where an error occurred is neglected. After the check, input a legal data again.
1–155 1–156 1–157	HOST port data error COM1 port data error COM2 port data error Possible cause: The flaming error, the parity error, the overrun error occurred to the data which was received at the port of HOST, or COM1, COM2. Remedy: Confirm whether or not a communications cable is connected right. Confirm the setting of a baud rate, parity, stop bit. The noise make are not on the communications cable.

Error code	Descriptions
1–158	Short of working AREA Possible cause: Because the free disk space of the RAM disk isn't adequate, the work area can not be secured. Remedy: Delete an unnecessary file in the RAM disk.
1–160	Compile error Possible cause: A program error has occurred during file selection. Remedy: Press the [F5] function key on the error display and confirm the error contents. Then, correct the relative part of the program.
1–164	PLC Backup data error Possible cause: The sequence data of PLC has been destroyed. Remedy: Reload the sequence program, using the TCPRGOS.
1–166	PLC Remote unit error Possible cause: [1] The setting of connecting the extension I/O unit is not identical with the user parameter setting. [2] The cable is connected improperly or broken. [3] The connected I/O unit will not operate normally. Remedy: [1] If the user parameter is set illegally, set the same parameter legally. [2] Check for the connected cable. If defective, replace it with a new one. [3] If the I/O unit is defective, contact our after-sale service agent in your territory.
1–169	PLC Undefine label Possible cause: The label used in the sequence data of PLC is not defined. Remedy: Correct the sequence program, using the TCPRGOS.

Error code	Descriptions
1–170	PLC Invalid command Possible cause: The command used in the sequence data of PLC is not defined. Remedy: Correct the sequence program, using the TCPRGOS.
1–173	PLC Overlap label Possible cause: The label used in the sequence data of PLC is defined repeatedly. Remedy: Correct the sequence program, using the TCPRGOS.

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